

UNIVERSITAT DE VALÈNCIA

Instituto de Ciencia Molecular (ICMol)

Doctoral Programme in Nanoscience and Nanotechnology

**Volatile Organic Memristive Devices
for Temporal Computing:
Compositional Control of Fading
Memory in Polymer–Electrolyte
Composites**

Doctoral thesis presented by

Carlos David Prado-Socorro

in partial fulfilment of the requirements for the degree of

Doctor of Philosophy (PhD)

Supervised by

Prof. Eugenio Coronado
Dr. Salvador Cardona-Serra

2026

Abstract

Data movement between separate memory and processing units accounts for an increasing share of the energy and latency of modern computing. One physical approach is to use device dynamics to store and transform temporal state. This thesis develops a *volatile organic memristive device*: a two-terminal element made from an ion-containing polymer–electrolyte composite whose conductance retains a decaying record of recent electrical history. The work evaluates this controlled *fading memory* as physical leaky integration for temporal computing.

The thesis first applies a set of synaptic measurements to a proof-of-concept Super Yellow/Hybrane/lithium-triflate device: current–voltage hysteresis, potentiation and depression, an excitatory post-synaptic current, distinguishable short- and long-term retention, and spike-timing-dependent plasticity. These responses are interpreted through the mobility asymmetry between the triflate anion and the lithium cation. The fabrication archive then shows a campaign-scale loss of reproducibility in the Hybrane line. Changing to a poly(ethylene oxide) host restores a usable switching window and leads to a fabrication-record methodology and an automated feature pipeline. On the PEO platform, a replicated composition series tunes the fading-memory time over roughly 2–20 s. Host, anion and cation substitutions remain a sample-limited survey, and the data do not support the hypothesised $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ ordering. Finally, compact behavioural models fitted to the measured dynamics

drive reservoir-computing simulations. A heterogeneous composition bank raises the linear memory capacity by a factor of 1.54 and the information-processing capacity from 7.1 to 11.6. Device-model banks classify stress on the WESAD corpus (leave-one-subject-out macro-F1 of 0.894 binary and 0.758 three-class), although the heterogeneity increment is not resolved for the sustained labels. On the CASE corpus, an equivalent linear multi-timescale filter, not the device bank, gains +0.176 in concordance correlation over a single-timescale filter; device-internal heterogeneity contributes only +0.005, within random-seed variation.

In the replicated PEO/lithium-triflate series, composition tunes fading memory. Mixed-composition banks broaden temporal memory in simulation. Because the device response relaxes over seconds, temporal and event-driven neuromorphic hardware provides the relevant reference class.

Resumen

El movimiento de datos entre unidades separadas de memoria y procesamiento representa una proporción creciente del consumo energético y la latencia de la computación moderna. Una estrategia física consiste en utilizar la dinámica del dispositivo para almacenar y transformar un estado temporal. Esta tesis desarrolla un *dispositivo memristivo orgánico volátil*: un elemento de dos terminales fabricado a partir de un compuesto polímero–electrolito con iones móviles cuya conductancia conserva un registro decreciente de su historia eléctrica reciente. El trabajo evalúa esta *memoria evanescente* controlada como integración física con fugas para la computación temporal.

La tesis aplica primero un conjunto de medidas sinápticas a un dispositivo de prueba de concepto Super Yellow/Hybrane/triflato de litio: histéresis corriente–voltaje, potenciación y depresión, corriente postsináptica excitatoria, retención a corto y largo plazo distinguibles, y plasticidad dependiente del tiempo de disparo. Estas respuestas se interpretan mediante la asimetría de movilidad entre el anión triflato y el catión litio. A continuación, el archivo de fabricación muestra una pérdida de reproducibilidad a escala de campaña en la línea de Hybrane. El cambio a una matriz de poli(óxido de etileno) restaura una ventana de conmutación utilizable y da lugar a una metodología de registro de fabricación y a una canalización automatizada de extracción de características. En la plataforma de PEO, una serie de composiciones replicada ajusta el tiempo de memoria evanes-

cente entre aproximadamente 2 y 20 s. Las sustituciones de matriz, anión y catión siguen siendo un estudio limitado por el tamaño muestral, y los datos no respaldan el orden catiónico hipotético $\text{Li}^+ > \text{Na}^+ > \text{K}^+$. Por último, modelos de comportamiento compactos ajustados a la dinámica medida alimentan simulaciones de computación de reservorio. Un banco heterogéneo de composiciones aumenta la capacidad de memoria lineal en un factor de 1,54 y la capacidad de procesamiento de información de 7,1 a 11,6. Los bancos de modelos de dispositivo clasifican el estrés en el corpus WESAD (macro-F1 de 0,894 binario y 0,758 de tres clases con validación dejando-un-sujeto-fuera), aunque no se resuelve el incremento atribuible a la heterogeneidad para las etiquetas sostenidas. En el corpus CASE, un filtro lineal multiescala, no el banco de dispositivos, obtiene una ganancia de +0,176 en correlación de concordancia frente a un filtro de una sola escala; la heterogeneidad interna del dispositivo aporta solo +0,005, dentro de la variación entre inicializaciones aleatorias.

En la serie replicada PEO/triflato de litio, la composición ajusta la memoria evanescente. Los bancos de composiciones mixtas amplían la memoria temporal en simulación. Como la respuesta del dispositivo se relaja en segundos, el hardware neuromórfico temporal y dirigido por eventos constituye la referencia pertinente.

Resum

El moviment de dades entre unitats separades de memòria i de processament representa una proporció creixent del consum energètic i la latència de la computació moderna. Una estratègia física consisteix a utilitzar la dinàmica del dispositiu per a emmagatzemar i transformar un estat temporal. Aquesta tesi desenvolupa un *dispositiu memristiu orgànic volàtil*: un element de dos terminals fabricat a partir d'un compòsit polímer–electròlit amb ions mòbils la conductància del qual conserva un registre decreixent de la seua història elèctrica recent. El treball avalua aquesta *memòria evanescent* controlada com a integració física amb fugues per a la computació temporal.

La tesi aplica primer un conjunt de mesures sinàptiques a un dispositiu de prova de concepte Super Yellow/Hybrane/triflat de liti: histèresi corrent–voltatge, potenciació i depressió, corrent postsinàptic excitatori, retenció a curt i llarg termini distingibles, i plasticitat dependent del temps de dispar. Aquestes respostes s'interpreten mitjançant l'asimetria de mobilitat entre l'anió triflat i el catió liti. A continuació, l'arxiu de fabricació mostra una pèrdua de reproductibilitat a escala de campanya en la línia de Hybrane. El canvi a una matriu de poli(òxid d'etilè) restaura una finestra de commutació utilitzable i dóna lloc a una metodologia de registre de fabricació i a una canalització automatitzada d'extracció de característiques. En la plataforma de PEO, una sèrie de composicions replicada ajusta el temps de memòria evanescent entre aproximadament

2 i 20 s. Les substitucions de matriu, anió i catió continuen sent un estudi limitat per la grandària mostral, i les dades no confirmen l'ordre catiònic hipotètic $\text{Li}^+ > \text{Na}^+ > \text{K}^+$. Finalment, models de comportament compactes ajustats a la dinàmica mesurada alimenten simulacions de computació de reservori. Un banc heterogeni de composicions augmenta la capacitat de memòria lineal en un factor de 1,54 i la capacitat de processament d'informació de 7,1 a 11,6. Els bancs de models de dispositiu classifiquen l'estrès en el corpus WESAD (macro-F1 de 0,894 binari i 0,758 de tres classes amb validació deixant-un-subjecte-fora), encara que l'increment atribuïble a l'heterogeneïtat no queda resolt per a les etiquetes sostingudes. En el corpus CASE, un filtre lineal multiescala, no el banc de dispositius, obté un guany de +0,176 en correlació de concordança respecte d'un filtre d'una sola escala; l'heterogeneïtat interna del dispositiu aporta només +0,005, dins de la variació entre inicialitzacions aleatòries.

En la sèrie replicada PEO/triflat de liti, la composició ajusta la memòria evanescent. Els bancs de composicions mixtes amplien la memòria temporal en simulació. Com que la resposta del dispositiu es relaxa en segons, el maquinari neuromòrfic temporal i dirigit per esdeveniments constitueix la referència pertinent.

List of Publications

The following output arose directly from the work presented in this thesis.

Peer-reviewed article

1. C. D. Prado-Socorro, S. Giménez-Santamarina, L. Mardegan, L. Escalera-Moreno, H. J. Bolink, S. Cardona-Serra and E. Coronado, “Polymer-Based Composites for Engineering Organic Memristive Devices,” *Advanced Electronic Materials* **8**, 2101192 (2022). DOI: 10.1002/aelm.202101192.

This article is the basis of chapter 2.

Contents

Abstract	iii
Resumen	v
Resum	vii
List of Publications	ix
List of Figures	xix
List of Tables	xxiii
List of Abbreviations	xxv
Motivation, Objectives and Structure of the Thesis	1
1 Introduction	7
1.1 Architectural Limits of Modern Computing	8
1.1.1 Von Neumann Architecture and Its Limitations	8
1.1.2 In-Memory and Event-Driven Computing Paradigms	11

1.2	The Biological Synapse as the Engineering Template . .	14
1.2.1	Neural Signal Transmission: A Brief Overview . .	14
1.2.2	Synaptic Plasticity: The Physical Basis of Learning	17
1.2.3	Why Hardware Emulation of the Synapse?	18
1.3	The Memristor: Theory and History	21
1.3.1	Memristive Systems as History-Dependent Devices	21
1.3.2	The First Physical Realisation: TiO ₂ Memristor .	23
1.3.3	Classification of Memristive Mechanisms	24
1.3.4	From Device Metrics to Fading Memory	28
1.3.5	Reservoir Computing with Volatile Memristors . .	30
1.4	Inorganic Memristors for Neuromorphic Computing . . .	31
1.4.1	Metal-Oxide Memristors	32
1.4.2	Phase-Change Memory	32
1.4.3	Spin-Transfer-Torque Magnetic Tunnel Junctions	33
1.4.4	General Limitations of Inorganic Memristors . . .	33
1.5	Organic Materials for Memristive Applications	34
1.5.1	Organic Mechanisms as Design Choices	37
1.5.2	Organic Precedents for the Composite Strategy .	40
1.6	Polymer Electrolyte Composites: The Enabling Strategy	45
1.6.1	Ion-Conducting Polymer Chemistry	45
1.6.2	LEC-Derived Composite Operation	50
1.6.3	The Ionic Response in Time and Frequency . . .	52
1.6.4	Rationale for the Composite Approach	53

2	Proof of Concept: Polymer-Based Composite Organic Memristive Devices	56
2.1	Introduction and Motivation	56
2.2	Materials Design and Composite Formulation	58
2.2.1	Design Rationale	58
2.2.2	Super Yellow: The Semiconducting Polymer . . .	60
2.2.3	Hybrane DEO750 8500: The Ion-Transport Polymer	60
2.2.4	Lithium Triflate: The Ionic Salt	61
2.2.5	Composite Formulation and Preparation	61
2.3	Device Architecture and Fabrication	61
2.3.1	Device Geometry: The Two-Terminal Vertical Structure	61
2.3.2	Fabrication and Active-Layer Morphology	63
2.4	Electrical Characterisation	64
2.4.1	Current-Voltage Hysteresis	64
2.4.2	Variable-Number-of-Pulses Potentiation	66
2.4.3	Variable-Delay Depotentiation: Short- and Long-Term Retention	70
2.4.4	Excitatory Post-Synaptic Current	72
2.4.5	Spike-Timing Dependent Plasticity	74
2.5	Theoretical Framework: Ion Transport and the Origin of Memristance	77
2.5.1	Lewis Acid-Base Chemistry and the Two Memory Regimes	78

2.6	Device Stability and Reproducibility	80
2.6.1	Ambient Shelf Stability	80
2.6.2	Within-Substrate Junction Repeatability	81
2.7	Comparison at the Time of Publication	82
2.8	Summary	83
3	Reproducibility, Degradation, and the Fabrication-Record Methodology	85
3.1	The Reproducibility Problem	85
3.2	Eliminating Candidate Causes	86
3.3	The Quantitative Degradation Signal	88
3.3.1	A Candidate Chemistry: Moisture-Driven Hydrolytic Aging of the <i>Hybrane</i> Stock	92
3.4	Change to a PEO Host	94
3.5	Methodological Outputs	95
3.6	Reconciliation of Stability and Reproducibility	97
4	Compositional and Chemical Control of Volatile Polymer-Electrolyte Memristive Dynamics	99
4.1	Introduction and Scope	99
4.2	Materials, Devices, and Measurement Protocol	100
4.3	Methods of Analysis	103
4.4	Composition Control of the Dynamics	104
4.5	Chemical Tuning of the Dynamics	108
4.6	Protocol-Dependent Timescale	113

4.7	Microscopic Corroboration: Impedance Spectroscopy . .	115
4.8	Electrode and Polarity Dependence	118
4.9	Discussion	123
4.10	Limitations and Future Work	125
4.11	Inputs to Temporal Computing	128
5	Data-Driven Temporal Computing with Polymer-Electrolyte Memristor Reservoirs	129
5.1	Introduction and Scope	129
5.2	Measured Behavioural Model	130
5.3	Reservoir Computing and the Common Metric Language	132
5.4	Benchmark Tier: Capacity and System Identification . .	133
5.5	Affective Computing	137
5.6	Physiological Temporal-Context Reconstruction	139
5.7	WESAD Affective Classification	140
5.8	Continuous Valence and Arousal	144
5.9	Continuous Monitoring Under Deployment Conditions .	146
5.10	Toward a Worn Device: Sensor Site and Operating Budget	149
5.11	Matching Device and Task Timescales	151
5.12	Limitations and Scope Conditions	153
5.13	Summary	155
6	Conclusions and Outlook	157
6.1	Summary of Contributions	157
6.2	Answers to the Thesis Questions	160

6.3	Benchmarking Against the Right Reference Class	161
6.4	Limitations of the Evidence Base	164
6.5	Outlook and Future Directions	166
6.6	Closing Remarks	168
A	Supporting Information for Chapter 2	169
A.1	Materials and Fabrication Detail	169
A.1.1	Component Notes	169
A.1.2	Solution Preparation and Film Deposition	170
A.2	Electrical Protocol Detail	172
A.2.1	Assay-to-Specimen Map	173
A.2.2	Read-Write-Wait Measurement	173
A.2.3	Extended Synaptic Protocols	174
A.2.4	Retention-Fit Parameters	175
A.3	Morphology and Stability Support	176
A.4	Ion-Transport Models for the Composite	178
A.4.1	Equilibrium Ion Distribution and Response to an Applied Field	178
A.4.2	Electrodynamic Model	179
A.4.3	Electrochemical Doping Model: Ohmic Injection .	180
A.4.4	Resolution of the Two Models	180
A.5	Historical Comparison at the Time of Publication	181
B	Supporting Information for Chapter 3	184
B.1	The Four Methodological Controls	184

C	Supporting Information for Chapter 4	186
C.1	Analysis Traceability and Aggregation	186
C.2	Film-Thickness and Fabrication Confound Audits	190
C.3	Per-Composition-Cell Summaries	191
C.4	Realised Device Palette	192
C.5	Impedance: Equivalent Circuit and Per-Cell Values . . .	193
C.6	Electrode-Contrast Per-Cell Values	195
C.7	The Gold-Electrode Negative-Polarity Dynamics Corpus	195
C.8	The Gold-Electrode Lock-in Frequency-Sweep Corpus . .	199
C.9	Significance of the Composition Gradient	204
C.10	Absolute Conductance Tests the Dilution Picture	206
C.11	Reconciling the Impedance and Fading-Memory Timescales	207
C.12	Within-Device Reproducibility	208
C.13	Heterogeneity as a Computational Resource	209
D	Supporting Information for Chapter 5	210
D.1	Parameter-Card Sourcing	210
D.2	Simulation and Read-out Settings	211
D.3	Measured Device-to-Device Scatter	211
D.4	Continuous-Monitoring Stress Test	213
D.5	Control Experiments	215
D.6	Deployment Envelope	219
D.7	Continuous Affect: CASE Protocol and Results	220

List of Figures

1.1	The von Neumann architecture and the cost of data movement	9
1.2	Two architectural responses to the von Neumann bottleneck	12
1.3	Anatomy of a neuron and the action-potential waveform	15
1.4	Anatomy and plasticity regimes of the chemical synapse .	16
1.5	Energy and density envelopes of synaptic-equivalent operations	19
1.6	The memristor as a constitutive element and its fingerprint	22
1.7	Mechanism gallery for the five memristive device families	26
1.8	Two complementary computational roles for memristive devices	30
1.9	Inorganic memristor families versus the soft-ionic composite route	36
1.10	Five mechanism families behind organic memristive behaviour	38
1.11	Historical map of organic memristive and synaptic devices	44
1.12	Ion-transport chemistry of the polyether-hosted composite	48
1.13	LEC operating models and their memristive reuse	51
1.14	The polymer-electrolyte composite device and its tunable timescale	55

2.1	Two-terminal device architecture	62
2.2	Current–voltage hysteresis	65
2.3	Pulse-driven conductance integration	69
2.4	Memory retention kinetics	71
2.5	Compact EPSC and STDP summary	76
3.1	Switching-window reduction at ~ 3 V	89
3.2	Ohmic drift and the methodology that isolates it	91
3.3	Annealing recovery of the switching window	94
3.4	PEO switching window and temporal reproducibility	96
4.1	Film-thickness control	102
4.2	Representative raw curves of the three measurements	105
4.3	Composition control of the memristive dynamics	106
4.4	Within-cell device-to-device heterogeneity	108
4.5	Composition control of pulse-driven potentiation	109
4.6	Microscopic corroboration of the chemistry axis: ATR and XRD	110
4.7	Chemistry dependence of the fading-memory time	111
4.8	UV-Vis: the <i>Super Yellow</i> band edge is invariant to the cation	114
4.9	The drive protocol sets the apparent fading-memory time	115
4.10	Impedance spectroscopy corroborates the ionic-transport mechanism	117
4.11	Electrode dependence: active silver versus inert gold	119

4.12	Negative-polarity dynamics of the gold/TMPE devices .	121
4.13	Frequency-domain response of the gold/TMPE devices .	123
4.14	Per-device design space	126
5.1	Memory capacity versus delay: heterogeneity broadens memory	134
5.2	Memory capacity and NARMA across the composition grid	136
5.3	Information-processing capacity: linear versus nonlinear .	137
5.4	Device timescales versus affective-signal bands	138
5.5	Physiological temporal-context reconstruction on WESAD	141
5.6	WESAD affective classification with device reservoirs . .	143
5.7	Continuous valence/arousal tracking on CASE	145
5.8	Continuous stress monitoring under sensor noise	148
5.9	Robustness to device-to-device scatter	153
A.1	Chemical components of the proof-of-concept composite .	171
A.2	Read–write–wait protocol	174
A.3	Extended EPSC and STDP supporting panels	175
A.4	Morphology and shelf-stability support	177
C.1	Realised device palette across the library	194
C.2	Absolute on-state conductance rises with PEO	207

List of Tables

- 1.1 Energy cost per operation in a 45 nm CMOS process . . . 10
- 1.2 Parameter-optimisation directions for the two device regimes 29
- 1.3 Typical operating parameters of the inorganic memristor families 35
- 1.4 Mechanism families behind organic memristive behaviour. 41
- 3.1 Methods-of-elimination summary 87
- 3.2 Quantitative degradation signal 90
- 4.1 Electrical measurement protocols 103
- 5.1 Affective signal timescales and device coverage 138
- 5.2 Physiological temporal-context reconstruction on WESAD 140
- 5.3 Matched WESAD literature comparison 141
- 5.4 Streaming three-class affect tracking on WESAD 143
- 5.5 Cross-corpus replication on the PhysioNet Non-EEG dataset 149
- 5.6 Stress detection: chest versus consumer wrist sensor . . . 150
- 5.7 Fading memory and timescale spread across the affective workload 152

6.1	Recent reference points for tunable physical reservoirs . .	163
A.1	Assay-to-specimen map for the proof-of-concept evidence	173
A.2	Published retention-fit parameters	176
A.3	Historical comparison with organic and hybrid memristive devices	182
C.1	Raw-to-summary trail for the comparative analysis . . .	187
C.2	Figure-level sources for Chapter 4	188
C.3	Per-cell fading-memory descriptors	192
C.4	Per-cell potentiation descriptors	193
C.5	Per-cell impedance summaries	195
C.6	Electrode-contrast per-cell metrics	196
C.7	Gold negative-polarity corpus, per-cation summary . . .	198
C.8	Lock-in day-23 bias series	203
C.9	Significance of the composition gradient	205
C.10	Factorial decomposition of the composition axes	206
D.1	Reservoir simulation and read-out settings	212
D.2	Measured device-to-device scatter of the fading-memory time	213
D.3	Classical exponential-moving-average control	216
D.4	Generic echo-state-network control	216
D.5	Coupling-sensitivity control	218
D.6	Reservoir deployment envelope versus a deep model . . .	219
D.7	Continuous valence/arousal on CASE	221

List of Abbreviations

Abbreviation	Meaning
ATR(-FTIR)	Attenuated total reflectance (Fourier-transform infrared spectroscopy)
CASE	Continuously Annotated Signals of Emotion (affective corpus)
CCC	Concordance correlation coefficient
CMOS	Complementary metal–oxide–semiconductor
DRAM	Dynamic random-access memory
EIS	Electrochemical impedance spectroscopy
ENODE	Electrochemical neuromorphic organic device
EPSC	Excitatory post-synaptic current
HSAB	Hard–soft acid–base (theory)
IPC	Information-processing capacity
ITO	Indium tin oxide
I–V	Current–voltage (characteristic)
KOTf	Potassium triflate (potassium trifluoromethanesulfonate)
LiOTf	Lithium triflate (lithium trifluoromethanesulfonate)

Abbreviation	Meaning
LOSO	Leave-one-subject-out (cross-validation)
LTM	Long-term memory
MC	(Linear) memory capacity
NaOTf	Sodium triflate (sodium trifluoromethanesulfonate)
NARMA	Nonlinear autoregressive moving average (benchmark task)
OTf	Triflate, CF_3SO_3^- (trifluoromethanesulfonate)
PEO	Poly(ethylene oxide)
RC	Reservoir computing
ReRAM	Resistive random-access memory
SI	Supporting information
SRAM	Static random-access memory
STDP	Spike-timing-dependent plasticity
STM	Short-term memory
SY	Super Yellow (PDY-132 poly(<i>p</i> -phenylene-vinylene) semiconductor)
T_g	Glass-transition temperature
TFSI	Bis(trifluoromethanesulfonyl)imide
TMPE	Trimethylolpropane ethoxylate (hyperbranched polyether host)
UV–Vis	Ultraviolet–visible (spectroscopy)
VFT	Vogel–Fulcher–Tammann (relaxation)
WESAD	Wearable Stress and Affect Detection (physiological corpus)

Abbreviation	Meaning
XRD	X-ray diffraction
1T1M	One-transistor-one-memristor (array cell)
τ_S, τ_L	Short- and long-term fading-memory time constants
β	Stretched-exponential (Kohlrausch) exponent

Motivation, Objectives and Structure of the Thesis

In many modern workloads, moving data between separate processing and memory units consumes more energy than the arithmetic operation itself. A single off-chip memory access can cost three to four orders of magnitude more energy than an arithmetic operation [1], and large machine-learning models and continuous sensor streams require frequent memory access [2]. As gains from transistor scaling slow, data movement accounts for an increasing share of system energy and latency [3]. This thesis examines one physical response: devices whose internal dynamics store and transform temporal state within the same substrate [4, 5].

The devices studied here are *volatile organic memristive devices*: two-terminal elements made from ion-containing polymer composites whose conductance retains a decaying record of recent electrical activity. Their relaxation is treated as a device function. A controlled decay makes each device a physical *leaky integrator*, while devices with different decay times form a substrate for *temporal* (reservoir-style) computing on time-varying signals. The thesis tests two linked propositions: composition controls the timescale in the replicated PEO/lithium-triflate series, and behavioural models fitted to the measured dynamics can perform temporal computations.

The thesis proceeds through four stages:

- **A fully characterised exemplar (Chapter 2).** A single proof-of-concept Super Yellow/Hybrane/lithium-triflate device is characterised with the complete synaptic measurement suite (current–voltage hysteresis, potentiation and depression, excitatory post-synaptic current, separated short- and long-term retention, and spike-timing-dependent plasticity) and given a mechanistic interpretation in terms of cation–anion mobility asymmetry.
- **A campaign-scale reproducibility and fabrication-record layer (Chapter 3).** The subsequent loss of reproducibility in the Hybrane devices is analysed across the fabrication archive, separating finished-device shelf stability from batch-scale supply reproducibility and motivating the switch to PEO. The same fabrication record and feature pipeline then become the methodological basis for the quantitative chapters.
- **A replicated composition series with a sample-limited chemical survey (Chapter 4).** A poly(ethylene oxide)-based corpus is built under a common protocol. The composition series is *replicated*; host, anion, and cation substitutions are sample-limited and are not read as material laws. In particular, the $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ cation ordering is treated as a hypothesis to be tested.
- **An in-silico temporal-computing demonstration from measured dynamics (Chapter 5).** Compact behavioural models extracted from the Chapter 4 measurements are evaluated with linear read-outs on benchmark and physiological tasks. Comparisons with homogeneous, memoryless, and linear-filter controls establish when a spread of timescales improves performance.

Open Questions and Objectives

Three groups of questions organise the work.

First, realising and understanding an exemplar. Can a fully reversible two-terminal organic memristive platform be made from a solution-processed polymer-electrolyte composite using standard organic-electronic fabrication? Can that common formulation and architecture support analogue switching, distinguishable short- and long-term memory, an excitatory post-synaptic current response, and spike-timing-dependent plasticity? Can the fast and slow memory components be explained by the composite's cation-anion mobility asymmetry?

Second, composition as the quantitative control variable. Within a common host and a common alkali-metal triflate family, how do the device dynamics depend on composition when the polymer-to-semiconductor and salt-to-semiconductor mass fractions are varied around a reference formulation? This composition dependence is the axis the thesis sets out to *replicate*. Host, anion, and cation substitutions pose hypotheses about the same dynamics, in particular whether moving the cation along the Li^+ , Na^+ , K^+ series shifts the measured fading-memory timescale as coordination chemistry suggests.

Third, utility for temporal computing. Can the volatile response of these devices be exploited for temporal computation? Can composition supply task-matched fading-memory timescales, while sample-limited chemistry and device-to-device spread populate a heterogeneous bank? Does measured variability broaden the computational response?

These questions translate into six ordered objectives:

1. Design, fabricate, and characterise a proof-of-concept two-terminal organic memristive device based on a Super Yellow, Hybrane, and lithium-triflate composite, processed from a single solution onto

- a standard ITO/composite/Ag stack, and demonstrate the basic memristive and synaptic functionality in a single platform.
2. Provide a mechanistic account of the device’s two memory regimes in terms of the displacement of mobile triflate anions and lithium cations within the oxygen-rich ion-conducting host, supported by hard–soft acid–base reasoning on the cation–oxygen interaction.
 3. Build a comparative experimental corpus on a poly(ethylene oxide)-based platform loaded with alkali-metal triflate salts, fabricated under a common protocol, and characterise it through three common dynamical measurements: I–V hysteresis, variable-number-of-pulses potentiation, and variable-delay depotentiation.
 4. Quantify the composition dependence through a replicated PEO and lithium-triflate concentration series, and survey host, anion, and cation substitutions on the same three measurements.
 5. Construct experimentally grounded behavioural models from those measurements, using composition-indexed fading-memory times, nonlinear pulse-update rules, device-to-device variability envelopes, and read-transfer behaviour.
 6. Demonstrate, in simulation, temporal-computing functionality from these models in a reservoir-computing framework: memory-capacity and NARMA benchmarks, WESAD physiological-context reconstruction, and affective-computing tasks across three corpora (WESAD stress, CASE continuous valence/arousal, and a PhysioNet Non-EEG cross-corpus replication).

Structure of the Thesis

The thesis is organised in six chapters that follow this objective list.

Chapter 1: Introduction. Provides the conceptual and materials background, moving from the limits of conventional computing, through the biological synapse and the memristor, to the polymer-electrolyte composite adopted as the device strategy for the rest of the thesis.

Chapter 2: Proof of Concept. Reports the Super Yellow, Hybrane, and lithium-triflate device. It is the only device in the thesis for which the complete synaptic suite is treated as primary evidence, and it isolates the cation–host coordination chemistry that the comparative work then tests. The chapter is built on Prado-Socorro et al. [6].

Chapter 3: Reproducibility, Degradation, and the Fabrication-Record Methodology. Documents the loss of reproducibility that ended the Hybrane fabrication line and motivated the switch to a PEO host. Under explicit controls (per-device weighting, a standard-protocol corpus, sweep-amplitude matching), it quantifies the reduction of the Hybrane switching window over successive batches, offers a moisture-driven hydrolytic-aging hypothesis for the host stock, and introduces the per-device fabrication record, the normalized characterization procedure, and the automated feature pipeline used in the remaining chapters.

Chapter 4: Compositional and Chemical Control. Reformulates the proof-of-concept architecture on a PEO host loaded with alkali-metal triflates. Its quantitative core is the replicated PEO/lithium-triflate composition grid; host, anion, and cation substitutions are surveyed at smaller sample sizes. All claims rely on the same three common measurements.

Chapter 5: Data-Driven Temporal Computing. Turns the experimental measurements into temporal-computing functionality. Compact

behavioural models extracted from the Chapter 4 dynamics drive reservoir-computing simulations: memory-capacity and NARMA-style benchmarks, WESAD physiological-context reconstruction, and affective-computing demonstrations across three corpora (WESAD stress classification, CASE continuous valence/arousal, and a PhysioNet Non-EEG cross-corpus replication), alongside homogeneous, memoryless, and linear-filter controls.

Chapter 6: Conclusions and Outlook. Brings together the device-chemistry axis and the application axis, benchmarks the work against temporal, reservoir, and event-driven neuromorphic hardware, and states the limitations of the evidence base.

Chapter 1

Introduction

Data movement between processing and memory imposes growing energy and latency costs on modern computing. Large machine-learning models and continuous sensor streams place particularly heavy demands on that movement [2]. Section 1.1 describes the architectural origin of this limit. The approach taken here uses the dynamics of a physical device to perform part of the computation. The biological synapse motivates this approach because a single element transmits a signal, modifies its strength according to recent activity, and retains that activity over several timescales [7].

This thesis examines organic memristive devices built from ion-containing polymer composites. It asks whether ion motion in a soft host can give a two-terminal electrical element the fading memory needed for temporal computation. Answering that question draws on solid-state electronics, polymer physical chemistry, and computational neuroscience.

1.1 Architectural Limits of Modern Computing

General-purpose digital computers use a stored-program architecture in which processing and memory are physically separate. On contemporary hardware, moving operands between those units can cost substantially more energy than the arithmetic operation [1, 8]. Smaller transistors and faster clocks do not remove this transfer cost. The resulting limits are architectural as well as technological: the *memory wall*, the *energy wall*, and the slowing of classical transistor scaling all arise from the separation between stored data and the circuitry that acts on it [3, 9].

1.1.1 Von Neumann Architecture and Its Limitations

The architectural model of modern digital computers is the *stored-program* concept set out by von Neumann in 1945 [10]: instructions and data share one memory and are fetched word by word by a central processing unit across a connecting bus. This is what makes a single machine general-purpose, and engineering refinements (registers, caches, branch prediction, memory controllers) have left the topology essentially unchanged. The same topology is the source of the limitation. Because the processor and the memory are *physically distinct* subsystems, every instruction fetch, operand load, and result write-back must cross the channel between them; no arithmetic can begin before its operands arrive, and no result is preserved until it is moved back. This is the *processor-memory dichotomy*.

Because only a bounded amount of data can cross that channel per cycle (the *von Neumann bottleneck* [11]), performance is set by the slowest interface, not by the arithmetic. Pipelining, multi-level caches, speculation, prefetching, and out-of-order issue all mitigate this, but they preserve the

separation between computation and storage, so they shift the bottleneck without removing it. Wulf and McKee [9] named the resulting imbalance the *memory wall*: with processor speed long outpacing DRAM latency and bandwidth, an ever-larger share of each cycle is spent waiting on memory. Modern microprocessors accordingly devote much of their silicon area, and an even larger fraction of their power, to caches, prefetch logic, memory controllers, and interconnect, whose only purpose is to keep the arithmetic units supplied with data [8]. The bandwidth and energy aspects of this imbalance are summarised in fig. 1.1.

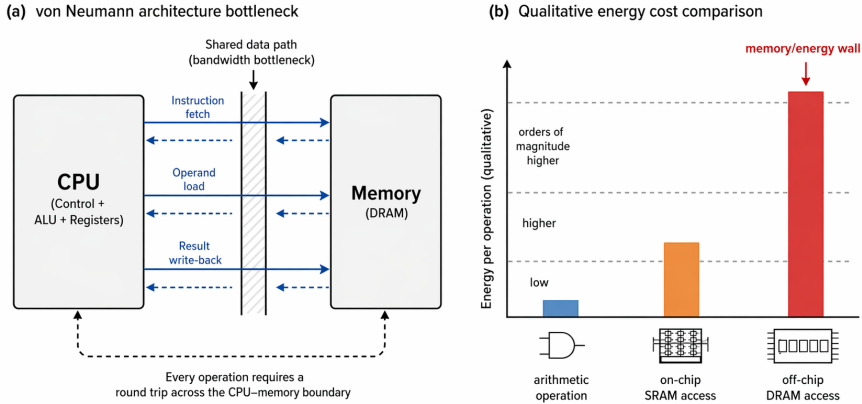


Figure 1.1: Schematic of the von Neumann architecture and the associated bandwidth and energy costs of data movement. (a) In the stored-program paradigm, the central processing unit and the memory unit are physically separated and communicate through a shared bus, so that every instruction fetch and every operand access must traverse that bus. (b) The energy cost of moving a word across this boundary exceeds, by orders of magnitude, the energy cost of the arithmetic operation performed on it, and this asymmetry has increased as technology has scaled [1].

Horowitz [1] compared the costs of elementary operations in a representative CMOS process: a 32-bit integer addition costs on the order of 0.1 pJ, an SRAM access costs tens of picojoules, and a DRAM access costs several hundred picojoules to nanojoules (table 1.1). The system-level contribution depends on how often each operation occurs. A neural-

network inference performs many multiply–accumulate operations, each requiring access to a stored weight, and training repeats these accesses across large datasets. Operand transport can therefore account for a substantial share of the energy budget. This *energy wall* arises from interconnect capacitance and from the total wire length of the memory subsystem. Spatial separation of computation and storage preserves these physical transfer costs.

Table 1.1: Approximate energy cost per operation in a representative 45 nm CMOS process, after Horowitz [1]. Memory accesses cost between one and four orders of magnitude more energy than the arithmetic they enable, and the asymmetry widens with the size of the memory addressed and with the distance the data has to travel.

Operation	Energy	Class
8-bit integer add	0.03 pJ	Arithmetic
32-bit integer add	0.1 pJ	Arithmetic
32-bit floating-point add	0.9 pJ	Arithmetic
32-bit floating-point multiply	3.7 pJ	Arithmetic
Register-file read	0.1 pJ	On-chip storage
SRAM read (8 kB)	5 pJ	On-chip storage
SRAM read (1 MB)	100 pJ	On-chip storage
Cross-chip wire (~ 10 mm)	100 pJ	Interconnect
DRAM read	1.3 nJ	Off-chip storage

For several decades, Moore’s increase in transistor counts [12] was accompanied by Dennard scaling [13], so successive generations provided more transistors with improved speed and energy efficiency. Dennard scaling ended around 2005, when leakage currents prevented proportional voltage scaling [3]. Transistor counts continued to rise, but performance and energy gains slowed, and power density became a limiting constraint (the *post-Dennard* or *dark-silicon* regime [14]). Process scaling alone now provides smaller reductions in the cost of the von Neumann model. At the same time, machine-learning workloads rely heavily on repeated matrix operations over memory-resident parameters [2]. Hennessy and Patterson [3] identify this period as an inflection point and argue for

domain-specific hardware that places memory and computation closer together.

1.1.2 In-Memory and Event-Driven Computing Paradigms

Two responses to this separation have emerged. The first keeps the digital workload largely intact but performs operations directly within, or very close to, the memory array; this route is described as *in-memory*, *near-memory*, or *processing-in-memory* computing. The second uses temporally structured event streams and treats the intrinsic dynamics of the physical device as part of the computation itself; this route is known as *event-driven* or *temporal* computing. The two responses, set against the von Neumann baseline they address, are summarised in fig. 1.2.

In-memory computing bypasses the classical *fetch–compute–write-back* cycle by executing operations directly inside the memory array, as in analogue matrix–vector multiplication on a resistive crossbar:

1. Each memory cell contains a conductance value that reflects a matrix entry.
2. Voltages are applied across the rows, representing the input vector.
3. The resulting column currents represent the sum of products, in accordance with Kirchhoff’s laws.

This changes the array from a passive storage device into an active computing tool. Demonstrations using metal-oxide resistive memories [15], phase-change materials [19, 20], and organic ionic devices [21–23] have achieved matrix–vector operations with energy efficiencies that surpass those of traditional digital accelerators, and similar principles can be

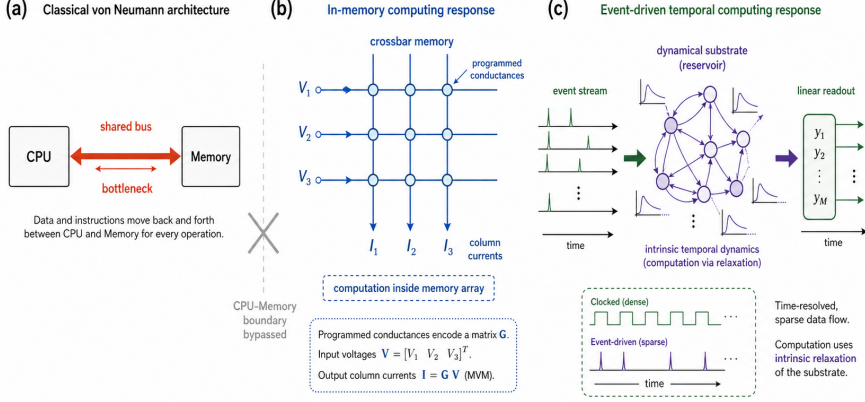


Figure 1.2: Two architectural responses to the von Neumann bottleneck. (a) In the classical stored-program architecture, every operation traverses a CPU–memory bus that fixes the bandwidth and energy ceilings of the system. (b) In the in-memory paradigm, computation is performed inside the memory array: a resistive crossbar with programmed conductances executes an analogue matrix–vector multiplication by Ohm’s and Kirchhoff’s laws, so the dot product no longer crosses the CPU–memory boundary [15, 16]. (c) In the event-driven paradigm, dense clocked data are replaced by a sparse stream of asynchronous events that drive a dynamical substrate whose intrinsic relaxation is itself the computation; only a linear readout layer downstream is trained [5, 17, 18].

extended to convolutional [24] and attention-like processes (the matrix-multiplication kernels at the core of transformer architectures [25]).

At the device level, however, this version of in-memory computing still requires memory properties. Each cell should hold a quasi-static conductance that encodes a trained weight, and which should be set once, read many times, and remain stable enough to meet the application’s numerical precision requirements. Non-volatility over seconds to years becomes a design goal.

Event-driven computing is motivated because many signals of interest carry information in time. Natural sounds, visual transients, motor commands, biomedical waveforms, and streaming sensor outputs are all

non-static examples. Their information is distributed across amplitude, waveform evolution, and event timing, so the computational system must evolve with the input. At the algorithmic level, this is the domain of spiking neural networks [26]. At the circuit level, it corresponds to event-driven asynchronous logic. At the device level, which is the level of this thesis, it requires materials whose intrinsic relaxation dynamics are matched to the signals they process.

Biological nervous systems handle sensorimotor and cognitive tasks at substantially lower power than current artificial systems. Relevant architectural features include sparse event-based signalling, synaptic responses that depend on recent activity, and electrochemical processes that store and modify state. Neuromorphic engineering applies selected principles from these systems, including sparse-event coding [27], the use of slow device dynamics as a computing element, and functional diversity supplied by device heterogeneity [4].

Boyd and Chua [28] showed that any time-invariant nonlinear operator with *fading memory*, in an appropriate function space, can be approximated by a finite Volterra series (a power-series expansion in time-shifted inputs). This means that a system whose current state retains a decaying record of past inputs can perform various temporal computations when paired with a trainable linear readout. This concept was later formalised through algorithms such as Jaeger’s *echo state network* [17, 29] and Maass, Natschläger, and Markram’s *liquid state machine* [18], which were combined under the umbrella of *reservoir computing* [30, 31]. In this framework, an untrained dynamical system (the reservoir) is driven by the input stream, with only the readout being trained. When the reservoir is realised in a physical substrate, the approach is known as *physical reservoir computing* [5], with implementations in optical, mechanical, spintronic, volatile-memristive, and organic electrochemical systems [32–35].

The relevant device metrics depend on the application. Trained-weight storage requires long retention. A reservoir instead requires a finite interval over which previous inputs influence the current output: the fading-memory timescale. Controlled variation between device timescales can then give a common readout access to temporal structure over a wider range than any individual device could cover [5].

1.2 The Biological Synapse as the Engineering Template

In neural circuits, a single physical element both transmits a signal and stores part of the history that shapes future transmissions: the synapse. The same molecular event that relays information from one neuron to the next also alters its future strength, so signalling and memory share a substrate. The device target is therefore synaptic signalling and plasticity, not full neuron emulation.

1.2.1 Neural Signal Transmission: A Brief Overview

A neuron is a polarised cell specialised to receive, integrate, and transmit electrical signals [36]. Dendrites collect inputs from other cells, the soma integrates them, and when the membrane potential at the axon hillock crosses a threshold the axon propagates an outgoing signal whose terminals release a chemical messenger onto downstream cells. A mature animal neuron typically receives 10^3 – 10^4 such contacts [37], and each is a *synapse*. How a neuron computes is therefore governed far more by the number and strength of its synapses, and by how they change over time, than by the soma itself.

The axonal signal is the *action potential*: a stereotyped, millisecond-scale

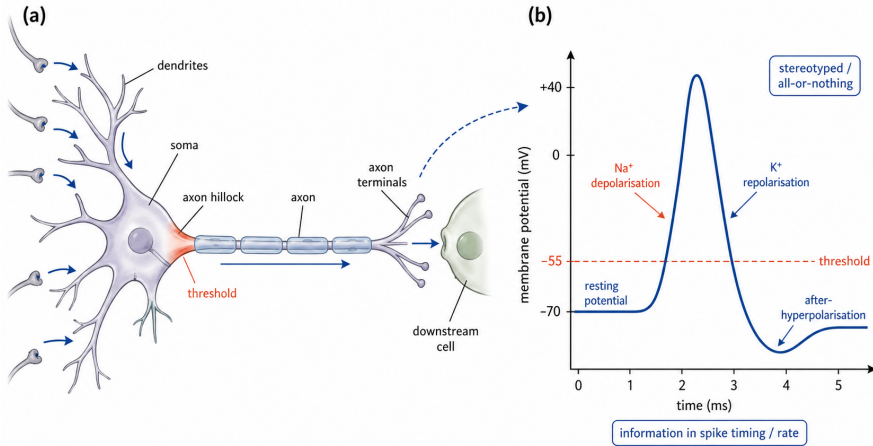


Figure 1.3: Anatomy of a neuron and the action-potential waveform. (a) A neuron collects synaptic inputs through its dendrites, integrates them at the soma, and fires an outgoing electrical event when the membrane potential at the axon hillock crosses a threshold; the resulting waveform propagates unidirectionally along the axon to the synaptic terminals, where it triggers chemical transmission onto downstream cells (the zoom shown in fig. 1.4). (b) The action potential is a stereotyped millisecond-scale depolarisation produced by the opening of voltage-gated Na^+ channels, followed by a K^+ -mediated repolarisation and a brief after-hyperpolarisation before the return to the resting potential; its all-or-nothing character means that information about the input is carried by the timing and rate of these events, not their amplitude, which is the cellular basis of the event-driven view of neural computation introduced in section 1.1.2.

depolarisation that is all-or-nothing, so that information is carried not in its amplitude but in the timing, rate, and coincidence of spike trains [38, 39]. Neural computation is in this sense event-driven (section 1.1.2) and arises from the pattern of signals arriving at many synapses. The neuron’s anatomy and the action potential waveform are summarised in fig. 1.3.

Synaptic transmission is largely chemical. An action potential arriving at the terminal opens voltage-gated Ca^{2+} channels, and the Ca^{2+} influx triggers neurotransmitter-filled vesicles (glutamate at excitatory cortical synapses) to fuse with the presynaptic membrane [40]. The transmitter

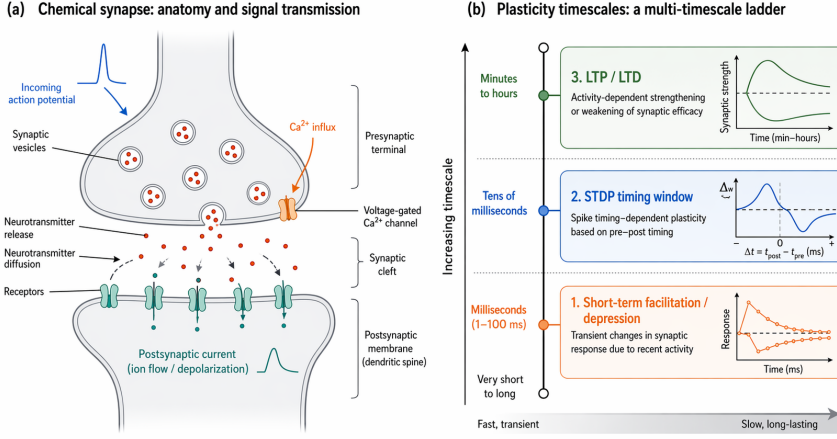


Figure 1.4: Anatomy and plasticity regimes of the chemical synapse. (a) A presynaptic terminal releases neurotransmitter vesicles into the synaptic cleft in response to a calcium-triggered action potential, and postsynaptic receptors convert the released signal into an excitatory postsynaptic current. (b) The same synapse supports a graded hierarchy of plasticity regimes (short-term facilitation and depression on millisecond timescales, spike-timing-dependent plasticity on tens of milliseconds, and long-term potentiation and depression on timescales from minutes to hours), each associated with distinct microscopic mechanisms.

diffuses across the ~ 20 nm cleft and binds postsynaptic receptors [37], producing an excitatory or inhibitory postsynaptic current (EPSC or IPSC) whose amplitude and kinetics define the *strength* of the synapse at that moment. The same molecular machinery carries the signal and sets the strength of future transmissions.

The synapse combines transmission, modulation, and memory: it passes a signal, adjusts its amplitude and time course, and stores changes produced by previous activity. This co-location provides a biological reference for electronic devices that combine state and computation. The relevant anatomy and plasticity regimes are summarised in fig. 1.4.

1.2.2 Synaptic Plasticity: The Physical Basis of Learning

Experience leaves a physical trace in the nervous system, stored mainly at synapses. Hebb's 1949 postulate, that a presynaptic cell which repeatedly helps fire a postsynaptic cell strengthens its connection to it [41] ("cells that fire together, wire together"), captures two features that matter for hardware: the rule is *local* to the two cells a synapse joins, and *sensitive to correlation*. Synaptic plasticity implements Hebbian or Hebbian-like behaviour across several timescales.

On the shortest timescales, *short-term plasticity* [42, 43] makes the immediate strength of a synapse depend on its recent activity. *Facilitation* is a transient increase in EPSC amplitude when a second spike follows the first after a short delay, usually attributed to residual presynaptic Ca^{2+} ; *depression* is a transient decrease caused by vesicle depletion or receptor desensitisation. With time constants from tens of milliseconds to several seconds, short-term plasticity makes the synapse a nonlinear, history-dependent temporal filter, so that identical spike sequences produce different responses depending on their timing.

On longer timescales, *long-term potentiation* (LTP) provides persistent, activity-dependent strengthening. First demonstrated by Bliss and Lømo in the rabbit hippocampus [44] and later shown to last days to weeks [45, 46], hippocampal LTP relies on NMDA receptors that open only when the postsynaptic membrane is already depolarised (a voltage-dependent Mg^{2+} block [47, 48]), making them Ca^{2+} -permeable coincidence detectors; the resulting Ca^{2+} influx drives additional AMPA receptors into the postsynaptic membrane [49, 50]. The complementary *long-term depression* (LTD) [51, 52] weakens synapses under low-frequency or anti-correlated activity, principally through reverse AMPA-receptor trafficking [53]. Together, LTP and LTD give the bidirectional, long-lasting amplitude

changes considered a primary physiological basis of learning.

Spanning these regimes, *spike-timing-dependent plasticity* (STDP), described in cortical pyramidal cells [54] and quantified in hippocampal cultures by Bi and Poo [55], makes the sign and magnitude of the change depend on the delay between pre- and postsynaptic spikes within a window of a few tens of milliseconds: a presynaptic spike before the postsynaptic one potentiates, the reverse depresses, and large delays in either direction produce no lasting change. The synapse thus holds a short memory of recent presynaptic activity, compares it with postsynaptic firing, and converts that timing relationship into a longer-lasting amplitude change.

Across these mechanisms, the synapse operates over several timescales: milliseconds for individual spikes, tens to thousands of milliseconds for short-term facilitation and depression, and minutes to hours and beyond for LTP and LTD. These partially overlapping molecular mechanisms led Abbott and Regehr [7] to describe the synapse as a computational unit. Both long-lasting and decaying state changes are relevant to hardware implementations.

1.2.3 Why Hardware Emulation of the Synapse?

The motivation for hardware synapses follows from section 1.1. In traditional computing, a software neural network retrieves data from a memory hierarchy physically separate from its arithmetic units. Each forward pass incurs data-movement costs that increase with the number of simulated synapses. A typical human brain contains approximately 10^{14} – 10^{15} synapses [56] and consumes about 20 W [57]. This comparison motivates hardware that co-locates analogue state and computation; biological synapses provide one reference architecture. The order-of-magnitude envelopes of these implementation regimes are compared in fig. 1.5.

This research programme is known as *neuromorphic engineering*, whose

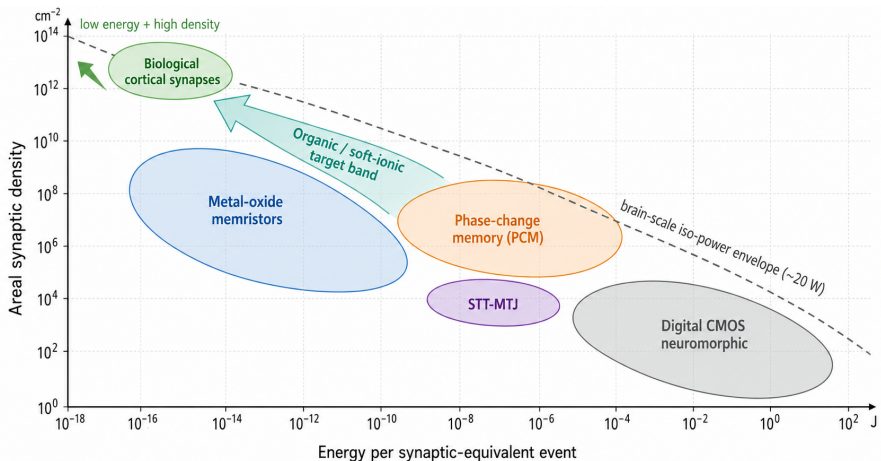


Figure 1.5: Approximate energy and density envelopes of synaptic-equivalent operations across implementation regimes. Each cluster represents a class of physical realisation: biological cortical synapses, digital CMOS neuromorphic implementations in which a synapse occupies an SRAM cell plus an arithmetic unit, the inorganic memristor families surveyed in section 1.4, and the soft-ionic composite target region forward-referenced to section 1.6. The diagonal reference line marks the brain-scale iso-power envelope (~ 20 W for $\sim 10^{14}$ – 10^{15} synapses) introduced at the start of section 1.2.3. Numerical bounds are conservative orders of magnitude drawn from the reviews cited in the surrounding text; individual devices may sit outside the envelopes shown.

programmatic origin is usually attributed to Mead [4]. It develops circuits in which device physics, topology, and dynamics implement selected computational characteristics of nervous systems. The synapse is a principal device-level reference.

A hardware synapse should satisfy several requirements:

- It should provide an analogue, multi-state conductance that can be set reversibly by electrical stimuli and read non-destructively with a small probe current.
- It should also be compact enough for large arrays, so that many thousands of elements can be addressed with a manageable number

of input and output lines [27, 58].

- Its response should be nonlinear in a meaningful way, for example by displaying a pulse-number or pulse-amplitude threshold below which little change is induced [16, 58].
- Finally, its dynamics should match the signals it is meant to process, whose relevant timescales span from milliseconds to seconds and beyond.

Two of these requirements deserve emphasis. The first is *analogue*, multi-state conductance: a binary element stores one bit and would need many devices per connection to represent a graded weight, whereas an analogue element encodes several distinguishable states in one device, and for ionic devices the write energy can be low because the state is set by moving ions over short distances instead of charging large digital circuits [22, 23]. The second is *reversibility*: a device that can be potentiated but not reliably depressed cannot follow the bidirectional rules of section 1.2.2 and so cannot fully replicate synaptic learning [16]. The resulting specification (analogue, reversible, multi-state, low-energy conductance changes set by electrical pulses) motivates the devices studied in the rest of the thesis.

Biological synapses combine quasi-stable memory through LTP and LTD with temporal filtering through short-term plasticity and STDP. Hardware can implement either a persistent or quasi-persistent programmable weight, as in crossbar arrays for in-memory computing [16], or a dynamic fading-memory element. This thesis focuses on the latter role through the reservoir-computing approach to event-driven temporal computing introduced in section 1.1.2 [5, 32–34].

This leads to a materials question. In biology, the timescales of short-term plasticity are determined by processes such as Ca^{2+} buffering, vesicle recycling, and receptor desensitisation, which vary across synapses [43]. A hardware implementation requires a corresponding means of adjusting

its electrical timescale. Memristive devices provide one two-terminal implementation of analogue, history-dependent behaviour.

1.3 The Memristor: Theory and History

A memristor or memristive device shows a current–voltage (I–V) response that depends on an internal state, which is influenced by the device’s previous electrical activity. The ideal form is the *memristor*; the broader experimental category is the family of *memristive devices*. A relaxing internal state remains consistent with the memristive-system definition. Its decay supplies the fading-memory behaviour used later in the thesis.

1.3.1 Memristive Systems as History-Dependent Devices

A memristor is, intuitively, a resistor with memory: a two-terminal element whose resistance depends on the charge that has previously passed through it. Chua [59] introduced it in 1971 on symmetry grounds, completing the set of fundamental passive elements (resistor, capacitor, inductor) with a fourth that relates charge $q = \int i \, dt$ to flux linkage $\varphi = \int v \, dt$. It is described by $v = M(q) i$, where the *memristance* $M(q)$ has the units of resistance but, unlike an ordinary resistor, carries a record of the device’s past.

If M is constant the element is an ordinary resistor; if M varies with q , the present response depends on the history of the device. Under periodic excitation this produces a *pinched hysteresis loop* in the i – v plane [60]: the loop passes through the origin (zero voltage gives zero current instantaneously) but its shape reflects the internal state left by prior stimulation. Pinched hysteresis is the standard experimental

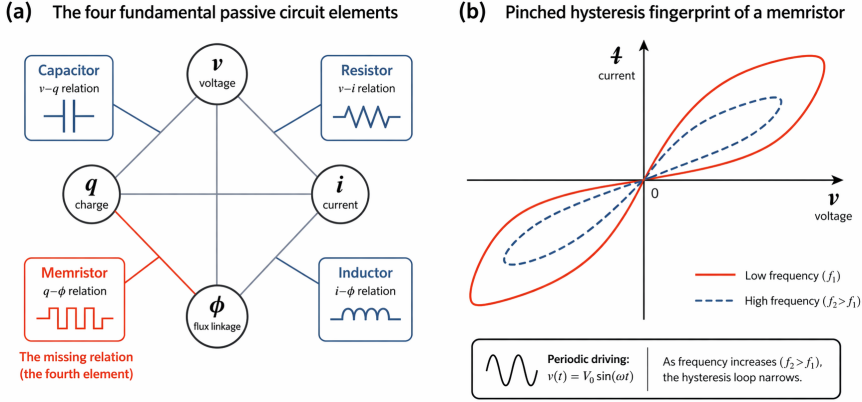


Figure 1.6: The memristor as a constitutive element and its experimental fingerprint. (a) The four fundamental two-terminal passive circuit elements, arranged by the pair of constitutive variables they relate: the resistor ($v-i$), the capacitor ($v-q$), the inductor ($i-\phi$), and the memristor ($q-\phi$), which completes the quadrant [59]. (b) A memristive device driven by a periodic voltage exhibits a pinched current–voltage hysteresis loop that narrows with increasing driving frequency and that passes through the origin at each zero-crossing; this loop is the canonical experimental fingerprint of memristive operation and is observed in all of the device families surveyed in the remainder of the chapter [60–62].

signature of memristive behaviour [61], and it is collected together with Chua’s constitutive argument in fig. 1.6.

Few real devices satisfy the ideal relation $\varphi = \hat{\varphi}(q)$. The broader notion, introduced by Chua and Kang [62] in 1976, is the *memristive system*: an input-driven internal state $\mathbf{x}$ evolves under generally nonlinear dynamics and gates the instantaneous electrical response. This model allows $\mathbf{x}$ to include relaxation terms that drive the device back toward equilibrium once the input is removed. A volatile device whose state decays is therefore still a memristive system, even though it is not an ideal memristor. This is the framework adopted for the polymer-composite devices reported here.

1.3.2 The First Physical Realisation: TiO_2 Memristor

For more than three decades after Chua’s proposal, the memristor remained mostly a theoretical construct. Thin-film metal-oxide resistive switches with hysteretic current–voltage characteristics had already been reported since the late 1960s [63–65], and many of them can now be recognised as memristive systems. At that time, however, they were not interpreted within Chua’s formalism. The field still lacked a standard physical device that connected the theoretical idea to a tangible thin-film structure.

In 2008, that connection was made possible by Strukov, Snider, Stewart and Williams [66] at HP Laboratories. Their device consisted of a thin titanium dioxide (TiO_2) layer sandwiched between two platinum electrodes. Within the film, a reduced TiO_{2-x} region acted as an oxygen-deficient conductive layer, while the neighbouring TiO_2 region served as a resistive barrier. When a voltage was applied, oxygen vacancies moved, shifting the boundary between these two regions [67]. Because the total resistance depends on the relative thickness of the conductive and resistive parts, it could be modulated continuously by moving this boundary back and forth. The boundary position is the internal state, oxygen-vacancy drift updates it, and the resulting i – v response reproduces the pinched hysteresis predicted by the theory.

This result transformed the memristor from a formal circuit element into a concrete experimental object and launched a large body of work on resistive switching and memristive devices. A two-terminal resistance that can be programmed by electrical pulses, tuned over several conductance values, and read at low voltage is close to the analogue weight element required by neuromorphic hardware [68]. Crossbar demonstrations of in-memory matrix–vector multiplication followed soon after [15, 16]. The

2008 TiO₂ device is, however, only one mechanism (oxygen-vacancy drift in a transition-metal oxide), while the memristive formalism is far broader.

1.3.3 Classification of Memristive Mechanisms

In device physics, the term *memristor* describes phenomenology independently of a particular microscopic mechanism. A device is “memristive” if its current–voltage response exhibits the pinched hysteresis that Chua described and if its internal state can be both set and read electrically. This can be achieved through many different physical processes, commonly grouped into broad mechanistic families [16, 68].

These taxonomies combine two different classification axes. The first is the *spatial pattern* of the state change, whether filamentary or interface/distributed, depending on whether the conductance change is localised to a narrow conductive path or extended across the active region. The second is the *physical nature* of the state variable itself: ionic distribution, structural phase, magnetic configuration, or redox population. The two axes are orthogonal: an ionic device can also be non-filamentary, so the families considered here locate a device along both axes.

Filamentary switching. This is historically the most visible family. When a voltage is applied, a narrow conductive path forms through a previously insulating thin film, which can later be ruptured or reformed. In *conductive-bridge* random-access memories (CBRAM), the filament is built from metal atoms displaced electrochemically from an active electrode (like silver or copper) through a solid electrolyte [69, 70]. In *valence-change* memories, which are generally based on binary transition-metal oxides, the filament is instead associated with oxygen vacancies or partially reduced metal-cation states inside the oxide lattice [67].

These devices offer high ON/OFF ratios, fast switching, and non-volatile retention; their main limitation is stochasticity, since filament formation,

positioning, and rupture vary across devices and cycles [71]. They therefore work well as binary or coarse multi-level memories but are poorly suited to fine analogue representation. Panel (a) of fig. 1.7 contrasts the CBRAM and VCM variants of these devices, while panels (b)–(e) display the other four families.

Interface (non-filamentary) switching. In these devices, conductance changes over a distributed region. Some implementations modulate the height or width of a Schottky barrier by accumulating or depleting mobile ions near a metal–semiconductor interface [72]. Other implementations, especially in mixed ionic–electronic systems, drive a distributed electrochemical doping of the semiconductor [21, 22]. Because the state change is spatially extended, these devices can support a quasi-continuous range of conductance values. The ON/OFF ratio is often smaller than that of filamentary devices, but variability is usually lower and the current tends to scale more predictably with device area. This is the closest *spatial-pattern* class for several organic electrochemical devices, including the polymer-composite devices studied here, whose state variable is itself described under the ionic family below.

Phase-change memories. The active material is a chalcogenide alloy, usually $\text{Ge}_2\text{Sb}_2\text{Te}_5$ or a related composition, whose resistance depends on whether the material is in a crystalline or amorphous configuration. Joule-heating pulses are then used either to melt and quench the material into the amorphous state, or to anneal it back toward the crystalline state [19, 20, 73]. Phase-change memories provide long retention, large ON/OFF ratios, and high endurance. Their write energies are relatively large, and devices tend toward binary or few-state operation unless specifically engineered for intermediate states. They are mature storage-class memories used in analogue in-memory computing demonstrations.

Spintronic (magnetic) devices. Their resistance depends on the magnetic configuration of a tunnel junction, typically determined by the

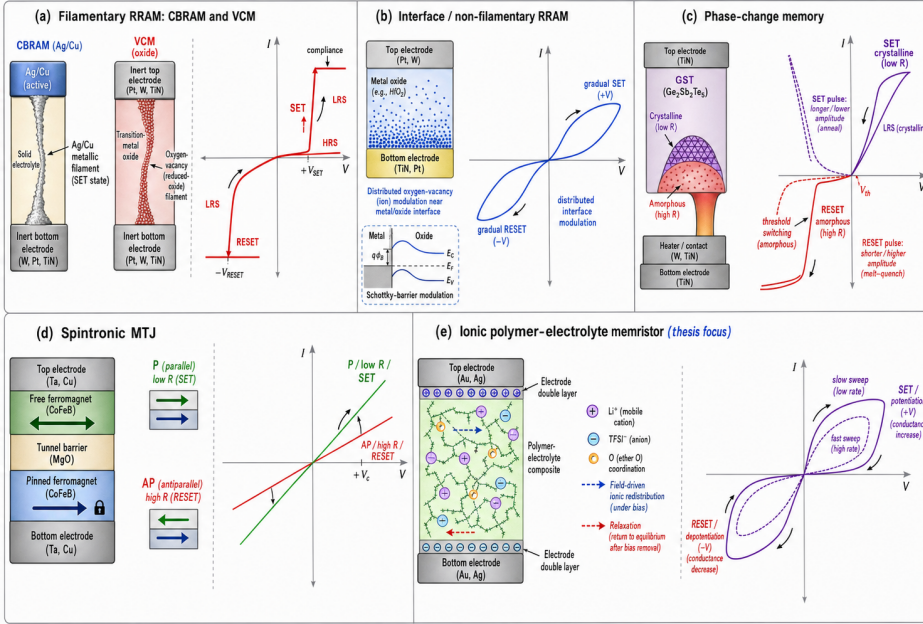


Figure 1.7: Mechanism gallery for the five memristive device families. (a) Filamentary devices, in their conductive-bridge (active-electrode metal filament) and valence-change (oxygen-vacancy filament) variants, switch abruptly between a high-resistance and a compliance-limited low-resistance branch. (b) Interface or non-filamentary devices switch through a spatially distributed change in the active layer, producing smooth, gradual, area-scaling hysteresis without an abrupt set or reset. (c) Phase-change devices toggle between amorphous and crystalline states of a chalcogenide alloy through Joule-heating melt-quench and anneal pulses, with a sharp threshold-switching event preceding the crystalline branch. (d) Spintronic magnetic tunnel junctions switch the relative orientation of two ferromagnetic layers, producing two nearly-linear branches separated by a small (order of two) ON/OFF ratio. (e) Ionic devices, in their polymer-electrolyte composite form most relevant to this thesis, redistribute mobile ions across the active layer under bias and relax back toward equilibrium when the bias is removed, producing continuous, sweep-rate-dependent hysteresis without abrupt transitions.

relative orientation of two ferromagnetic layers [74]. Spintronic devices switch very quickly at low write energy and form the basis of spin-transfer-torque magnetic random-access memory. Their limitations for analogue synaptic use are a small ON/OFF ratio (often of order two) and few stable intermediate states. Even so, their intrinsic stochastic dynamics have been put to use as a computational resource within appropriately designed architectures [75, 76].

Ionic memristors. This family is the one most relevant to this thesis. Their internal state is the distribution of one or more mobile ionic species inside a solid matrix. The host can be an inorganic oxide, a chalcogenide, a halide perovskite, a mixed ionic–electronic conductor, or a polymer electrolyte. Likewise, the mobile species can be a small cation such as Li^+ , Na^+ , or Ag^+ , a proton, or a small anion; in halide perovskites it is typically a migrating halide vacancy, which has been used to build energy-efficient memristive and synaptic devices [77]. The state variable is a chemical distribution of mobile species, distinguished from structural, morphological, or magnetic configurations of the host [70]. Its kinetics are governed by ionic transport, coordination, and relaxation within the matrix.

Ionic systems are intrinsically history-dependent: the distribution of mobile species at any moment reflects the electric fields previously applied. They also relax once the field is removed, as intermolecular forces and concentration gradients drive the biased distribution back toward equilibrium, so volatility is intrinsic to their operation. The associated timescales are set by ion transport and binding, both tunable chemically through the host matrix and the mobile species, the link between chemistry and relaxation that leads this thesis to the organic polymer-electrolyte subfamily.

1.3.4 From Device Metrics to Fading Memory

Memristive devices are often compared through a compact set of practical figures of merit: high- and low-resistance states, ON/OFF ratio, set/reset voltage, write energy, switching speed, retention, endurance, update linearity and symmetry, and device-to-device or cycle-to-cycle variability [16, 68, 71]. These quantities are useful only after the computational role of the device has been fixed.

For analogue in-memory computing, the device stores a *trained weight*: a conductance programmed into a crossbar cell and read repeatedly during matrix–vector multiplication [15, 16]. Long retention, high endurance, low variability, and linear symmetric updates are therefore valuable because they preserve numerical accuracy. Judged by that standard, an organic or polymer-composite device with finite retention can look weak.

For temporal computing, the same finite retention becomes the operating principle. The state of a fading-memory device should relax on a timescale matched to the input signal [18, 28]; nonlinearity can enrich the reservoir state, and device-to-device differences can supply a spread of temporal kernels. Figure 1.8 sketches the two modes schematically, and table 1.2 sets the corresponding parameter targets side by side.

Variability must also be separated into two kinds. Intra-device instability reduces repeatability under identical conditions. Controlled inter-device heterogeneity can broaden a reservoir’s response by supplying different time constants and nonlinear transfer functions [5]; the same heterogeneity complicates trained-weight arrays.

Table 1.2: Direction in which each device parameter should be optimised under the trained-weight regime (section 1.3.4) and under the fading-memory regime (section 1.3.5). For several parameters, the optimum direction reverses between regimes, so a single device cannot in general satisfy both simultaneously.

Parameter	Trained-weight optimum	Fading-memory optimum
Retention	As long as possible (s-years)	Matched to the input-signal timescale
Switching speed	As fast as possible	Matched to the input event rate
Write energy	Minimised (programmed once, read many)	Minimised within the fast-event budget
Conductance update linearity	Linear and uniform	Less critical; nonlinearity exploited as a feature
Update symmetry ($\pm$ pulses)	Symmetric	Less critical; asymmetry can be exploited
Cycle-to-cycle variability	Minimised	Minimised (intra-device drift remains a defect)
Device-to-device heterogeneity	Minimised	Useful as reservoir diversity [5]

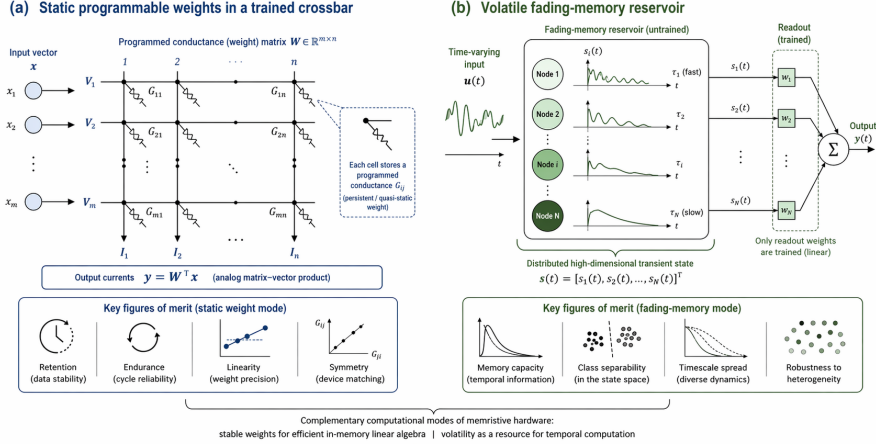


Figure 1.8: Two complementary computational roles for memristive devices. (a) A trained crossbar holds programmed conductances as non-volatile weights and performs matrix–vector multiplication by Ohm’s and Kirchhoff’s laws; here retention, endurance, linearity and symmetry are the relevant figures of merit. (b) A reservoir of volatile fading-memory nodes, driven by a time-varying input, projects temporal signals into a high-dimensional dynamical state; only a linear readout layer is trained, and the relevant figures of merit become memory capacity, class separability, and the spread of available time constants.

1.3.5 Reservoir Computing with Volatile Memristors

Over the past decade, volatile memristive devices have been used as physical *leaky integrators*: their state rises under stimulation, decays after the stimulus is removed, and thereby carries a temporary record of recent input. Physical reservoir computing provides the computational setting for that behaviour [5].

In a volatile-memristor reservoir, the input is encoded as a stream of voltage pulses applied to an array of devices. Their internal states evolve in response to the stimuli and relax between them, so the array provides a nonlinear mapping of recent input history into a high-dimensional feature space. A linear readout trained on that space can then tackle

temporal classification, regression, or prediction tasks. Demonstrations have been reported in diffusive metal-filament devices [32, 33], dynamic oxide memristors [34], and organic electrochemical systems [35].

Coincidence detection and multi-timescale filtering are closely related uses of the same physics. Pulses that arrive within the relaxation time of a device can accumulate, giving timing sensitivity and STDP-like responses [78]; banks of devices with different relaxation constants can process one input in parallel, with fast devices capturing sharp transients and slow devices integrating longer trends [27, 79]. In this thesis these ideas are folded into the reservoir-computing framework: coincidence appears as one effect of the fading state, and the filter-bank idea becomes a heterogeneous reservoir whose spread of time constants is supplied by composition, chemistry, and device-to-device variation.

Memory capacity measures how well the present reservoir state reconstructs past inputs at different delays; class separability on benchmark tasks and robustness to measured variability provide complementary tests [80]. These metrics are *task-matched*: they test whether the physical timescales of the device align with the temporal structure of the signal, which HRS, LRS, ON/OFF ratio, and retention alone do not capture.

The useful subset for this thesis is therefore narrower than the full memristor literature: ionic memristive systems whose state is a chemical distribution of mobile species, whose relaxation can be tuned, and whose volatility can be used as fading memory.

1.4 Inorganic Memristors for Neuromorphic Computing

The inorganic families that dominate neuromorphic hardware are metal-oxide resistive memories, phase-change memories, and spin-transfer-torque

magnetic tunnel junctions. Each is technologically important, and together they have made memristive hardware a credible post-von-Neumann substrate. Other inorganic candidates (ferroelectric tunnel junctions and FeFETs, Mott-insulator devices, and threshold-switching selectors) are narrower in analogue synaptic operation and are left outside this comparison.

1.4.1 Metal-Oxide Memristors

The most developed inorganic family is resistive switching in transition-metal-oxide thin films, in the VCM and CBRAM variants of section 1.3.3. After the TiO_2 device [66], the field expanded to HfO_2 , TaO_x , NiO , and WO_x , many already familiar to silicon processing. Compatibility with back-end-of-line CMOS enabled large crossbar demonstrations [15, 69, 70]. Reported non-volatile devices reach retention of years, nanosecond switching, endurance beyond 10^9 cycles, and sub-10 nm dimensions [16, 68]. Diffusive metal-filament variants can also operate as volatile devices, with Ag or Cu clusters dissolving after programming [32–34, 81]. For analogue synaptic operation, filament stochasticity produces device-to-device and cycle-to-cycle variation in set/reset voltage and achieved conductance [71]. Nonlinear, asymmetric updates and a limited number of reliable conductance levels constrain the accuracy of in-place training on metal-oxide crossbars [16].

1.4.2 Phase-Change Memory

Phase-change memory encodes its state in the proportion of crystalline and amorphous material in a chalcogenide volume (section 1.3.3) [19]. It is a mature storage-class technology used in analogue in-memory computing. Demonstrations include mixed-precision arithmetic for solving linear

systems [73] and neural-network training with PCM as the long-term weight store [82, 83]. Its thermal–structural mechanism imposes several costs: reset requires local melting, the quenched amorphous phase drifts with time and requires compensation [20], and crystallisation kinetics are tuned through alloy and geometry. These properties suit high-retention, high-endurance trained weights. They offer fewer chemical controls over fading-memory timescales than the soft-matter platform pursued here.

1.4.3 Spin-Transfer-Torque Magnetic Tunnel Junctions

A spin-transfer-torque device is the magnetic tunnel junction (MTJ) of section 1.3.3, here with a thin MgO barrier and switched between parallel and antiparallel magnetisation by a spin-polarised current through a shared read/write port [74]. STT–MTJs switch in nanoseconds at femtojoule energies, exceed 10^{12} cycles, and integrate in the CMOS back end [84]. Their intrinsic stochasticity has been used for probabilistic computing [75, 76]. For analogue synaptic operation, the parallel/antiparallel resistance ratio is small (often near two), stable intermediate states are few, and increasing their number requires more elaborate stacks [84, 85]. Retention and write energy trade off through thermal stability and switching current [84], while the crystalline ferromagnetic stack requires fabrication methods distinct from solution processing. These properties make STT–MTJs fast, non-volatile, few-state stochastic elements.

1.4.4 General Limitations of Inorganic Memristors

The three families differ in mechanism and maturity but offer limited control over the specific variables studied in this thesis. Their design parameters are primarily geometric, process-related, or stoichiometric

within narrow windows [16, 71, 79]. They do not generally permit direct substitution of mobile ions with different binding chemistry or of hosts with different segmental motion. Volatile implementations exist through diffusive filaments [32–34, 81], amorphous-phase drift [20], and superparamagnetic fluctuations [75]. Their relaxation times are set by vacancy drift, crystallisation, or magnetic energy landscapes, so adjustment through a single chemical composition variable is limited. Vacuum deposition, high-temperature annealing, and top-down lithography suit dense CMOS-integrated crossbars but restrict substrate and process temperature [15, 82, 83]. Solution-processable layers allow direct reformulation and a wider range of substrates [79]. Deliberate composition changes may also separate controlled inter-device heterogeneity from intra-device instability more clearly than these established platforms do [5, 16, 20, 71].

This thesis therefore examines an organic ion-mediated platform that combines analogue response, chemical control of relaxation, and solution processing. Figure 1.9 summarises the qualitative comparison, while table 1.3 provides the corresponding quantitative parameters.

1.5 Organic Materials for Memristive Applications

The inorganic platforms each solve part of the problem, but none of them can be tuned by chemistry alone: reshaping the switching physics requires redesigning a rigid multilayer stack, not changing a molecule, a host, a salt, or a mixing ratio [23, 79]. Organic and polymeric materials lift that constraint. A semiconducting polymer can be tuned through backbone, side chains, and molecular weight [86]; an ion-transport host through its chelating groups, topology, and glass-transition temperature [87, 88]; and a mobile ion through charge, radius, and Lewis acidity [89]. Most such

Table 1.3: Typical operating parameters for the inorganic memristor families. Diffusive (volatile) metal-filament devices are listed separately because they are deliberately operated in the fading-memory regime, not the non-volatile one. Numerical ranges are conservative orders of magnitude drawn from the indicated reviews; individual devices may sit outside them. “cont.” = quasi-continuous conductance.

Family	Switching	Retention	ON/OFF	Endurance	Energy/write	Analogue states
VCM [67, 68, 71]	1–100 ns	years	10^{-10^3}	$10^9\text{--}10^{12}$	0.1–10 pJ	4–8
CBRAM [16, 68]	1–100 ns	years	$10^4\text{--}10^6$	$10^8\text{--}10^{10}$	0.1–10 pJ	2–4
PCM [19, 20, 73, 82]	10–100 ns	years (drifts)	$10^2\text{--}10^3$	$10^8\text{--}10^{10}$	10–100 pJ	8–16
STT-MTJ [75, 76]	1–10 ns	years	2–3	$> 10^{12}$	0.1–1 pJ	2
Diffusive (volatile) [32–34]	$\mu\text{s--ms}$	ms–s	10^{-10^3}	$\sim 10^6$	0.1–10 pJ	cont. (volatile)

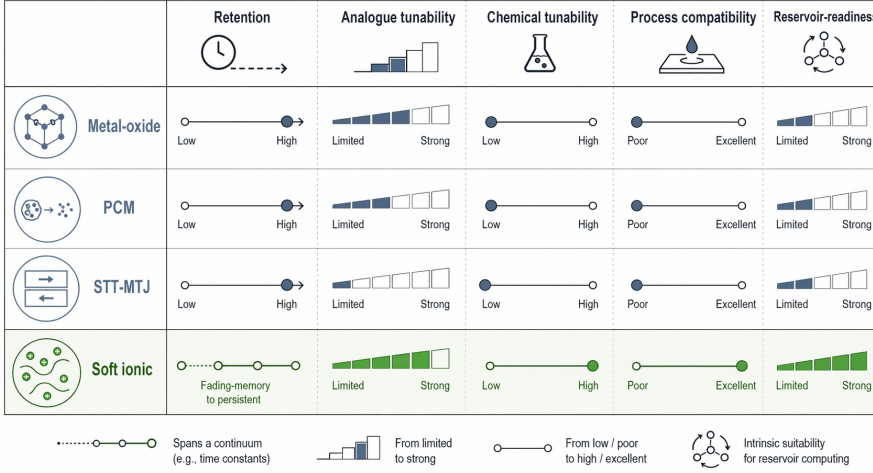


Figure 1.9: Comparison of inorganic memristor families with the soft-ionic composite route pursued in this thesis. For each family, qualitative assessments are shown along five axes: retention (the range of fading-memory to non-volatile behaviour that has been demonstrated), analogue tunability (the number of reliably programmable intermediate states), chemical tunability (the extent to which composition acts as an independent handle on the switching physics), substrate compatibility (vacuum/high-temperature workflows versus solution-processable and mild-temperature workflows), and reservoir-readiness (suitability as a fading-memory node in a temporal-computing scheme).

active layers are also solution-processable and processed below 150 °C, so a new formulation can be prepared by changing component concentrations in solution, and the mild processing is compatible with flexible, conformal, and biological substrates, as the stretchable artificial afferent nerve of Kim et al. [90] and the *nerve*tronic systems reviewed by Park and co-workers [91] illustrate.

Two consequences matter most here. First, organic switching can spread over many molecular sites, which makes graded analogue states more accessible than in a single-filament mechanism [21, 22]. Second, relaxation timescales can be engineered *chemically*: ion transport in a polymer electrolyte depends on how strongly the mobile ion binds the host, on

host segmental mobility, and on the number of mobile species [87, 92], so timescales from milliseconds to many seconds (the range of biological short-term plasticity and slow temporal signals [42, 43]) can be reached within one material family by compositional change. Cost, reproducibility, biocompatibility, and lifetime must be established case by case. The target in this thesis is specifically a reversible mixed ionic–electronic platform whose chemistry controls the fading-memory state.

1.5.1 Organic Mechanisms as Design Choices

As with their inorganic counterparts, the term *organic memristor* names a material family, not a single mechanism. The relevant comparison is what each mechanism (each with its own state variable, reversibility, timescale, and degree of analogue control) offers for a chemically tunable, ion-mediated, two-terminal synaptic element.

Charge trapping and detrapping. Deep or shallow traps (molecular defects, embedded nanoparticles, or boundaries between incompatible polymer phases) are filled and emptied under bias, and the trap occupation, through its local space-charge environment, sets the conductance. Trapping is common in disordered organic semiconductors and operates at modest voltages, but its analogue control is limited: the state is often binary [93], the conductance can depend non-monotonically on occupation, the trap identity is hard to establish, and the retention (set by trap depth relative to thermal energy) is not easily tuned by composition. Panel (a) of fig. 1.10 sketches this trap-occupation mechanism.

Redox switching. A redox-active species (a discrete molecule in a polymer host, a redox polymer, or a unit on a conjugated backbone) is oxidised or reduced under bias, and the distribution of oxidation states sets the conductance. The state variable is then a chemical oxidation state with experimentally accessible potentials, reversible over

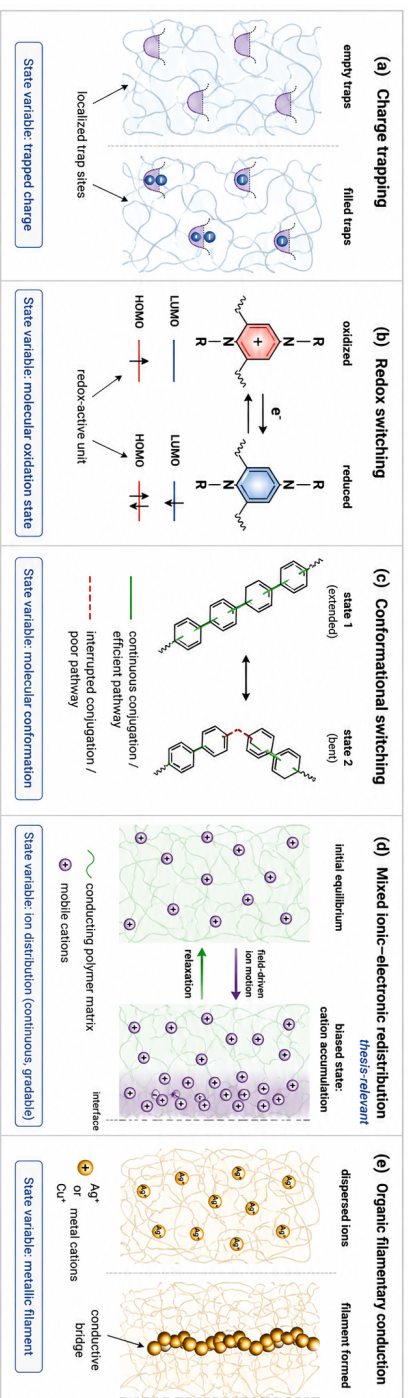


Figure 1.10: Molecular-level representation of the five mechanism families behind organic memristive behaviour. (a) Charge trapping: localised trap sites associated with defects, embedded nanoparticles, or interphase boundaries in a disordered polymer matrix, filled and emptied under bias. (b) Redox switching: a redox-active moiety alternating between its oxidised and reduced states, with the local electronic structure changing accordingly. (c) Conformational switching: a molecular unit toggling between two well-defined geometric or isomeric states with different conjugation or transport pathways. (d) Mixed ionic–electronic redistribution: a population of mobile cations within an electronically conducting polymer matrix, displaced and accumulated near an interface under bias and relaxing back toward equilibrium when the bias is removed. (e) Organic filamentary conduction: metallic cations (typically Ag^+ or Cu^+) from an active electrode migrating through the organic matrix to form a conductive bridge.

many cycles in favourable cases; important examples include redox-active conjugated-polymer systems [94–96] and early electrochemical devices based on coordination complexes [97]. Not every redox process is cleanly reversible, though (side reactions, dimerisation, and slow degradation limit endurance), and purely redox devices tend to be bistable unless the response is distributed over many sites.

Conformational (structural) switching. In this case, the active layer contains a molecular species whose geometry, isomerisation state, or aggregation state can be modulated by an electric field, local heating, or coupled electrochemical processes. Examples include photochromic and electrochromic switches, as well as bistable molecular architectures whose conductance depends on molecular conformation [95, 96]. The state of these systems is related to a well-defined structural change, so they offer a high level of molecular design precision. However, the synthesis can be challenging, and the switching timescale is fixed by a particular molecular rearrangement.

Ionic redistribution in mixed ionic–electronic systems. This is the family most relevant to this thesis. The active layer combines an electronic conductor (usually a conjugated polymer) with mobile ions held in an ion-conducting host. Under bias the ions move and accumulate near interfaces, modulating electronic transport through interfacial doping, electrostatic band bending, or electrochemical doping; the conductance change is gradual, multi-state, and directly tied to the amount and arrangement of displaced ionic charge [23]. The same physics underlies light-emitting electrochemical cells (LECs) [98–104], where redistribution forms a doped junction, and organic electrochemical transistors (OECTs) [105], where a gate drives the ion exchange. In the two-terminal memristive mode the driving bias itself plays the role of the gate, and the residual ionic distribution left after the bias is removed is what carries the memory of the stimulus.

The state variable is thus the distribution of mobile ions in the host, and relaxation after bias (governed by ion transport, the polymer binding environment, salt concentration, and morphology) sets the fading-memory process. The practical costs are that ionic motion is slower than electronic switching and that side reactions with the electrodes must be suppressed through chemistry and contact choice [106].

Organic filamentary conduction. Metallic species (typically Ag^+ or Cu^+ from an active electrode) migrate through the organic matrix and form conductive bridges similar to those found in inorganic CBRAM [69, 70]. These devices inherit the strengths of filamentary switching, such as large ON/OFF ratios and low switching voltages, but also its weaknesses, like stochastic filament formation and intra-device instability [107, 108].

The five families and their trade-offs are collected in table 1.4.

1.5.2 Organic Precedents for the Composite Strategy

The relevant precedents form a set of partial solutions to the same materials problem, not a chronological checklist: how to combine ion-mediated conductance control, analogue reversibility, and a simple device geometry. The earliest two-terminal organic memristive devices are usually taken to be the polyaniline (PANI) electrochemical junctions of Erokhin, Berzina, and co-workers [109–112], in which a PANI channel interfaced with a poly(ethylene oxide)-based electrolyte is doped electrochemically by ionic redistribution under bias. This work already paired a conjugated polymer with an ion-conducting polyether host (the materials logic extended throughout this thesis) and framed the device from the start as an adaptive, analogue-tunable element supporting history-dependent conductance changes.

Table 1.4: Mechanism families behind organic memristive behaviour.

Family	State variable	Reversibility	Analogue gradation	Timescale tunability	Refs.
Charge trapping	Trap occupation	Partial	Limited (often binary)	Low	[93]
Redox	Oxidation state of a moiety	Variable	Bistable to few-state	Low–medium	[94, 97]
Conformational	Molecular geometry / isomer	Good (per design)	Bistable / few-state	Low (per molecule)	[95, 96]
Mixed ionic–electronic	Mobile-ion distribution	Good	Continuous	High (host chemistry)	[21–23]
Filamentary (organic CBRAM)	Metallic filament	Limited (stochastic)	Few states	Low	[107, 108]

The LEC route is especially close to the platform used here. Zakhidov’s *light-emitting memristor* [97], in which an ionic $[\text{Ru}(\text{bpy})_3]^{2+}$ complex supplied both the emitter and the mobile ions, showed that redistribution in a polymer host yields memristive behaviour, while resistive switching in simple films such as poly(vinyl alcohol) [93] confirmed that the phenomenon is accessible in cheap matrices but typically binary and mechanistically opaque. The best analogue control came from the electrochemical-transistor route: Gkoupidenis’ PEDOT:PSS OECTs reproduced short-term plasticity, paired-pulse facilitation, and long-term potentiation and depression [105, 113], Xu’s core–sheath nanowire transistors [114] and the Van de Burgt–Fuller organic electrochemical neuromorphic device (ENODE) [21, 22] delivered linear, symmetric updates through ion exchange between electrolyte and channel [23], and halide-perovskite-gated OECTs extended the accessible timescales [115]. Each such synaptic element, however, requires a third terminal and a contacted electrolyte volume, which makes a self-contained two-terminal device an attractive complement for the reservoir paradigms of section 1.3.5.

The two-terminal line developed in parallel. Liu’s biomimicking redox organic memristor [94] demonstrated coexisting short- and long-term memory in a passive two-terminal device, but also showed partial irreversibility in the redox route: the long-term regime sometimes involved permanent chemical change that limited controlled state evolution [116]. Molecular memristors, especially Goswami’s metal-coordinated azo-aromatic layers [95, 96], achieved contrast, endurance, and multi-state operation through a demanding synthesis; stretchable and flexible devices [90, 91, 117, 118] demonstrated mechanical compliance; and perovskite–organic hybrids [77, 119] and organic electrochemical reservoir computing [35] broadened the field further, the latter showing that mixed ionic–electronic devices can serve as volatile physical-reservoir elements.

By the early 2020s the field had matured into a subfield consolidating

around shared physical principles [23, 120], with the individually demanding features each demonstrated (ion-mediated operation in two- and three-terminal forms, analogue symmetric programming, short-to-long-term transitions, mechanical compliance, molecular precision) but not yet combined in one simple platform. The three-terminal route bought reversibility and analogue control at the cost of architectural complexity, the two-terminal redox route simplicity at the cost of full reversibility, and the molecular route precision at the cost of synthesis. In that 2022 publication context, the specific gap addressed by the proof-of-concept chapter was a two-terminal, reversible, multi-state, ion-mediated device with interpretable chemistry built from standard organic-electronic components. The route pursued here fills that gap with a composite of a conjugated semiconducting polymer, an ion-conducting polymer host, and a dissociated alkali-metal salt. These parallel developments are collected in fig. 1.11.

Contemporary physical-reservoir computing spans several node, input, output, and network architectures, so comparison requires those layers to be stated separately [121]. Since the proof-of-concept publication, organic hardware has advanced from electrochemical networks to gate- and wavelength-tuned vertical transistors with measured temporal characteristics from 0.005 s–13.2 s [122], and to reconfigurable organic electrochemical transistors that implement both reservoir and read-out functions [123]. Electrical control of relaxation is also no longer unique to multi-terminal structures: a two-terminal HfO_2 memristor has used an inter-pulse base voltage to tune its fitted relaxation time over more than two orders of magnitude [124]. The present thesis therefore makes no priority claim for the first tunable two-terminal physical reservoir. Its narrower contribution is replicated composition control of seconds-scale fading memory in a solution-processed, self-contained two-terminal organic composite, followed by model-based rather than wired-reservoir task evaluation.

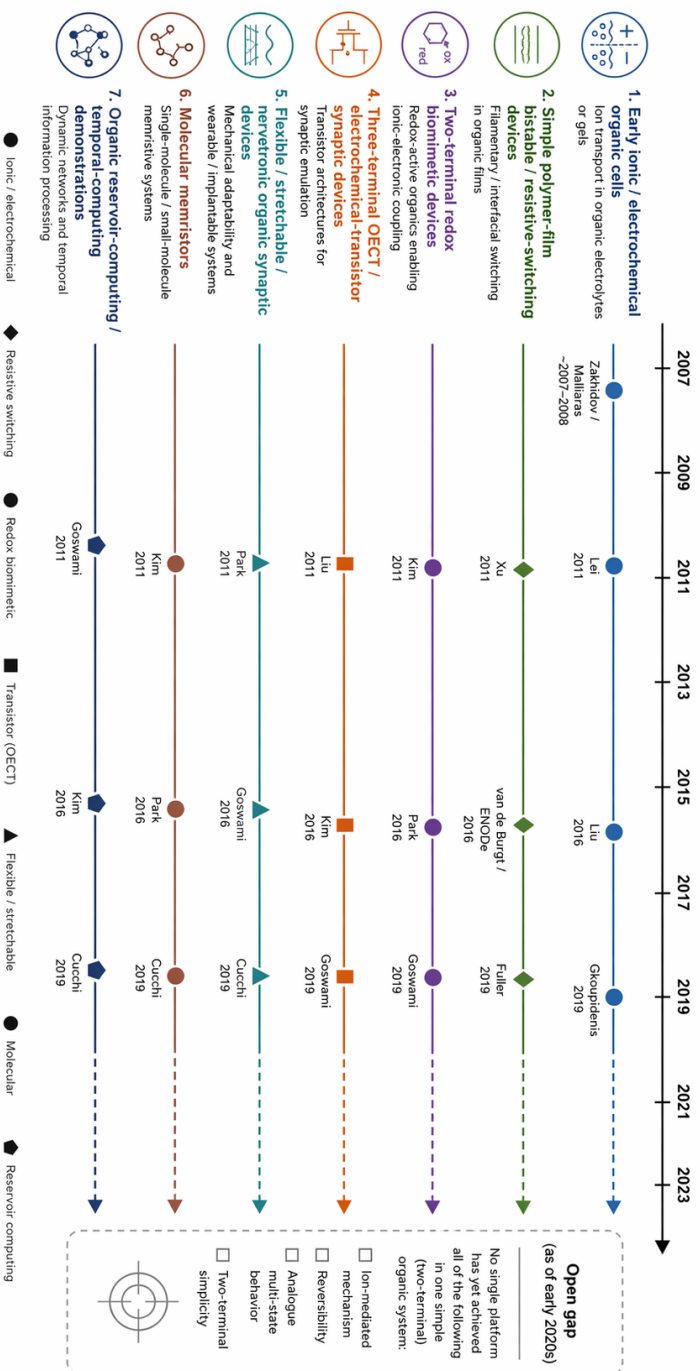


Figure 1.11: Historical map of organic memristive and organic synaptic devices, arranged by mechanism family and by year of representative publication. The map frames the literature context of the 2022 proof-of-concept publication: by the early 2020s, several demanding features of a neuromorphic organic device had each been demonstrated, but the specific combination targeted here was a self-contained two-terminal platform combining reversibility, multi-state analogue behaviour, and an interpretable ion-mediated mechanism.

1.6 Polymer Electrolyte Composites: The Enabling Strategy

The composite platform follows directly from the mixed ionic–electronic precedents above. It places a semiconducting conjugated polymer, an oxygen-rich ion-conducting polymer host, and a dissociated alkali-metal salt in one solution-processed active layer, then operates that layer in a standard two-terminal stack. Electronic transport, ion solvation, and mobile-charge supply are therefore assigned to separate components that can be changed independently.

1.6.1 Ion-Conducting Polymer Chemistry

An ion-conducting polymer is a macromolecular host in which dissolved or dispersed ions can move under an applied electric field without the help of a liquid solvent phase. Poly(ethylene oxide) (PEO) established the model system: alkali-metal salts were first shown to form solid complexes with PEO in 1973 [125], ionic conductivity was measured in those complexes in 1975 [126], and the resulting solid polymer electrolytes became practical ion conductors through the late 1970s and early 1980s [87, 88, 92]. Their relevance here is control of cation coordination, segmental motion, host crystallinity, and composition (the same variables that set memristive relaxation), not battery performance.

The polyether host has a high density of ether oxygens along the backbone or side chains. In PEO, those oxygens appear regularly along the $-\text{CH}_2-\text{CH}_2-\text{O}-$ repeat unit, and the chain conformation in the amorphous phase allows each cation to coordinate several ether oxygens from one or more polymer chains [92, 127]. Hybrane-type hyperbranched polyester-amide hosts arrange the oxygen density differently, through ether and ester groups attached as side chains on a branched polymer core [128], but the

coordination chemistry is of the same kind.

In both hosts, the oxygen atoms play the same chemical role. They act as hard Lewis bases that solvate alkali cations, stabilise salt dissociation, and define the local coordination environment of the cation. The anion, in contrast, is usually held more weakly (mainly through electrostatic compensation) and tends to be more mobile than the cation within the same host [129]. This is especially the case for soft, charge-delocalised anions such as triflate (CF_3SO_3^-) or bis(trifluoromethanesulfonyl)imide (TFSI, $\text{N}(\text{SO}_2\text{CF}_3)_2^-$).

Ion transport in these hosts is usually described through three related microscopic phenomena:

- From the *vehicular* side, a cation and its solvation shell move together through the polymer matrix, so long-range transport is coupled to polymer motion [130].
- From the *hopping* or site-to-site side, the cation instead transfers between coordination sites through a local rearrangement of oxygens, without macroscopic motion of the host. This last view is conceptually close to Grotthuss-type transport in proton conductors [131], even though the chemistry of the transfer itself is different.
- Finally, from the *segmental-motion-assisted* side, short polymer segments reorganise around the cation, creating and destroying coordination sites that allow effective diffusion [132, 133].

These three phenomena are not mutually exclusive, although the weight of each contribution during transport depends on host chemistry, cation identity, salt concentration, and temperature [87, 92].

The macroscopic ionic mobility of a polymer electrolyte generally correlates with the segmental mobility of the host, which is set by the glass

transition temperature T_g and by whether the host remains amorphous during operation. Below T_g , segmental motion is essentially frozen and ionic conduction drops. Above T_g , it is activated, and ionic transport becomes measurable, with the temperature dependence in this regime typically following the Vogel-Fulcher-Tammann form characteristic of fragile glass-formers [87, 134].

Even above T_g , however, not all of the host contributes equally. Crystalline regions contribute little to conduction and can even trap ions in ordered environments [127, 133]; the textbook picture is that conduction proceeds essentially through the amorphous fraction, although it has since been shown that specific crystalline stoichiometric phases can themselves support measurable ionic motion [135]. Low-crystallinity or fully amorphous hosts are therefore preferred for ion transport. A low T_g and a low crystalline fraction can be pursued through copolymerisation, branching, or plasticisation; hyperbranched architectures such as the Hybrane family are attractive here because branching itself suppresses crystallinity [136]. For room-temperature memristive operation the host should remain amorphous and mobile under ambient conditions, ideally without added plasticisers.

The strength of cation binding depends on cation identity, which hard–soft acid–base (HSAB) reasoning [89] makes explicit. Ether and ester oxygens in polyether and polyester-amide hosts are hard Lewis bases: their donor pairs are localised, their polarisability is low, and their coordination to alkali cations is primarily electrostatic. Among the alkali cations, Li^+ is the hardest Lewis acid (small, with a high charge density), and should therefore interact most strongly with hard oxygen donors. Na^+ and K^+ are progressively softer, with larger radii, lower charge densities, and weaker electrostatic coupling to the same oxygen environment.

In a common polyether host and at fixed composition, the residence time of the cation at a coordination site might be expected to follow the

coordination-strength order $\text{Li}^+ > \text{Na}^+ > \text{K}^+$. This provides a hypothesis for the devices' fading-memory timescale: harder cations should relax more slowly and softer cations more quickly. The experimental chapters test this hypothesis. In the archive used here, composition is the replicated quantitative variable, while cation order is not robust across host and anion. The coordination geometry, the three transport mechanisms, and the cation–oxygen binding hierarchy that motivate the hypothesis are summarised in fig. 1.12.

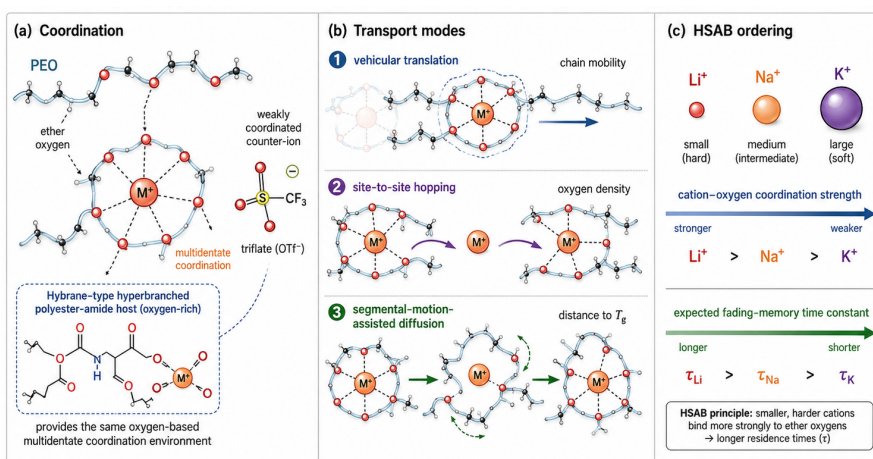


Figure 1.12: Molecular-level chemistry of ion transport in the polyether-hosted polymer-electrolyte composite. (a) A short poly(ethylene oxide) segment coordinates an alkali cation through four to six ether oxygens, with the triflate anion held nearby as a weakly coordinated charge-compensating counter-ion; the same coordination chemistry applies to the hyperbranched polyester-amide hosts used in chapter 2 [92, 127, 129]. (b) Long-range ionic transport in this host proceeds by three coupled contributions: vehicular translation of the cation together with its solvation shell, site-to-site hopping between coordination pockets through local oxygen-cage rearrangement, and segmental-motion-assisted diffusion in which short polymer segments reorganise to create and destroy coordination sites [130, 132, 133]. (c) Within a common polyether host, hard-soft acid-base reasoning ranks the alkali cations along the $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ series in decreasing cation–oxygen coordination strength [89]. This ordering motivates the cation-timescale hypothesis tested later, but the thesis does not assume it as a host- and anion-independent law.

A second set of design parameters is provided through composition. A polymer electrolyte is defined by the host, ion identity, and salt-to-host ratio; in a composite device, the polymer-to-semiconductor ratio also contributes. The salt-to-host ratio sets the number and spatial density of occupied coordination sites, while the host-to-semiconductor ratio sets the volume available for ionic transport and the connectivity of the electronic phase. The two ratios are therefore treated as independent composition axes in the later chapters.

These considerations motivate the host choices made in this thesis. Hybrane DEO750 8500 is a hyperbranched polyester-amide with a high density of ether and ester oxygens. It provides an amorphous, low-crystallinity ion-transport host and is used in chapter 2 as the host of the proof-of-concept Super Yellow (SY)/Hybrane/lithium triflate (LiOTf, LiCF_3SO_3) composite. The synaptic responses were measured on assay-specific junctions sharing this formulation and architecture; they do not form a device-population study.

PEO is the host used for the broader comparative work. Like Hybrane, its ether oxygens coordinate alkali-metal cations, segmental motion enables transport, and composition can be varied while the electronic semiconductor is held fixed. PEO also has an extensive solid-polymer-electrolyte literature covering its phase diagrams, coordination geometries, conduction mechanisms, and salt-dependent dynamics [87, 92, 126, 127, 133]. The replicated composition series therefore uses a host whose structural and transport behaviour is independently constrained. A hyperbranched polyether, trimethylolpropane ethoxylate (TMPE), provides a comparison between linear and branched polyether architectures. Host, anion, and cation substitutions are surveyed at smaller sample sizes and treated as illustrative contrasts rather than material laws.

1.6.2 LEC-Derived Composite Operation

Polymer light-emitting electrochemical cells (LECs) provide an established example of stable, reversible coupling between ionic redistribution in a polymer host and electronic transport in a conjugated semiconductor [137]. An LEC is a simple two-terminal sandwich [98]: a conjugated light-emitting polymer is mixed from a common solution with an ion-conducting polymer and a dissociated salt. Under bias, cations migrate toward the cathode and anions toward the anode, forming doped regions that inject carriers into the undoped bulk, and the resulting dynamic p-i-n profile produces electroluminescence in the intrinsic region between the doped edges [101].

Two models were historically used to describe this operation. In the electrochemical-doping model [98, 138], the dominant effect is bulk doping of the conjugated polymer; in the electrodynamic model [99, 100], it is the lowering of charge-injection barriers by ion accumulation at the contacts. Later experiments [101, 103] and a unifying framework [102] showed these to be limiting cases of the same ion redistribution under different injection regimes, not contradictory mechanisms. The two models, their reconciliation, and the bias-off configuration that the memristive composite reuses are compared on a common cathode–anode position axis in fig. 1.13; studies of electrochemical side reactions [106] additionally identify the degradation pathways that must be suppressed.

Two points are relevant here. First, LECs operate over useful windows with lifetimes of hundreds to thousands of hours [104, 139], provided the contacts and operating protocol support repeated charge injection and withdrawal without rapid destructive side reactions [140–142]. Second, the electrolyte supplies mobile ions whose redistribution modulates the conjugated polymer. In an LEC, sustained bias maintains a p-i-n profile for emission. In a memristive device, a pulse writes an ionic distribution

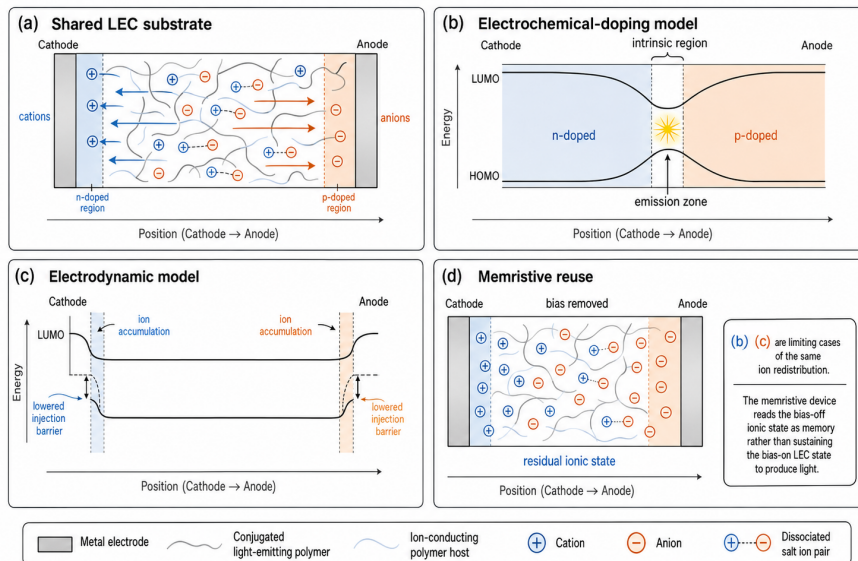


Figure 1.13: Light-emitting electrochemical cell operating models and their memristive reuse, drawn on a common cathode–anode position axis. (a) Shared substrate: under sustained bias, cations drift toward the cathode and anions toward the anode, forming doped regions at the two contacts. (b) Electrochemical-doping model: the dominant effect of the ionic redistribution is *bulk doping* of the conjugated polymer, producing a spatially wide p-i-n profile with emission localised at a narrow intrinsic region. (c) Electrodynamic model: the dominant effect of the same redistribution is *injection-barrier lowering* at the metal/polymer interfaces, producing thin ion-accumulation layers at the contacts and an essentially flat-band bulk. The two pictures were later reconciled as limiting cases of the same redistribution under different injection regimes. (d) Memristive reuse: with the bias removed, the residual ionic distribution left behind by the previous bias is the *stored internal state* of the device.

and the residual bias-off distribution provides the stored state. Ionic redistribution can therefore serve as the internal state variable of a two-terminal memory device.

The most direct precedent is the Hybrane-based composite that underlies the proof-of-concept device of chapter 2. Mardegan and co-workers [104] demonstrated stable LECs based on Hybrane hyperbranched polyester-

amide hosts blended with a conjugated light-emitting polymer, reporting long operational lifetimes under continuous bias. Their work validates the semiconducting-polymer / ion-conducting-host / salt architecture, the solvent system, and the processing protocol that chapter 2 then reuses under pulsed bias to operate the same composite as a memristive device.

1.6.3 The Ionic Response in Time and Frequency

The ionic redistribution that stores the device state has a characteristic relaxation time, set by the host segmental mobility discussed in section 1.6.1. This timescale has two equivalent descriptions. In the time domain, a written ionic profile decays toward equilibrium over a period that the proof-of-concept and comparative chapters measure in seconds (chapters 2 and 4). In the frequency domain, it appears as a cutoff. At low drive frequencies, ionic redistribution produces the pinched hysteresis of fig. 1.6. Above the redistribution rate, the junction approaches a passive dielectric response and the hysteresis loop narrows [61, 62]. For a slow polymer-electrolyte host, this crossover falls in the sub-kilohertz range and overlaps the frequency bands of peripheral physiological signals examined in chapter 5.

The standard small-signal probe of this frequency response is electrochemical impedance spectroscopy (EIS). A small alternating voltage is applied across a range of frequencies and the complex impedance recorded; plotted as a Nyquist or Bode spectrum, it separates the bulk resistance to ion transport through the host from the capacitance of the interfacial layers that accumulate at the contacts. An equivalent-circuit model assigns these features to discrete circuit elements: for polymer light-emitting electrochemical cells and the related composites used here, the circuit of Munar and co-workers resolves a bulk ionic resistance together with geometric and chemical (double-layer) capacitances [143]. That bulk

ionic resistance is the equilibrium counterpart of the driven relaxation (a more ion-rich, more mobile matrix has both a lower ionic resistance and a faster fading memory), so impedance spectroscopy gives an independent, small-signal test of the link between composition and timescale that large-signal switching alone cannot supply [87, 129]. Chapter 4 uses both probes, equilibrium impedance spectra and biased frequency sweeps, as microscopic corroboration of the composition control established from the driven dynamics.

1.6.4 Rationale for the Composite Approach

The composite separates electronic transport, ion solvation and transport, and mobile-charge supply across three components [98]: a conjugated polymer, an oxygen-rich host, and a dissociated salt. Salt, cation, and blend ratio can therefore be changed independently of the rest of the stack [23]. The salt fraction controls the identity and density of the mobile ions that write the internal state. By the HSAB argument of section 1.6.1, harder alkali cations bind ether oxygens more strongly and should be slower to displace, whereas charge-delocalised anions such as triflate interact weakly and redistribute more readily [129]. This mobility asymmetry provides a proposed origin for the device memory: a faster component near the contacts and a slower, more tightly bound component can retain effects of stimulation over multiple timescales [109, 111, 112]. Cation identity is thus one hypothesis for the slow response; its effect must be tested within each host, anion, electrode, and drive protocol.

Composition provides a second control variable. The salt fraction sets the number of mobile ions and hence the achievable size of the conductance change [87, 88], while the host fraction sets the volume and connectivity of the segmentally mobile phase in which they rearrange; in principle these vary the amplitude and the rate of the response separately, though

the comparative chapter treats their independence as an experimental question. The three components share a common solvent, so changing the formulation requires only different solution ratios on the same electrode geometry. This makes a replicated composition grid practical, while host, anion, and cation substitutions remain lower-power chemical contrasts.

Devices with the same polymers but different blend ratios should exhibit a spread of ionic relaxation times, with sample-limited chemistry and device-to-device variation adding further diversity. In a common circuit, these devices form a bank of temporal kernels with distinct decay constants [5, 80]. This spread complicates precise trained-weight storage but can cover a range of fading-memory timescales [32–35]. The application chapter evaluates this proposal quantitatively from the measured device behaviour.

The experimental chapters test this rationale first with the Hybrane/LiOTf proof-of-concept device of chapter 2, and then with the broader PEO-based corpus in which composition is varied as the replicated axis and chemistry is surveyed as illustrative side evidence. Figure 1.14 summarises this concept used throughout the thesis, linking the two-terminal stack to the qualitative timescale ladder originally expected from cation and composition chemistry.

From Background to Device

The resulting device target is a solution-processed composite of a conjugated semiconducting polymer, an oxygen-rich ion-conducting host, and a dissociated alkali-metal salt. It is operated as a two-terminal element whose residual ionic distribution forms the internal state. Chapter 2 introduces the proof-of-concept SY/Hybrane/LiOTf device, its memristive and synaptic response, and the ion-transport mechanism proposed to explain its two retention regimes.

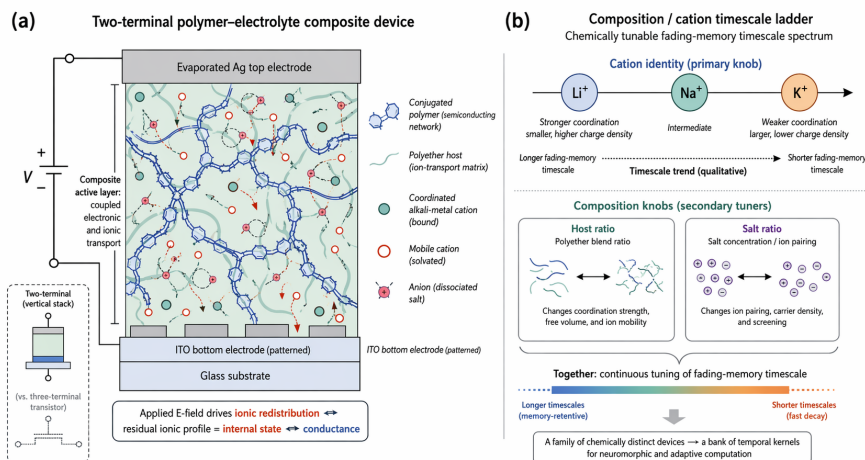


Figure 1.14: The thesis-specific polymer-electrolyte composite device and its chemically tunable dynamical timescale. (a) Vertical two-terminal stack: a solution-processed composite active layer, combining a conjugated semiconducting polymer network with an oxygen-rich ion-conducting host, coordinated alkali-metal cations, and a dissociated salt, is sandwiched between a patterned ITO bottom electrode and an evaporated silver top electrode; the internal state of the device is set by the residual ionic distribution that remains after a driving bias is removed. (b) Timescale ladder *as anticipated at the design stage*, with cation identity drawn as the primary control and the host and salt mass fractions as secondary ones, placing the fading-memory time along the $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ coordination order. The panel records the starting hypothesis, not the result. Chapter 4 inverts this hierarchy: composition (the PEO and salt mass fractions) is the replicated quantitative control of the timescale, whereas the cation proves the weakest and least cleanly resolved variable, and its anticipated $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ ordering is not borne out (the triflate retention series runs the reverse).

Chapter 2

Proof of Concept: Polymer-Based Composite Organic Memristive Devices

2.1 Introduction and Motivation

This chapter reports the proof-of-concept polymer-electrolyte memristive platform introduced at the end of chapter 1. The composite combines a semiconducting poly(phenylene vinylene) (PPV) derivative (*Super Yellow*, SY), an oxygen-rich hyperbranched polyester-amide ion-conducting host (*Hybrane*[®] DEO750 8500), and a dissociated lithium salt (lithium triflate, LiOTf, LiCF₃SO₃), all processed from a common solvent into a standard two-terminal (2-T) sandwich on indium tin oxide (ITO) and silver. The experiments test whether this common formulation and architecture support current–voltage hysteresis, potentiation and depression, excitatory post-synaptic current, distinct retention regimes, and spike-timing-dependent plasticity. The assays were not all performed on one physical junction: the available specimen mapping is stated in table A.1.

The chapter also proposes a cation–host coordination mechanism that is tested across cations and compositions in chapter 4.

At the start of this work the literature already contained several partial demonstrations of these functions in different architectures (reviewed in section 1.5.2): ion-mediated analogue switching in multi-layer polyaniline junctions [109, 111, 112], bistable ionic redistribution in a light-emitting memristor [97], tunable short-to-long-term transitions in a partially-irreversible redox device [94, 116], and reversible analogue control in three-terminal organic electrochemical transistors [21, 22, 105, 114]. Each covered part of the specification developed in section 1.2.3 (analogue, reversible, multi-state, low-energy conductance changes set by electrical pulses), but their combination in one self-contained two-terminal organic platform with an interpretable ion-mediated mechanism remained, at that point, an open problem.

The active-layer logic adopted here is the composite logic of section 1.6.4. The three functional roles (electronic transport, ionic transport, and mobile-charge supply) are assigned to three different components, so that each role can be tuned chemically without redesigning the rest of the stack. *Super Yellow* supplies the electronic-conductance pathway. *Hybrane* supplies an amorphous, low-crystallinity ion-transport host with a high density of hard oxygen donors. LiOTf supplies a hard alkali cation paired with a soft, charge-delocalised anion, so that the cation–host and anion–host interactions are asymmetric in the hard–soft acid–base sense of Pearson [89]. The mechanistic account developed in section 2.5 attributes the two memory regimes to this asymmetry.

The measurements fall into two groups. The first comprises three dynamical protocols: current–voltage hysteresis characteristic of memristive switching, potentiation as a function of the number of applied voltage pulses, and depotentiation as a function of the delay before read-out. These three measurements recur, on the same 2-T device geometry,

throughout the comparative work of Chapters 4 and 5. The second group comprises extended synaptic protocols specific to this chapter: excitatory post-synaptic current (EPSC) responses with four distinguishable readout states; the resolution of the depotentiation measurement into short- and longer-lived memory regimes, denoted short-term memory (STM) and long-term memory (LTM) following the convention of the artificial-synapse literature [23, 90]; and spike-timing dependent plasticity (STDP) with a time constant in the biologically realistic range [55, 58]. The proposed interpretation is developed in section 2.5: the fast component is associated with displacement of the weakly bound triflate anion, while the slow component is associated with displacement of the lithium cation from its hard oxygen coordination environment. This interpretation yields the cation-substitution prediction tested in Chapter 4.

Within the thesis the chapter is the only one in which the full synaptic suite is the main evidence; only the three dynamical measurements named above carry forward to the comparative corpus.

2.2 Materials Design and Composite Formulation

2.2.1 Design Rationale

The operating principle of the device is the same one developed in section 1.6.4: ionic migration within a solid-state polymer matrix under an applied electric field redistributes mobile ions near the electrodes, which in turn modifies the energy barriers for charge injection and the doping level of the semiconducting polymer, and therefore the electronic conductance of the device. The composite active layer is built to satisfy three functional requirements simultaneously:

1. **Electronic-conductance pathway:** a semiconducting polymer provides the medium through which the electronic current flows, and whose conductivity is susceptible to modulation by ionic redistribution and electrochemical doping.
2. **Ion-transport medium:** a polymer electrolyte provides a structural environment in which dissolved ions remain mobile under an applied field, yet are retarded enough by intermolecular interactions that, once the field is removed, the conductance change persists for seconds (long enough to be read out) before fading back toward equilibrium.
3. **Ion source:** a dissociated salt provides the mobile charge carriers (a cation and an anion) whose differential response to the field is proposed to underlie the distinct memory regimes reported in section 2.4.

The composition adopted here is built around a deliberate asymmetry in Lewis acid–base character between the two ions of the salt. LiOTf is chosen because Li^+ is a hard Lewis acid with a high charge-to-radius ratio ($\lambda = z/r \approx 1.3 \text{ \AA}^{-1}$ – $1.7 \text{ \AA}^{-1}$ across the four- to six-fold oxygen coordination that Li^+ adopts in polyether hosts, with r_{Li^+} in the range $0.59 \text{ \AA}$ – $0.76 \text{ \AA}$ for the corresponding Shannon effective radii [144]), whereas the triflate anion CF_3SO_3^- is a soft Lewis base with a diffuse, low-charge-density anionic centre. The ion-transport polymer, *Hybrane*, contains a high density of ether and ester oxygen atoms that act as hard Lewis bases. Under the hard–soft acid–base (HSAB) reasoning of Pearson [89] already used in section 1.6.1, hard acids associate preferentially with hard bases. The Li^+ cations are therefore tightly coordinated by the oxygen atoms of *Hybrane*, while the CF_3SO_3^- anions interact only weakly with the host. Under bias, the weakly coordinated triflate anions are displaced at low voltage, while displacement of the strongly coordinated lithium cations

requires a substantially higher field. This differential mobility is proposed as the molecular origin of the two memory regimes (short-term memory and long-term memory) reported below.

2.2.2 Super Yellow: The Semiconducting Polymer

Super Yellow (commercial PDY-132, Merck) is the semiconducting PPV-family copolymer in the blend. Its role in the proof-of-concept device is to provide the electronic-conductance path whose injection barriers and local doping state can be modulated by mobile ions. The relevant material facts for the main argument are its solution processability in cyclohexanone, its established use in organic light-emitting devices [86], and frontier levels of approximately $E_{\text{HOMO}} = -5.4 \text{ eV}$ and $E_{\text{LUMO}} = -3.0 \text{ eV}$ [145]. The full copolymer name and component-level notes are collected in section A.1.

2.2.3 Hybrane DEO750 8500: The Ion-Transport Polymer

Hybrane is an oxygen-rich hyperbranched polyester-amide [128]. In this chapter it is used as the ion-transport host: ether and ester oxygen atoms provide hard Lewis-base coordination sites for alkali cations, while the hyperbranched architecture suppresses crystallisation and keeps the blend amorphous. Ion motion is therefore treated as segmental-motion-assisted hopping between local coordination environments, in the polymer-electrolyte picture introduced in section 1.6.1. The relevant migration activation scale is of order 50 kJ mol^{-1} for polymer electrolytes near room temperature [87]; the longer discussion of host structure and transport is in section A.1.

2.2.4 Lithium Triflate: The Ionic Salt

LiOTf (lithium trifluoromethanesulfonate, LiCF_3SO_3) supplies the mobile ions. It is compatible with the cyclohexanone-processed *Super Yellow*/*Hybrane* blend and is widely used in polymer-electrolyte and LEC systems [98, 103]. Mechanistically, its value is the contrast between Li^+ , a small hard Lewis acid, and CF_3SO_3^- , a charge-delocalised soft anion. The chapter uses that asymmetry to interpret the two relaxation regimes, while the detailed ion-pairing rationale is kept in section A.1.

2.2.5 Composite Formulation and Preparation

The active layer uses a mass ratio of *Super Yellow*:*Hybrane*:LiOTf = 1:0.30:0.09, processed from cyclohexanone under dry nitrogen. After mixing, the final solution concentrations are 8.72 mg mL^{-1} (*Super Yellow*), 2.62 mg mL^{-1} (*Hybrane*), and 0.78 mg mL^{-1} (LiOTf). The ratio is the optimised formulation reported in the published proof-of-concept work [6]: higher salt contents degraded film homogeneity, whereas lower salt contents reduced the available mobile-carrier density. Detailed solution preparation and optimisation notes are given in section A.1. The same formulation had also been validated in related *Super Yellow*/*Hybrane*/lithium-salt LECs with long electroluminescent operation [104].

2.3 Device Architecture and Fabrication

2.3.1 Device Geometry: The Two-Terminal Vertical Structure

The device adopts a vertical sandwich geometry, in which the active layer is placed between a bottom transparent electrode and a top metallic

electrode (see fig. 2.1). This 2-T architecture is the simplest memristive configuration: it has no gate electrode and needs no transistor to address the device, in contrast to the *three-terminal* (3-T) organic synaptic transistors discussed in section 1.5.2 [105, 113, 114].

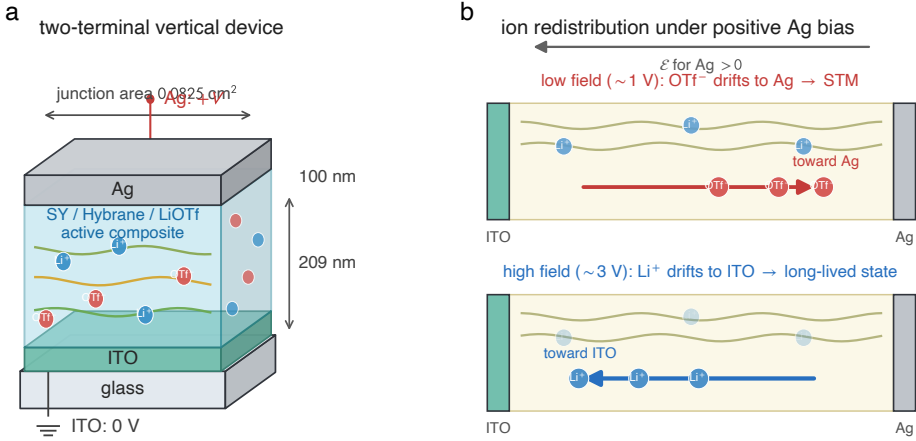


Figure 2.1: Schematic of the two-terminal vertical synaptic device and active-layer mechanism. The device stack is ITO-glass / *Super Yellow*/*Hybrane*/LiOTf composite / Ag; the composite separates the electronic pathway, ion-transport host, and mobile-ion supply. The proposed mechanism assigns the faster STM response to weakly coordinated triflate redistribution and the longer-lived state to displacement of strongly coordinated Li^+ . Schematic generated from the device geometry and mechanism reported in Ref. [6].

The 2-T design reduces fabrication complexity: the device is made by three depositions (ITO substrate, spin-coated active layer, evaporated metal), which makes the same stack compatible with the comparative corpus of Chapter 4. The coexistence of short-term memory and long-term memory without an independent gating signal also supports attribution of the measured memory to the active layer rather than to a gate field acting on a separate channel.

The bottom electrode is a pre-patterned layer of ITO deposited on borosil-

icate glass. ITO combines optical transparency with adequate electronic conductance ($\sigma \approx 10^4 \text{ S cm}^{-1}$) [146], allowing optical inspection of the active layer and characterisation of the device stack by techniques such as UV-Vis spectroscopy and electroluminescence imaging. The work function of ITO ($\Phi_{\text{ITO}} \approx 4.7 \text{ eV}$) is well matched to the ionisation potential of PPV-family semiconductors [147], facilitating hole injection into the active layer.

The top electrode is silver (Ag), deposited by thermal evaporation. Ag was selected over gold for reasons of cost and practical feasibility; its work function ($\Phi_{\text{Ag}} \approx 4.3 \text{ eV}$) creates a modest electron injection barrier with the *Super Yellow* conduction band, consistent with asymmetric injection conditions often observed in LEC-type structures [99]. The junction active area defined by the shadow mask geometry is 0.0825 cm^2 .

2.3.2 Fabrication and Active-Layer Morphology

ITO-patterned glass substrates were cleaned by sequential ultrasonication, dried under nitrogen, and transferred into a nitrogen glovebox. The composite solution was spin-coated at 2000 rpm for 60 s and annealed at 75°C for 3 h. Contact profilometry gave an active-layer thickness of 209 nm at this spin speed, and tapping-mode AFM showed smooth films with RMS roughness in the range 1 nm–3 nm across the investigated processing window [6]. The spin-speed optimisation, profilometry method, AFM parameters, and supporting morphology evidence are reported in section A.3.

The top electrode was 100 nm of Ag deposited by thermal evaporation through a shadow mask. The mask defines a junction area of 0.0825 cm^2 . Detailed cleaning and evaporation parameters are collected in section A.1; the main point for the chapter is that the complete stack is fabricated by solution processing plus one metal evaporation step.

2.4 Electrical Characterisation

Electrical measurements used a Keithley 2450 SourceMeter in a custom probe station. Unless otherwise stated, measurements were made under nitrogen with voltage applied to the Ag top electrode relative to grounded ITO. Protocol-level details are given in section A.2.

The characterisation follows the two-family distinction of section 2.1. Sections 2.4.1 to 2.4.3 report the three dynamical measurements carried forward to Chapters 4 and 5; sections 2.4.4 and 2.4.5 then report the device-specific extended protocols, the EPSC multi-state readout and spike-timing dependent plasticity. The short- and long-term memory regimes (short-term memory/long-term memory) are read off the variable-delay depotentiation measurement and so appear with it in section 2.4.3.

2.4.1 Current-Voltage Hysteresis

A defining electrical response of a memristive device is a current–voltage (I – V) hysteresis loop pinched at zero voltage [59–61]. At $V = 0$, the net current is zero regardless of device history, while the *slope* of the I – V curve at zero voltage (i.e., the instantaneous conductance) depends on the history of applied voltage.

Figure 2.2 shows the I – V curves obtained by applying successive triangular voltage sweeps to the device at a sweep rate of 0.25 V s^{-1} , with the voltage ramped from 0 V to a maximum of 1.2 V and back to 0 V (single polarity, positive). A waiting period of 10 s was observed between successive sweeps to allow partial ionic relaxation. Ten sequential sweeps are shown.

The curves show four features:

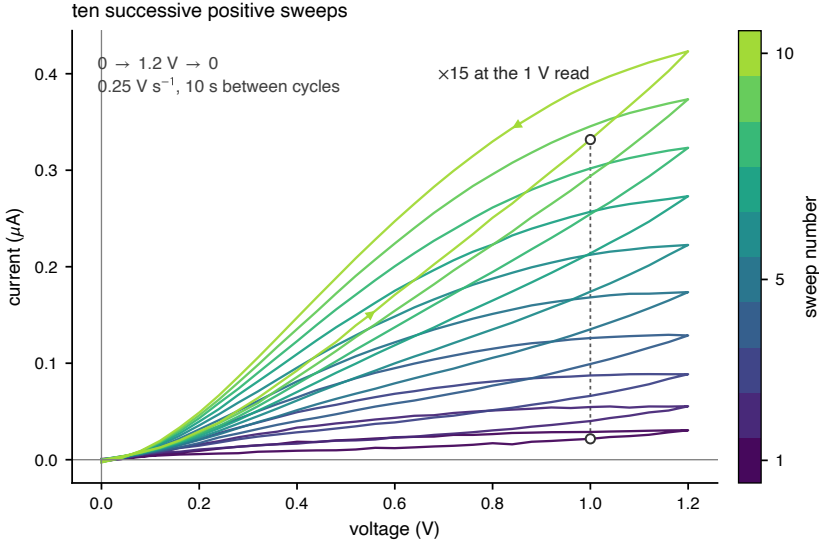


Figure 2.2: Current–voltage hysteresis of the composite memristive device. Ten successive positive triangular sweeps (0 V–1.2 V, 0.25 V s⁻¹, 10 s between cycles) fan out in near-uniform increments, the forward current at 1 V growing roughly 15-fold from the first to the tenth cycle. The clockwise loop closes at the origin (pinched hysteresis) without abrupt switching. Representative SY/Hybrane/LiOTf device (NM_v025) of the published 1:0.30:0.09 formulation, reproducing the hysteresis of Ref. [6].

Progressive, well-resolved increase of conductance. The current at any fixed voltage *increases* with each successive sweep, in well-separated increments that persist across the sequence. At 1 V the forward current grows from 0.022 μA on the first sweep to 0.33 μA on the tenth (an on/off ratio of approximately 15 across ten cycles). The increase is +85 % from cycle 1 to cycle 2 and +13 % from cycle 9 to cycle 10. These distinguishable intermediate levels demonstrate graded analogue operation. They are attributed to progressive ionic redistribution: each cycle produces a slightly more asymmetric configuration and a lower-resistance state.

Hysteresis without current rectification. In all cycles, the return sweep (decreasing voltage) produces lower current than the forward sweep

(increasing voltage) at the same voltage, generating the characteristic clockwise hysteresis loop for positive voltage sweeps. This is distinct from simple rectification (diode behaviour): a diode would produce the same forward and reverse I - V curves; the hysteresis here reflects a time-dependent internal state variable (the ionic configuration) that lags the applied voltage.

Absence of abrupt switching. Unlike digital resistance switching (SET/RESET transitions), in which the resistance changes abruptly by orders of magnitude at a threshold voltage, the conductance here evolves continuously and smoothly with each cycle. This *analogue* gradation matters for the synaptic role outlined in section 1.2.3, since it allows the device state to be tuned to one of many distinguishable intermediate levels instead of being confined to two (or a few) binary states.

Behaviour under negative polarity. Application of negative voltage sweeps (not shown individually in the main figure) produces a qualitatively different hysteresis: the current decreases with successive negative cycles instead of increasing. This has been attributed to electrochemical side-reactions in the vicinity of the ITO electrode under large negative bias, which generate a thin insulating layer at the ITO/active layer interface analogous to the behaviour reported in MEH-PPV-based LECs [103, 106]. This effect is not detrimental to memristive operation when devices are used in the single-polarity positive regime.

2.4.2 Variable-Number-of-Pulses Potentiation

Potentiation by a train of write pulses is the second of the three dynamical measurements introduced at the start of this section. Two complementary experiments are reported: a reversible potentiation/depression cycle that

establishes the analogue, multi-state character of the conductance, and a measurement of the conductance change as a function of the number of applied pulses.

Reversible Potentiation and Depression

The conductance can be set to a specific value by applying a controlled number of voltage pulses, and decreased back toward the initial value by pulses of opposite polarity, as the potentiation/depression experiment in fig. 2.3a shows.

A pulsed voltage protocol was applied: 50 consecutive write pulses of 1 V amplitude and 0.1 s duration, followed by 50 write pulses of -2 V. The conductance was monitored at the peak of each pulse (i.e., at the 1 V and -2 V levels, respectively). This protocol is analogous to the repeated pre-synaptic stimulation of a biological synapse and its subsequent inhibition.

The conductance is continuously potentiated from its initial value by more than 200 % during the 1 V pulse train, and subsequently depotentiated toward the initial value during the -2 V train. The potentiation and depression curves are smooth and monotonic, without abrupt transitions. Return to a value near the initial level after the depression protocol is consistent with reversible ionic redistribution within the measured protocol.

The energy cost of each synaptic event (one write pulse of 1 V, 0.1 s) can be estimated from the average current during the pulse. The resulting energy per event is approximately 50 nJ, or approximately $6 \text{ fJ } \mu\text{m}^{-2}$ of device area (the 0.0825 cm^2 junction). This value is consistent with those reported for organic synaptic devices operating in the picojoule-to-nanojoule range [77], and comparable to the picojoule-to-nanojoule switching energies reported for inorganic memristive devices [68].

Quantitative Pulsed Read–Write Protocol

To quantify the potentiation as a function of pulse number, and to separate the act of writing from the act of reading, a non-perturbing read protocol was adopted; the same protocol is reused for the depotentiation measurement of section 2.4.3. The read–write–wait sequence, detailed in section A.2.2, comprises three steps:

1. **Read.** A reading pulse ($V_{\text{read}} = 0.5 \text{ V}$) is applied to establish the initial conductance G_0 . This reading voltage is intentionally low: the activation threshold for ionic redistribution in this system was determined by independent experiments to lie above 0.7 V [6], so $V_{\text{read}} = 0.5 \text{ V}$ measures the conductance state without perturbing it.
2. **Write.** A train of N_{pulses} write pulses (V_{write} , duration 0.1 s each) is applied.
3. **Wait and read.** After a waiting time t_{wait} under open-circuit (high-impedance) conditions, a second reading pulse is applied to measure the final conductance G_f .

The conductance change is reported as the percentage ratio $\Delta G = (G_f/G_0) \times 100$.

Between experiments, the device is fully reset by connecting the junction to a short circuit for 300 s , which discharges the ionic double layers and restores the homogeneous equilibrium ion distribution. The suitability of this reset protocol was verified by confirming that G_0 returns to its original value within the measurement uncertainty before each new experiment.

Potentiation as a Function of Pulse Number

Figure 2.3b shows the dependence of ΔG on N_{pulses} for $V_{\text{write}} = 1 \text{ V}$ and $t_{\text{wait}} = 150 \text{ ms}$. The conductance ratio increases monotonically with

N_{pulses} , reaching values in excess of 10³% for $N_{\text{pulses}} > 10^3$. The relationship is approximately linear in log-log space for $N_{\text{pulses}} < 100$ and saturates at large pulse numbers, consistent with a scenario in which the degree of ionic asymmetry that can be built up under a given write voltage is bounded by the total number of mobile ionic carriers.

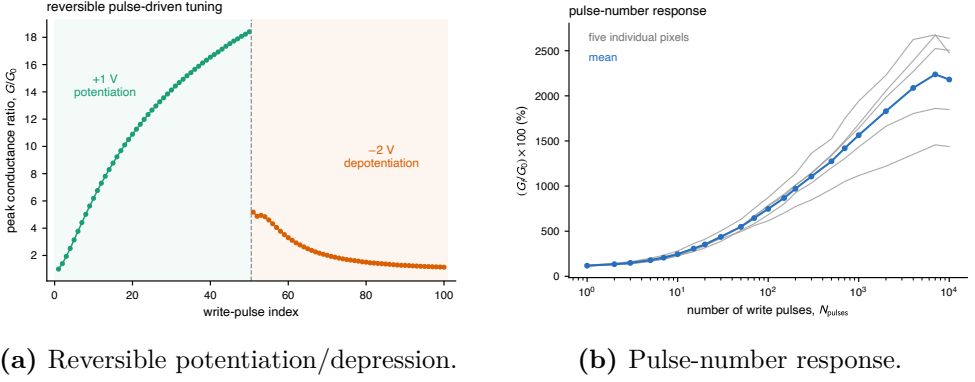


Figure 2.3: Pulse-driven conductance integration in the composite memristive platform. **(a)** Fifty consecutive 1 V write pulses continuously potentiate junction L5 on NM_v055; subsequent -2 V pulses depotentiate it toward the initial state. **(b)** Dependence of the conductance ratio $(G_f/G_0) \times 100\%$ on the number of write pulses for $V_{\text{write}} = 1$ V and $t_{\text{wait}} = 150$ ms. Thin grey lines are five crossbar junctions on the same NM_v055 substrate; the bold line is their mean. The spread therefore measures within-substrate junction variability, not device- or batch-level reproducibility. Data regenerated from the local archive associated with Ref. [6].

Within this archived pulse-train dataset, the low-pulse-count response varies by approximately 10 % across junctions on one substrate: for the five pixels in fig. 2.3b, the median relative standard deviation is 8.9 % and the largest value is 11.8 % over the points with $N_{\text{pulses}} < 100$. The dispersion increases modestly at large pulse numbers. These values establish within-substrate junction repeatability only; they do not estimate independent-device or batch variability.

2.4.3 Variable-Delay Depotentiation: Short- and Long-Term Retention

The third dynamical measurement carried through the thesis probes how a written conductance state fades once the drive is removed: the depotentiation of the device measured as a function of the delay before read-out. This protocol determines whether the device can emulate short-lived and longer-lived synaptic states on experimentally accessible timescales. In the artificial-synapse literature these regimes are commonly labelled short-term memory and long-term memory; in the present system, the longer-lived regime remains on the sub-minute scale and should therefore be understood operationally, not as a claim of indefinitely persistent storage.

The measurement reuses the pulsed read–write protocol of section 2.4.2, with the waiting time t_{wait} now the swept variable and the pulse number held fixed. Figure 2.4 shows the dependence of ΔG on t_{wait} for three excitation conditions: (i) $V_{\text{write}} = 1\text{ V}$, $N_{\text{pulses}} = 10$ (STM, blue); (ii) $V_{\text{write}} = 1\text{ V}$, $N_{\text{pulses}} = 50$ (STM, red); and (iii) $V_{\text{write}} = 3\text{ V}$, $N_{\text{pulses}} = 10$ (LTM, black).

For short waiting times ($t_{\text{wait}} < 500\text{ ms}$ for STM conditions, $t_{\text{wait}} < 2\text{ s}$ for LTM conditions), the conductance first *increases* slightly relative to the value immediately after the write pulse train. This counterintuitive behaviour is *not* ascribed to ionic inertia: ion transport in the polymer matrix is strongly overdamped, so any momentum imparted by the field relaxes on a timescale orders of magnitude shorter than the observed enhancement. It is instead attributed to the continued redistribution of ions after the drive is removed. At the end of the pulse train the ionic configuration and its self-consistent space-charge field have not yet reached their quasi-steady profile; the residual internal field, together with the slow advance of the electrochemically doped region near the

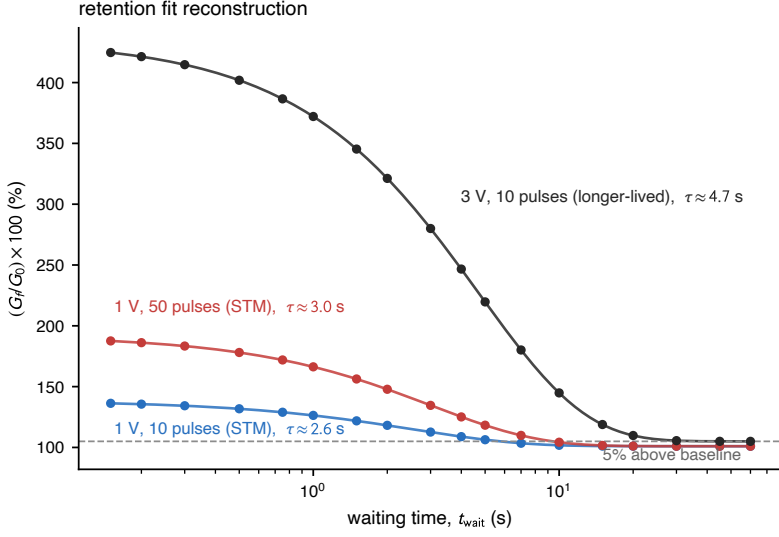


Figure 2.4: Memory retention kinetics. Conductance ratio $(G_f/G_0) \times 100\%$ versus waiting time t_{wait} for STM conditions (1 V; 10 and 50 pulses, blue/red) and the longer-lived regime denoted LTM (3 V, 10 pulses, black). Solid lines are standard-exponential reconstructions of the form in eq. (2.1); the fit parameters are summarised in section A.2.4. Data and fit parameters from Ref. [6].

electrode, continues to raise the conductance for a brief interval before the diffusion-driven relaxation sets in. An analogous transient enhancement following a stimulus train has been described as “post-tetanic potentiation” in biological synaptic preparations [42, 43].

For longer waiting times, the expected exponential relaxation of conductance is observed as the ionic configuration relaxes back toward equilibrium by thermal diffusion. The fit record summarised in section A.2.4 reports least-squares fits in IGOR Pro 8 to a standard exponential:

$$\frac{G_f(t_{\text{wait}})}{G_0} = R_\infty + A \exp\left(-\frac{t_{\text{wait}}}{\tau}\right) \quad (2.1)$$

where τ is the characteristic relaxation time, A is the decaying amplitude,

and R_∞ is the long-waiting-time conductance ratio. Equivalently, the published proof-of-concept fit corresponds to a stretched-exponential model with the stretch exponent fixed at $\beta = 1$. No free β , confidence interval, or goodness-of-fit statistic was reported for these three traces, so they establish only the two experimentally resolved relaxation times, not a fitted distribution of relaxation barriers; the free-exponent Kohlrausch descriptor is reserved for chapter 4, where the DELAYTIME archive is refitted consistently across the corpus.

The fitted characteristic times are $\tau_S = 2.57$ s for the 1 V, 10-pulse STM trace, $\tau_S = 3.01$ s for the 1 V, 50-pulse STM trace, and $\tau_L = 4.73$ s for the 3 V, 10-pulse LTM trace. The corresponding offsets and amplitudes are tabulated in section A.2.4. The retention time (defined operationally as the waiting time at which G_f/G_0 returns to within 5 % of the baseline, $G_f/G_0 < 1.05$) is 10 s–15 s for STM and exceeds 45 s for the longer-lived regime labelled LTM. At event intervals shorter than these windows, the device state can retain information from multiple sequential inputs. The measured retention remains far shorter than that of a conventional non-volatile memory.

The molecular mechanism underlying the STM/LTM distinction is discussed in detail in section 2.5.

2.4.4 Excitatory Post-Synaptic Current

The EPSC measurement is a standard protocol for artificial synaptic devices, quantifying the number and stability of distinguishable conductance states accessible to the device [23, 90]. The protocol employed here closely mimics the biological experiment in which a patch-clamped neuron is subjected to a defined sequence of pre-synaptic inputs and the post-synaptic current is monitored.

The measurement protocol is as follows (fig. 2.5a):

1. **Initialisation.** The device is initialised to its ground state S_0 by applying a fixed read-out bias $V_0 = 1$ V for 1 s. This bias differs in purpose from the sub-threshold probe ($V_{\text{read}} = 0.5$ V) used for the non-perturbing read-out of sections 2.4.2 and 2.4.3: the EPSC states are defined operationally by the current response at a single common read-out bias against which all states S_0 – S_4 are referenced, not by a passive probe required to leave the state untouched.
2. **Excitation.** A short positive write pulse of 2 V for 50 ms is applied; the device is subsequently held at 1 V for 1 s, during which the conductance spontaneously relaxes to a new excited steady state S_1 . The same sequence is repeated to produce a second excited state S_2 .
3. **Inhibition.** Two negative write pulses of -2.5 V for 50 ms are applied, producing inhibitory states S_3 and S_4 .

The four excited states are readily distinguishable from each other and from the ground state. Defining the EPSC readout ratio $R_n = I_{S_n}/I_{S_0}$ as the signed current response of state S_n normalised to that of the ground state S_0 under the fixed read-out bias used in the experiment, the values obtained are: $R_1 = 7.33$, $R_2 = 15.55$, $R_3 = -3.72$, $R_4 = -16.97$. (The ratio is formed from signed currents, not conductances, so the inhibitory states carry a negative sign; conductance itself remains non-negative throughout.) The large absolute ratios (up to ~ 16 -fold increase or ~ 17 -fold decrease relative to S_0) provide a wide margin for reliable state identification in a practical device, even in the presence of noise. The four-state sequence is a representative demonstration, not an upper bound: given the continuous analogue tunability established in section 2.4.2, many more intermediate levels could in principle be resolved by the same protocol within the available margin.

The states S_1 and S_2 are reached by *excitatory* (positive polarity) stim-

ulation, while S_3 and S_4 are reached by *inhibitory* (negative polarity) stimulation, in direct analogy with long-term potentiation (LTP) and long-term depression (LTD) at biological synapses. The coexistence of both excitatory and inhibitory states in the same 2-T device, without an independent gate terminal, distinguishes the present platform from the 3-T organic implementations discussed in section 1.5.2.

2.4.5 Spike-Timing Dependent Plasticity

Spike-timing dependent plasticity is a timing-dependent Hebbian learning rule: it implements Hebb’s local correlation principle [41] in a form that was later observed experimentally in biological preparations [54, 55]. Here the timing delay is defined as $\Delta t = t_{\text{post}} - t_{\text{pre}}$. With this convention, if the presynaptic neuron fires shortly *before* the postsynaptic neuron (causal ordering, $\Delta t < 0$), the synapse is potentiated (LTP); if the presynaptic neuron fires shortly *after* the postsynaptic neuron (anti-causal ordering, $\Delta t > 0$), the synapse is depressed (LTD). The magnitude of the change decays exponentially with $|\Delta t|$. This temporal asymmetry is essential for the synaptic encoding of causal relationships in neural circuits and underlies the ability of STDP-based learning rules to implement unsupervised feature extraction and pattern recognition [78].

Experimental Protocol

The STDP experiment requires the emulation of pre- and postsynaptic spike waveforms by voltage pulses applied at the two electrodes of the device. In a 2-T device, both pulses must be applied at the same physical terminal (the Ag electrode), so the experimentally measured total voltage $V(t)$ is the algebraic sum of the presynaptic and postsynaptic voltage waveforms separated by the delay Δt :

$$V(t) = V_{\text{pre}}(t) + V_{\text{post}}(t + \Delta t) \quad (2.2)$$

Each spike waveform consists of a unipolar triangular pulse with maximum amplitude V_{sp} and duration 1.2 s. The delay Δt was varied from -600 ms to 600 ms. The amplitude $V_{sp} = 1.55$ V was chosen to be large enough to activate ionic migration (above the threshold of approximately 0.7 V) while remaining below the onset of electrochemical degradation of the active layer (conservatively estimated at 2 V for the pristine device).

For $\Delta t < 0$ (pre fires before post), the two spike waveforms partially overlap in time with additive polarity, producing a composite pulse whose peak voltage exceeds V_{sp} and which is directed in the positive sense. The ions then redistribute toward the ITO electrode, the conductance increases, and the device records a net synaptic potentiation.

For $\Delta t > 0$ (post fires before pre), the overlap produces an anti-causal summation. Depending on the exact waveforms, the effective net voltage can be of reduced amplitude or of reversed polarity relative to the causal case, so the ionic redistribution is either weaker or reversed and the conductance decreases, in the form of a net synaptic depression.

STDP Curve and Fitting

Figure 2.5b shows the measured STDP function: the percentage conductance change ΔG plotted against the timing delay Δt . Under the sign convention above, the measured kernel is asymmetric Hebbian: negative Δt (causal) produces positive ΔG (potentiation), and positive Δt (anti-causal) produces negative ΔG (depression). Each branch (potentiation for $\Delta t < 0$, depression for $\Delta t > 0$) is fitted separately, as a function of the absolute delay $|\Delta t|$, by a decaying exponential:

$$\Delta G(\Delta t) = A_{\pm} \exp\left(-\frac{|\Delta t|}{\tau_{\pm}}\right) + G_0 \quad (2.3)$$

where $A_{\pm}$ and $\tau_{\pm}$ are the branch-specific amplitude and time constant (+ for the potentiation branch, – for the depression branch) and G_0 is a constant offset. The best-fit values of $\tau_{\pm}$ for both the potentiation and depression branches lie in the range 85 ms–90 ms. This is in close quantitative agreement with the characteristic timescale of biological STDP, approximately 100 ms [55, 58]. The device therefore reproduces the STDP function with biologically realistic time constants.

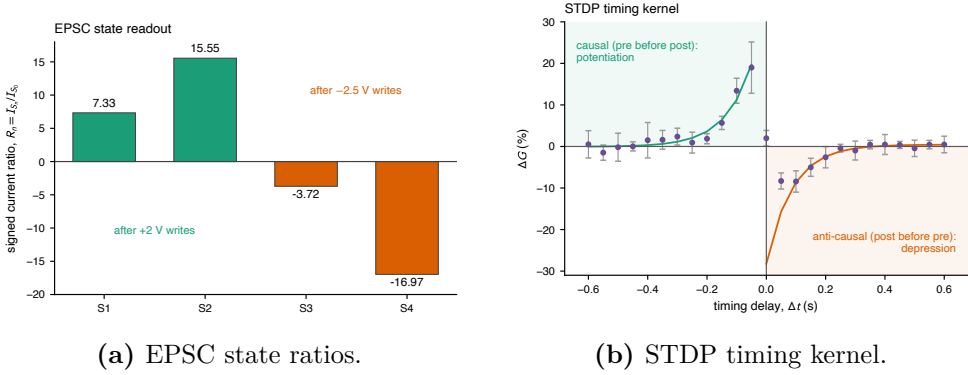


Figure 2.5: Compact summary of the extended synaptic measurements. **(a)** Signed EPSC readout ratios $R_n = I_{S_n}/I_{S_0}$ for the four demonstrated states, reconstructed from the published values; the recovered archive does not resolve their specimen identifier. **(b)** Spike-timing dependent plasticity: mean percentage conductance change ΔG across five junctions on the NM_v055 substrate as a function of the pre/post spike-timing delay Δt , with each branch fitted to the decaying exponential of eq. (2.3); error bars are one standard deviation across junctions. Data and reported state ratios from Ref. [6]; STDP curve regenerated from the local NM_v055 Day19 archive.

That the learning rule is *asymmetric Hebbian* reflects the specific voltage waveform design, the stated definition of Δt , and the polarity conventions of the electrode configuration. Different waveform designs or an opposite sign convention would be expected to produce differently labelled

STDP kernels, offering an additional degree of tunability in the design of hardware neural network architectures built from these devices.

2.5 Theoretical Framework: Ion Transport and the Origin of Memristance

The injection and bulk-transport physics of the composite are described by two limiting-case models inherited from the polymer light-emitting electrochemical cell (LEC) literature: an *electrodynamic* (injection-limited) model [99, 100] and an *electrochemical-doping* (ohmic) model [98, 101, 138], which van Reenen *et al.* [102] reconciled as two regimes of the same field-driven ion redistribution. The equilibrium ion distribution, the thermionic-injection description, and the two models in full are set out in section A.4; the focus here is on what they imply for the two observed memory regimes.

For the present SY/Hybrane/LiOTf system, the *Super Yellow* polymer has moderate hole mobility and a moderate injection barrier from ITO, so both mechanisms may contribute simultaneously. The interpretation adopted here associates the ED mechanism with the rapid conductance changes at short timescales (milliseconds to seconds), driven primarily by the fast-moving triflate anions, and the ECD mechanism with the larger, slower changes at longer timescales (seconds to tens of seconds), driven by the slower-moving Li^+ cations; on this reading the hierarchy of timescales is the physical origin of the STM/LTM distinction. The device-level measurements of section 2.4 do not by themselves separate the two contributions: a clean discrimination would require their temperature, film-thickness, or sweep-rate dependence, which is not pursued here. The assignment is therefore a working interpretation, not an established fact. Its falsifiable consequence is that the slow timescale should track the

cation–host coordination strength; Chapter 4 tests this by varying the cation at fixed host chemistry.

2.5.1 Lewis Acid-Base Chemistry and the Two Memory Regimes

The chemical mechanism proposed here for the two memory regimes is the same hard–soft acid–base reasoning developed in sections 1.6.1 and 1.6.4, applied to the specific salt and host of this device.

Within the HSAB framework, the strength of the interaction between a Lewis acid and a Lewis base in a coordination environment is set by the matching of their hardness or softness [89]. The hard–hard interaction between Li^+ (a small, hard Lewis acid with charge density $z/r \approx 1.3 \text{ \AA}^{-1}$ – $1.7 \text{ \AA}^{-1}$) and the ether oxygen atoms of *Hybrane* (hard Lewis bases) is governed predominantly by electrostatic forces, and is therefore thermodynamically strong and geometrically specific. In polyether environments Li^+ is coordinated by four to six oxygen donors (five in the crystalline complex $\text{PEO}_3\text{:LiCF}_3\text{SO}_3$: three ether oxygens and two triflate oxygens), with oxygen–lithium distances of $1.9 \text{ \AA}$ – $2.1 \text{ \AA}$ [127]. This local site *binding* should not be confused with the *migration* activation barrier that governs hopping: the latter is the saddle-point barrier between adjacent sites (of order 50 kJ mol^{-1} , $\approx 20 k_{\text{B}}T$; section 2.2.3) and is therefore smaller than the full coordination well, while still being substantially larger than $k_{\text{B}}T$.

The interaction between the CF_3SO_3^- triflate anion and the host oxygens is, by contrast, much weaker. The diffuse, low-charge-density triflate anion does not engage in strong electrostatic interactions with the hard-base oxygen atoms, and the residual interaction is governed mainly by weak van der Waals forces [129, 148]. The energy barrier for displacement of a triflate anion from one interstitial site to the next is therefore expected

to be substantially lower than the barrier for a Li^+ cation.

This differential activation barrier translates directly into a differential ionic mobility under an applied field, and hence into different voltage thresholds for significant ionic redistribution:

- A modest voltage of 1 V applied across a 209 nm film generates a field of approximately 4.8 MV m^{-1} . In the interpretation adopted here, this is sufficient to drive the more weakly bound triflate-anion redistribution observed experimentally [6]. The fast anionic redistribution produces a rapid conductance change, which then relaxes on the timescale of triflate diffusion through the host. This is the short-term memory regime, with retention 10 s–15 s, driven primarily by triflate relaxation.
- A voltage of 3 V across the same film generates approximately 14.4 MV m^{-1} . This higher field is assigned to the onset of a slower cation contribution, consistent with the stronger Li^+ coordination expected in oxygen-rich polyether hosts [127, 129]. The cation contribution sets in, produces a delayed and stronger ionic redistribution, and relaxes on the much longer timescale of cation re-coordination. This is the long-term memory regime, with retention exceeding 45 s, driven primarily by cation redistribution.

This account yields a specific, experimentally testable prediction. Replacing Li^+ with a monovalent cation of larger ionic radius (Na^+ or K^+) should reduce the charge density, and therefore the hardness, of the Lewis acid. The cation–oxygen coordination with the polyether host should weaken accordingly, and the slow fading-memory timescale of the device should shorten as the cation is moved along the $\text{Li}^+ \rightarrow \text{Na}^+ \rightarrow \text{K}^+$ series. The prediction is tested in Chapter 4 on the main comparative corpus of the thesis: a PEO-based polymer-electrolyte series loaded with

lithium, sodium and potassium triflate salts (PEO/LiOTf, PEO/NaOTf, PEO/KOTf) at fixed polymer and salt mass fractions, complemented by a dedicated PEO/LiOTf concentration sub-study. The prediction is addressed there as a statement about the shift of the measured variable-delay depotentiation timescale when the cation is varied at fixed host chemistry, not as a claim about the full synaptic suite reported in the present chapter.

2.6 Device Stability and Reproducibility

2.6.1 Ambient Shelf Stability

Two distinct notions of device longevity must be kept separate. *Cycling endurance* (the number of write/erase cycles a device sustains before its performance parameters drift outside specified tolerances) governs suitability for the millions-to-billions of weight updates of on-chip training; it is not characterised for the present devices and is left for future work. *Shelf (calendar) stability* (the retention of performance during storage and intermittent operation under ambient conditions) is what is reported here, the more immediately relevant metric at the proof-of-concept stage, where the question is whether the unencapsulated composite survives realistic handling at all.

The devices reported here were characterised at regular intervals over a period of two weeks (~ 300 h) after fabrication, without encapsulation, while stored in ambient air at room temperature. The monitored metrics were the baseline conductance G_0 , the maximum achievable conductance change $\Delta G_{\max}$, and the STM retention time τ_S ; the supporting stability record is collected in section A.3.

All three metrics remain within 20 % of their initial values over the full

two-week measurement period [6]. This stability compares favourably with other organic memristive devices reported in the literature, which typically exhibit significant degradation within hours to days under ambient conditions [119]. The enhanced stability is attributed to the solid-state nature of the composite: unlike devices based on liquid electrolytes or gel polymers, the Hybrane matrix provides a mechanically cohesive environment that protects the ions from evaporation and the polymer films from delamination, consistent with the stability of related Hybrane-based LEC blends [104].

2.6.2 Within-Substrate Junction Repeatability

Reproducibility is a persistent challenge for memristive devices, particularly those based on filamentary switching in inorganic materials, where the stochastic nucleation of conducting filaments produces large device-to-device and cycle-to-cycle variability [71]. Three evidence units must be distinguished here: repeated cycles on one junction, different junctions on one patterned substrate, and independently fabricated devices or batches. The archived proof-of-concept data quantify only the first two.

The strongest repeatability statement available from the archived proof-of-concept data is the one attached to the pulse-number experiment in fig. 2.3b. That dataset contains five crossbar junctions on the NM_v055 substrate, measured under the same 1 V write and 150 ms wait protocol. The shaded band is one standard deviation across those five junctions. For $N_{\text{pulses}} < 100$, the median RSD is 8.9 % and the largest RSD is 11.8 %; the appropriate claim is therefore within-substrate junction repeatability at about the 10 % level in the low-pulse-count regime, not an independent-device reproducibility estimate.

The EPSC readout and STDP measurements are retained as protocol demonstrations, but the recovered archive does not contain a normalized

EPSC population table comparable to the PULSES and DELAYTIME tables used later in the thesis. They are therefore not used to support an independent EPSC RSD claim here. Likewise, the published proof-of-concept data establish same-substrate junction repeatability and two-week shelf stability; they do not constitute a powered device-to-device or batch-to-batch reproducibility study. Campaign-scale reproducibility (a separate metric on a separate axis) and its resolution by a change of ion-transport host are taken up in chapter 3.

2.7 Comparison at the Time of Publication

The historical comparison table reproduced in table A.3 places this platform alongside representative organic and hybrid memristive devices available at the time of the 2022 publication. Two combinations stand out as unrepresented in that set. The first is that the same fully reversible 2-T formulation and architecture support both short-term memory- and long-term memory-labelled regimes: the closest prior report, Liu *et al.* [94], achieved short-to-long-term transitions but through irreversible redox chemistry [116], whereas here the conductance returns to G_0 after the reset protocol. The second is STDP in a 2-T architecture, where the timing rule emerges from the superposition of pre- and postsynaptic pulses at the Ag terminal, not from the gate of a 3-T device [90, 105, 115]. On the operating-regime axes the platform is competitive in unencapsulated shelf lifetime (~ 300 h, roughly an order of magnitude beyond comparable organic devices [119]) but not in energy per event (~ 50 nJ, above CMOS and inorganic-memristor budgets [27, 68]); the full table and a per-axis discussion are given in section A.5.

The comparison above retains the device-level metrics conventional in the organic memristive literature of the early 2020s: reversibility, retention window, energy per event, and operational lifetime. Subsequent work has

expanded the field through organic electrochemical neurons and neuro-morphic circuits [149, 150] and the wider organic-iontronic memristor field [151], so the table locates the 2022 result historically. Within the thesis argument those metrics play a narrower role: they establish that the platform is reversible, dynamically addressable, and stable enough to support the comparative study of Chapter 4 (a PEO-based triflate composition and cation sweep, characterised through the same three common dynamical measurements) and the temporal-computing reinterpretation of Chapter 5. The extended synaptic protocols reported here remain specific to this SY/Hybrane/LiOTf device.

2.8 Summary

The fully reversible 2-T organic memristive platform supports two sets of synaptic measurements with the same polymer-composite formulation and stack. I–V hysteresis, variable-number-of-pulses potentiation, and variable-delay depotentiation carry through the rest of the thesis. EPSC, the separated short-term memory/long-term memory reading of the retention data, and STDP remain specific to this proof-of-concept formulation.

A ternary *Super Yellow/Hybrane*/LiOTf composite satisfies the three functional requirements of section 2.2.1 (electronic pathway, ion-transport medium, ion source), fabricated as a 2-T ITO/composite/Ag sandwich by solution processing and one evaporation step, with a 209 nm active layer below 3 nm RMS roughness. On this common platform the three dynamical measurements carried forward to the comparative corpus were established: analogue I–V hysteresis (forward current at 1 V growing ~ 15 -fold over ten cycles in near-uniform increments); fully reversible pulse potentiation/depression exceeding 200 % at ~ 50 nJ per event, growing past 10³% for $N_{\text{pulses}} > 10^3$; and variable-delay depotentiation resolving

two regimes (short-term memory, $\tau_S = 2.57\text{ s} - 3.01\text{ s}$, retention $10\text{ s} - 15\text{ s}$; long-term memory, $\tau_L = 4.73\text{ s}$, retention $> 45\text{ s}$, from $\beta = 1$ fits). The extended protocols add a four-state EPSC read-out (signed ratios -17 to $+16$) and an asymmetric-Hebbian STDP window ($\tau = 85\text{ ms} - 90\text{ ms}$, biologically realistic [55, 58]). The short-term memory/long-term memory distinction is attributed to the Lewis acid–base mobility asymmetry between the weakly coordinated triflate anion and the strongly coordinated Li^+ cation in the oxygen-rich host, with the injection and bulk-transport physics described by the electrodynamic and electrochemical-doping models in their reconciled form [102] (section A.4). The published shelf record spans $\sim 300\text{ h}$ unencapsulated in ambient conditions (cycling endurance not characterised), while the NM_v055 pulse data show $\sim 10\%$ low-pulse-count variability across five same-substrate junctions (median RSD 8.9%); campaign-scale reproducibility is taken up in chapter 3.

These results establish the SY/Hybrane/LiOTf formulation and 2-T architecture as the proof-of-concept platform and motivate testing cation–polyether coordination in a replicated device corpus. Chapter 4 performs that test on a PEO-based corpus. Composition is the replicated quantitative axis, while host, anion, and cation substitutions remain sample-limited; the data do not establish a host- and anion-independent Li/Na/K law. The EPSC, separated short-term memory/long-term memory retention, and STDP measurements reported here serve only as formulation-level lithium priors in the later chapters. Chapter 5 instead uses behavioural models fitted to the comparative corpus to test the computational value of the measured fading memory.

Chapter 3

Reproducibility, Degradation, and the Fabrication-Record Methodology

3.1 The Reproducibility Problem

Chapter 2 established the memristive response of the *Super Yellow/Hybrane*/LiOTf formulation across assay-specific junctions and reported two-week ambient shelf stability. In the subsequent fabrication campaign, memristive hysteresis became progressively harder to obtain. Successive batches of nominally identical devices shifted from the low-current, high-area, multi-level response of the proof-of-concept platform toward high conductivity, a narrow switching window, and no usable potentiation. This chapter analyses that loss of reproducibility and the subsequent change of host before chapter 4 examines a replicated composition grid.

Individual-device shelf stability and campaign-scale reproducibility measure different processes. The proof-of-concept shelf record retained its

characteristics over days to weeks, as did other finished *Hybrane* devices. The fabrication line nevertheless changed between batches over the following year. This places the degradation in the supply or fabrication process over calendar time, while the finished devices remained stable over the measured shelf interval; section 3.3 quantifies the distinction.

The fabrication and characterisation notes identify devices NM_v026–031 (2021-04-22) as a change point, recorded at the time as “the first devices that changed electrical behaviour”. The contemporaneous hypotheses remained open (“could be evaporation rate . . . the metal used for evaporation, and possible chemical degradation of substances”) and were still unresolved in a note dated 2022-01-31, after nine months of investigation. The analysis uses this documented change point as a pre/post boundary and does not assume a smooth trend. It tests the candidate causes, quantifies the degradation under four controls, and evaluates the change to PEO.

3.2 Eliminating Candidate Causes

The team tested the candidate causes one at a time in batches with the other variables held fixed. Table 3.1 lists each cause, the batches that tested it, and the resulting verdict. This process also motivated the fabrication-record library used later in the thesis, since each test required the process variables to be recorded per device.

Device NM_v067 (2021-10-21) records that a second experimenter, working independently, “has the same problems with devices”. This cross-person result excludes individual technique as the cause and indicates a shared material or process source. The same note proposed the ITO substrate, but the substrate study did not support that explanation. The *Super Yellow* semiconductor was also tested because an old bottle was

Table 3.1: Methods-of-elimination summary for the degradation investigation. Candidate causes were tested in deliberate batches with variables recorded in the fabrication record. After the exclusions, the hyperbranched *Hybrane* host stock remains the candidate examined quantitatively in section 3.3.

Candidate cause	Crossed as (batches)	Verdict
Operator / technique	cross-person check (v067): “Lorenzo has the same problems with devices”	Eliminated. The same result occurred with a second experimenter, excluding one operator’s technique.
Illumination (light vs dark)	v015–019, vt001–006, v041–044, v053–056	Excluded (no systematic recovery of the window).
Storage / measurement atmosphere (glovebox / air / vacuum)	v015–019, v088–099	Excluded.
Baby-chamber transport (full 2 × 2 on Au)	v083–087, v038–040	Excluded.
Substrate (ITO) lot / colour	v079–082; substrate-difference study	Not borne out (a one-off ITO guess by a colleague did not survive the substrate comparison).
Solvent (cyclohexanone) lot	v079, v091	Excluded.
Evaporation rate / silver holder	v045–048, v106–113	Recovery <i>attempted</i> by reverting to the “year-ago” holder; not reproducibly restored.
Annealing regime	late hot-anneal batches	<i>Partial</i> window recovery (section 3.3); consistent with the chemical hypothesis but insufficient as a process correction.
<i>Super Yellow</i> stock (old vs new)	v045–048 (“more viscous ... maybe bonds degradation in the old one”)	Eliminated as primary (old vs new <i>Super Yellow</i> gave no significant change, and fresh <i>Super Yellow</i> continued to be purchased while the window loss persisted).
<i>Hybrane</i> host stock	residual after the above	Remaining candidate , examined quantitatively and mechanistically below.

noticeably “more viscous . . . maybe because of bonds degradation in the old one”. The old-versus-new *Super Yellow* comparison produced no significant change, and devices retained the reduced window while fresh *Super Yellow* was used. After illumination, atmosphere, transport, substrate, solvent, evaporation, and semiconductor were tested and excluded, the hyperbranched *Hybrane* host stock remained the candidate consumable that had aged between batches. The remainder of the chapter tests that attribution quantitatively and proposes a candidate chemistry.

3.3 The Quantitative Degradation Signal

The device archive provides a quantitative test of the degradation recorded in the laboratory notes. The analysis controls for the sampling and protocol changes within the campaign and is reproduced in full by `scripts/bridge_hybrane_peo_reproducibility.py`; it mirrors the slicing of the group’s interactive feature-explorer tool.

Four controls, each shown to matter. The analysis rests on four controls, each verified to change the conclusion if dropped: *per-device weighting* (the pixels-per-device count falls with date, Spearman $\rho = -0.80$, so every feature is reduced to one value per device before any test); a *standard-protocol corpus* ($n \approx 69$ devices, lead mass ratios, excluding the invasive recovery experiments); *sweep-amplitude matching* (the window features are read in the ~ 3 V stratum where they are expressed, the conductivity features in a tightly matched ~ 1.2 V band); and a *pre/post-inflection step test* fixed at the documented NM_v026 inflection, with every conductivity trend reported with and without the deliberately-reverted recovery batches. The four controls are stated in full, with the supporting correlations, in section B.1.

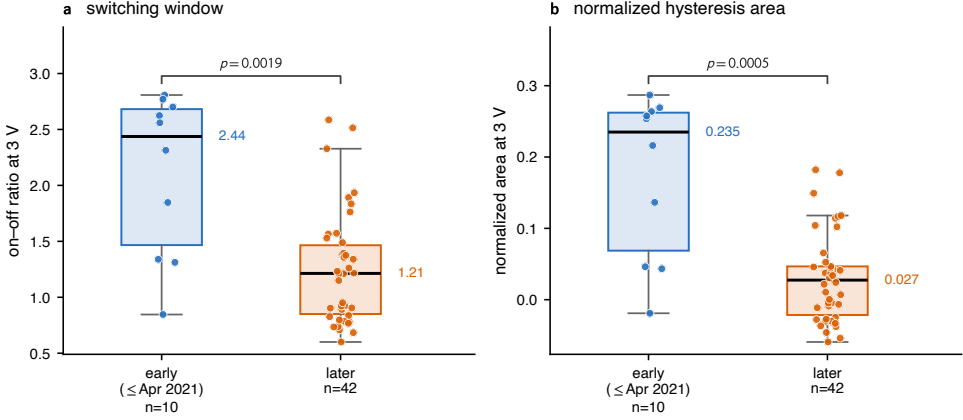


Figure 3.1: Switching-window reduction across the campaign on the standard *Super Yellow/Hybrane*/LiOTf/Ag corpus, read at the ~ 3 V stratum where the window is expressed, one value per device. (a) On–off ratio and (b) normalized hysteresis area, both for devices fabricated before the NM_v026 change point ($\leq$ Apr 2021) versus later. Boxes are quartiles with the median annotated beside each box; points are individual devices; the bracket gives the two-sided Mann–Whitney p . The on–off ratio decreases by roughly a factor of two and the area by an order of magnitude. Reproduced by `scripts/bridge_figures.py` (F1).

Principal result: the switching window decreases at ~ 3 V. On the standard corpus, per device, the switching window measured in the ~ 3 V stratum decreases across the change point (fig. 3.1, table 3.2). The median on–off ratio falls from 2.44 (early, $n = 10$) to 1.21 (later, $n = 42$; Mann–Whitney $p = 0.0019$); the median normalized hysteresis area falls from 0.235 to 0.027, an order of magnitude ($p = 0.0005$); and the potentiation, measured as the per-cent change in the maximum-voltage current over a sweep, changes from +17.9 % to -2.1 % ($p = 0.013$). The on–off ratio, hysteresis area, and potentiation thus identify the same loss of switching function. The magnitude agrees with the team’s recollection of an early on–off near 2.5 falling to the 1–1.7 range later, now tied to the documented change point.

Table 3.2: The degradation as a multi-feature signal on the standard corpus, per device. *Window reduction* is read at ~ 3 V as an early-vs-later step at the NM_v026 change point (Mann–Whitney). *Ohmic drift* is read at matched ~ 1.2 V as a per-device Spearman correlation with fabrication date, reported with all dates and with the recovery tail ($\geq 2022-03$) dropped; the correlation increases when the reverted batches are removed, as expected if they are partial recoveries. Health flags are included and discussed in the text.

Window reduction @ ~ 3 V (early $\leq$ Apr 2021, $n = 10$ vs later, $n = 42$)			
Feature	early	later	Mann–Whitney
on–off ratio	2.44	1.21	$p = 0.0019$
normalized area	0.235	0.027	$p = 0.0005$
% change in max-V current	+17.9 %	–2.1 %	$p = 0.013$
Ohmic drift @ matched ~ 1.2 V (per-device Spearman vs date, $n = 64$)			
Feature	all dates	drop recovery tail	
current at max V	$\rho = +0.45$ ($p = 0.0002$)	$\rho = +0.55$ ($p < 10^{-4}$)	
hysteresis area ($V \cdot \mu A$)	$\rho = +0.57$ ($p < 10^{-4}$)	$\rho = +0.69$ ($p < 10^{-4}$)	
current difference at on–off	$\rho = +0.51$ ($p < 10^{-4}$)	$\rho = +0.63$ ($p < 10^{-4}$)	
Health flags (per-device, ~ 1.2 V, $n = 64$)			
is_broken <i>decreases</i> ($\rho = -0.40$, $p = 0.001$), a handling metric, <i>not</i> degradation;			
is_saturated (fails to potentiate) <i>rises</i> ($\rho = +0.27$, $p = 0.03$).			

Mechanism: ohmic drift at matched ~ 1.2 V. Conductivity provides a complementary signal, read in the tightly matched ~ 1.2 V band so that the sweep-amplitude confound is held fixed. The absolute current rises with fabrication date: per device, the current at maximum voltage correlates with date at $\rho = +0.45$ ($p = 0.0002$), the raw hysteresis area at $\rho = +0.57$, and the current difference at the on–off point at $\rho = +0.51$ (all $n = 64$; fig. 3.2a, table 3.2). These correlations increase when the deliberately reverted recovery batches ($\geq 2022-03$) are dropped (to $\rho = +0.55$, $+0.69$, and $+0.63$ respectively), as expected if those late batches are partial recoveries. Because the amplitude is held within a narrow band, the trend indicates more ohmic conduction independently of sweep range. The reduction in relative hysteresis and the increase in

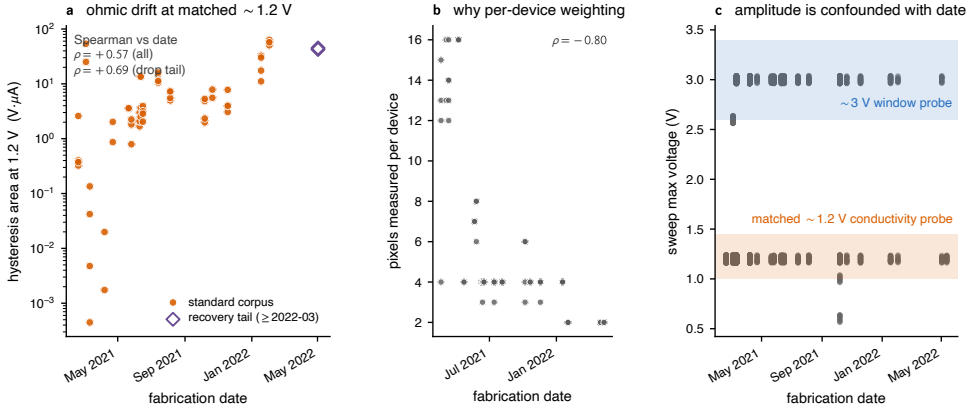


Figure 3.2: Mechanism and methodology. (a) Per-device hysteresis area read at the matched ~ 1.2 V band rises with fabrication date (Spearman $\rho = +0.57$ over all dates, strengthening to $+0.69$ when the recovery-tail batches, open diamonds, $\geq 2022-03$, are dropped; the statistics use all $n = 64$ devices, while the log axis cannot display the few non-positive areas). (b) The number of pixels measured per device falls over the campaign ($\rho = -0.80$), which is why every statistic is computed per device rather than per pixel. (c) Sweep amplitude is itself confounded with date: the matched ~ 1.2 V conductivity probe and the ~ 3 V window probe (shaded) are the two strata used throughout, never pooled. Reproduced by `scripts/bridge_figures.py` (F2).

absolute current are two measurements of the same degradation. The magnitudes are reported as correlations and medians because absolute current depends on sweep amplitude.

Interpretation of the health flags. The fraction of broken devices would mischaracterize the degradation. The curve-level `is_broken` flag (erratic or open-circuit behaviour caused by contact or handling problems) decreases over the campaign ($\rho = -0.40$, $p = 0.001$) as handling and contacting improved. The `is_saturated` flag (a curve whose maximum current fails to exceed the previous one, indicating loss of potentiation) rises with date ($\rho = +0.27$, $p = 0.03$). The two flags are reported separately because they measure different processes.

3.3.1 A Candidate Chemistry: Moisture-Driven Hydrolytic Aging of the *Hybrane* Stock

The available record supports attribution to the stock but cannot establish a chemical mechanism. No electrochemical impedance spectra were recorded on a *Hybrane* device; no gel-permeation chromatography, nuclear-magnetic-resonance, or Karl-Fischer analysis was performed on the reagent stock; and the aged stock no longer exists in its degraded state. The following mechanism is therefore a hypothesis accompanied by experiments that could test it.

The hypothesis. *Hybrane* is a hyperbranched polyester-amide [128, 136]: ester- and amide-rich, and strongly hygroscopic. The leading hypothesis is that, over roughly a year in the bottle, the stock takes up atmospheric water and slowly hydrolyses (chain scission), and that this process could account for the measured electrical signatures without requiring a separate finished-device ageing mechanism.

- **Ohmic drift** (current $\uparrow$ at matched 1.2 V): absorbed water together with polar scission products raises the baseline ionic and leakage conductivity and plasticises the matrix.
- **Window reduction** (on-off $\rightarrow \sim 1.2$, area $\rightarrow \sim 0.03$, $\% \Delta I \rightarrow -2\%$): in the metastable-polarisation model of chapter 2, memristance requires a matrix that retains a field-displaced ionic profile. A plasticised, lower-molecular-weight, water-laden host would relax more rapidly, reducing the loop area and potentiation across successive sweeps.
- (*secondary*) acidic hydrolysis products could partially dope the PPV-type *Super Yellow*, adding a fixed, irreversible conductance, a

second route to the “high-conductivity, undesirable” state recorded in the notes.

Internal support. Three observations are consistent with the hypothesis but do not establish it. First, among late devices, annealing at $\geq 150^\circ\text{C}$ increases the median on–off ratio at $\sim 3\text{ V}$ to 1.95 ($n = 6$), compared with 1.22 for standard 75°C late devices ($n = 44$; one-sided Mann–Whitney $p = 0.024$), near the early-standard reference of 1.74 (fig. 3.3). Removal of absorbed water and film densification could account for this recovery. The test remains limited by six elevated-temperature devices, co-varying process changes, and measurement-pin issues in two devices. Second, the combination of ohmic drift, window reduction, and loss of potentiation is consistent with a water-plasticised, ion-leaky film. Third, ester and amide linkages make this material class susceptible to hydrolytic chain scission, and its polar, hyperbranched architecture is hygroscopic [128, 136].

Confirming experiments (future work). The hypothesis would be tested directly by Fourier-transform infrared spectroscopy (loss of the ester $\text{C}=\text{O}$ band, growth of $-\text{OH}$) and gel-permeation chromatography (a molecular-weight drop) on *retained aged versus fresh* stock; by Karl-Fischer titration or thermogravimetry for water content; and by impedance spectroscopy to show that the ionic resistance fell. None were run on the *Hybrane* corpus, so the hypothesis stays open, and may never fully close, because it concerns a consumable that no longer exists in its aged state.

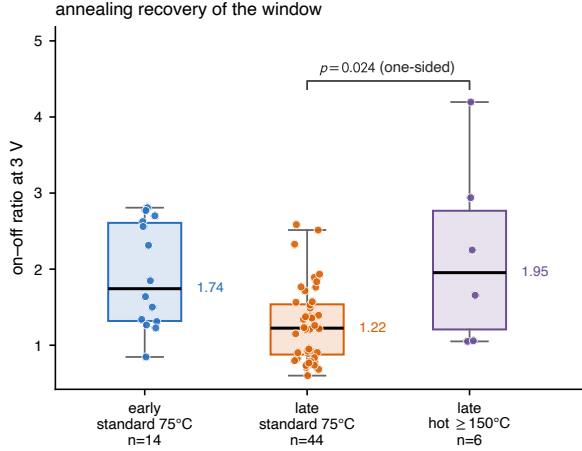


Figure 3.3: Annealing-recovery test of the hydrolysis hypothesis (*Hybrane*/Ag, on-off ratio at ~ 3 V, per device). Among late devices, annealing at $\geq 150^\circ\text{C}$ increases the median from 1.22 to 1.95, near the early-standard reference of 1.74; one-sided Mann–Whitney $p = 0.024$. Interpretation is limited by $n = 6$ high-temperature devices, co-varying process changes, and pin issues in two devices. Reproduced by `scripts/bridge_figures.py` (F4).

3.4 Change to a PEO Host

The fabrication programme changed the ion-transport host after attempts to recover the *Hybrane* line. Replacing hyperbranched *Hybrane* with linear poly(ethylene oxide) (PEO), the host used in the comparative study of chapter 4, produced a larger switching window and multi-year reproducibility. Laboratory notes first record PEO in use by 2021-12 (NM_v083).

Switching-window comparison. At a matched ~ 3 V sweep, comparing valid (non-broken, non-saturated) devices per device, the PEO host gives a larger switching window than contemporaneous *Hybrane* devices: median normalized area 0.26 (*Hybrane*) $\rightarrow$ 0.42 (PEO), Mann–Whitney $p = 0.018$; median on-off ratio 2.45 $\rightarrow$ 4.91 (fig. 3.4a,b).

Sustained temporal reproducibility. The PEO/Ag platform retained a usable switching window over roughly two and a half years (2022–2024, $n = 46$ devices): the per-device on–off ratio at $\sim 3\text{ V}$ had a median of 2.61, with yearly medians of 3.3/2.6/1.9 and no significant downward trend over time. The *Hybrane* platform had a median on–off ratio of 1.35 after its stock-associated window reduction (fig. 3.4c). The PEO record shows no corresponding supply-degradation trend over the measured interval.

Meaning of “reproducible”. Here, reproducibility refers to the absence of a stock-associated reduction between batches over several years. It does not imply superior within-batch device uniformity: at matched $\sim 3\text{ V}$, the device-to-device coefficients of variation differ only modestly (PEO ≈ 0.38 vs *Hybrane* ≈ 0.47). The evidence for PEO is its larger switching window and the absence of the supply-degradation trend over the multi-year record. Changing the suspected *Hybrane* reagent removed the trend after *Super Yellow* had been eliminated as its source (section 3.2).

3.5 Methodological Outputs

The investigation required four tools that were not available at the start of the campaign: a fabrication record, a normalized characterization procedure, an automated feature pipeline, and trend-discovery software. The same tools produce the quantitative results in the comparative and temporal-computing chapters. Five outputs are retained from this work.

- **A fabrication-record standard.** The investigation produced a per-device library schema for storage atmosphere, illumination, transport, reagent and solvent lots, operator, evaporation profile, and free-text fabrication and characterisation notes. This record

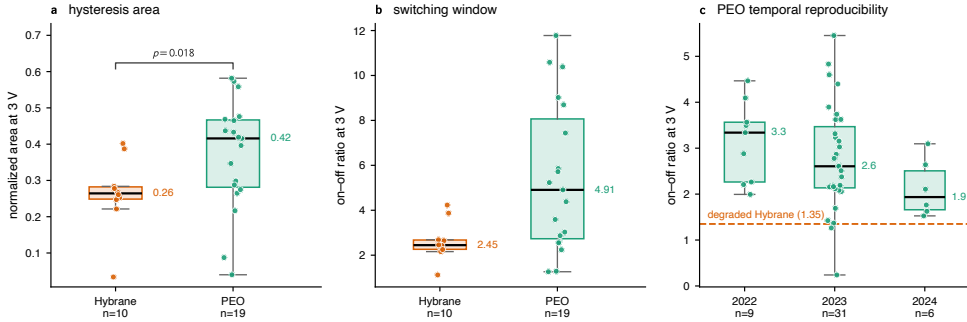


Figure 3.4: Switching-window response after changing to a PEO host (silver electrode, ~ 3 V, per device). (a) Normalized hysteresis area and (b) on-off ratio are larger for PEO than for contemporaneous valid *Hybrane* devices ($p = 0.018$ for area). (c) The per-device PEO on-off ratio remains at a usable level (median ≈ 2.6) across 2022–2024 without a significant downward trend; the dashed line gives the degraded-*Hybrane* level (1.35). Reproduced by `scripts/bridge_figures.py` (F3).

enabled the exclusions in section 3.2 and is retained for every device in the archive.

- **A normalized electrical-characterization procedure.** Section 3.3 shows that sweep amplitude affects apparent conductivity and whether a switching window is expressed. The campaign therefore converged on fixed, documented drive and read protocols that support comparisons between devices and batches.
- **An automated feature pipeline.** The raw instrument traces are reduced to per-device, per-pixel, and per-curve feature tables by a dedicated pipeline; every feature used in this chapter’s panel, and in chapter 4 and the temporal-computing chapter, is computed there.
- **Trend-discovery and curation tooling.** An interactive feature explorer identified the reported trends, whose slices are reproduced by the four controls of section 3.3; a curation tool produces the screened device set used in the analysis.

- **Figure generation.** A thesis-figure pipeline renders the quantitative figures from the same feature tables, so that every plotted number is traceable to the archive.

The comparative study of chapter 4 and the data-driven temporal-computing analysis use this same apparatus.

3.6 Reconciliation of Stability and Reproducibility

The shelf-stability result of chapter 2 and the campaign-scale loss of reproducibility concern the two different measurements defined in section 3.1.

- The published proof-of-concept shelf record was stable on the measured $\sim$ two-week ambient scale [6]. A separate multi-day study likewise shows that finished *Hybrane* devices retain their characteristics over weeks to months. These measurements associate the campaign-scale change with the supply or fabrication process, while finished devices remained stable over the measured intervals.
- The *batch-scale reproducibility of the Hybrane platform over the fabrication campaign* is a different metric, and it declined over the following year as quantified in section 3.3.

Together, the records describe short-interval shelf stability within the proof-of-concept platform and a fabrication line over a year. The exact specimen roster of the published shelf record is not resolved in the recovered archive, and the campaign-scale reproducibility axis was not measured in the proof-of-concept study.

After the switch to PEO, replicated measurements of the volatile dynamics became possible. Chapter 4 uses the *Super Yellow*/PEO/LiOTf system

on silver as a composition grid, measured with the fabrication-record standard, normalized protocols, and feature pipeline developed during the reproducibility investigation.

Chapter 4

Compositional and Chemical Control of Volatile Polymer-Electrolyte Memristive Dynamics

4.1 Introduction and Scope

Chapter 2 established the memristive response of the *Super Yellow/Hybrane*/LiOTf formulation and 2-T architecture across assay-specific junctions. This chapter measures how changes in composition, electrolyte chemistry, electrode, and drive protocol alter the volatile dynamics of the same three-component architecture in a broader device corpus.

Three families of measurement recur on the same 2-T geometry throughout this chapter and the next: (i) current–voltage hysteresis, which reports the steady-state switching window; (ii) potentiation as a function of the number of applied write pulses, which reports how the conductance is built up; and (iii) depotentiation as a function of the delay before read-out,

which reports how a written state fades. Introduced as a named set in chapter 2, they are the shared, abundant basis of the comparative work. The richer proof-of-concept synaptic protocols of chapter 2 (excitatory post-synaptic current, separated short- and long-term memory, and spike-timing dependent plasticity) are *not* repeated here; where invoked, they serve only as formulation-level prior evidence.

Scope of the comparisons. The *Super Yellow*/PEO/LiOTf composition grid contains typically two to four nominally identical devices per cell and supports quantitative comparisons. After data-quality screening, the host, anion, and cation contrasts contain at most one or two devices per matched cell and are used only as an illustrative survey. The electrode comparison is likewise illustrative and is available only at the lead composition (section 4.8). Drive protocol is analysed separately because it changes the apparent fading-memory timescale and therefore limits comparisons between protocols.

4.2 Materials, Devices, and Measurement Protocol

The devices retain the architecture of chapter 2: a vertical two-terminal sandwich of ITO (bottom, transparent), a solution-processed composite active layer, and a thermally evaporated silver top electrode, with a junction area of $8.25 \times 10^{-2} \text{ cm}^2$. *Super Yellow* is retained as the semiconducting component in every device of this chapter, so that the electronic-transport role is held fixed and the variation is confined to the ion-transport phase.

Two design axes are varied. The *composition* axis varies the mass fraction of the ion-transport polymer poly(ethylene oxide) (PEO) and of the lithium triflate salt (LiOTf) relative to the semiconductor. The *chem-*

istry axis substitutes the ion-transport host, the salt anion (triflate or bis(trifluoromethanesulfonyl)imide, TFSI), and the alkali cation (Li^+ , Na^+ , K^+ , as the triflate or TFSI salt). The comparative work replaces the hyperbranched polyester-amide *Hybrane* host of the proof-of-concept device (chapter 2) with two hosts used here at the same level: linear PEO and the hyperbranched polyether trimethylolpropane ethoxylate (TMPE), so that the host axis contrasts a linear against a branched polyether at matched chemistry. Every device reported *quantitatively* uses the silver electrode. A separate gold-electrode corpus at the lead composition provides a third, illustrative axis (section 4.8). It compares an active with an inert electrode, measures drive-polarity dependence on gold, and provides an independent-electrode check on the chemistry comparisons.

Control for film thickness. Adding ion-transport polymer thickens the solution-processed film, so most higher-PEO cells were spun faster to reduce thickness differences across the grid (1500 rpm–3500 rpm). This compensation was incomplete and was not applied uniformly to every comparison batch. In the silver composition series ($n = 30$), median thickness rises from ≈ 227 nm at PEO 0.3 to ≈ 298 nm at PEO 1.2 (Pearson $r \approx 0.68$; fig. 4.1). Several controls separate this covariation from the composition result. The reported responses are dimensionless ratios or timescales; activation voltage remains within 2.2 V–2.4 V across the full $2.6\times$ thickness range ($r \approx -0.06$); and a 38% within-cell thickness spread leaves the fading-memory time essentially unchanged. The partial correlation between thickness and $\log t_{1/2}$, controlling for PEO, is $\approx +0.05$, whereas the association with composition, controlling for thickness, is ≈ -0.42 . Thus, thickness covaries with polymer loading, while the dynamics retain an association with composition after thickness control. Full statistics and the audit of other fabrication variables are given in section C.2.

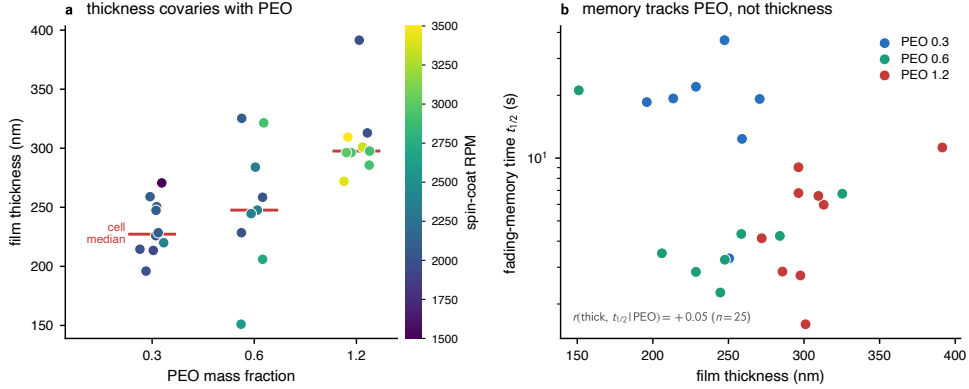


Figure 4.1: Film thickness and fading-memory time in the silver composition series ($n = 30$). (a) Film thickness rises with PEO fraction (red bars are cell medians); higher-PEO cells were spun faster (colour) to reduce this difference, although residual covariation remains. (b) Within each PEO band, $t_{1/2}$ is approximately flat against thickness; the partial correlation between thickness and $t_{1/2}$, controlling for PEO, is ≈ 0 .

Drive and read protocols. Because the comparisons below turn on them, the electrical protocols are stated explicitly, and they are *not* identical across the three measurements (table 4.1). Hysteresis uses a triangular voltage sweep. The potentiation (N -pulse build-up) was acquired with a 3 V write and 1.5 V read at a variable pulse count ($N \leq 1000$); the depotentiation (fading-memory decay) used a fixed train of 30 write pulses at a 4 V write and 2 V read. The inter-pulse interval is effectively common (0.104 s for potentiation, 0.103 s for depotentiation), but the *write amplitude differs* between the two (3 V vs 4 V), a point that matters when the two are later composed (section 4.10, Chapter 5). The read voltage is above the ionic activation threshold established in chapter 2, a point returned to in section 4.6. An older 2022 device generation used a 6 V write with a 3 V read; the modern and old amplitudes are never pooled, and the same-device contrast between two amplitudes is itself the drive-protocol result (section 4.6).

Table 4.1: The three common measurements use distinct drive protocols (modern *Super Yellow*/PEO/LiOTf/Ag corpus). Potentiation and depotentiation differ in write amplitude, so the chapter never pools them and Chapter 5 composes them only as an explicit assumption.

Measurement	Write	Read	Pulse train
Hysteresis (HYST)	triangular sweep	—	—
Potentiation (PULSES)	3 V	1.5 V	variable $N \leq 1000$, 0.104 s
Depotentiation (DELAYTIME)	4 V	2 V	30 pulses, 0.103 s
Protocol overlay (v114)	3 V, 6 V	1.5 V, 3 V	30 pulses

4.3 Methods of Analysis

Hysteresis curves are summarised by the on–off conductance ratio and by a normalised loop area; potentiation curves by the per-decade growth of the conductance ratio with pulse number, by the peak ratio, and by whether the response turns over (decreases) at the highest pulse counts; and depotentiation curves by a fading-memory time constant.

For the depotentiation (fading-memory) decay, the comparative archive is refitted with the more general Kohlrausch (stretched-exponential) function

$$\Delta G(t_{\text{wait}}) = \Delta G_0 \exp \left[- \left(\frac{t_{\text{wait}}}{\tau} \right)^\beta \right] + \Delta G_\infty, \quad (4.1)$$

with characteristic relaxation time τ , stretch exponent $0 < \beta \leq 1$, initial change ΔG_0 , and residual ΔG_∞ [152]. Two practical points follow from the data. First, the eight-to-thirteen-point decay curves often under-constrain β and τ jointly; the fitted τ is therefore reported only where the fit is well identified, and a robust, model-free half-enhancement time $t_{1/2}$ (the delay at which the conductance enhancement has fallen to half its initial value) is used as the primary timescale. Where both are available they agree. Both

are *apparent* timescales determined by the material and by the amplitude used to write and read the state. Section 4.6 establishes this protocol dependence; the composition series and chemistry comparisons therefore each use a fixed protocol. Second, fitted β values lie predominantly in the range 0.6–0.9, consistent with stretched relaxation and the heterogeneous distribution of activation barriers invoked in chapter 2.

Data-quality control and aggregation. Two screening layers are applied before any number is reported: erratic or non-physical measurements are flagged and excluded, and a per-device visual review stores per-curve verdicts (used as measured, cleaned, or discarded) in a curation registry applied automatically by the analysis scripts. All timescales are recomputed after this screening, because the instrument-side fit is not robust to anomalous points. Summaries then follow a conservative curve→pixel→device→cell ladder of medians, and the representative curves of fig. 4.2 are drawn by script from devices near their cell medians, not hand-picked. The full raw-to-summary trail, the aggregation ladder, and the stretched-exponential identification thresholds are documented in section C.1.

4.4 Composition Control of the Dynamics

The quantitative core of this chapter is the *Super Yellow*/PEO/LiOTf composition grid, in which the PEO mass fraction is varied over 0.3, 0.6 and 1.2 and the salt mass fraction over 0.045, 0.09 and 0.18, with two to four nominally identical silver-electrode devices per cell. Each measurement was acquired under its own fixed protocol (table 4.1); the composition contrasts below are therefore always drawn *within* a single measurement, never by pooling the 3 V potentiation with the 4 V depotentiation. Before the grid is summarised, fig. 4.2 shows what the three

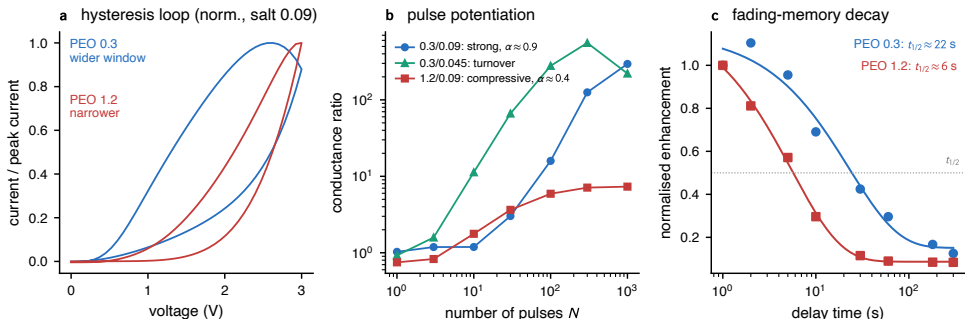


Figure 4.2: Representative raw curves of the three common measurements, on devices chosen near their cell medians (*Super Yellow*/PEO/LiOTf/Ag). (a) Hysteresis loops, each normalised to its own peak current so the comparison is the loop *openness* (the switching window), not absolute conductance: the low-PEO device (PEO 0.3, blue) opens a wider window than the high-PEO one (PEO 1.2, red), which carries more current but switches over a narrower range. (b) Pulse potentiation (conductance ratio vs pulse number, log-log): a strong near-linear accumulation ($\alpha \approx 0.9$), a low-salt cell that *turns over* and then declines, and a weak, compressive high-PEO response ($\alpha \approx 0.4$). (c) Fading-memory decay (enhancement vs delay, normalised to the first read) with stretched-exponential fits and the half-enhancement level dotted: the low-PEO device crosses $t_{1/2}$ far later (≈ 22 s) than the high-PEO one (≈ 6 s).

underlying measurements actually look like on representative devices, so that the on-off ratio, the growth exponent, the turnover, and the fading-memory time can be read off real curves before they are reduced to per-cell numbers. Three trends, mutually consistent and each visible across the grid, define the result.

First, the steady-state switching window contracts as the PEO fraction rises. At fixed salt mass fraction 0.09, the median on-off ratio falls from ≈ 3.3 at PEO 0.3 to ≈ 2.4 at PEO 1.2, and the normalised loop area falls correspondingly from ≈ 0.33 to ≈ 0.18 ; the activation voltage is, by contrast, essentially flat across the whole grid (2.2 V–2.4 V), indicating that composition tunes how *much* the conductance moves, not the voltage at which it begins to move. The widest window in the grid sits in the

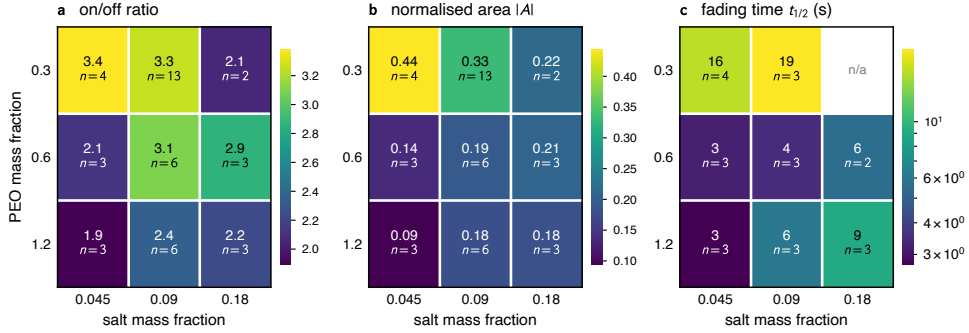


Figure 4.3: Composition control of the steady-state and dynamical response of *Super Yellow*/PEO/LiOTf devices. The switching window (a, b) and the fading-memory time (c) both decrease as the PEO mass fraction increases at fixed salt content.

low-PEO row, with the largest loop area at the low-salt corner (PEO 0.3, salt 0.045: area ≈ 0.44); the per-sweep on–off ratio is strongly right-skewed there (cell median ≈ 3.4 but individual sweeps reaching ~ 40), so robust cell medians, not means, are reported throughout to avoid letting a few outlying sweeps define a cell.

Second, potentiation weakens, and changes shape, as the composition is varied (fig. 4.5). Two axes act distinctly. The PEO fraction controls the *strength* of pulse integration: the per-decade growth of the conductance ratio (the log–log slope α) falls with PEO content, most cleanly at the lowest salt, where it drops from $\alpha \approx 1.1$ (near-linear accumulation) at PEO 0.3 to $\alpha \approx 0.3$ (strongly compressive) at PEO 1.2; and the peak conductance ratio, i.e. the usable dynamic range, is largest in the low-PEO, low-salt corner ($\approx 480\times$ at PEO 0.3/salt 0.045) and falls to a few-fold at high PEO. The salt fraction, by contrast, controls *how far* the response can be driven before it saturates: at the lowest salt (0.045) every cell *turns over* (the conductance ratio peaks and then declines) by $N \sim 100\text{--}300$ pulses, whereas at the highest salt (0.18) the response grows monotonically through the largest counts tested ($N \approx 1000$), with

salt 0.09 intermediate. Two caveats are kept in view: the α trend is noisier at the higher salts, where the sparsely sampled PEO 0.3/salt 0.18 cell ($n = 2$, curation-recovered) departs from it; and the turnover sets a practical ceiling on the usable pulse count. Read constructively, these responses are a composition-tunable, predominantly compressive and (in the low-salt cells) non-monotone integration of the input pulse train, the nonlinear input encoding that Chapter 5 builds on.

Third, the fading-memory timescale varies with composition over roughly 2 s–20 s. The longest retention occurs in the low-PEO row (at PEO 0.3, $t_{1/2} \approx 19$ s at salt 0.09, with an identified $\tau \approx 25$ s); increasing the PEO fraction to 0.6 or 1.2 shortens the timescale to a few seconds. A single curation-recovered device at the still-lower PEO 0.15 level ($t_{1/2} \approx 22$ s) is consistent with this trend, though it lies outside the replicated grid and is not counted toward it. The independently computed instrument-side single-exponential time constant reproduces the same ordering. Within each cell, the fading-memory time constant varies by $\sigma(\ln \tau) \approx 0.85$ between nominally identical devices at fixed composition; film thickness, spin speed, anneal, and device age do not explain this spread (section C.2; quantified in section D.3). Chapter 5 models the between-cell and within-cell distributions as a heterogeneous bank of fading-memory elements; the range across the whole device library is mapped in section C.4. The same composition axis appears in potentiation strength: the growth exponent α falls monotonically with PEO fraction (cell medians $\approx 0.9 \rightarrow 0.7 \rightarrow 0.35$), with within-cell variation around each median.

Across the replicated grid, increasing the proportion of the ion-transport phase shortens the fading-memory time and is accompanied by smaller switching and potentiation responses, while salt content mainly controls when the response saturates. The PEO gradient is statistically established for the fading-memory time (per-device Spearman $\rho = -0.52$, permutation $p = 0.009$); the window and potentiation descriptors move

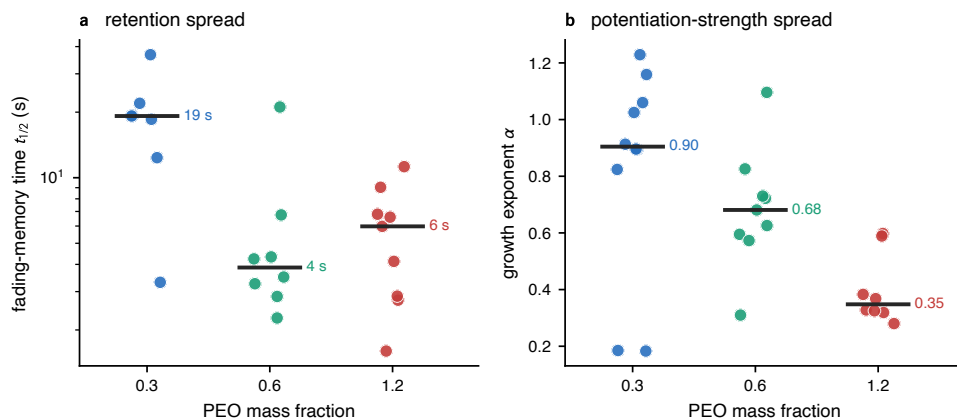


Figure 4.4: Per-device distributions underlying the cell medians of fig. 4.3 (*Super Yellow*/PEO/LiOTf/Ag; black bar is the cell median). (a) Fading-memory time $t_{1/2}$ and (b) potentiation growth exponent α , each against PEO fraction. The PEO fraction sets the broad level, while within-cell device-to-device variation is comparable to the steps between cells. Chapter 5 evaluates both sources of variation.

in the same direction but remain marginal ($p \approx 0.06$ – 0.07). A two-way decomposition finds no significant interaction between PEO and salt, and bootstrap confidence intervals on every cell median are given in section C.9. This replicated composition result is the quantitative core of the chapter; the per-cell values are tabulated in section C.3. If a more ion-rich matrix relaxes a displaced ionic profile more readily, its equilibrium ionic resistance should also fall. Section 4.7 tests that prediction by impedance spectroscopy.

4.5 Chemical Tuning of the Dynamics

Beyond composition, the identity of the electrolyte components shifts the dynamics further. These comparisons are reported as an illustrative survey: each matched cell (fixed composition, electrode, and protocol) contains at most one or two screened devices, so the magnitudes below

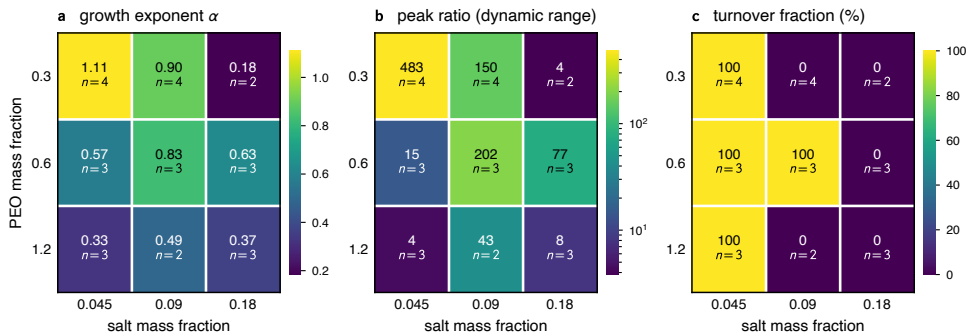


Figure 4.5: Potentiation (pulse-integration) response across the *Super Yellow*/PEO/LiOTf composition grid: (a) the log–log growth exponent α , (b) the peak conductance ratio (usable dynamic range, log scale), and (c) the fraction of devices whose response turns over within the measured range ($N \leq 1000$). Increasing the PEO fraction lowers α and the dynamic range; increasing the salt fraction suppresses turnover, sustaining growth to higher pulse counts. Per-cell median, with the device count n annotated in each cell.

are indicative and the sample size is given in every case. They are complemented here by structural and optical probes on *separate* sample sets (blend films and scratched powders at the lead composition, 0.3/0.09) examined by ATR-FTIR, X-ray diffraction, and UV-Vis absorption, which probe ion association, host backbone order, crystallinity, and the semiconductor’s electronic gap directly, independently of how a device is driven (figs. 4.6 and 4.8). These too are single specimens per chemistry and are read illustratively.

Host. Substituting the linear polyether PEO with the hyperbranched polyether TMPE, at fixed lithium-triflate chemistry and composition, shortens the fading-memory time by roughly a factor of six (PEO/LiOTf: $t_{1/2} \approx 20$ s–25 s, three screened silver devices; TMPE/LiOTf: ≈ 4 s, one screened silver device with an identified decay). This comparison has three PEO devices but only one TMPE device. The lithium pair from the designed host $\times$ cation batch could not be used because the PEO/LiOTf

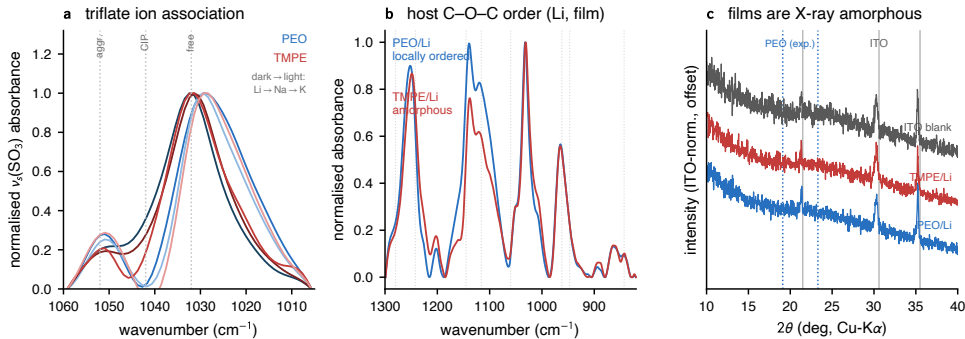


Figure 4.6: Microscopic corroboration of the chemistry axis (electrode-free *Super Yellow*-blend films/powders, 0.3/0.09, one specimen per chemistry). (a) ATR-FTIR triflate ion-association band $\nu_s(\text{SO}_3)$, whose position reports the ion-association state (spectroscopically free near 1032 cm^{-1} , contact ion pair near 1042 cm^{-1} , aggregate above; reference positions dotted [148]; PEO hosts in blues, TMPE hosts in reds, darker to lighter for Li/Na/K): it lies at 1029 cm^{-1} – 1032 cm^{-1} in every host $\times$ cation sample, with no cation-resolved shift. (b) ATR-FTIR C–O–C envelope: the linear PEO host (blue) shows a resolved multiplet, indicating retained *local* (conformational) ether-backbone order, whereas the hyperbranched TMPE host (red) shows a single broad, amorphous band [148]. (c) X-ray diffraction of PEO/Li and TMPE/Li with a bare-ITO blank: all patterns show the crystalline ITO substrate reflections (grey lines; 21.5° , 30.6° and 35.5°), with no sharp PEO crystalline doublet (blue dotted; 19.1° and 23.3°) or crystalline salt phase. The composite films are therefore X-ray amorphous, and the PEO order in (b) is short-range [127, 148]. ATR spectra are normalised after a local/rubber-band baseline and XRD is normalised to the ITO peak; intensities are not comparable across specimens because sampling and thickness differ, so only band positions and shapes are interpreted.

device was broken. The analysis therefore compares the TMPE device with the nearest-matched silver PEO/LiOTf devices (same operator, fresh semiconductor, glovebox), which differ in fabrication date.

An independent microscopic correlate of the host substitution is visible in the polyether backbone itself, and at two length scales. In ATR-FTIR the linear PEO film shows a resolved C–O–C stretching multiplet, with the weak order-sensitive marker bands near 843 cm^{-1} , 962 cm^{-1} and

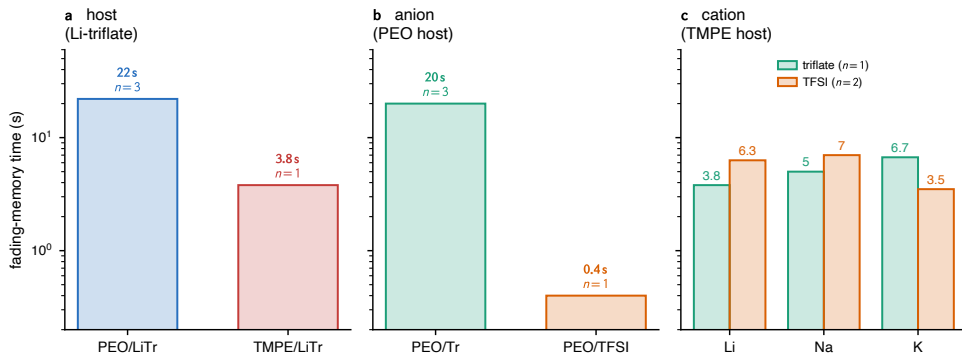


Figure 4.7: Illustrative effect of electrolyte chemistry on the fading-memory time constant (*Super Yellow*, 0.3/0.09, silver, matched 4 V/2 V protocol). Host (PEO vs TMPE, Li-triflate) and anion (triflate vs TFSI, PEO host) produce the largest shifts. The cation panel shows the same three cations in the TMPE host under each anion: potassium is longest in triflate and shortest in TFSI, so the cation produces no consistent, host- and anion-independent ordering. Sample sizes are small (n as labelled); see text.

1242 cm^{-1} , indicating retained *local* (conformational) order of the ether backbone, whereas the hyperbranched TMPE film collapses these into the single broad band of an amorphous polyether (fig. 4.6b) [148]. At the longer length scale of crystallites, however, X-ray diffraction on the same sample set finds *neither* host crystalline in the composite: every pattern is dominated by the ITO substrate (confirmed against a bare-substrate blank), with no sharp PEO crystalline reflections and no segregated crystalline salt phase (fig. 4.6c), so the films are X-ray amorphous and PEO retains short-range order without forming long-range crystallites [127, 148]. A host that is locally more ordered, even while globally amorphous, set against a fully amorphous TMPE, is consistent with the lower segmental mobility and slower ionic relaxation that the *longer* PEO fading-memory time would require: in polyether electrolytes ion transport is carried by segmental motion of the host above its glass transition, so the relaxation of a displaced ionic profile is rate-limited by that segmental mobility [88, 132,

133]. With one specimen per host this is offered as a structural correlate, not a demonstrated cause. The host ordering is moreover not specific to the silver electrode: on the independent gold-electrode corpus at the same composition, PEO again retains far longer than TMPE and supports a markedly wider switching window and stronger potentiation (section 4.8), so the host effect survives a change of electrode. That the salt forms no crystalline phase in any chemistry additionally supports reading these blends as homogeneous, salt-dissolved ion-conducting media.

Anion. Replacing triflate with the more charge-delocalised TFSI anion in the PEO host reduces the fading-memory time from ≈ 20 s to approximately 1 s. This direction is consistent with the electrolyte literature: charge delocalisation in TFSI weakens ion pairing and lowers salt lattice energy, promoting dissociation and plasticisation of the polyether host and increasing ionic mobility [129, 153]. The TFSI decays also tend toward a compressed shape, whereas the triflate decays are stretched, suggesting different relaxation-time distributions. This comparison is weakly controlled: the single PEO/TFSI device belongs to a later fabrication generation and also differs in spin speed and salt filtration. The approximately twenty-fold difference is therefore reported as illustrative.

Cation. The data do not support the hypothesised $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ ordering of fading-memory time derived from the hard-soft acid-base argument of chapter 2 [89]. Two cation series use the same TMPE host, composition, electrode, and protocol, and differ in the salt anion (fig. 4.7). With triflate, the timescale increases from Li^+ (≈ 3.8 s) to Na^+ (≈ 5.0 s) and K^+ (≈ 6.7 s); each value comes from one screened device with $R^2 \geq 0.93$. With TFSI, potassium is shortest (≈ 3.5 s), while lithium and sodium are longer (≈ 6 s–7 s, two devices per cation). Thus, cation identity shifts the timescale by roughly a factor of two within a

fixed anion, but the ordering changes with the anion and within-cation variation is comparable to between-cation differences. These sample sizes cannot establish a host- and anion-independent ordering. A powered test would require a dedicated series at fixed host, anion, protocol, and electrode with at least three replicates per cell (section 4.10).

Independent probes provide additional constraints. The triflate symmetric SO_3 stretch $\nu_s(\text{SO}_3)$ lies at 1029 cm^{-1} – 1032 cm^{-1} in all six host $\times$ cation films and powders, a spread at or below instrumental resolution with no cation ordering (fig. 4.6a). UV-Vis absorption of *Super Yellow*/TMPE films places the *Super Yellow* π – π^* peak [86] at 446 nm for Li^+ , Na^+ , and K^+ (fig. 4.8). These results indicate no resolved cation effect on ion-pairing state or semiconductor optical gap; neither measurement tests the ordering of retention directly. On gold electrodes, the TMPE/triflate steady-state window is also flat across the three cations (on–off ratio ≈ 1.3 – 1.5 , no monotone trend; section 4.8).

In these illustrative comparisons, host and anion substitutions produce larger shifts in fading-memory time than cation substitution.

4.6 Protocol-Dependent Timescale

A methodological result constrains every comparison above. Because the read voltage is above the ionic activation threshold and write amplitude controls the displaced ionic profile, the *drive protocol* influences the apparent fading-memory time. In one device measured under both protocols, the fitted relaxation time increases from $\approx 4.6\text{ s}$ under a 3 V write and 1.5 V read to $\approx 15.5\text{ s}$ under a 6 V write and 3 V read (fig. 4.9).

One interpretation is that a larger write amplitude displaces more ionic charge and, at the largest amplitudes, may introduce electrode-level electrochemical processes, producing a longer-lived written state. The

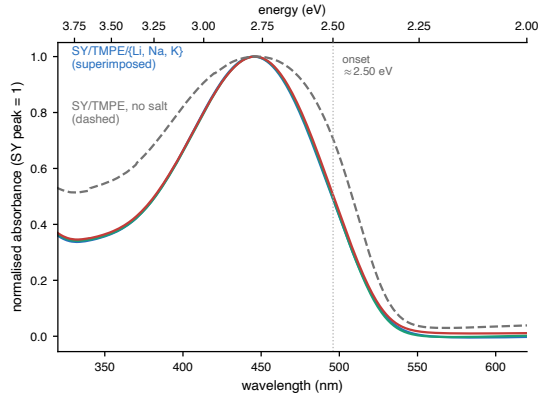


Figure 4.8: UV-Vis absorption of *Super Yellow*/TMPE blend films (lead composition 0.3/0.09; Au-electrode 2025 generation; one specimen per chemistry), normalised to the *Super Yellow* π - π^* peak. The peak (≈ 446 nm) and the absorption onset (≈ 2.50 eV) are superimposed for Li^+ , Na^+ , and K^+ ; the salt-free control (dashed) coincides at the band edge (its differing blue side and lower absolute absorbance are thickness/scattering, not electronic). The cation does not shift the semiconductor’s optical gap. Only band *positions* are read because absorbance magnitudes are thickness-confounded.

supra-threshold read may also perturb the state during interrogation. The composition result of section 4.4 remains internally comparable because it uses one fixed protocol. Chemistry comparisons spanning device generations are more exposed to protocol differences and are reported as illustrative. The apparent volatile timescale is therefore co-determined by material and drive conditions; cross-device or cross-chemistry comparisons must match the write/read protocol, electrode, and composition.

The same lever reappears, independently, on the gold-electrode corpus of section 4.8. There, matched-amplitude negative pulse trains sweep the pulse count and the inter-pulse interval together, with the relaxation recorded in the same run, and both halves of the protocol dependence are observed directly: potentiation grows steeply as the inter-pulse interval shortens (rank correlation $\rho \approx -0.94$ to -0.99 , $p < 0.001$, in three inde-

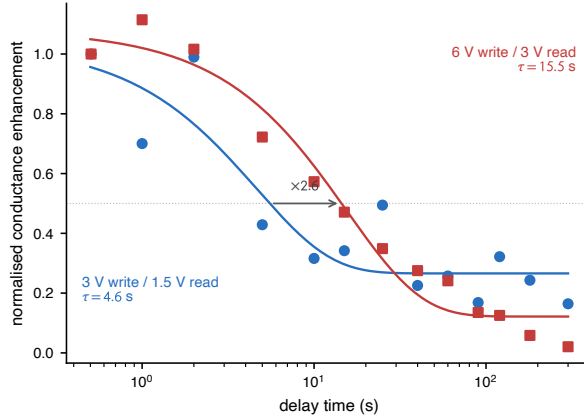


Figure 4.9: The apparent fading-memory time is set in part by the drive protocol. The same device, measured under a larger write/read amplitude, shows a markedly longer apparent relaxation time: the fitted single-exponential time constant rises from $\tau \approx 4.6$ s to ≈ 15.5 s, and the half-enhancement crossing (dotted, arrow) shifts by $\approx 2.8\times$.

pendent series of 9–11 matched runs), and the depth of the written state co-sets the subsequent relaxation time ($\rho \approx +0.46$ between potentiation ratio and decay half-time, $p = 0.015$, $n = 28$ runs). The drive–retention coupling that this section infers from a two-protocol contrast is therefore seen *within single experiments* on a second electrode (section C.7).

4.7 Microscopic Corroboration: Impedance Spectroscopy

Every result so far rests on the *driven* response (pulses and decays). Small-signal impedance spectroscopy offers an independent test of the mechanism proposed for the composition trend: if increasing the PEO fraction shortens the fading-memory time because a more ion-rich matrix lets the displaced ionic profile relax more readily, then the *equilibrium* ionic resistance should fall along the same composition axis. The archive

contains such a measurement, acquired on the same devices: 0 V-DC spectra (40 mV AC, 1 Hz– 1×10^6 Hz) on 26 silver-electrode *Super Yellow*/PEO/LiOTf devices spanning the composition grid.

Each spectrum is summarised in two separate ways. The first is *model-free*: the real part of the impedance at the apex of the main Nyquist semicircle, $Z_{\text{real}}^{\text{apex}}$, which fixes the resistance scale of the spectrum without committing to any circuit. The second is *model-based*: the bulk ionic resistance R_{ion} of an equivalent circuit adapted from the polymer light-emitting electrochemical cell analysis of Munar *et al.* [143], reported only as corroboration of the model-free quantity because the per-spectrum circuit fits are genuinely ill-conditioned. The circuit topology, its two automated fits, and the per-cell values of both observables are detailed in section C.5.

Both observables fall steeply and monotonically with PEO fraction (fig. 4.10a). The model-free $Z_{\text{real}}^{\text{apex}}$ drops by more than three orders of magnitude across the grid (cell medians from $\sim 10^8 \Omega$ at PEO 0.15–0.3 to $\sim 10^5 \Omega$ at PEO 1.2), with a rank correlation against PEO of -0.69 per device ($n = 26$) and -0.83 on cell medians; the fitted R_{ion} gives the same picture (-0.84 on cell medians). This is the direction the mechanism requires: more ion-transport polymer, lower ionic resistance.

The EIS resistance scale and the *independently* measured fading-memory time fall along the *same* composition axis (fig. 4.10b): both decline together as the PEO fraction rises, and across cells $\log Z_{\text{real}}^{\text{apex}}$ covaries with $\log t_{1/2}$ at $r \approx +0.74$ ($+0.61$ per device, $n = 23$). Three qualifications bound this reading:

- The PEO fraction is the *common cause* of both quantities, so this correlation confirms that the same composition axis moves the equilibrium ionic resistance and the driven retention together; it is not evidence of a second, independent factor.

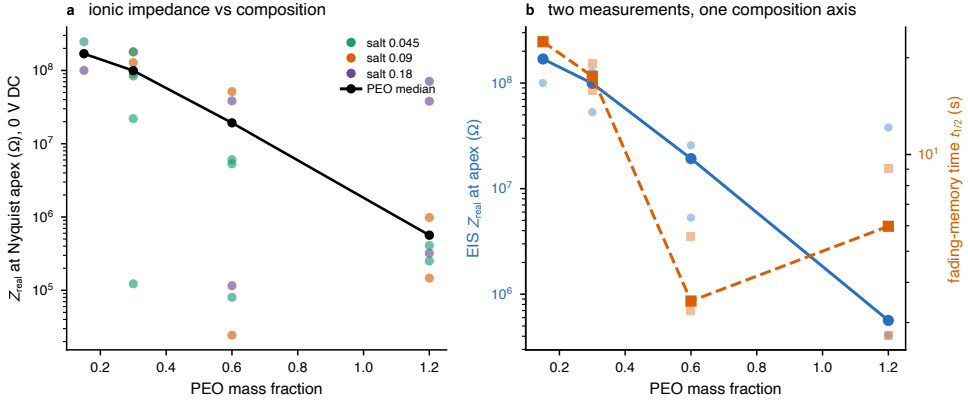


Figure 4.10: Equilibrium (0 V-DC) impedance spectroscopy corroborates the composition control of the dynamics (silver *Super Yellow*/PEO/LiOTf grid). (a) The impedance resistance scale (here the model-free real impedance at the Nyquist apex) falls by more than three orders of magnitude as the PEO fraction rises (points are devices, coloured by salt; the black line is the PEO median). (b) The equilibrium impedance scale (left axis) and the *independently* measured fading-memory time $t_{1/2}$ (right axis) decline along the *same* PEO composition axis (cell-level points with per-PEO median trends): the two are co-driven by composition, not independent variables.

- The resistance *scale* carries the composition signal, whereas the EIS relaxation *frequency* does not: the apex frequency corresponds to a sub-second RC time and does *not* track the seconds-long fading-memory time. This separates the small-signal ionic-transport bottleneck reported by impedance spectroscopy from the slower large-signal relaxation that determines the fading-memory response. The two timescales are nonetheless reconcilable in *magnitude*: the sub-second apex RC is the geometric-capacitance response, and scaling it by the chemical-to-geometric capacitance ratio measured independently from the low-frequency branch ($\gtrsim 33$) predicts a large-signal relaxation of several seconds, inside the observed $t_{1/2}$ range (section C.11).

- The effect is not a film-thickness artefact: $Z_{\text{real}}^{\text{apex}}$ spans $\sim 10^4\times$ while the film thickness spans only $2.6\times$, and the PEO dependence survives controlling for thickness (partial correlation -0.60 ; section 4.2).

Impedance spectroscopy thus supplies, for the ionic-resistance half of the mechanistic picture below, the independent microscopic support that the driven measurements alone cannot.

4.8 Electrode and Polarity Dependence

Every result so far uses the silver top electrode. The archive also contains 50 gold-electrode devices, treated here as a third illustrative axis. Every gold device uses the single lead composition (PEO or TMPE host, 0.3/0.09); this corpus therefore contains no information about the composition series of section 4.4, which remains a silver-only result. It instead contrasts electrochemically active silver with inert gold at matched chemistry and provides an independent-electrode check on the chemistry comparisons of section 4.5.

At matched chemistry and composition, and under the canonical positive-polarity protocols used everywhere above, the electrode shifts all three behavioural families, and does so coherently (fig. 4.11). In the lead PEO/LiOTf cell, replacing silver by gold narrows the steady-state switching window (median on–off ratio $3.3 \rightarrow 2.0$, normalised loop area $0.33 \rightarrow 0.12$), weakens and flattens potentiation (peak conductance ratio $\approx 124\times \rightarrow 29\times$, log–log slope $\alpha \approx 0.9 \rightarrow 0.45$), and raises the activation voltage ($\approx 2.4\text{ V}–2.7\text{ V}$), yet *lengthens* the fading-memory time, from $t_{1/2} \approx 19\text{ s}$ on silver (three devices) to $\approx 87\text{ s}$ on the two clean gold devices ($R^2 \approx 0.97–0.99$, $\beta \approx 0.6$). The same window-and-potentiation contraction holds in the TMPE/LiOTf cell (on–off $3.8 \rightarrow 1.3$; peak $54\times \rightarrow 1.9\times$),

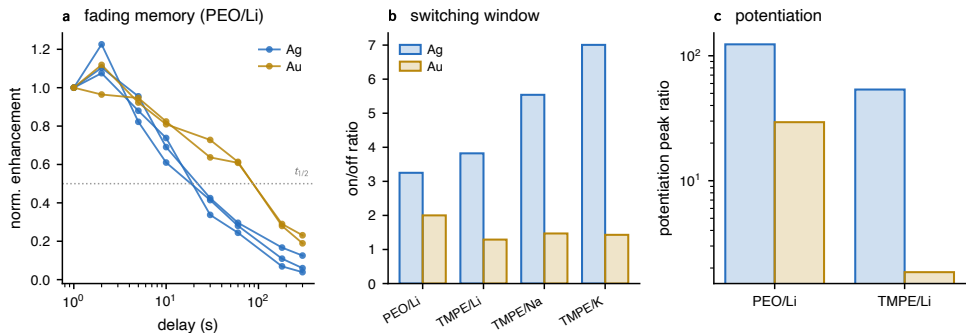


Figure 4.11: Illustrative electrode contrast at the lead composition (0.3/0.09), under the canonical positive-polarity protocols (for the negative-polarity behaviour of the gold devices see fig. 4.12). (a) Fading-memory decays of PEO/LiOTf devices, normalised to the first read: the gold curves (gold) cross half-enhancement later than the silver ones (blue), i.e. gold retains *longer*. (b) The steady-state switching window (on-off ratio) is narrower on gold for every chemistry, and is flat across the TMPE cation series on gold (no Li/Na/K ordering). (c) Pulse-potential dynamic range (peak ratio, log scale) is far smaller on gold. The inert electrode thus gives a weaker, higher-threshold, but more retentive response. Small n per cell, no human curve curation on the gold side, and a generation offset between the corpora bound the comparison (see text).

again at positive drive. The per-cell metrics for every chemistry on both electrodes, including the host-free and salt-free negative controls, are tabulated in section C.6.

This difference is consistent with the contrast between active and inert electrodes. Silver can inject and electrodeposit mobile Ag^+ , the basis of electrochemical-metallisation switching, and may therefore add a large volatile component: a wider window, steeper potentiation, lower threshold, and faster relaxation [69, 70]. Gold is noble and inert, leaving the polymer-electrolyte ionic mechanism to produce a smaller, higher-threshold, and more persistent response. The higher activation voltage on thinner gold devices supports an interfacial electrode contribution over a film-thickness explanation [106].

Negative-polarity response of the gold devices. Additional campaigns on the same Au/TMPE/triflate generation measured negative bias. These data were recovered from outside the curated archive and analysed with a dedicated parser; their sources, validation against catalogued sweeps, and per-file tables are documented in section C.7. The devices are strongly polarity-asymmetric: within a junction, bipolar sweeps carry $\approx 2.5\text{--}42\times$ more current at -3 V than at $+3\text{ V}$ (fig. 4.12a), with the same direction across the catalogued negative-going sweeps. Under negative drive, the gold devices show a rate-dependent switching window. Across sweep rates of $\sim 0.003\text{--}0.4\text{ V s}^{-1}$, the on-off ratio rises as the sweep slows (rank correlations down to -0.99 , $p < 0.001$), from ≈ 1 at the standard rate to $20\text{--}31$ in the slowest -3 V sweeps (fig. 4.12b). Constant-bias holds locate an activation threshold between -1.5 and -2.5 V : current decays below this range and grows by one to three orders of magnitude above it over tens to hundreds of seconds. Negative pulse trains potentiate by one to two orders of magnitude at a matched protocol (-5 V pulses, -2.25 V read: cation medians $\approx 40\text{--}230\times$), while positive pulses depress the written state. The response is therefore bidirectionally programmable under this negative-polarity protocol.

These campaigns add fading-memory measurements for the TMPE host on gold. Sparse delayed reads after the pulse trains, with the device floating between reads, follow strongly stretched exponentials ($\beta \approx 0.33\text{--}0.5$) with model-free half-decay times of $\approx 0.4\text{ s--}21\text{ s}$ across the three cations (fig. 4.12c). These values are on the same seconds scale as TMPE on silver and below the $\approx 87\text{ s}$ measured for PEO on gold, providing a gold-electrode comparison of host retention. Under the matched negative protocol, all three cations have overlapping potentiation ranges and relaxation times, with substrate-to-substrate spread comparable to any cation difference. The concentration of the response in one bias direction after removal of the active Ag^+ pathway is consistent with an interfacial role for the

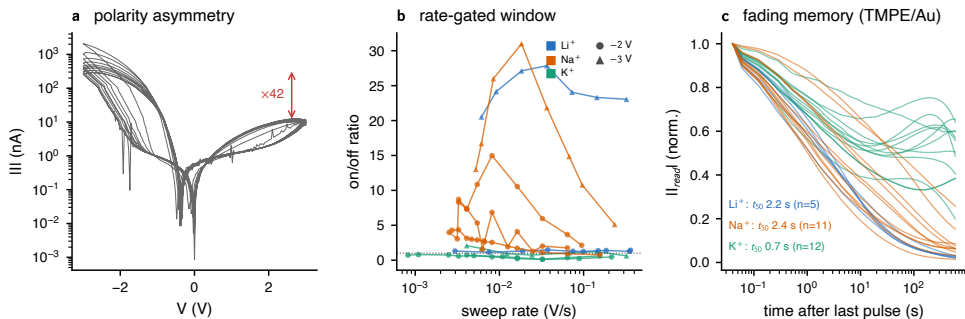


Figure 4.12: Negative-polarity response of the gold/TMPE/triflate devices (illustrative tier; section C.7). (a) Exemplar bipolar sweep (± 3 V, ten cycles, Na^+ device): the negative branch carries $\approx 42\times$ the current of the positive one. (b) On–off ratio against sweep rate for all step-count series (colour = cation, marker = amplitude): slow sweeps give ratios of 20–31, while the standard fast protocol gives ≈ 1 . (c) Sparse-read relaxation after -5 V pulse trains, normalised to the first read, with per-cation median half-decay times. The times are on the seconds scale, as for TMPE on silver, and below those of PEO on gold.

dissimilar ITO and gold contacts and their accumulated ion profiles, but the electrical data do not establish that mechanism.

The second electrode reproduces two observations from section 4.5: PEO retains longer than TMPE, and the switching window has no transferable cation ordering. The gold corpus also provides negative controls absent from the silver set. Host-free (semiconductor-plus-salt) and salt-free (semiconductor-plus-host) films show neither a switching window (on–off ≈ 1.0) nor potentiation (peak ≈ 1), indicating that both the ion-transport polymer and dissolved salt are required for the memristive response.

Frequency-domain response: a bias-programmed low-frequency amplifier. A third corpus measures the same Au/TMPE/triflate chemistry in the frequency domain: 549 lock-in sweeps from 1 Hz to 500 kHz on a fresh batch, with one substrate per cation tracked from day 1 to day 23. The junction is placed in series with a 4 nF capacitor and driven by a small

sinusoid superimposed on a programmable DC offset. The apparatus, channel decode, and per-file tables are documented in section C.8. Above ~ 10 Hz, the unbiased junction behaves as a passive linear dielectric: the transfer ratio follows the capacitive divider and is amplitude-independent to within a median 0.2 % from 0.5 to 2.4 V_{pk}. This boundary is consistent with the rate dependence in fig. 4.12b. In the same plateau, junction capacitance increases $\text{Li} < \text{Na} < \text{K}$ by $\approx 1.4\times$ across the same-batch trio, compared with an $\approx 1\%$ pixel spread. Same-recipe profilometry makes a thickness origin less likely, but one substrate per salt limits this result to a candidate dielectric cation effect; profilometry on matched specimens would test it directly. Under DC bias, the fundamental-frequency transfer below ~ 10 Hz increases from $|H| \approx 0.5$ without bias to ≈ 18 at ± 2.5 V (fig. 4.13). The bias-modulated ionic conductance transfers energy from the DC offset into the signal band. This gain occurs for all three cations at every age tested, is confined to the sub-10 Hz band, and is nearly symmetric in bias polarity (section C.8). A biased junction therefore operates as a voltage-programmable low-pass amplifier over a band that overlaps the physiological signals analysed in Chapter 5.

All three gold corpora share the same illustrative-tier bounds, which the SI states for each corpus (sections C.7 and C.8). Each uses the single lead composition on one or two substrates per cation; the curves are uncurated; and the supra-threshold negative read makes decay times protocol-specific, allowing only order-of-magnitude comparison with silver. The 2024–2025 gold generation is also partly confounded with electrode because the silver chemistry was fabricated earlier and the co-fabricated silver batch was not measured under the decay protocol. Electrode and polarity are therefore treated as illustrative effects and cross-electrode checks on the chemistry axis. A matched-generation batch split between silver and gold is required for a quantitative electrode comparison.

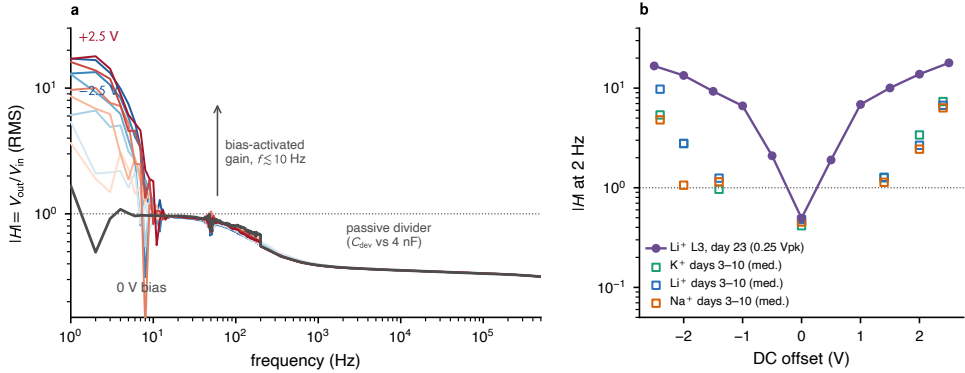


Figure 4.13: Frequency-domain response of the gold/TMPE/triflate devices (illustrative tier; section C.8). (a) Transfer ratio $|H|(f)$ of one Li^+ junction in series with 4 nF (day 23, 0.25 Vpk) at eleven DC offsets from 0 (grey) to ± 2.5 V (blue negative, red positive). Above ~ 1 kHz, all curves follow the passive capacitive divider; below ~ 10 Hz, bias produces a fundamental-frequency gain of up to ≈ 18 , increasing with bias magnitude and nearly symmetric in sign. (b) $|H|$ at 2 Hz against DC offset for the day-23 series (line) and per-cation medians of earlier sessions (squares; days 3–10, 0.4 Vpk–1.4 Vpk). Gain increases with bias and occurs for all three cations.

4.9 Discussion

Composition is the only replicated axis for control of the volatile dynamics. Increasing the ion-transport-polymer fraction shortens the fading-memory time and is accompanied by smaller switching and potentiation responses. Host and anion substitutions shift the timescale over a wider range, but those comparisons are illustrative. Cation identity also shifts the dynamics, although the direction changes with host, anion, and protocol at the available sample sizes; the predicted $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ order is therefore not transferable across the dataset.

The electrode acts jointly with drive polarity. Relative to active silver, inert gold weakens switching and potentiation and lengthens retention under the canonical positive drive. Under negative drive, the same gold

devices show a full, rate-gated, bidirectionally programmable response (section 4.8). The frequency measurements locate the corresponding spectral boundary: above ~ 10 Hz the gold junctions behave as passive linear capacitors at every tested amplitude, whereas a DC bias produces programmable low-frequency gain below that boundary.

The hard–soft acid–base model of chapter 2 remains a qualitative interpretation. It anticipates the cation ordering in the TFSI family, where the most weakly coordinating cation relaxes fastest, but the triflate family runs in the opposite direction. The present data therefore do not support a quantitative cation law.

Mechanistic interpretation. The mechanism remains a hypothesis. Increasing the PEO fraction does not produce the decrease in absolute on-state conductance expected from simple dilution of the *Super Yellow* percolation pathway; instead, on-state conductance rises by more than three orders of magnitude across the grid (section C.10). A rising conductance floor is consistent with the contracting window: the ion-rich phase may add a parallel, weakly switchable path, with off-state conductance rising faster than on-state conductance. Faster relaxation of the displaced ionic profile could account for the accompanying decrease in fading-memory time. Salt content may set the number of mobile, chargeable states available before saturation, consistent with the later turnover at higher salt fraction.

Host and anion effects may arise from cation–ether coordination, ion pairing, and segmental mobility [129, 133], which set the distribution of local activation barriers behind stretched-exponential relaxation [152]. Equilibrium impedance supports the ionic-resistance part of this interpretation: it falls along the composition axis and tracks the fading-memory time (section 4.7). ATR associates the host effect with local ether-backbone order, retained in PEO and absent in amorphous TMPE, while XRD

shows that every composite remains X-ray amorphous and that the salt is dissolved rather than segregated. Across the cation series, ATR shows no corresponding ion-association fingerprint and UV-Vis shows no shift of the *Super Yellow* optical gap (section 4.5). These measurements are consistent with electrolyte-controlled ion transport rather than a change in the *Super Yellow* electronic backbone.

The anion and barrier-distribution links have not been established microscopically. Testing them requires temperature-dependent relaxation measurements and chemistry-resolved impedance (section 4.10); the hard-soft acid-base interpretation therefore remains qualitative.

The measured spread of fading-memory times can populate a physical reservoir with distinct temporal responses [5, 18, 28]. Across compositions and nominally identical devices, the timescale spans roughly an order of magnitude. A two- σ criterion on the cell-median span of $\Delta \ln t_{1/2} \approx 2.1$, relative to within-cell scatter, resolves approximately two to five composition bands (section C.13). Each junction reproduces its switching window to $\approx 15\%$ sweep-to-sweep (section C.12), substantially less than the diversity across the bank. Figure 4.14 places every screened silver device in a fading-memory ($t_{1/2}$) versus dynamic-range (peak ratio) plane. The PEO fraction sets the broad region occupied by each device, so one composition selects a paired memory/nonlinearity operating point; the remaining within-composition scatter supplies additional device-to-device diversity. Chapter 5 evaluates both sources of variation.

4.10 Limitations and Future Work

The principal limitation is statistical power outside the composition axis. Only the *Super Yellow*/PEO/LiOTf composition grid is replicated; the host, anion, and cation comparisons rest on one or two screened devices

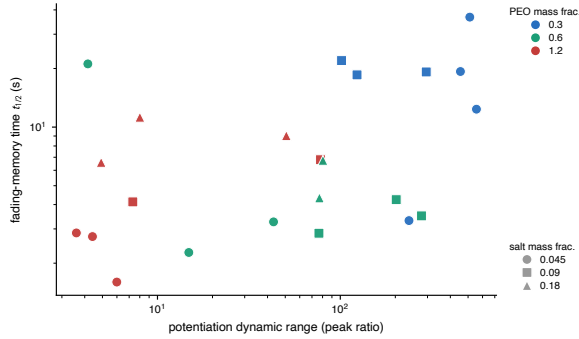


Figure 4.14: Per-device design space of the *Super Yellow*/PEO/LiOTf/Ag composition grid: fading-memory time $t_{1/2}$ against potentiation dynamic range (peak ratio), coloured by PEO fraction and marked by salt fraction (one point per device, median over its pixels). The PEO fraction selects the broad region a device occupies: low PEO (blue) in the long-memory, high-dynamic-range corner, high PEO (red) in the opposite one, so memory and nonlinearity are *paired*, not independently tunable. The substantial within-composition scatter is the genuine device-to-device heterogeneity that Chapter 5 exploits.

per matched cell and remain illustrative. Five consequences follow:

1. The cation hypothesis is not tested here at adequate power. The cation does shift the dynamics, but not as predicted: the triflate retention series runs $K > Na > Li$, the reverse of the hypothesised $Li^+ > Na^+ > K^+$, and re-orders again with the anion, so the hypothesised ordering is not what the data show and no transferable cation law can be fixed at one or two devices per cell.
2. The supra-threshold read and the amplitude dependence of the apparent timescale (section 4.6) mean that absolute retention values are protocol-specific.
3. The electrode is reported as an illustrative axis (section 4.8), but it remains partly confounded with fabrication generation, and device ageing is an uncontrolled variable that was not modelled.

4. Pulse integration and state relaxation were characterised *separately and at different write amplitudes*: the potentiation curves used a 3 V write and the fading-memory decays a 4 V write (table 4.1), at an effectively common inter-pulse interval. On the lead silver composition, the archive contains no experiment that varies input timing to probe their interaction directly. A matched-amplitude experiment exists on the gold/TMPE corpus (section 4.8), but it uses a different host, electrode, polarity, and read, so it establishes the direction of coupling rather than the lead-cell parameters.
5. The fading-memory protocol measured a single junction per device, so repeat-decay reproducibility of $t_{1/2}$ is not characterised at the population level (only one device carries two junctions; section C.12). Within-device reproducibility is established only for the steady-state switching window, from its repeated sweeps.

Because drive amplitude affects the apparent timescale (section 4.6), combining the 3 V potentiation shape with the 4 V decay in Chapter 5 requires an explicit modelling assumption that the regimes are compatible. A matched-amplitude experiment that varies both pulse count and inter-pulse interval would test this assumption. Its gold/TMPE negative-polarity counterpart shows coupling between the two measurements (section 4.6); the lead-composition silver experiment remains future work.

A dedicated cation series in one host and anion, at a fixed protocol and electrode, with a sub-threshold read voltage (as in chapter 2) and at least three replicates per cell, would turn the illustrative chemistry survey into a powered comparison. The impedance study of section 4.7 could also be extended beyond the replicated PEO/LiOTf axis at 0 V DC. The archive contains unmodelled bias-resolved spectra up to 3 V DC and chemistry-varied devices; analysing them would test the proposed host,

anion, and barrier-distribution effects. A matched-generation batch split between silver and gold top electrodes would isolate the electrode effect from fabrication generation.

Temperature-dependent relaxation measurements are needed to test the proposed role of segmental mobility and distributed activation barriers. A $\tau(T)$ series yielding an Arrhenius or Vogel–Fulcher–Tammann activation energy would connect the host and anion differences to a segmental-relaxation barrier and test the temperature dependence of the stretch exponent β . Encapsulation and systematic ageing measurements would address the remaining stability questions.

4.11 Inputs to Temporal Computing

In the replicated *Super Yellow*/PEO/LiOTf grid, composition tunes the fading-memory time over roughly 2s–20s, with the switching window and potentiation strength moving in the same direction. Illustrative host and anion contrasts extend the measured range. Cation identity also shifts the dynamics, but the available data do not establish the predicted $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ order as a transferable law. Drive protocol co-sets the apparent timescale and must remain fixed in quantitative comparisons.

Chapter 5 uses the fading-memory times and device-to-device variability measured in this chapter. It also uses the pulse-update curves, including their saturation and turnover. Because the PEO fraction changes both the fading-memory time and the input nonlinearity α (fig. 4.14), each composition defines a paired operating point rather than two independent settings. Salt fraction provides a partly separate control of dynamic range and turnover. The simulations therefore compare banks drawn from one composition with banks spanning several compositions.

Chapter 5

Data-Driven Temporal Computing with Polymer-Electrolyte Memristor Reservoirs

5.1 Introduction and Scope

Chapters 2 and 4 established and quantified the slow, volatile response of these polymer-electrolyte composites. Here, behavioural models represent the measured response and are evaluated on temporal-computing tasks. A reservoir maps a time-varying input onto a high-dimensional dynamical state from which a linear read-out can recover temporal information.

The central comparison is between a bank drawn from one composition and a bank spanning several measured timescales. The latter should help when a target depends on several lags, but need not improve a task dominated by one timescale. Wearable *affective computing* (the recognition of emotional and physiological state) provides a relevant

test because its signals contain both fast and slow components. The evaluation proceeds from reservoir benchmarks to physiological streams, WESAD classification, continuous valence/arousal tracking, and tests with sensor noise and wrist-band data.

All results in this chapter are *in silico*; no new devices were fabricated. Each node is a compact behavioural model parameterised by the fits in chapter 4 (section 5.2), and every read-out is a single linear ridge regression. Demonstration A tests whether one composition suffices for a single-timescale feature. Demonstration B tests whether a composition-diverse bank improves multi-timescale tasks.

5.2 Measured Behavioural Model

Each composition cell is reduced to a compact behavioural model and used as one reservoir node. The model is fitted only to the three common measurements of chapter 4: hysteresis, pulsed potentiation, and delayed depotentiation. Its parameters therefore derive from measured device data.

Each composition cell is summarised by a *parameter card*: the write nonlinearity ϕ_c from the potentiation curves, the fading-memory law λ_c from the depotentiation curves, and an associated device-to-device spread. The fading memory is the stretched-exponential (Kohlrausch) retention identified in chapter 4,

$$\lambda_c(\Delta t) = \exp\left[-(\Delta t/\tau_c)^{\beta_c}\right],$$

where τ_c and β_c are the per-cell relaxation time and stretch exponent; for the few cells without an identified stretched fit, a single exponential reproducing the model-free half-life is used instead ($\tau_c = t_{1/2}/\ln 2$, $\beta_c = 1$). The write nonlinearity is the compressive, saturating build-up of

conductance with pulse number measured in chapter 4, parametrised by its log-log growth exponent α_c and its peak enhancement.

Driven by a rate-coded input $u_n \in [0, 1]$ sampled at interval Δt , a single device node updates as

$$x_n^{(i)} = \lambda_i x_{n-1}^{(i)} + \left[\max(w_i u_n, 0) \right]^{\alpha_i}, \quad \lambda_i = \exp \left[- (\Delta t / \tau_i)^{\beta_i} \right],$$

so that the node integrates a compressive write of its masked input and then leaks according to the measured retention λ_i over the sample interval; w_i is the reservoir input mask. The internal state is read sub-threshold and is assumed to be state-preserving, as in the proof-of-concept device of chapter 2. Every read-out is a single *linear* ridge regression on the device states.

The model has two measurement-related constraints. First, write nonlinearity and fading memory were measured at different write amplitudes (3 V for potentiation and 4 V for depotentiation; table 4.1). Combining them in one node assumes that the two regimes are compatible and probably underestimates the coupling between drive and retention. Second, both measurements used a single, effectively common inter-pulse cadence, so the model does not resolve input timing below that cadence. The gold/TMPE negative-polarity corpus of chapter 4 measures integration and relaxation within single runs and finds both interval-dependent potentiation and written-depth-dependent relaxation. Because it uses another host, electrode, polarity, and read protocol, it does not provide parameters for the banks used here. The experiment needed for the lead composition is specified in section 5.12. Per-cell parameter cards and all simulation and read-out settings are tabulated in section D.2.

5.3 Reservoir Computing and the Common Metric Language

The reservoir principle. Reservoir computing uses a fixed dynamical system whose state is connected to a trained linear read-out [17, 18, 30]. A physical reservoir requires fading memory, so its state retains recent input history, and a nonlinear response that maps inputs into features accessible to the read-out. Chapter 4 measures both properties in these devices: stretched-exponential depotentiation supplies fading memory, and compressive potentiation supplies nonlinearity. Memristive and organic implementations have previously used the same combination [5, 32, 34, 35]. This chapter evaluates the polymer-electrolyte devices as another implementation.

Architecture. The reservoir is a spatial bank of N *independent* device nodes, each represented by the behavioural model of section 5.2 and driven by a randomly masked copy of the input. With no inter-node recurrence, differences between banks arise from their intrinsic timescales, device-to-device scatter, and input masks rather than engineered connectivity. For Demonstration A, the same single-composition node could instead be deployed as a time-multiplexed delay-feedback reservoir with virtual nodes along a delay line [154]; the spatial bank is used throughout for consistency.

Common metrics. Linear memory capacity MC_k and its total [155] quantify the past input recoverable at each lag. NARMA-10 [156] provides a nonlinear system-identification benchmark, and the information-processing-capacity framework [80] separates linear and nonlinear contributions (fig. 5.3). Affective tasks are scored by class-balanced macro-F1. Robustness is tested by injecting the measured device-to-device and

cycle-to-cycle spread into the bank (fig. 5.9).

5.4 Benchmark Tier: Capacity and System Identification

Reservoir suitability is first evaluated without an application-specific task. Linear memory capacity measures fading memory, while information-processing capacity separates its linear and nonlinear components. The NARMA-10 system-identification task is used only to rank compositions, not to claim absolute performance. All three benchmarks use a random rate-coded input, N device nodes, and a linear ridge read-out.

Statistical units. Random input and bank seeds sample the algorithm’s sensitivity to stochastic initialisation; they are not independent devices, experiments, or subjects. Seed-only benchmark comparisons therefore report means, standard deviations, and sign consistency without inferential p -values. Inferential paired tests are reserved for independent held-out subjects in the WESAD and CASE analyses, or for independently fabricated devices in the experimental chapters. The complete distinction is recorded in table D.1.

Memory capacity. The linear memory capacity of Jaeger measures how much of the past input a reservoir can reconstruct: MC_k is the squared correlation between a ridge reconstruction of the input k steps ago and its true value, and the total is the sum over all lags, bounded above by the number of nodes N [155]. Figure 5.1 compares two banks of equal size ($N = 24$): a *homogeneous* bank of jittered copies of the lead cell (PEO 0.3/0.09, differing only by device-to-device scatter) and a *heterogeneous* bank spanning the composition grid (τ from ≈ 3 to ≈ 26 s).

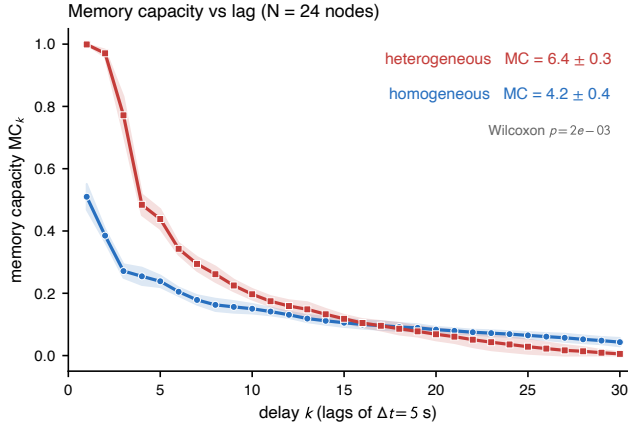


Figure 5.1: Linear memory capacity MC_k as a function of delay k for a homogeneous bank (all PEO 0.3/0.09 nodes) versus a heterogeneous composition bank ($\tau \approx 3\text{--}26\text{ s}$). The heterogeneous bank broadens recoverable memory across lags, raising total memory capacity from 4.2 to 6.4 (a $1.54\times$ increase, positive in all ten seed pairs), with the gain concentrated at short-to-intermediate lags. Shaded bands are ± 1 SD over ten random input/bank seeds. Seeds quantify algorithmic sensitivity rather than experimental replication. $N = 24$ leaky nodes, random uniform input, linear (ridge) read-out; in-silico devices built from the Chapter 4 parameter cards.

The heterogeneous bank broadens the memory profile and raises total capacity from 4.2 to 6.4 (means over ten random input/bank seeds), a factor of 1.54 ± 0.10 . Single-step recall rises from $MC_1 = 0.51$ to 1.00, with most of the gain at short-to-intermediate lags. The increase occurs in all ten seed pairs; this is a sensitivity result, not a population-level significance test. Because the nodes are independent, the increase comes from their timescale distribution, device scatter, and input masks.

Composition sweep and the single-node optimum. The same benchmarks, run on single-composition banks, rank the compositions against one another and so test the Demonstration-A claim that one cell suffices for a single-timescale task. Figure 5.2 shows that the lead cell PEO 0.3/0.09 attains the highest single-composition memory capac-

ity (3.98 ± 0.19 , mean $\pm$ SD over ten seeds), while the neighbouring PEO 0.3/0.045 cell is marginally better on the NARMA-10 system-identification benchmark [156] (NRMSE 0.587 ± 0.019 against the lead's 0.616 ± 0.031 , a close second). Both optima fall in the low-PEO row that chapter 4 identifies as carrying the longer fading-memory times, so the benchmark ranking is consistent with the composition result, not an artefact of the simulation. This supports the Demonstration-A statement: for a single-timescale feature, the lead composition is the memory-capacity optimum and a heterogeneous bank is unnecessary. The NARMA-10 numbers carry only *relative* weight: the absolute error is high (≈ 0.6 – 0.8) because a bank of independent, non-recurrent nodes is a weak NARMA-10 reservoir, so the stronger evidence here comes from the capacity measures above (figs. 5.1 and 5.3) and NARMA-10 serves only to corroborate the composition ranking, a limitation a recurrent or delay-feedback architecture would relax.

Comparison with randomly distributed timescales. A generic echo-state bank of the same size tests whether the measured timescales are individually important or whether their range is sufficient. A random bank spanning the same range as the compositions ($\tau \approx 3$ – 26 s) has a memory capacity of 6.7 ± 0.3 , compared with 6.4 ± 0.3 for the device bank (ten matched seeds; mean difference $+0.28$). Expanding the random range to 0.5 – 60 s raises the capacity to 7.6 ± 0.3 (mean difference $+1.14$, positive in all ten seed pairs). The measured τ values are therefore not uniquely optimal; the relevant property is the range of timescales. The broader-range control also indicates that extending this range, for example through the drive-amplitude dependence measured in chapter 4, could increase capacity further.

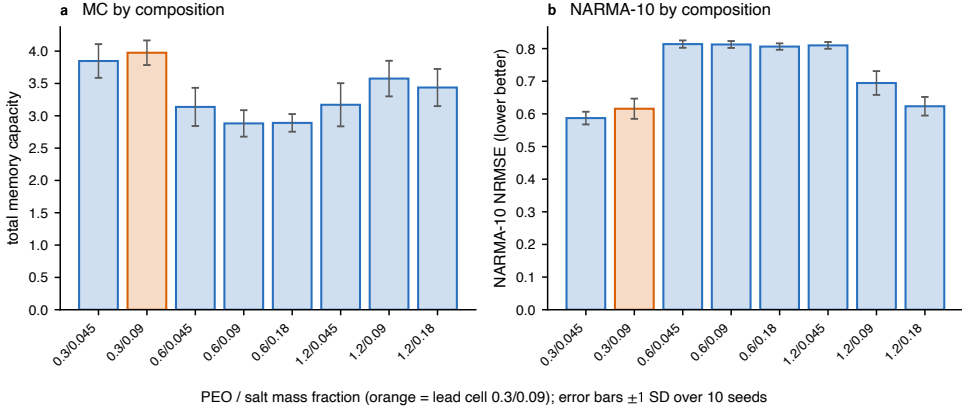


Figure 5.2: (a) Total linear memory capacity and (b) NARMA-10 reconstruction error (NRMSE, lower is better) for single-composition reservoirs across the PEO/LiOTf grid. The lead cell PEO 0.3/0.09 (orange) attains the highest memory capacity; the neighbouring 0.3/0.045 cell is marginally better on NARMA-10. Both optima fall in the low-PEO row anticipated by chapter 4. Only the *relative* composition ranking is claimed: absolute NARMA-10 NRMSE is high because a bank of independent, non-recurrent leaky nodes is a weak NARMA-10 reservoir.

Linear and nonlinear capacity. Memory capacity counts only linearly recoverable information. The information-processing-capacity framework of Dambre et al. decomposes total capacity into orthogonal polynomial degrees: degree-one terms give linear capacity, while degree-two terms measure nonlinear self and cross-lag transformations; the total is bounded by the node count N [80]. Figure 5.3 reports this decomposition. The homogeneous bank has a nonlinear capacity of 2.7 within a total of 7.1, showing that compressive potentiation contributes a nonlinear response in addition to fading memory. The heterogeneous bank raises the mean total capacity to 11.6, with the increase present in all ten seed pairs and in both components. Almost all nonlinear capacity comes from degree-two self terms; cross-lag products are negligible, as expected for an independent, non-recurrent bank. The total remains below the N bound for both banks.

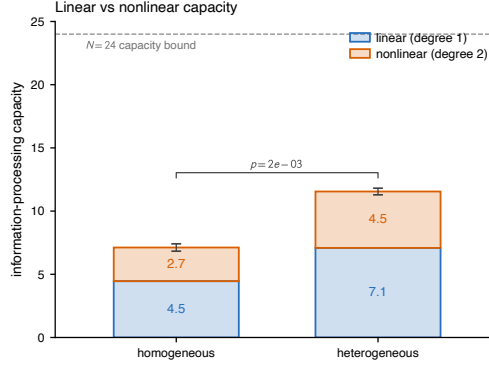


Figure 5.3: Information-processing capacity [80] split into linear (degree-one) and nonlinear (degree-two) parts for homogeneous and heterogeneous banks ($N = 24$, ten seeds). Both banks have nonzero degree-two capacity, indicating a nonlinear contribution from the compressive write. Heterogeneity raises mean total capacity from 7.1 to 11.6, with a positive difference in all ten seed pairs. The dashed line is the $N = 24$ upper bound; error bars are ± 1 SD on the total over seeds, which quantify algorithmic sensitivity.

5.5 Affective Computing

Affective computing uses physiological signals to infer emotional and physiological state [157]. The device timescales, from a few seconds to tens of seconds, overlap the phasic electrodermal, respiration, and heart-rate-variability bands without time rescaling (table 5.1). These signals also contain several timescales at once: phasic electrodermal responses, respiration and heart-rate-variability rhythms over seconds, and tonic levels drifting over minutes [158–160]. Figure 5.4 places the measured relaxation time of each composition on the same axis as these physiological bands and shows the minutes-scale range not reached by composition alone.

Extending the model to tonic timescales. Raising the write amplitude lengthens fading memory in the measurements of chapter 4. Adding modelled, drive-boosted minutes-scale nodes nearly doubles recall beyond

Table 5.1: Characteristic timescales of the peripheral affective signals used in this chapter, and the composition cells whose fading-memory window covers them [158–160]. The grid of chapter 4 spans the full range, with the lead PEO 0.3 cells serving the slow LF-HRV band.

Affective signal	Timescale	Device coverage
Phasic EDA (SCR)	$\sim 1\text{--}10\text{ s}$	low–mid PEO ($t_{1/2}$ 3–19 s)
Respiration	$\sim 3\text{--}5\text{ s}$	mid PEO
HRV, HF band	$\sim 2.5\text{--}7\text{ s}$	mid PEO
HRV, LF band	$\sim 7\text{--}25\text{ s}$	PEO 0.3 ($\tau \approx 19\text{--}26\text{ s}$)
Tonic SCL	minutes	slowest / drive-boosted

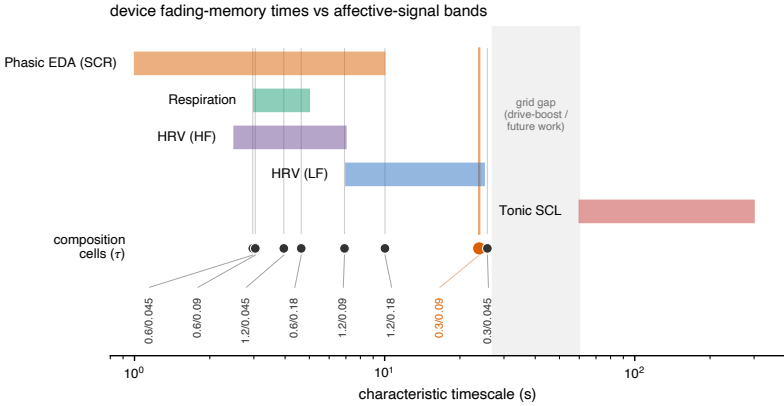


Figure 5.4: Each PEO/LiOTf composition cell is placed at its measured fading-memory time τ (markers, lower track) on the same logarithmic axis as characteristic timescale bands of peripheral affective signals (coloured bars; table 5.1). The composition grid ($\tau \approx 3\text{--}26\text{ s}$) overlaps the phasic-EDA, respiration, and HRV bands, with the lead cell PEO 0.3/0.09 (orange) in the slow LF-HRV band. Composition alone does not reach the tonic, minutes-scale band (shaded gap); testing that range would require drive-amplitude measurements or a slower cell.

75 s and raises total memory capacity from 6.4 to 7.0. On the physiological streams, reconstruction at delays up to 120 s improves by +0.019 in held-out R^2 , with a positive difference in all ten seed pairs (section D.5). This is a modelled extrapolation because no minutes-scale cell was measured. A drive-amplitude experiment at these timescales is required to test it.

The experiments that follow use the Wearable Stress and Affect Detection corpus (WESAD) of Schmidt et al. [161]: fifteen subjects wearing a chest sensor that records electrodermal activity, respiration, temperature, and an electrocardiogram, with periods labelled baseline, stress, or amusement. WESAD is used in two complementary ways: as a source of real physiological streams whose multi-lag temporal context the reservoir must encode (section 5.6), and as a downstream affective-classification task (section 5.7).

5.6 Physiological Temporal-Context Reconstruction

The benchmarks of section 5.4 show that a spread of device timescales broadens memory on a random input. This section tests the same property on the multi-timescale physiological signals introduced in section 5.5, isolating temporal-context encoding before turning to affect labels (section 5.7).

The reservoir is run continuously over each WESAD session at $\Delta t = 1$ s on four chest channels (electrodermal activity, respiration, temperature, and an electrocardiogram-derived heart rate), and a linear read-out reconstructs each channel at delays of 1, 3, 8, 20 and 45 s. These delays span fast phasic and beat-to-beat context, respiration- and HF-HRV-scale context, and tonic, LF-HRV-scale context [158, 159]. Evaluation is leave-one-subject-out over all fifteen subjects, with $N = 48$ nodes per bank.

Table 5.2 reports the held-out reconstruction quality. A memoryless bank gives $R^2 = 0.674$, compared with 0.673 for instantaneous input. A single-timescale memory raises R^2 to about 0.74, and the heterogeneous composition bank reaches 0.756: +0.084 over instantaneous input and

Table 5.2: Leave-one-subject-out held-out R^2 for multi-lag physiological context reconstruction on WESAD ($N = 48$ nodes; means over ten bank seeds where applicable). Bands are reconstruction delays: fast = 1–3 s, mid = 8 s, slow = 20–45 s. Fast and slow homogeneous banks give their highest scores in their corresponding bands. Across the full context vector, the heterogeneous bank exceeds the best homogeneous bank by +0.012 R^2 , with a positive difference in all fifteen subjects (paired Wilcoxon $p < 10^{-4}$).

Bank	overall R^2	fast	mid	slow
Instantaneous	0.673	0.715	0.670	0.632
Memoryless	0.674	0.721	0.667	0.630
Homogeneous, fast	0.741	0.815	0.735	0.669
Homogeneous, slow	0.744	0.787	0.724	0.711
Heterogeneous	0.756	0.815	0.737	0.708

+0.012 over the best same-size homogeneous control. This difference is positive for all fifteen subjects (paired Wilcoxon signed-rank $p < 10^{-4}$). In the band-resolved results, the fast homogeneous bank performs best at the shortest delays and the slow homogeneous bank at the longest. The heterogeneous bank gives the highest single-bank score across the full delay range (fig. 5.5).

5.7 WESAD Affective Classification

The downstream task is affective-state classification from the WESAD physiological streams with a leave-one-subject-out linear read-out. The analysis reports the contribution of timescale heterogeneity with uncertainty, including when the increment does not differ from zero. The study that introduced WESAD reports three-class chest macro-F1 ≈ 0.72 with engineered window features and conventional classifiers under the same leave-one-subject-out protocol [161]; the present streaming model reaches 0.758 under that matched metric and subject split (table 5.3). Accuracy-only results or studies using different validation splits are excluded from

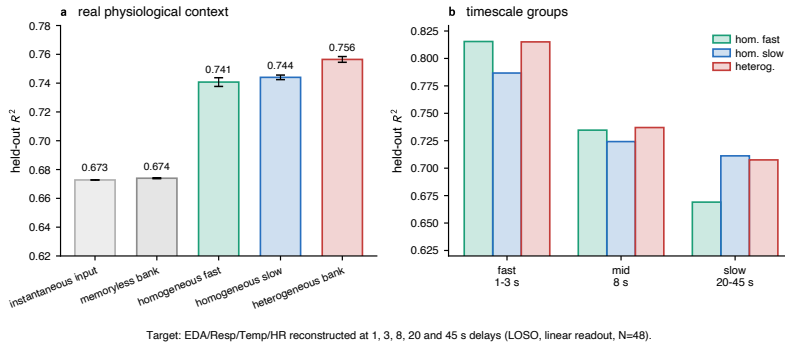


Figure 5.5: Multi-lag physiological context reconstruction on real WESAD streams (chest EDA, respiration, temperature, and ECG-derived heart rate). A linear read-out reconstructs each channel at delays of 1, 3, 8, 20 and 45s from the reservoir state (leave-one-subject-out, $N = 48$). The heterogeneous composition bank gives the best single-bank held-out R^2 (0.756) across the full fast-to-slow context vector, ahead of the best same-size homogeneous control (0.744) and well ahead of instantaneous input (0.673). Fast and slow homogeneous banks each win only their own band.

Table 5.3: Matched comparison on three-class WESAD chest signals. Both rows report class-balanced macro-F1 under leave-one-subject-out validation over the same fifteen subjects. The models and temporal formulations still differ, so the table provides a scale reference rather than a superiority claim. Accuracy-only and differently split studies are omitted.

Study	Features and read-out	Validation	macro-F1
Schmidt et al. (2018) [161]	engineered window features, AdaBoost/RF	15-subject LOSO	≈ 0.72
This work	streaming device-model states, linear ridge read-out	15-subject LOSO	0.758

this numerical comparison because they estimate a different quantity.

Demonstration A: window-level electrodermal classification. Binary stress-versus-baseline classification from electrodermal activity, read out at the window level from the lead composition alone, reaches a leave-one-subject-out macro-F1 of 0.894 (fig. 5.6a). A static baseline using

the mean, standard deviation, last value, and slope of the same signal reaches 0.899. The 60-second-window task therefore does not demonstrate a contribution from temporal memory or timescale heterogeneity. This motivates the streaming formulation below.

Demonstration B: streaming three-class tracking. The reservoir runs continuously over each session, and the affective class (baseline, stress, or amusement) is read out at every step. This formulation uses the evolving reservoir state without a static window summary. Table 5.4 decomposes the result into four matched-size banks. Under the same leave-one-subject-out protocol, a majority-class predictor reaches macro-F1 0.23 and a stratified-random predictor 0.33, compared with 0.758 for the heterogeneous reservoir. Bank comparisons are assessed across the fifteen independent held-out subjects, preventing within-subject temporal autocorrelation from inflating the test. Relative to instantaneous input, a same-size memoryless bank contributes +0.028, a homogeneous fading-memory bank adds +0.016, and timescale heterogeneity adds +0.005. The fading-memory gain over a same-size memoryless bank is +0.014 to +0.020. The heterogeneity increment is $+0.005 \pm 0.011$, within seed variation; it is positive in seven of ten seeds, and remains in $[+0.002, +0.007]$ across sampling intervals from 0.5 to 4s without separating from zero. Per-class F1 is 0.90 for baseline, 0.92 for stress, and 0.55 for amusement. Amusement has the least labelled data (roughly 5.6 ks, compared with 17.6 ks for baseline and 10.0 ks for stress) and the lowest autonomic arousal. Its score remains low across all banks, so this class does not distinguish memory configurations.

Why heterogeneity does not improve the label scores. The affect labels are slowly varying sustained states, and their discriminative memory is dominated by the slow tonic band already covered by the

Table 5.4: Leave-one-subject-out macro-F1 for streaming three-class WESAD tracking (means $\pm$ SD over ten bank seeds; matched input masks). Each increment is the gain over the row above: fading memory contributes a small, consistent gain, while the timescale-heterogeneity increment is within noise (*ns*).

Bank	macro-F1	increment
Instantaneous (no memory)	0.709	—
Memoryless (dimensionality only)	0.737	+0.028
Homogeneous (memory, single τ)	0.753 ± 0.009	+0.016
Heterogeneous (memory, τ spread)	0.758 ± 0.006	+0.005 (<i>ns</i>)

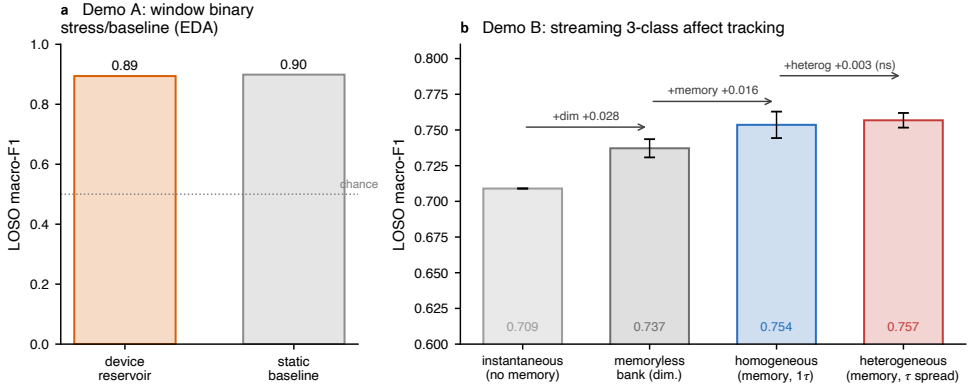


Figure 5.6: Affective classification with in-silico device reservoirs (leave-one-subject-out, linear read-out). (a) Demonstration A, window-level binary stress-versus-baseline from EDA: the single lead composition matches a static feature baseline, i.e. the 60 s-window task is quasi-static and requires no fading memory. (b) Demonstration B, streaming three-class affect tracking (baseline/stress/amusement) decomposed as instantaneous $\rightarrow$ +dimensionality $\rightarrow$ +fading memory $\rightarrow$ +timescale heterogeneity. Fading memory contributes a small but consistent gain; the timescale-heterogeneity increment ($+0.005 \pm 0.011$) is within seed noise. Error bars are ± 1 SD over 10 bank seeds.

lead composition ($\tau \approx 19\text{--}26$ s). A wider timescale distribution improves performance when a target requires several lags simultaneously, as in the memory-capacity and physiological-context tasks (fig. 5.1 and table 5.2). The sustained label task does not show that requirement. Under this pro-

tol, fading memory helps classification, but the additional heterogeneity increment is not resolved.

5.8 Continuous Valence and Arousal

The WESAD tasks of section 5.7 use categorical labels dominated by sustained states. Continuous valence (pleasantness) and arousal (activation) provide a dimensional target with fast phasic components and slow tonic drift [157]. This third corpus tests whether a spread of timescales improves prediction of that continuous target.

The Continuously Annotated Signals of Emotion corpus (CASE) of Sharma et al. [162] records thirty subjects who rated valence and arousal continuously on a joystick (range [0.5, 9.5], 20 Hz) while watching emotional videos, with synchronised physiology at 1 kHz. The reservoir is run over the eight emotional videos at $\Delta t = 1$ s on the same slow channel set as WESAD (electrodermal activity, respiration, skin temperature, and an electrocardiogram-derived heart rate), with robust scaling per subject. The state is reset and a short washout is removed at each video boundary to prevent persistence across stimuli. A linear ridge read-out predicts valence and arousal under leave-one-subject-out evaluation, scored by the concordance correlation coefficient (CCC) [163].

Every fading-memory bank outperforms the instantaneous read on the CASE task (the heterogeneous bank by +0.079 mean CCC, positive in 24 of 30 subjects, paired Wilcoxon $p < 10^{-4}$). A multi- τ filter bank raises CCC from 0.23 for a single timescale to 0.41 (+0.176, 27/30 subjects, $p < 10^{-6}$). Timescale spread therefore improves this continuously annotated target, consistent with the memory-capacity and physiological-context results of sections 5.4 and 5.6 and unlike the sustained WESAD labels.

On this smooth regression, a linear multi- τ filter uses the timescale spread

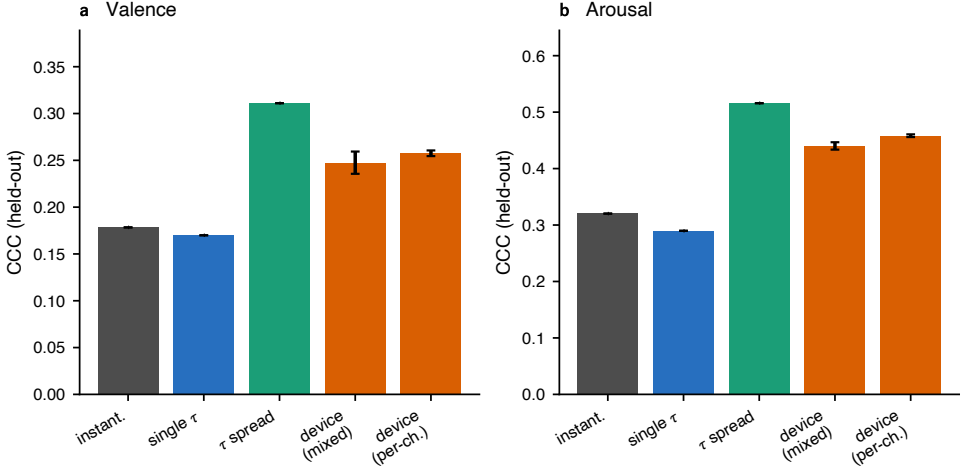


Figure 5.7: Continuous affect tracking on the CASE corpus [162] (thirty subjects; held-out concordance correlation coefficient, leave-one-subject-out, linear read-out; mean $\pm$ SD over ten bank seeds). (a) Valence and (b) arousal. A linear multi- τ filter bank increases CCC by +0.176 over a single- τ filter, with a positive difference in 27/30 subjects ($p < 10^{-6}$). Device banks with mixed and per-channel input masks also use a spread of timescales but score below the linear filter on this regression task.

more effectively than the device model. A per-channel device bank reaches CCC 0.358, compared with 0.413 for the linear bank (-0.047 , $p < 10^{-2}$ across the thirty held-out subjects); random cross-channel mixing in the standard bank reduces the score by a further ≈ 0.014 . The device models therefore do not outperform an equivalent digital filter on this task. Their proposed advantage is passive material-level filtering at the core-power estimate of section 5.10. The increment from device-internal heterogeneity is not resolved across held-out subjects ($+0.005$, $p = 0.18$; subject values are seed-averaged), because the jittered homogeneous bank already contains a timescale spread (section D.7).

5.9 Continuous Monitoring Under Deployment Conditions

The sustained WESAD states can often be classified from the instantaneous signal level, as shown by the static-feature baseline. Wearable measurements also contain sensor noise and motion artefacts, particularly in wrist-worn electrodermal and photoplethysmographic signals. Fading memory implements temporal low-pass filtering and may therefore reduce sensitivity to transient artefacts. The following experiments test that effect directly.

Figure 5.8 tests a continuous binary stress detector (stress versus baseline plus amusement, read out at every step, leave-one-subject-out, with the same linear read-out used throughout). On clean streams, all banks reach macro-F1 ≈ 0.91 – 0.92 , comparable to the $\approx 93\%$ binary benchmark for the corpus [161]. With increasing sensor noise and motion-artefact bursts, the instantaneous and memoryless controls fall from 0.914 to 0.840 at $\sigma = 0.4$ of the signal range, while the fading-memory banks remain near 0.92. At $\sigma = 0.4$, the heterogeneous bank exceeds the instantaneous read by +0.088 for all fifteen subjects (paired Wilcoxon $p < 10^{-4}$). On clean streams, fading memory reduces the false-alarm rate from 0.046 to 0.026. Homogeneous and heterogeneous banks have similar robustness, associating the result with their shared fading memory. Near label transitions, the increase from 0.747 to 0.775 is not resolved across subjects, and detection latency is similar across banks because it is set by the prediction-smoothing horizon.

Accelerometer-gated motion artefact. Because the preceding corruption is synthetic, a second test gates its amplitude with the measured chest accelerometer. At each step, injected amplitude follows measured movement rather than being distributed uniformly. At the strongest

gating, the instantaneous read falls from macro-F1 0.914 to 0.898, while the memory bank remains near 0.92. The +0.032 difference is positive across subjects (paired $p = 0.04$), and a classical moving average gives a similar +0.036 difference (section D.4). The effect is smaller than under the synthetic injection, consistent with a mechanically stable chest strap and a largely seated protocol. Wrist measurements are examined in section 5.10.

Classical-filter control. An exponential moving average applied to the same raw channels tests whether the noise robustness requires the device model. A single time constant matched to the lead cell reaches macro-F1 0.927 at $\sigma = 0.4$, compared with 0.840 for the instantaneous read and 0.916 for the device bank (+0.091 over instantaneous, all fifteen subjects, $p < 10^{-4}$). Spreading the moving-average timescales adds +0.005 (*ns*), and the device’s compressive nonlinearity adds -0.008 relative to that linear filter bank (*ns*; table D.3). These controls attribute the robustness to fading-memory filtering; the device-specific nonlinearity provides no resolved increment. The proposed device contribution is to perform this filtering through material relaxation, with no trained dynamics and at the core-power estimate of section 5.10. The multi-timescale benchmarks of sections 5.4 and 5.6 distinguish the heterogeneous bank; this single-timescale stress detector does not.

Cross-corpus replication. The robustness test was repeated on the PhysioNet Non-EEG corpus of Birjandtalab et al. [164]: twenty subjects wearing an Affectiva Q wrist sensor for electrodermal activity and skin temperature, with pulse-oximeter heart rate, during relaxation and cognitive, emotional, and physical stress phases. Cognitive and emotional stress are compared with relaxation; the physical-exertion phase is held out because its elevated heart rate represents a different mechanism.

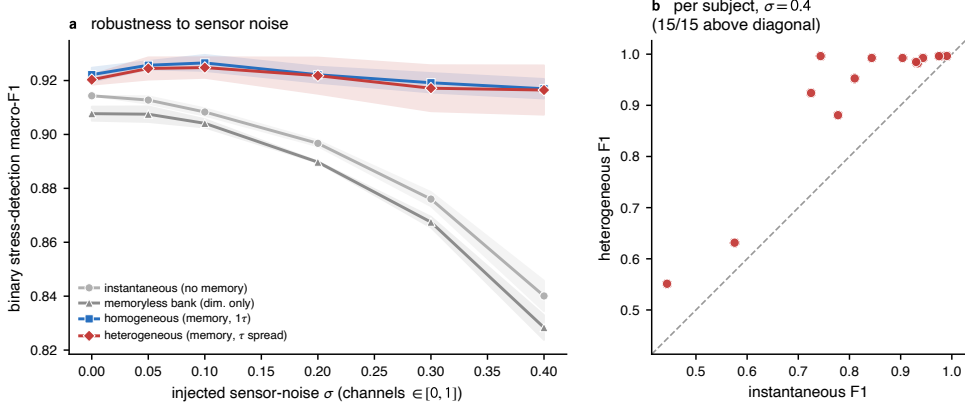


Figure 5.8: Continuous stress monitoring with device reservoirs (binary stress vs. baseline-plus-amusement, per-step, leave-one-subject-out, linear read-out). (a) Detection macro-F1 against the magnitude σ of injected sensor noise and motion-artefact bursts. Fading-memory banks remain near 0.92, while instantaneous and memoryless controls decrease with noise; bands are ± 1 SD over seeds. (b) At $\sigma = 0.4$, the heterogeneous memory bank exceeds the instantaneous read for all fifteen subjects (paired Wilcoxon $p < 10^{-4}$). Homogeneous and heterogeneous banks have similar scores, associating this result with fading memory rather than timescale spread.

With the same detector, clean macro-F1 is ≈ 0.65 – 0.68 , compared with ≈ 0.91 on WESAD, and the clean-stream memory difference is not resolved. Under injected noise, the instantaneous detector decreases from 0.651 to 0.516, while the fading-memory bank remains near 0.66. At $\sigma = 0.4$, the difference is $+0.130$, positive in nineteen of twenty subjects (paired Wilcoxon $p < 10^{-5}$; table 5.5). This independent corpus reproduces the noise-response ordering with a different sensor and population.

Table 5.5: Noise-response comparison on the independent PhysioNet Non-EEG stress corpus [164] (twenty subjects; Affectiva Q wrist electrodermal activity and temperature plus pulse-oximeter heart rate; cognitive and emotional stress versus relaxation; leave-one-subject-out, linear read-out). At $\sigma = 0.4$, the heterogeneous-minus-instantaneous difference is $+0.130$, positive in 19/20 subjects ($p < 10^{-5}$). Homogeneous and heterogeneous banks again have similar scores.

Detector	macro-F1 (clean)	macro-F1 ($\sigma = 0.4$)
Instantaneous (no memory)	0.651	0.516
Homogeneous (memory)	0.682	0.657
Heterogeneous (memory)	0.667	0.646

5.10 Toward a Worn Device: Sensor Site and Operating Budget

The robustness of the previous section is a laboratory statement until two further questions are answered: whether the advantage survives a consumer-grade sensor, and what the device would cost to run. Both are answered from the same measured ingredients (the WESAD wrist recordings and the device constants of chapter 2) without any new assumption.

Sensor site. WESAD records each session with both a chest strap and an Empatica E4 wrist band. The wrist band has fewer and noisier channels: skin conductance, skin temperature, and photoplethysmography-derived heart rate, without respiration. On the wrist data (table 5.6), the clean macro-F1 of the instantaneous detector falls from 0.914 to 0.881, whereas the fading-memory reservoir changes from 0.920 to 0.922. The memory advantage is consequently larger on the wrist than on the chest ($+0.041$ versus $+0.006$ on clean streams and $+0.069$ under injected noise). Fading memory also reduces the wrist false-alarm rate from 0.062 to 0.040.

Table 5.6: Continuous binary stress-detection macro-F1 on the WESAD chest strap versus the Empatica E4 wrist band (leave-one-subject-out, linear read-out). Moving to the noisier consumer sensor degrades the instantaneous (static) detector but not the fading-memory reservoir, so the memory advantage is larger on the wrist; σ is the injected sensor-noise level of fig. 5.8.

Detector	Chest		Wrist E4	
	clean	$\sigma = 0.4$	clean	$\sigma = 0.4$
Instantaneous (no memory)	0.914	0.840	0.881	0.847
Memory reservoir (het)	0.920	0.916	0.922	0.916

Operating budget. The device dynamics are fixed, and only a linear read-out is fitted by one closed-form ridge solve. The trained component is an $(N+1) \times K$ weight matrix: 75 weights for the $N = 24$, three-class banks and 147 for the $N = 48$ banks. The deep models commonly used for wearable affect contain 10^4 – 10^6 trainable weights, two to four orders of magnitude more. Using the measured proof-of-concept energy of ≈ 50 nJ per write event (≈ 6 fJ μm^{-2} over a 0.0825 cm² junction), charging $N = 24$ nodes once per second corresponds to ≈ 1.2 μW . The linear read-out requires NK multiply–accumulates, well under a nanojoule per step. These estimates use unoptimised large-area devices; smaller junctions would reduce the event energy.

The estimate covers only the reservoir core: device charging and read-out operations. It excludes the sensor analogue front-end, analogue-to-digital conversion, and microcontroller overhead and is not a full system-power budget. The device response remains faster than the sub-hertz physiological signals considered here, and its relaxation performs the temporal filtering during the write/read cycle. The assumptions are tabulated in section D.6.

5.11 Matching Device and Task Timescales

Performance depends on the overlap between device and task timescales (fig. 5.4). In the single-composition sweep, the lead cell gives the highest memory capacity for the seconds-scale benchmark (fig. 5.2). For the multi-lag physiological task, no single cell covers the full delay range, and the heterogeneous bank gives the highest single-bank score (table 5.2).

Effect of timescale breadth. Table 5.7 separates the effect of fading memory from that of timescale breadth. Both homogeneous and heterogeneous memory banks outperform memoryless and instantaneous baselines, with gains of up to +0.08 macro-F1 under sensor noise (sections 5.9 and 5.10). Heterogeneity provides an additional gain when several lags are required: memory capacity increases by a factor of 1.54 in simulation (positive in all ten seed pairs; fig. 5.1), and physiological-context reconstruction improves by +0.012 held-out R^2 over the best homogeneous bank (15/15 subjects, $p < 10^{-4}$; table 5.2). A multi- τ filter similarly improves continuous valence/arousal tracking by +0.176 CCC (27/30 subjects, $p < 10^{-6}$; section 5.8). The seed result is a sensitivity summary; the subject-level results support population inference. When one slow timescale dominates, as in the sustained stress labels and single-channel monitoring task, the best-matched homogeneous bank is as good as or marginally better than the heterogeneous bank. A composition-diverse bank is therefore useful for an unknown mixture of tasks: it loses at most about 0.01 macro-F1 on the single-timescale tasks considered here and gains more on the multi-timescale tasks. The available variation comes from composition and device-to-device scatter; chemistry and drive amplitude provide additional illustrative axes in chapter 4.

Table 5.7: Comparison of instantaneous, homogeneous, and heterogeneous banks across the affective tasks in this chapter. Both memory banks exceed the instantaneous baseline, with the largest differences under sensor noise. The heterogeneous bank gives the highest physiological-context score, while the best-matched homogeneous bank gives the highest score for sustained labels and single-channel monitoring; homogeneous–heterogeneous differences on the classification tasks remain below about 0.01 macro-F1. Bold marks the highest score per row. Scores from tables 5.2 and 5.4 and sections 5.9 and 5.10.

Affective task	Metric	Instantaneous	Homogeneous	Heterogeneous
Physiological context (multi- τ)	R^2	0.673	0.744	0.756
Three-class affect tracking	macro-F1	0.709	0.753	0.758
Stress monitoring, clean	macro-F1	0.914	0.922	0.920
Stress monitoring, $\sigma=0.4$	macro-F1	0.840	0.917	0.916
Wrist stress, clean	macro-F1	0.881	0.930	0.922
Wrist stress, $\sigma=0.4$	macro-F1	0.847	0.918	0.916

Operating point and integration. The models apply in the measured sub-kilohertz regime, which covers the physiological signals used here but not fast logic. The assumed sub-threshold, state-preserving read sets the read-voltage and read-disturb limits for a circuit implementation. Figure 5.9 reports total memory capacity as device-to-device scatter is varied. The heterogeneous bank retains its advantage across the tested range, including the measured $\sigma(\ln \tau) \approx 0.85$. This value is estimated from silver PEO/LiOTf devices within each composition cell and is not explained by spin speed, anneal, film thickness, or device age (section D.3). At zero scatter, a bank of identical nodes has little memory capacity because its states are redundant. A homogeneous bank therefore depends on within-composition scatter, whereas the heterogeneous bank also contains composition-dependent timescale differences.

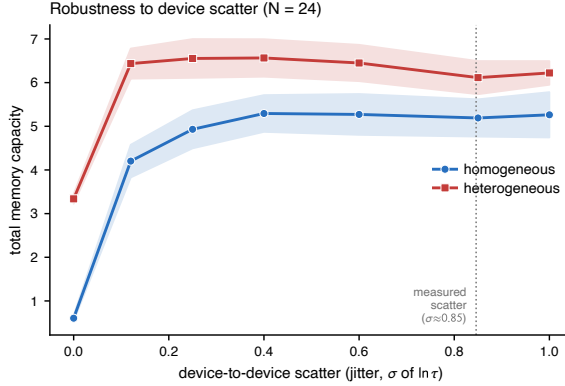


Figure 5.9: Total memory capacity versus injected device-to-device scatter (jitter on τ , β , and α) for homogeneous and heterogeneous banks ($N = 24$, ten seeds; shaded bands ± 1 SD). The heterogeneous bank retains a higher capacity across the tested range, including at the measured scatter (dotted line: $\sigma(\ln \tau) \approx 0.85$, derived in section D.3). At zero scatter, identical node states are redundant and the homogeneous bank has little memory capacity.

5.12 Limitations and Scope Conditions

The results are subject to the following scope conditions.

In silico. The demonstrations are simulations. Each node is the measured $\phi \otimes \lambda \otimes f$ behavioural model of section 5.2, not a bench reservoir, and every figure caption names the data behind it; the step from a fitted per-device model to a wired hardware reservoir is left to future work.

The composition assumption. As detailed in section 5.2, write non-linearity and fading memory were measured separately, at different write amplitudes and one inter-pulse cadence. Combining them assumes that the two regimes are compatible and probably underestimates the coupling between drive and retention. In a sensitivity analysis, each node’s retention was rescaled with its write strength across a wide coupling

range. The heterogeneous-to-homogeneous capacity ratio remained between 1.15 and 1.53 (section D.5). The gold/TMPE negative-polarity measurements provide one empirical coupling estimate: write depth and decay half-time have a rank correlation of $\rho = +0.46$ across 28 single-run traces. Calibrating the sensitivity model to this value gives $\kappa^* = +0.18$, at which the capacity ratio is 1.45 and the heterogeneous bank is better in all ten paired random-seed runs (section D.5). That corpus also shows interval-dependent potentiation, but its host, electrode, and polarity differ from the lead-cell model. A matched-amplitude experiment on the lead composition that varies pulse count and inter-pulse interval together would remove this assumption.

Operating regime and read-out. The model is identified only in the measured sub-kilohertz regime; fast temporal tasks are outside its scope. Every read-out is a single ridge regression because a nonlinear read-out could supply the task nonlinearity independently of the reservoir. The read is assumed to be sub-threshold and state-preserving.

Heterogeneity axes. The quantitative heterogeneous banks span the replicated PEO/LiOTf composition grid. The chemistry and drive-amplitude axes of chapter 4, being sample-limited ($n \leq 2$ per matched cell), enter only as illustrative additional controls; Demonstration B is therefore a principle demonstration of composition heterogeneity, not a powered comparison across chemistries. The composition cells that define these banks index *composition*, not film thickness: although spin-coating speed and residual film thickness covary with the PEO fraction, the parameter cards inherited here were shown to be governed by composition, not thickness (fig. 4.1, section 4.2), so the heterogeneity exploited in this chapter is compositional in origin.

Corpora and negative controls. The affective results use WESAD as the main corpus. Noise robustness is replicated on the PhysioNet Non-EEG dataset (section 5.9), and continuous valence and arousal are tested on CASE (section 5.8). A broader multi-corpus study remains future work. Two comparisons do not show a device-specific improvement: timescale heterogeneity does not separately improve final-label classification when one slow timescale dominates (section 5.7), and the nonlinear device encoding does not exceed a linear multi- τ filter on smooth continuous-affect regression (section 5.8). By contrast, the heterogeneous bank improves the multi-timescale memory-capacity and physiological-context tasks.

5.13 Summary

Behavioural models fitted to the measurements of chapter 4 support temporal-computing simulations with a linear read-out. A heterogeneous bank raises memory capacity by a factor of 1.54 and improves multi-lag physiological-context reconstruction by +0.012 held-out R^2 over the best homogeneous bank. On WESAD, the models reach leave-one-subject-out macro-F1 scores of 0.894 for binary stress and 0.758 for three-class streaming classification. Under injected sensor noise, fading-memory banks remain near 0.92 macro-F1 while the instantaneous baseline falls to 0.84 (+0.088, all fifteen subjects, $p < 10^{-4}$). A classical moving average matches this robustness, attributing it to fading-memory filtering. The model’s nonlinearity provides no resolved increment on this task. The proposed material implementation would realise that filter at an estimated reservoir-core power of about a microwatt, with tens to hundreds of trained linear-read-out weights.

The additional value of a timescale distribution depends on the task. It is resolved for random-input memory, physiological-context recon-

struction, and continuous valence/arousal tracking, all of which depend on several lags. It is not resolved for the sustained WESAD labels or single-timescale stress monitoring. Composition-dependent relaxation and within-composition scatter provide the distributions used in these simulations.

The next experimental steps are a matched-amplitude pulse protocol that varies pulse count and cadence, a replicated study of chemistry and drive-amplitude heterogeneity, and a wired device bank that tests the behavioural-model predictions in hardware.

Chapter 6

Conclusions and Outlook

This thesis investigated whether the volatility of an organic polymer-electrolyte memristive device can support temporal computation. Chapters 2 to 4 establish the device response, diagnose its reproducibility, and measure how composition and chemistry change its dynamics. Chapter 5 evaluates behavioural models of those dynamics on temporal-computing tasks. This chapter summarises the results, their limitations, and the experiments needed to test them in hardware.

6.1 Summary of Contributions

The contributions fall into four parts: a proof-of-concept device, a reproducibility and fabrication-record study, a comparative composition and chemistry study, and temporal-computing simulations. Their evidentiary strength differs because only the composition grid is replicated across devices.

A characterised synaptic platform (chapter 2). A reversible two-terminal organic memristive platform combines *Super Yellow*, *Hybrane*,

and LiOTf in a solution-processed active layer between ITO and Ag electrodes. Across assay-specific junctions sharing that formulation and architecture, it supports analogue I–V hysteresis (an 85 % conductance increase from the first to the second sweep at 1 V, growing ~ 15 -fold over ten cycles), reversible potentiation and depression exceeding 200 % at an energy cost of order 50 nJ per event, and two distinguishable retention regimes. Standard-exponential fits reported in the published SI give $\tau_S \approx 2.5\text{ s}$ – 3 s for short-term memory and $\tau_L \approx 4.7\text{ s}$ for long-term memory. The formulation also supports a four-state excitatory post-synaptic current read-out and an asymmetric Hebbian STDP window with a time constant of 85 ms–90 ms. Mobility differences between weakly coordinated triflate and strongly coordinated lithium in the oxygen-rich host provide a hard–soft acid–base interpretation of the two retention regimes. The assays were independently peer reviewed [6], but they do not constitute a population study; table A.1 gives the specimen mapping.

A reproducibility diagnosis and a fabrication-record methodology (chapter 3). Successive batches of nominally identical *Hybrane* devices changed from low-current, high-area, multilevel behaviour to a high-conductivity response without a usable switching window. Per-device weighting, restriction to a standard protocol, sweep-amplitude matching, and recovery-tail sensitivity confirmed a multi-feature change: the on–off ratio fell from 2.44 to 1.21, normalised hysteresis area from 0.235 to 0.027, and potentiation at 3 V changed from +17.9 % to –2.1 %. Conduction also became more ohmic at a matched 1.2 V. The change followed fabrication date rather than device age, because finished *Hybrane* devices retained their characteristics over weeks. The evidence points to ageing of the *Hybrane* stock, with *Super Yellow* excluded, under a moisture-driven hydrolysis hypothesis supported by an annealing-recovery test. Replacing *Hybrane* with poly(ethylene oxide) increased the normalised window area from 0.26 to 0.42 at 3 V and maintained usable switching across roughly

two and a half years. The study also produced the fabrication-record standard, normalised characterisation procedure, automated feature pipeline, and trend-analysis tools used in the later chapters.

Compositional and chemical control of the dynamics (chapter 4). The replicated *Super Yellow*/PEO/LiOTf/Ag grid shows that composition tunes fading-memory time over roughly 2 s–20 s; higher ion-transport-polymer content gives a weaker, faster-relaxing response. Host (PEO versus TMPE) and anion (triflate versus TFSI) extend the range in the illustrative chemistry survey. Cation identity also shifts the dynamics, but the triflate series runs opposite to the hypothesised $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ order and changes again with the anion. The available data therefore do not establish a transferable cation law. Drive amplitude changes the apparent timescale, so quantitative comparisons require a matched protocol. Equilibrium impedance falls along the composition axis and tracks fading-memory time. ATR, XRD, and UV-Vis associate the host effect with short-range ether order in an X-ray-amorphous, salt-dissolved film without a shift in the *Super Yellow* optical gap. Because the PEO fraction changes both fading memory and input nonlinearity, each composition defines a paired operating point for the simulations of chapter 5.

A temporal-computing demonstration from measured dynamics (chapter 5). Compact behavioural models ($\phi \otimes \lambda \otimes f$) fitted to the chapter 4 measurements were evaluated with linear read-outs. A heterogeneous composition bank raises total linear memory capacity from 4.2 to 6.4 (a factor of 1.54 ± 0.10 , positive in all ten seed pairs) and total information-processing capacity from 7.1 to 11.6. Seeds quantify algorithmic sensitivity rather than experimental replication. The nonzero nonlinear capacity of the homogeneous bank shows that the compressive write contributes a usable nonlinear response. The heterogeneous bank

improves multi-lag reconstruction of physiological streams. On WESAD, the models reach leave-one-subject-out macro-F1 scores of 0.894 for binary stress and 0.758 for three-class streaming classification. On CASE, a multi-timescale filter improves continuous valence/arousal tracking by +0.176 concordance correlation over a single timescale (27/30 subjects). Timescale diversity does not separately improve final-label classification when one slow timescale dominates. These results are simulations of measured device models, not measurements from a wired reservoir.

6.2 Answers to the Thesis Questions

The results answer the three questions posed in chapter 1.

First: realising and understanding an exemplar. *Can a reversible two-terminal organic memristive platform made from a solution-processed polymer-electrolyte composite support analogue switching, potentiation and depression, distinct retention regimes, EPSC, and STDP responses, and can cation–anion mobility asymmetry account for the retention regimes?* The common formulation and architecture of chapter 2, fabricated by solution processing and thermal evaporation, support these responses across assay-specific junctions. Differential displacement and relaxation of triflate and lithium provide a mechanism consistent with the two retention regimes. Because the extended assays are not a replicated device-population comparison, the mechanism remains a hypothesis rather than a population-level law.

Second: composition as the quantitative control, with chemistry as a sample-limited survey. *How do the dynamics depend on composition, and does moving the cation along the Li^+ , Na^+ , K^+ series shift the timescale as coordination chemistry suggests?* The replicated PEO/LiOTf

grid shows a monotonic composition dependence: the ratio tunes fading-memory time over roughly an order of magnitude, with switching window and potentiation strength moving in the same direction. Cation identity shifts the dynamics, but the triflate retention series runs opposite to the predicted $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ order and changes again with the anion. Sample size and protocol/electrode confounds prevent a transferable cation law. Coordination chemistry remains a qualitative interpretation rather than a quantitative result.

Third: utility for temporal computing. *Can the volatile response be used for temporal computation, can composition supply task-matched timescales, and does measured variability broaden the response of a heterogeneous bank?* In the simulations of chapter 5, composition supplies fading-memory times that overlap peripheral affective-signal bands. A heterogeneous bank broadens memory and information-processing capacity, and device-to-device variation contributes additional state diversity. The benefit depends on the task: the lead composition is sufficient for a single-timescale target, and the heterogeneous increment is not resolved for the sustained WESAD labels.

6.3 Benchmarking Against the Right Reference Class

Because the device state relaxes over seconds, temporal and event-driven neuromorphic systems provide the relevant reference class [5, 26, 27]. Non-volatile DRAM, SRAM, and crossbar ReRAM serve different storage functions. Current physical-reservoir comparisons must also separate the physical node, input encoding, output layer, and network architecture [121]; task scores from different layers and protocols are not a common

leaderboard.

Against physical reservoirs. Physical reservoir computing has been demonstrated in dynamic memristors [32, 34], organic electrochemical devices [35], and delay-coupled single nodes [154]. More recent systems add direct electrical tuning in two-terminal oxide devices [124], gate- and wavelength-controlled organic transistor dynamics [122], and an organic electrochemical reservoir with a physical device read-out [123]. These systems require fading memory and nonlinearity [5, 18]. The present devices have measured fading memory, while the fitted models have nonzero nonlinear information-processing capacity. Their sub-kilohertz response overlaps many physiological and environmental signals without time rescaling. Composition changes the fading-memory time over 2 s–20 s, and the corresponding heterogeneous model bank broadens memory capacity by a factor of 1.54.

Against organic neuromorphic devices. Much of organic neuromorphic electronics targets linear, symmetric, non-volatile conductance updates for weight storage, as in the ENODE family [21–23]. The devices studied here use ion-relaxation timescales for temporal processing and are compared with ionic and electrolyte-gated organic neuromorphic systems [35, 113].

Placement and evidence boundary. The experimentally supported claim is that composition tunes seconds-scale fading memory in this two-terminal organic platform. The computational claim is that banks of behavioural models fitted to those measurements can support the reported benchmarks and physiological tasks. The application scores are *in silico* and do not establish the performance of a wired array; nor is priority claimed for a first tunable two-terminal reservoir. The simulated reservoir

Table 6.1: Recent directly relevant reference points for tunable physical reservoirs. Temporal ranges are the values reported under each study’s own extraction protocol; tasks, read-outs, and metrics differ, so the table calibrates scope rather than ranking performance. “Physical” means that measured device states formed the reservoir; it does not imply that every input/output operation was analogue or on-chip.

Study	Node and terminals	Tuning handle	Reported temporal range	Reservoir evidence and task scope
This thesis	<i>Super Yel-low</i> /PEO/LiOTf organic composite; two-terminal	PEO/salt composition; drive amplitude	2 s–20 s across the replicated composition grid	Single-node dynamics measured; heterogeneous banks and WESAD/CASE tasks simulated; no wired reservoir
Gao et al. (2024) [122]	Organic vertical field-effect transistor; three-terminal optoelectronic	Input wavelength and gate bias	0.005 s–13.2 s with optical input and gate bias	Physical grouped-reservoir states; image recognition and time-series prediction
Han et al. (2025) [123]	Solid-electrolyte OECT; three-terminal	Ethylene-glycol content selects volatile or non-volatile operation	No common fitted- τ range reported	Physical reservoir and physical device read-out; three-class human-activity classification
Lee et al. (2025) [124]	Pt/HfO ₂ /TiN memristor; two-terminal	Inter-pulse base voltage	Fitted τ : 0.92 ms to 123 ms; inputs 0.14 ms–100 ms	Physical reservoir states; N-MNIST classification and Mackey–Glass prediction

contains independent, non-recurrent nodes; its absolute NARMA-10 errors are high and its capacities remain below the node-count bound. Recurrent or delay-feedback connectivity could raise these values. The evidence chain therefore ends at measured dynamics plus model-based task validation; the missing end-to-end test is a fabricated, composition-diverse reservoir under the same tasks.

6.4 Limitations of the Evidence Base

The principal limitations are as follows.

1. **The extended synaptic protocols are proof-of-concept rather than population-level.** EPSC, separated short-term memory/long-term memory retention, and STDP exist only for the chapter 2 *Hybrane* formulation; some specimen identifiers are not resolved in the recovered archive, and the available multi-junction data share one substrate. They function as formulation-level lithium priors and checks in the later chapters and were not extended to comparative status across the PEO family.
2. **Only composition is replicated.** The quantitative core is the *Super Yellow*/PEO/LiOTf/Ag composition grid. The host, anion, and cation comparisons rest on one or two screened devices per matched cell and are illustrative. The electrode contrast is also illustrative and partly confounded with fabrication generation; its negative-polarity measurements are single-composition and uncured, and its frequency-domain measurements rest on one batch with substrate-confounded cations. No powered material law, including a $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ ordering, is claimed.
3. **Protocol-specific timescales.** Because the drive amplitude co-sets the apparent fading-memory time, absolute retention values

are protocol-specific, and the supra-threshold read used in the comparative corpus differs from the sub-threshold read of the proof-of-concept device.

4. **The composition assumption behind the models.** Pulse integration and state relaxation were characterised separately, at different write amplitudes (potentiation at 3 V, decay at 4 V) and one inter-pulse cadence. Combining them in the behavioural models assumes that the two regimes are compatible. A coupling-sensitivity control uses the direct estimate from the gold/TMPE corpus ($\rho = +0.46$, mapping to $\kappa^* = +0.18$); the heterogeneous bank retains its capacity advantage at that value. A matched protocol is still needed for the lead composition.
5. **The application results are in silico and only partly cross-corpus.** Every temporal-computing result is a simulation of measured device models with a linear read-out; no wired reservoir was built. The physiological-context and affective-label results use WE-SAD as the main corpus, and heterogeneity does not separately improve its sustained labels. Noise robustness replicates on the PhysioNet Non-EEG corpus. Continuous valence and arousal are tested on CASE, where the device’s nonlinear encoding does not exceed a linear multi- τ filter with the same timescale spread. Broader multi-corpus validation remains future work.
6. **Material stability remains open.** Devices were characterised unencapsulated on a ~ 300 h ambient shelf scale; cycling endurance, read-disturb under continuous pulsing, and long-run drift under encapsulation were not systematically studied, and device ageing was not modelled in the comparative corpus.

6.5 Outlook and Future Directions

The immediate experiments address the model assumptions and sample-limited comparisons; longer-term work concerns stability and hardware integration.

Matched pulse protocol. A matched-amplitude pulse train should vary pulse count and inter-pulse interval together for one composition, electrode, and read protocol. The gold/TMPE negative-polarity corpus of chapter 4 already records integration and relaxation within single runs and finds interval-dependent potentiation and drive-depth-dependent relaxation. Repeating this measurement on the lead silver composition with a sub-threshold read would supply the missing joint parameters for the behavioural model.

Replicated chemistry survey. A cation series in one host and anion, at fixed protocol and electrode, with a sub-threshold read and at least three replicates per cell, would test the $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ hypothesis at adequate power. The silver TMPE retention follows the opposite order, $\text{K} > \text{Na} > \text{Li}$; the gold steady-state window is flat; and the gold negative-polarity data resolve no order. The lock-in corpus contains one cation-ordered candidate: a $\approx 1.4\times$ $\text{K} > \text{Na} > \text{Li}$ junction-capacitance trend at one substrate per salt (chapter 4). Profilometry of the existing substrates and a replicated series could test whether this trend is chemical or geometric. Chemistry- and bias-resolved impedance and a generation-controlled silver-versus-gold split would strengthen the microscopic and electrode comparisons.

Test the degradation chemistry. The moisture-driven hydrolytic-aging hypothesis for the *Hybrane* stock (chapter 3) cannot be tested directly because the aged consumable no longer exists in its degraded

state. FTIR, GPC, Karl-Fischer/TGA, and EIS on retained aged-versus-fresh stock would test the hypothesis in future material campaigns. The fabrication-record methodology now records reagent identity and history.

Engineer the host and salt for finer timescale control. Modifying the polyether host and salt could extend fading memory toward the minutes-scale tonic band and separate memory time from the input nonlinearity that composition currently changes with it.

Stabilise for long-run inference. Encapsulation and systematic measurements of ageing, endurance, and read disturb are required to assess continuous operation and remove uncontrolled ageing from later comparisons.

Build the hardware reservoir. A wired bank of composition-diverse PEO/LiOTf devices with an on-chip linear read-out would test the simulations directly. Integration toward a 1T1M-style array and operation on an affective stream would measure whether the predicted benefit of timescale heterogeneity survives in hardware.

Physiological sensor interfaces. Solution processing, mechanical flexibility, biocompatible chemistry, and seconds-scale dynamics make these devices candidates for skin-conformal sensing and inference. In the frequency-domain measurements of chapter 4, a DC-biased junction acts as a voltage-programmable low-pass amplifier below 10 Hz. A hardware experiment could test whether one junction can combine pre-amplification and temporal filtering at a physiological sensor interface.

6.6 Closing Remarks

This thesis shows that composition tunes the seconds-scale fading memory of polymer-electrolyte memristive composites and that behavioural models fitted to those measurements can process temporal signals. The replicated evidence supports composition-dependent control; host, anion, cation, and electrode effects remain illustrative, and the simulations do not replace a wired reservoir. The next experimental priorities are a matched-amplitude pulse protocol, a replicated chemistry series, and a hardware bank of composition-diverse devices.

Appendix A

Supporting Information for Chapter 2

This appendix gathers the detailed experimental information and supporting evidence for the proof-of-concept platform of chapter 2. The main chapter retains the evidence needed to assess the formulation-and-architecture claim; the material collected here records the specimen mapping, supporting methods, protocol details, and auxiliary measurements.

A.1 Materials and Fabrication Detail

A.1.1 Component Notes

The semiconducting component is *Super Yellow*, the trivial name for the high-molecular-weight PPV copolymer sold as PDY-132 by Merck. The formal copolymer name is:

poly[{2,5-di(3',7'-dimethyloctyloxy)-1,4-phenylenevinylene}-co-
{3-(4'-(3'',7''-dimethyloctyloxy)phenyl)-1,4-phenylenevinylene}-

co-

{3-(3'-(3',7'-dimethyloctyloxy)phenyl)-1,4-phenylenevinylene}}].

The conjugated phenylene-vinylene backbone supplies electronic transport, while the branched alkoxy side chains support solubility and reduce crystallisation-driven phase separation in blends [6, 145]. In ion-containing *Super Yellow/Hybrane* systems, related LEC devices have shown electroluminescence lifetimes exceeding 1600 h, supporting the chemical compatibility of this formulation family [104].

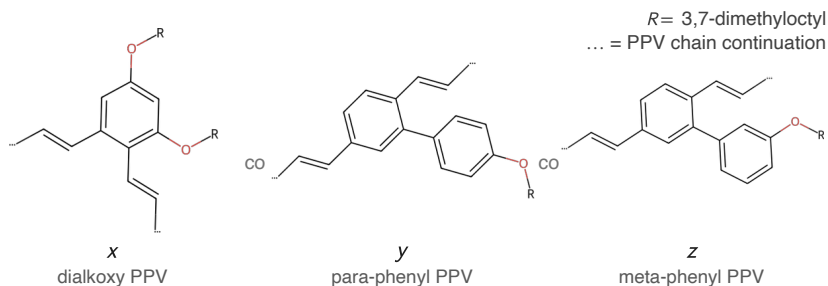
Hybrane is a hyperbranched polyester-amide produced by Polymer Factory (Sweden) [128]. The hyperbranched architecture creates an irregular, amorphous macromolecule rich in ether and ester oxygens. These oxygen atoms provide hard Lewis-base coordination sites for alkali-metal cations. Because ion motion is coupled to polymer segmental motion, transport in this class of host follows the polymer-electrolyte picture described in section 1.6.1, including non-Arrhenius temperature dependence near the glass-transition regime [88, 165].

LiOTf is a polymer-electrolyte salt widely used in batteries and LECs [98, 103]. Its processing role is to dissolve with *Super Yellow* and *Hybrane* in cyclohexanone without visible phase separation at the selected ratio. Its mechanistic role is to provide a small, hard Li^+ cation (0.59 Å–0.76 Å for four- to six-fold oxygen coordination [144]) and a charge-delocalised triflate anion. The resulting cation/anion asymmetry is the chemical basis for the STM/LTM interpretation in section 2.5.1.

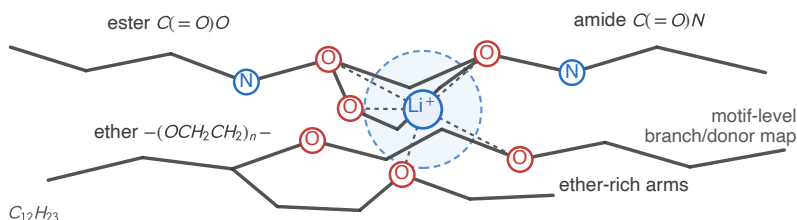
A.1.2 Solution Preparation and Film Deposition

The three stock solutions were prepared separately in cyclohexanone at 15 mg mL⁻¹ (*Super Yellow*), 10 mg mL⁻¹ (*Hybrane*), and 5 mg mL⁻¹ (LiOTf). The *Super Yellow* solution was stirred at 50 °C overnight.

a Super Yellow (PDY-132): statistical PPV copolymer



b Hybrane DEO750 8500: hyperbranched polyester-amide ion host



c Lithium triflate (LiOTf): mobile-ion source



Figure A.1: Chemical components of the proof-of-concept composite. **(a)** *Super Yellow* is the commercial PDY-132 PPV-family statistical copolymer; R denotes the branched 3,7-dimethyloctyl side chain. **(b)** *Hybrane* DEO750 8500 is represented as an oxygen-rich hyperbranched polyester-amide motif, highlighting ether, ester, and amide donor sites rather than a monodisperse molecular formula. **(c)** LiOTf is shown as the lithium salt of triflate, LiCF_3SO_3 . The selected active-layer formulation is *Super Yellow:Hybrane:LiOTf* = 1:0.30:0.09 by mass.

The solutions were then mixed to give the final mass ratio *Super Yellow:Hybrane:LiOTf* = 1:0.30:0.09, corresponding to final concentrations of 8.72 mg mL^{-1} , 2.62 mg mL^{-1} , and 0.78 mg mL^{-1} , respectively. Solution handling was performed under dry nitrogen to limit moisture uptake.

ITO-patterned glass substrates were ultrasonicated successively in aqueous Mucosol detergent solution, deionised water, and 2-propanol, each for 15 min. After drying under nitrogen, substrates were transferred immediately into a nitrogen glovebox (oxygen and water below 1 ppm). The active layer was spin-coated at 2000 rpm for 60 s, then annealed at 75°C for 3 h. The anneal removes residual cyclohexanone, allows chain relaxation, and stabilises the mixed polymer-electrolyte morphology.

The top Ag electrode was thermally evaporated in the glovebox through a shadow mask defining the 0.0825 cm^2 junction. Ag was deposited at 0.1 nm s^{-1} – 0.5 nm s^{-1} to a total thickness of 100 nm, monitored by quartz crystal microbalance. The slow initial rate was used to limit metal penetration into the soft polymer layer, which would otherwise risk filamentary shorts [166].

A.2 Electrical Protocol Detail

Electrical measurements used a Keithley 2450 SourceMeter connected to a custom probe station and controlled through Keithley TestScript Builder. Unless noted otherwise, measurements were carried out under nitrogen, with voltage applied to the Ag top electrode relative to the grounded ITO bottom electrode.

A.2.1 Assay-to-Specimen Map

The proof-of-concept claim concerns one formulation and device architecture, not one physical junction carrying every assay. Table A.1 records the finest specimen identity recoverable from the local archive and the published supporting information. A crossbar “junction” or “pixel” is one ITO/active-layer/Ag overlap; several junctions share a fabricated substrate and therefore are not independent device or batch replicates.

Table A.1: Assay-to-specimen map for the Chapter 2 evidence. “Published summary” indicates that the numerical result is recoverable from Ref. [6], but the local archive does not resolve the physical specimen identifier.

Assay	Recovered evidence unit	Supported inference
I–V hysteresis	One junction, NM_v025	Representative analogue hysteresis of the common formulation; no population estimate.
Pulse potentiation/depression	One junction (L5), NM_v055	Reversible pulse-protocol demonstration.
Pulse-number response	Five junctions, NM_v055, Day 8	Within-substrate junction spread; not independent-device or batch reproducibility.
STM/LTM retention	Three published traces; specimen ID unresolved	Existence and timescale of two protocol-defined retention regimes; no population estimate.
EPSC state sequence	Published state ratios; specimen ID unresolved	Four-state protocol demonstration; no population estimate.
STDP timing kernel	Five junctions, NM_v055, Day 19	Within-substrate mean timing response and junction spread.
Shelf stability	Published time series; specimen roster unresolved	Calendar stability over the reported interval; not cycling endurance or batch stability.

A.2.2 Read–Write–Wait Measurement

The variable-pulse and variable-delay measurements use a common read–write–wait sequence (fig. A.2). First, a low reading pulse, $V_{\text{read}} = 0.5 \text{ V}$,

establishes the initial conductance G_0 . This voltage is below the independently determined ionic-redistribution threshold of approximately 0.7 V [6]. Second, a train of N_{pulses} write pulses of duration 0.1 s perturbs the ionic configuration. Third, after a waiting time t_{wait} under open-circuit conditions, a second read pulse measures G_f . The reported response is $\Delta G = (G_f/G_0) \times 100\%$. Between experiments, the device was reset by shorting the junction for 300 s, and G_0 was checked before the next measurement.

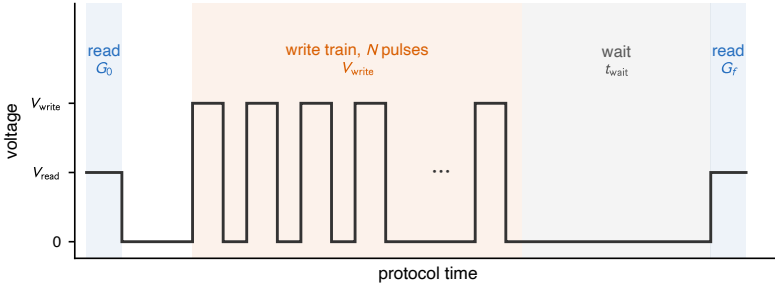


Figure A.2: Read–write–wait protocol used for the pulse-number and retention measurements. A low read pulse establishes G_0 , a train of write pulses perturbs the ionic configuration, a controlled open-circuit waiting time allows relaxation, and a second read pulse measures G_f . Generated from the protocol reported in Ref. [6].

A.2.3 Extended Synaptic Protocols

The EPSC experiment defines states operationally by their signed current response under a common 1 V read-out bias. The device is first held at 1 V for 1 s to define S_0 . Two 2 V, 50 ms pulses generate excited states S_1 and S_2 , and two -2.5 V, 50 ms pulses generate inhibitory states S_3 and S_4 . The signed ratios $R_n = I_{S_n}/I_{S_0}$ are reported in section 2.4.4.

For STDP, pre- and postsynaptic spikes are emulated by triangular voltage waveforms applied to the two-terminal device. The experimentally applied

voltage is the algebraic sum

$$V(t) = V_{\text{pre}}(t) + V_{\text{post}}(t + \Delta t), \quad (\text{A.1})$$

with $\Delta t = t_{\text{post}} - t_{\text{pre}}$. Each spike waveform has maximum amplitude $V_{sp} = 1.55 \text{ V}$ and duration 1.2 s ; Δt was varied from -600 ms to 600 ms . Negative Δt produces potentiation and positive Δt produces depression under the sign convention used in section 2.4.5. The applied-voltage logic of both protocols is shown in fig. A.3.

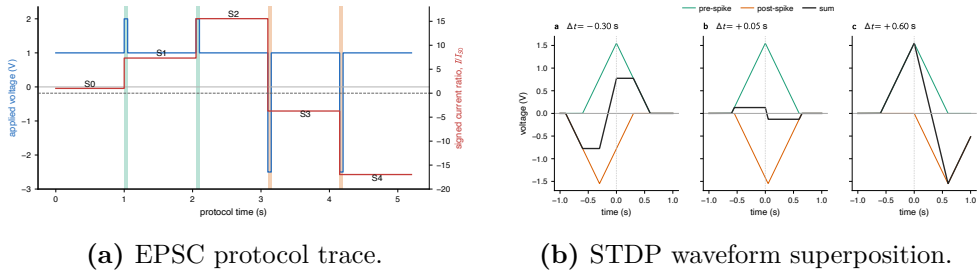


Figure A.3: Supporting protocol panels for the extended synaptic measurements. **(a)** Reconstructed EPSC trace showing the applied-voltage sequence and the signed current ratios reported for states S_0 – S_4 . **(b)** Example pre/post triangular spikes and their algebraic sum for three timing offsets. These panels document the protocol logic; the compact evidence used in the main chapter is reported in fig. 2.5.

A.2.4 Retention-Fit Parameters

Table A.2 lists the standard-exponential fit parameters reported for the proof-of-concept retention traces. The fits use $G_f/G_0 = R_\infty + A \exp(-t_{\text{wait}}/\tau)$, with $\beta = 1$ fixed.

Table A.2: Published standard-exponential fit parameters for the Chapter 2 retention traces. No free stretch exponent, confidence interval, or goodness-of-fit statistic was reported for these proof-of-concept fits.

Condition	τ (s)	R_∞	A
1 V, 10 pulses (STM)	2.57	1.01	0.374
1 V, 50 pulses (STM)	3.01	1.01	0.91
3 V, 10 pulses (LTM)	4.73	1.05	3.30

A.3 Morphology and Stability Support

The spin speed was selected from a 500 rpm–3000 rpm optimisation [6]. Low speeds produced films that were too thick and less uniform; high speeds made the film thin enough for substrate roughness to matter. The selected fabrication condition used 2000 rpm, giving a 209 nm active layer measured by contact profilometry across a masked step edge with an Ambios XP-1 profilometer.

Surface topography was characterised by tapping-mode AFM using a Bruker Dimension Icon instrument, a silicon probe with resonant frequency 300 kHz and force constant 40 N m⁻¹, and a scan rate of 0.6 Hz. The AFM images showed smooth films with RMS roughness of 1 nm–3 nm over the investigated spin-speed range [6]. The absence of large phase-separated features supports the reading of the active layer as a homogeneous polymer-electrolyte composite at the scale relevant to the vertical junction.

Ambient shelf stability was assessed for approximately two weeks (~ 300 h) on unencapsulated devices stored in air at room temperature [6]. The monitored quantities were baseline conductance G_0 , maximum conductance change $\Delta G_{\max}$, and STM retention time τ_S . All three remained within 20 % of their initial values over the measurement window. This is a shelf/calendar stability result, not a cycling-endurance measurement. Figure A.4 summarises the film thickness, AFM roughness, and shelf-stability

metrics together.

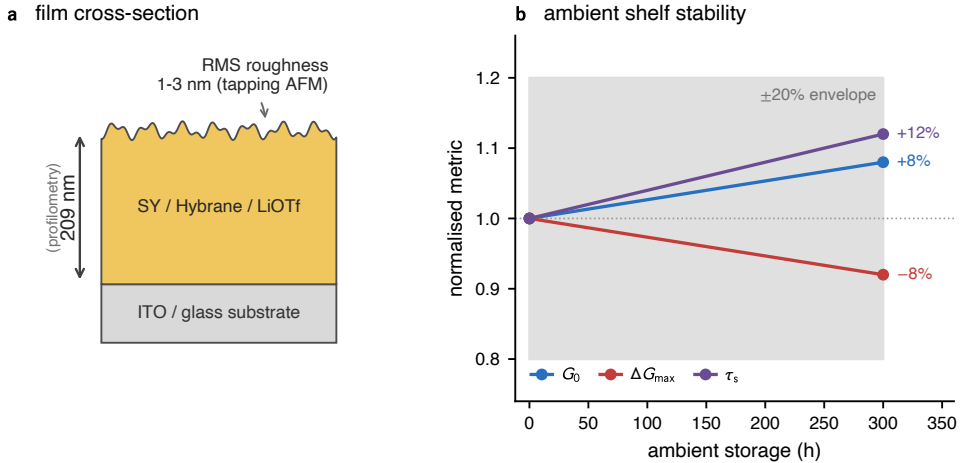


Figure A.4: Supporting morphology and shelf-stability summary for the Chapter 2 device family. The selected processing condition gives a 209 nm active layer; tapping-mode AFM reports 1 nm–3 nm RMS roughness across the investigated spin-speed window; and the ambient shelf-stability metrics remain within a 20 % envelope over approximately 300 h. Compiled from the supporting measurements reported in Ref. [6]; this is not a cycling-endurance result.

Encapsulation outlook. The ambient stability is expected to improve further with encapsulation, standard practice in organic electronics: a thin SiO_2 or Si_3N_4 barrier deposited by atomic-layer or chemical-vapour deposition would suppress the oxygen and water ingress that are the two primary degradation pathways [167]. That even unencapsulated devices survive two weeks establishes the baseline robustness of the composite.

A.4 Ion-Transport Models for the Composite

This section details the injection and bulk-transport descriptions summarised in section 2.5. The two limiting-case models are inherited from the polymer light-emitting electrochemical cell (LEC) literature and reconciled by van Reenen *et al.* [102] as two regimes of the same ion redistribution.

A.4.1 Equilibrium Ion Distribution and Response to an Applied Field

In the absence of an applied voltage, the ions (Li^+ cations and CF_3SO_3^- anions) are distributed homogeneously throughout the active layer. The equilibrium distribution is not strictly uniform at the atomic scale (each Li^+ ion occupies a local coordination site within the Hybrane matrix, and the triflate anions occupy interstitial positions between the coordinated cations), but on the scale of the film thickness (209 nm), the macroscopic charge density is uniform and the net electric field within the film is zero (apart from any built-in field arising from the electrode work function difference), as in the polymer-electrolyte picture inherited from LECs [98, 99, 102].

When a voltage is applied across the device, an electric field $\mathcal{E} = V/d$ (where d is the film thickness) is established. In the absence of free charge carriers that could screen this field, it would be uniform throughout the film. However, the mobile ions respond to the field by migrating: cations drift toward the cathode (ITO at positive bias on Ag), and anions drift toward the anode (Ag). This migration leads to the build-up of a charged double layer at each electrode interface on a timescale determined by the ionic mobility and the film thickness. The resulting non-uniform

ionic charge distribution concentrates the field in a thin screening length near each electrode, while the bulk of the film is field-free [99, 100], with consequences for charge injection described by the two models below.

A.4.2 Electrodynamic Model

In the electrodynamic (ED) model, first formulated by deMello *et al.* [99, 100] to describe LECs, charge injection at the electrode–organic interface is the rate-limiting step for current flow. In this injection-limited regime, the current density is governed by thermionic emission over the Schottky barrier at the metal–polymer interface,

$$J \propto T^2 \exp\left(-\frac{\Phi_B}{k_B T}\right), \quad (\text{A.2})$$

where Φ_B is the effective barrier height, T the absolute temperature, and k_B the Boltzmann constant. The ionic double layer at the electrode interface reduces Φ_B by bending the polymer bands: positive ionic charge near ITO lowers the conduction-band edge there, while negative ionic charge near Ag raises the valence-band edge, both reducing the effective injection barrier. In the memristive context, conductance changes therefore arise from changes in the interface ion concentration: a higher concentration of positive ions near ITO means lower Φ_B , higher hole injection, and higher conductance, and the ions relax back toward equilibrium by thermal diffusion once the field is removed, the physical basis of the memory decay.

A.4.3 Electrochemical Doping Model: Ohmic Injection

The electrochemical doping (ECD) model, formulated by Pei *et al.* [98] and developed by subsequent authors [101, 138], instead takes charge injection to be ohmic (not rate-limiting), so the majority of the voltage drop occurs in the bulk and the current is limited by bulk conductivity. The defining feature is that ion accumulation *electrochemically dopes* the conjugated polymer: cations accumulate near the cathode, stabilising injected electrons (*n*-doping), and anions near the anode, stabilising injected holes (*p*-doping); the doped regions are far more conductive than the undoped polymer. The degree of doping (and hence the conductance) is set by the extent of ionic migration, which is governed by the history of applied voltage, so the conductance records the cumulative ionic-displacement history.

A.4.4 Resolution of the Two Models

The prolonged LEC-community debate over the relative validity of the ED and ECD models was resolved by van Reenen *et al.* [102], who showed both to be correct in different limits: the ED model when injection is strongly barrier-limited (low-conductance semiconductor, large electrode work-function mismatch), the ECD model when injection is ohmic (high-conductance semiconductor, small mismatch). A device may operate in the ED regime at early times, before significant ionic redistribution, and transition toward the ECD regime as the double layers build up. The working interpretation adopted for the present platform (ED with the fast triflate-driven response, ECD with the slow Li^+ -driven response) is stated in section 2.5.

A.5 Historical Comparison at the Time of Publication

Table A.3 places the device alongside representative organic and hybrid memristive devices available at the time of the 2022 publication, on the axes of section 1.3.4: device structure, accessible conductance states, short- and long-term memory, STDP, and reversibility. It is a historical positioning, not a 2026 state-of-the-art claim.

Two combinations not represented in the comparison set at the time stand out. The first is that the same 2-T formulation and architecture support both short-term memory- and long-term memory-labelled regimes while remaining fully reversible. Of the prior reports, Liu and co-workers [94] comes closest, with tunable short- to long-term transitions in a 2-T structure; the redox route, however, modifies the active layer irreversibly at each cycle, precluding practical cyclic operation [116]. The present platform reaches comparable short-term memory/long-term memory functionality with full reversibility (conductance returns to G_0 after the reset protocol). The second is STDP in a fully reversible 2-T architecture: the earlier organic STDP demonstrations [90, 105, 115] were all 3-T with a gate supplying the timing rule, whereas here the rule emerges from the superposition of pre- and postsynaptic pulses at the Ag terminal.

The remaining axes summarise the operating regime rather than a competitive claim. The ambient shelf lifetime (about 300 h unencapsulated) exceeds most comparable unencapsulated organic memristive devices by roughly an order of magnitude [119]; cycling endurance, not characterised here, is where mature inorganic platforms reach $> 10^{10}$ cycles [16]. The energy per synaptic event (about 50 nJ) is consistent with other organic ionic devices [77] but higher than the picojoule budgets of dedicated CMOS neuromorphic hardware [1, 27] and of the best inorganic memristors [68], reflecting the lower switching speed and larger area of this proof-of-concept

Table A.3: Historical comparison of key performance metrics of the present work with representative organic and hybrid memristive devices available at the time of the 2022 publication. “2-T” = two-terminal; “3-T” = three-terminal; “STM” = short-term memory; “LTM” = long-term memory; “STDp” = spike-timing dependent plasticity; “—” = not reported or not applicable.

Reference	Material	Structure	States	STM	LTM	STDp	Reversible
Zakhidov 2010 [97]	[Ru(bpy) ₃] ²⁺ /PF ₆ [−]	2-T	2	—	—	No	Partial
Lei 2014 [93]	PVA film	2-T	2	—	—	No	No
Lin 2016 [94]	[EV(ClO ₄)]/(BTPA-F)	2-T	cont.	Yes (tunable)	Yes	No	No
Gkoupidenis 2015 [105]	PEDOT:PSS (OECT)	3-T	cont.	Yes	No	Yes	Yes
Xu 2016 [114]	PEDOT:PSS (OECT)	3-T	cont.	Yes	Yes	No	Yes
Kim 2018 [90]	Elastic organic composite	3-T	cont.	Yes	No	Yes	Yes
Goswami 2017 [95]	Molecular switch	2-T	2	—	Yes	No	Yes
Seo 2019 [115]	P3HT organic FET	3-T	cont.	Yes	No	Yes	Yes
This work [6]	SY/Hybrane/LiOTf	2-T	cont.	Yes (10–15 s)	Yes (>45 s)	Yes	Full

A.5. HISTORICAL COMPARISON AT THE TIME OF PUBLICATION

device; area scaling toward the $\sim 100 \text{ nm}^2$ inorganic regime would close much of that gap.

Appendix B

Supporting Information for Chapter 3

This appendix records the methodological backbone of the degradation analysis of section 3.3. The bridge chapter retains the result (the window collapse and the ohmic drift) and the figures; the four controls that make that result defensible against the confounds peculiar to this dataset are stated in full here. The analysis is reproduced by `scripts/bridge_hybrane_peo_reproducibility.py`.

B.1 The Four Methodological Controls

The degradation analysis is built on a methodological backbone of four controls; each was verified to change the conclusion if dropped.

1. **Per-device weighting (mandatory).** The number of pixels measured per device collapses from sixteen to two over the campaign and is strongly anti-correlated with date (Spearman $\rho = -0.80$, $p \approx 1 \times 10^{-15}$, $n = 65$; fig. 3.2a inset). Pixel-pooled statistics are

therefore biased toward the early, densely measured devices. Every feature is reduced to *one value per device* (the median over that device’s freshest-day curves) before any test.

2. **Standard-protocol corpus only** ($n \approx 69$ devices, span 2021-02 to 2022-05): *Super Yellow/Hybrane/LiOTf* on silver, annealed at 75 °C with no second stage, at the lead mass ratios (*Hybrane* 0.3, *LiOTf* 0.09). This excludes the invasive experiments (above all the high-temperature anneal batches that partially recovered devices a year in) which would otherwise contaminate the feature-versus-date story.
3. **Sweep-amplitude matching, with each feature read where it is expressed.** A $0 \rightarrow +X \rightarrow 0$ loop excites the device more at larger X , so it conducts more even when read back at the same voltage; amplitude is confounded with both date and material (fig. 3.2b). The *window* features (on–off ratio, normalized area) only open up at high amplitude (at ~ 1.2 V they sit near unity for every device, early or late, and *cannot* show a collapse), so they are read in the ~ 3 V stratum, while the *conductivity* features are read in a tightly matched ~ 1.2 V band.
4. **Pre/post-inflection contrast and recovery-tail sensitivity.** Because the notes locate the behavioural inflection at NM_v026 (2021-04-22), the window collapse is tested as a step, early ($\leq$ Apr 2021) versus later (Mann–Whitney), instead of as a whole-campaign correlation. Every conductivity trend is additionally reported with and without the deliberately-reverted recovery batches ($\geq 2022-03$).

The quantitative outcome of applying these controls (the window collapse at ~ 3 V and the matched-amplitude ohmic drift) is reported in table 3.2 and discussed in section 3.3.

Appendix C

Supporting Information for Chapter 4

This appendix gathers the analysis traceability, the confound audits, the fit methodology, and the per-composition-cell quantitative tables that underlie chapter 4. The main chapter retains the result and the figures; the material here records how the numbers were produced and lists the per-cell values behind the heatmaps and design-space plots.

C.1 Analysis Traceability and Aggregation

Data-quality control. Two screening layers are applied before any number is reported, both recorded so that the analysis is reproducible. Individual measurements that are erratic or non-physical are flagged and excluded. In addition, a per-device visual review of the recorded curves was carried out; the resulting verdicts, whether a curve is used as measured, cleaned by removing identified anomalous points, or discarded, together with the exact points retained, are stored in a curation registry and applied automatically by the analysis scripts, instead of being made

Table C.1: Raw-to-summary trail for the three common measurements used in chapter 4. The same recorded trail underlies the figures of section 4.4 and the behavioural models of chapter 5.

Stage	Artifact	Role in the analysis
Raw measurement	DEVICES_LAB_DATA/.../D1_all.txt or split D1_I/V/T.txt	Keithley current, voltage, and time traces for each device day, measurement type, and crossbar pixel.
Fabrication metadata	DATABASE/UPDATED_DEVICES_LIBRARY.csv	Canonical device composition, electrode, spin speed, annealing, storage, measurement atmosphere, and operator metadata.
Extracted features	DEVICES_HYST_*, DEVICES_PULSES_*, DEVICES_DELAYTIME_*	Per-curve, per-pixel, and per-device feature tables generated by the archive extraction pipeline.
Exclusion and curation	FILTERED_DEVICES.csv; handouts/ch4_png_qa_curation.csv	Red-flag exclusion list plus per-curve visual QA verdicts; retained points are applied by script, not selected separately for each figure.
Recomputed descriptors	handouts/ch4_decay_fits.csv; handouts/ch4_pulse_descriptors.csv	Post-curation descriptors: $t_{1/2}$, identified τ, β where stable, pulse-growth exponent α , peak ratio, and turnover.
Cell summaries and figures	handouts/ch4_decay_by_cell.csv; handouts/ch4_pulses_by_cell.csv; scripts/ch4_comparative_figures.py	Cell medians, device counts, representative curves, heatmaps, swarm plots, and design-space figures.

ad hoc. The fitted timescale obtained directly from the instrument-side pipeline was found not to be robust to such anomalous points, so all timescales reported in chapter 4 are recomputed after this screening. Table C.1 records the full raw-to-summary trail underlying these screened descriptors.

The aggregation ladder. The aggregation is deliberately conservative. HYST raw points are first reduced to sweep-level features in DEVICES_HYST_CURVE_INFO.csv; broken or FILTERED rows are re-

Table C.2: Figure-level sources for the Chapter 4 figures whose detailed file paths are not repeated in the main captions. Script names are relative to the thesis repository; raw measurements reside in the experimental archive paths listed by the scripts.

Output	Reproduced by	Principal inputs and audit trail
Figure 4.1	<code>scripts/thickness_rpm_audit.py</code>	<code>DEVICES_PROFILOMETRY_STATS.csv</code> joined to the screened Chapter 4 descriptors; audit in <code>handouts/14_thickness_rpm_confound_audit.md</code> .
Figures 4.2 to 4.5 and 4.14	<code>scripts/ch4_comparative_figures.py</code>	<code>handouts/ch4_decay_fits.csv</code> , <code>handouts/ch4_pulse_descriptors.csv</code> , <code>handouts/ch4_decay_by_cell.csv</code> , and <code>handouts/ch4_pulses_by_cell.csv</code> ; curve curation in <code>handouts/ch4_png_qa_curation.csv</code> .
Figures 4.7 and 4.9	<code>scripts/ch4_comparative_figures.py</code>	Chemistry-landscape values from the screened descriptor tables; the protocol overlay uses the same-device v114 matched-amplitude comparison recorded in the Chapter 4 audit trail.
Figure 4.6	<code>scripts/ch4_iratr.py</code> ; <code>scripts/ch4_xrd.py</code>	ATR band extraction in <code>handouts/ch4_iratr_bands.csv</code> ; assessments in <code>handouts/16_iratr_assessment.md</code> and <code>handouts/17_xrd_assessment.md</code> .
Figure 4.8	<code>scripts/ch4_uvvis.py</code>	2025 <i>Super Yellow</i> /TMPE cation-series UV-Vis spectra; assessment in <code>handouts/18_uvvis_assessment.md</code> .
Figure 4.10	<code>scripts/ch4_eis.py</code>	0 V-DC impedance descriptors from the experimental database, joined to <code>handouts/ch4_decay_by_cell.csv</code> ; per-cell output in <code>handouts/ch4_eis_by_cell.csv</code> .
Figure 4.11	<code>scripts/ch4_electrode.py</code>	Silver-versus-gold metrics in <code>handouts/ch4_electrode_by_cell.csv</code> ; audit in <code>handouts/19_electrode_a_u_assessment.md</code> .
Figure C.2	<code>scripts/ch4_rigor.py</code>	On-state conductance from <code>DEVICES_HYST_PIXEL_INFO</code> ; gradient/factorial/bootstrap, EIS timescale bridge, reproducibility and heterogeneity-resource statistics in <code>handouts/ch4_gradient_stats.csv</code> , <code>ch4_cell_ci.csv</code> , <code>ch4_factorial.csv</code> , <code>ch4_percolation.csv</code> , <code>ch4_eis_timescale.csv</code> , <code>ch4_cycling.csv</code> , <code>ch4_heterogeneity_resource.csv</code> .

moved; device summaries are medians across the surviving sweeps and pixels; and composition-cell values are medians across devices. PULSES and DELAYTIME follow the same curve/pixel/device/cell logic, but with protocol-specific descriptors: the pulse curves are summarised by α , peak ratio, and turnover, while the decay curves are summarised primarily by $t_{1/2}$, with τ and β retained only when the fit passes the identification thresholds in `scripts/ch4_dynamics_fits.py`. The representative curves in fig. 4.2 are selected by the figure script from devices near the relevant cell medians, so they are illustrative examples of the same summarized population, not hand-picked evidence outside the pipeline.

Stretched-exponential identification. The depotentiation decays are fitted with the Kohlrausch form of eq. (4.1). The eight-to-thirteen-point decay curves often under-constrain β and τ jointly, so the fitted τ is reported only where the fit is well identified, and the model-free half-enhancement time $t_{1/2}$ is used as the primary timescale; where both are available they agree (table C.3). The stretch exponent is constrained to the physical Kohlrausch range $0 < \beta \leq 1$ of eq. (4.1): an unconstrained fit occasionally returns $\beta > 1$ (a *compressed* exponential, which does not correspond to a distribution of activation barriers), and such cells are instead pinned at $\beta = 1$, i.e. a near-undispersed simple-exponential relaxation, not reported as a spurious stretched fit. Fitted β values then lie predominantly in 0.6–0.9, i.e. genuinely stretched relaxation consistent with the heterogeneous distribution of activation barriers invoked in chapter 2, with the most ion-rich cells (high PEO, high salt) approaching $\beta = 1$, a narrower barrier distribution, as expected when ionic transport dominates. The intra-device disorder implied by the stretch and the device-to-device heterogeneity $\sigma(\ln \tau)$ of section D.3 are related, not independent: the relaxation-time spread implied by the median stretch is comparable in magnitude to the measured inter-device scatter (section C.13), so the same barrier-distribution width underlies both.

C.2 Film-Thickness and Fabrication Confound Audits

Film thickness is a controlled covariate. In most batches, the spin-coating speed was raised for higher-PEO cells to partly equalise thickness (spanning 1500 rpm–3500 rpm). The compensation was deliberate but incomplete: median film thickness still rises from ≈ 227 nm at PEO 0.3 to ≈ 298 nm at PEO 1.2, giving a Pearson $r \approx 0.68$ between PEO fraction and thickness (fig. 4.1). Four observations show that thickness is nonetheless not the operative variable:

1. The reported metrics are dimensionless ratios and timescales (on–off ratio, log–log slope, relaxation time), in which an absolute series-resistance or geometric effect of thickness cancels.
2. The activation voltage is flat (2.2 V–2.4 V, $r \approx -0.06$ against thickness) across a $2.6\times$ thickness range, so the switching threshold is an intrinsic ionic property, not a bulk-field effect that would scale with film thickness.
3. Within the single lead cell (PEO 0.3/salt 0.09) a 38% device-to-device thickness spread (196 nm–271 nm) leaves the fading-memory time essentially unchanged ($t_{1/2} \approx 18$ s–22 s).
4. Most directly, once composition is held fixed the thickness correlation with $\log t_{1/2}$ vanishes (partial $r \approx +0.05$), while the PEO effect survives after controlling for thickness (partial $r \approx -0.42$); the same holds for the potentiation metrics.

Thickness covaries with PEO as a downstream consequence of polymer loading, but the dynamics track composition. The full join and statistics are reproduced by `scripts/thickness_rpm_audit.py` and `handouts/14_thickness_rpm_confound_audit.md`.

Other fabrication variables are held constant or crossed. The full per-device fabrication record was audited to confirm that the composition trends are not driven by some other processing variable. Across the silver composition grid the principal process variables are held *constant*: the annealing temperature (75 °C, no second stage), the electrode (100 nm silver, evaporated by the same operator), the semiconductor loading, the solvent, and, which matters for a hygroscopic polyether, storage *and* measurement under a glovebox atmosphere, so ambient humidity is controlled out. The variables that do change across the grid (fabrication batch and date, the operator who prepared the solutions, the semiconductor lot, and the device age at measurement) are *crossed* with composition, not confounded with it: each composition cell is populated from several independent batches, and within any single batch every composition level was fabricated together and measured on the same day. The composition trends are accordingly reproduced *within* individual single-day batches, including one in which the spin speed was held fixed across all compositions, so that the trend appears with no thickness compensation at all. The full column-by-column audit is recorded in `handouts/15_fabrication_confound_audit.md` and reproduced by `scripts/fabrication_confound_audit.py`.

C.3 Per-Composition-Cell Summaries

Tables C.3 and C.4 list the per-cell values behind the heatmaps of section 4.4 and the design space of fig. 4.14. Each row is a composition cell; values are medians over the screened devices in the cell, with the device count n given so that the replicated Li^+ composition series is distinguished from the single-device Na^+/K^+ chemistry points.

Table C.3: Fading-memory (DELAYTIME) descriptors per composition cell, silver electrode, matched 4 V/2 V protocol. $t_{1/2}$: model-free half-enhancement time (s). R_{60} : retention at 60 s. τ, β : Kohlrausch parameters (median over the n_{id} devices with an identified fit). Em-dash: not identified.

Cation	PEO/salt	n	$t_{1/2}$ (s)	R_{60}	n_{id}	τ (s)	β
Li ⁺	0.15/0.18	1	22.4	0.27	0	—	—
Li ⁺	0.3/0.045	4	15.8	0.20	3	25.8	0.64
Li ⁺	0.3/0.09	3	19.2	0.28	2	23.9	0.83
Li ⁺	0.6/0.045	3	3.3	0.06	1	3.0	0.97
Li ⁺	0.6/0.09	3	3.5	0.02	1	3.0	0.61
Li ⁺	0.6/0.18	2	5.5	0.04	2	4.6	0.61
Li ⁺	1.2/0.045	3	2.7	0.13	0	—	—
Li ⁺	1.2/0.09	3	6.0	0.09	2	6.7	0.91
Li ⁺	1.2/0.18	3	9.0	0.13	3	10.0	0.87
Na ⁺	0.3/0.09	1	24.0	0.42	0	—	—
K ⁺	0.3/0.09	1	8.0	0.22	0	—	—

C.4 Realised Device Palette

The quantitative analyses of chapter 4 are deliberately confined to the replicated silver PEO/LiOTf composition series. It is nonetheless useful to see the full set of behaviours the device family actually realises, because it is this realised spread, not any single optimised cell, that the temporal chapter treats as a computational resource. Figure C.1 places every silver device that carries both a measured switching window and a measured fading timescale ($n = 54$, all hosts, cations and anions; FILTERED-flagged and broken pixels removed; device medians) in a single window-versus-retention map. The map is descriptive only: the chemistry axes are illustrative (section 4.5) and no comparative claim is drawn from it. Two features are worth noting. The populated region spans more than an order of magnitude in both axes, so the substrate offers a broad palette of (window, τ) operating points. And the lead composition is a cluster of replicate devices spread over $\tau \approx 0.4\text{ s} - 20\text{ s}$, not an isolated point, the visual counterpart of the within-cell device-to-device scatter quantified in

Table C.4: Potentiation (PULSES) descriptors per composition cell, silver electrode, 3 V/1.5 V protocol. α : log-log growth exponent. Peak: peak conductance ratio ($\times$). N_{peak} : pulse count at peak. Turnover: fraction of devices whose response turns over within $N \leq 1000$.

Cation	PEO/salt	n	α	Peak ($\times$)	N_{peak}	Turnover
Li ⁺	0.15/0.045	1	0.06	1.4	100	100%
Li ⁺	0.3/0.045	4	1.11	482.6	300	100%
Li ⁺	0.3/0.09	4	0.90	149.9	1000	0%
Li ⁺	0.3/0.18	2	0.18	3.8	1000	0%
Li ⁺	0.6/0.045	3	0.57	14.8	100	100%
Li ⁺	0.6/0.09	3	0.83	202.3	300	100%
Li ⁺	0.6/0.18	3	0.63	76.9	1000	0%
Li ⁺	1.2/0.045	3	0.33	4.4	100	100%
Li ⁺	1.2/0.09	2	0.49	42.6	1000	0%
Li ⁺	1.2/0.18	3	0.37	8.0	1000	0%
Na ⁺	0.3/0.09	1	0.27	10.0	1000	0%
K ⁺	0.3/0.09	1	0.86	103.2	300	100%

section D.3.

C.5 Impedance: Equivalent Circuit and Per-Cell Values

Equivalent-circuit model. Beyond the model-free Nyquist-apex resistance used in section 4.7, each spectrum was also summarised by the bulk ionic resistance R_{ion} of an equivalent circuit adapted from the polymer light-emitting electrochemical cell analysis of Munar *et al.* [143], in which a series cable term is placed in parallel with (i) the semiconductor electronic branch (effectively open at 0 V), (ii) a geometric constant-phase element, and (iii) an ionic branch of two interfacial double layers in series with R_{ion} . Two cautions apply. First, the archive holds two automated fits of this circuit; the freely-fitted one (with per-parameter error estimates) tracks composition appreciably better than a more tightly bounded variant, and

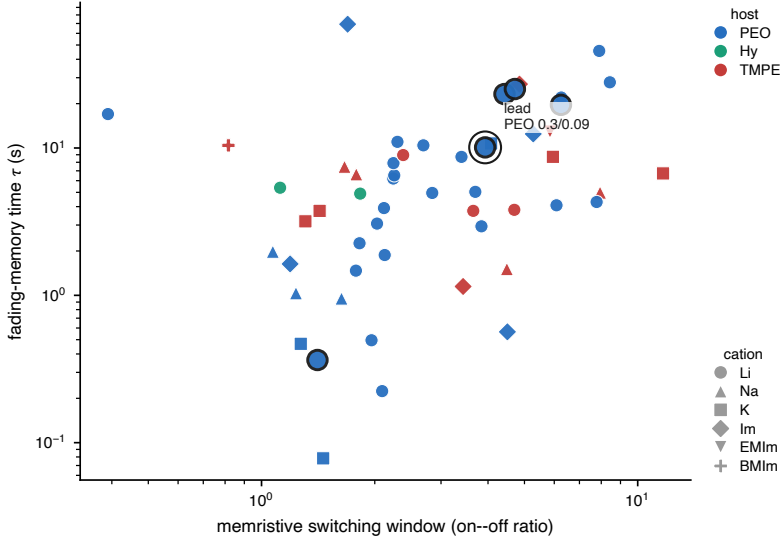


Figure C.1: Measured window-versus-retention palette of the silver device library ($n = 54$ devices with both a clean hysteresis window and a fading-memory fit, all chemistries). Colour encodes the ion-conductor host, marker encodes the salt cation; the lead PEO 0.3/0.09 cell is ringed. The cloud spans over a decade in each axis, and the lead-cell replicates alone span $\tau \approx 0.4\text{ s} - 20\text{ s}$. Descriptive overview only; the quantitative claims of chapter 4 rest on the replicated composition series, not on this map.

only the former is used. Second, per-spectrum circuit fits are genuinely ill-conditioned, an independent refit confirms that several do not converge robustly, so R_{ion} is reported only as corroboration of the model-free quantity, never as a precise number. Both observables nonetheless fall steeply and monotonically with PEO fraction (table C.5): the model-free $Z_{\text{real}}^{\text{apex}}$ at rank correlation -0.83 against PEO on cell medians, the fitted R_{ion} at -0.84 .

Table C.5: Equilibrium (0 V-DC) impedance per composition cell, silver *Super Yellow*/PEO/LiOTf. $Z_{\text{real}}^{\text{apex}}$: model-free real impedance at the Nyquist apex. R_{ion} : freely-fitted bulk ionic resistance (corroborative only). $t_{1/2}$: independently measured fading-memory time, repeated from table C.3 for the co-decline of fig. 4.10b. Em-dash: no matched decay.

PEO	salt	$Z_{\text{real}}^{\text{apex}} (\Omega)$	$R_{\text{ion}} (\Omega)$	$t_{1/2} (\text{s})$
0.15	0.045	2.5×10^8	2.3×10^7	—
0.15	0.09	1.7×10^8	8.7×10^6	—
0.15	0.18	1.0×10^8	8.4×10^6	22.4
0.3	0.045	5.3×10^7	4.0×10^5	15.8
0.3	0.09	1.3×10^8	1.0×10^6	19.2
0.3	0.18	9.9×10^7	4.9×10^5	—
0.6	0.045	5.3×10^6	1.4×10^5	3.3
0.6	0.09	2.6×10^7	4.6×10^4	3.5
0.6	0.18	1.9×10^7	8.9×10^4	5.5
1.2	0.045	4.1×10^5	1.9×10^2	2.7
1.2	0.09	5.6×10^5	1.1×10^4	6.0
1.2	0.18	3.8×10^7	1.5×10^5	9.0

C.6 Electrode-Contrast Per-Cell Values

Table C.6 lists the per-cell behavioural metrics behind the silver-versus-gold contrast of section 4.8, including the host-free and salt-free negative controls that confirm both the ion-transport polymer and the dissolved salt are required for the memristive response.

C.7 The Gold-Electrode Negative-Polarity Dynamics Corpus

This section documents the origin, validation, and per-protocol results of the negative-polarity measurement campaigns behind the polarity findings of section 4.8 and the single-run drive–retention measurements cited in section 4.6. Reproduction: `scripts/uncat_au_tmpe_audit.py` (parser, waveform classifier,

Table C.6: Per-cell behavioural metrics by top electrode (active silver vs inert gold), at the lead composition 0.3/0.09, under the canonical positive-polarity protocols. On-off and normalised loop area from HYST; V_{act} activation voltage (V); peak ratio and α from PULSES; $t_{1/2}$ from DELAYTIME (s). Em-dash: no clean decay/fit at that cell. †: no positive-protocol decay exists, but the negative-polarity corpus supplies the missing retention measurement under a different protocol (-5 V write, -2.25 V sparse read): model-free half-decay medians Li^+ 2.2 s ($n = 5$), Na^+ 2.4 s ($n = 11$), K^+ 0.7 s ($n = 12$); see section C.7. “none” rows are host-free or salt-free negative controls.

Chemistry	Elec.	on-off	area	V_{act}	peak	α	$t_{1/2}$
PEO/Li/OTf	Ag	3.25	0.33	2.38	123.5	0.89	19.2
PEO/Li/OTf	Au	2.00	0.12	2.74	29.4	0.45	87.2
TMPE/Li/OTf	Ag	3.82	0.32	4.18	53.7	0.33	3.7
TMPE/Li/OTf	Au	1.29	0.07	3.61	1.9	0.17	†
TMPE/Na/OTf	Ag	5.54	0.30	3.26	497.9	1.23	3.9
TMPE/Na/OTf	Au	1.47	0.11	6.39	—	—	†
TMPE/K/OTf	Ag	7.01	0.41	3.10	237.5	1.19	5.2
TMPE/K/OTf	Au	1.43	0.06	5.86	—	—	†
TMPE/Na/TFSI	Ag	1.76	0.10	2.38	25.1	0.46	—
Hy/Li/OTf	Ag	1.41	0.06	1.61	25.2	0.39	—
none/Li/OTf	Ag	0.99	0.00	1.07	—	—	—
TMPE/none	Ag	2.95	0.20	3.18	73.0	0.96	—
none/none	Au	1.03	0.01	1.29	2.8	—	—

per-file inventory) and scripts/uncat_au_tmpe_findings.py (analyses); per-file tables in handouts/uncat_inventory.csv and handouts/uncat_findings_*.csv; a full audit record in handouts/28_uncategorized_au_tmpe_audit.md.

Provenance. The corpus comprises eight dated campaign folders (2025-01-23 to 2025-02-21) stored outside the curated archive layout, holding 356 raw files of which 349 parse as standard Keithley source-meter dumps (three comma-separated rows: current, voltage, time; acquisition cadences matching the instrument’s NPLC settings; 1 mA compliance). All campaigns sit on the Au/TMPE(0.3)/triflate(0.09) generation, one salt per

campaign (Li, Na, K triflate). The January campaign is identified at file level: its contents are byte-identical to the catalogued folder of device NM_v316 (TMPE/LiOTf/Au, 3000 rpm, 90 °C 3 h, glovebox-stored and measured), which it duplicates and extends. The February campaigns exist only in this corpus and are assigned to the same generation by chemistry, recipe, and measurement date (most plausibly the catalogued NaTr, KTr, and LiTr substrates of late 2024, then $\sim 2\text{--}3$ months old); their exact substrate identities are not documented. One recovered file is the measurement script itself, which fixes the relaxation protocol exactly: $-5\text{ V}/0.02\text{ s}$ pulse packets at a -2.25 V inter-pulse level, applied until a $\sim 600\text{ s}$ stabilisation window elapses, then single-point -2.25 V reads after exponentially spaced delays ($0.075\text{ s}\text{--}30\text{ s}$) with the source at high impedance between reads.

Validation. Four consistency checks tie the corpus to the catalogued archive. (i) File format, acquisition cadences, and compliance match the instrument configuration of the curated raw data. (ii) Parsing the catalogued positive- and negative-going sweeps of the same generation (90 sweeps, v296–v316) with the same code reproduces the negative-over-positive response asymmetry in direction and magnitude, and the uncategorized sweeps agree with the catalogued on–off values at matched (fast) sweep rate, the large windows appear only at slow rate, which the catalogued protocol never probed. (iii) Driven positively, the corpus reproduces the weak gold response of table C.6. (iv) The only polarity-inverted measurements are the two sessions whose folder notes document physically reversed wiring; they are excluded from all asymmetry statistics.

Per-protocol results. Table C.7 summarises the corpus by protocol class and cation. Beyond the headline numbers of section 4.8: constant-bias holds bracket the activation threshold between -1.5 and -2.5 V

Table C.7: Per-cation summary of the gold/TMPE/triflate negative-polarity corpus (medians; illustrative tier). Window: on-off ratio of the slowest -3 V sweeps versus the fastest. Potentiation: read-current enhancement at the matched protocol (-5 V pulses, -2.25 V read), compliance-saturated runs excluded. t_{50} : model-free half-decay of the sparse-read relaxation (drop $> 30\%$, non-zero read). Kohlrausch β range over clean fits.

Cation	window (slow/fast, -3 V)	potentiation	n	t_{50} (s)	β
Li ⁺	$\approx 22/\approx 23$	$233\times$	6	2.2 ($n = 5$)	$0.33\text{--}0.44$
Na ⁺	$\approx 15/\approx 8$	$48\times$	12	2.4 ($n = 11$)	$0.33\text{--}0.43$
K ⁺	$\approx 1.9/\approx 0.9$	$40\times$	15	0.7 ($n = 12$)	$0.15\text{--}1.0$

in every campaign (current decaying below it, growing $10\text{--}3000\times$ above it, with 63%-rise times of tens to hundreds of seconds); potentiation–depression runs show deep, fast depression by positive pulses (final-to-peak read ratios of $0.002\text{--}0.05$ at $\pm 4\text{--}5$ V); and 17 files touch the 1 mA compliance ceiling (concentrated in the strongest K trains) and are flagged in the inventory and excluded from the potentiation statistics. The decay fits bound the stretch exponent to $\beta \in (0, 1]$ and are accompanied by a model-free half-decay time; the table reports the latter.

Bounds. The corpus is single-composition ($0.3/0.09$ throughout), spans one or two substrates per cation, is uncured, and mixes protocol variants; the February substrate identities are inferred; the supra-threshold negative read makes the decay times protocol-specific (comparable to the silver delay-time values in order of magnitude only); and device age at measurement differs between the January (days 1–6) and February (months-old) campaigns. Accordingly, every statement drawn from it in the main text is illustrative, and no cation ordering is claimed.

C.8 The Gold-Electrode Lock-in Frequency-Sweep Corpus

This section documents the origin, measurement decode, and quantitative findings of the lock-in frequency-sweep corpus behind the frequency-domain results of section 4.8. Reproduction: `scripts/lockin_au_tmpe_audit.py` (filename parser, per-file metrics, QA panels) and `scripts/lockin_au_tmpe_findings.py` (analyses L1–L5); per-file tables in `handouts/lockin_manifest.csv`, `handouts/lockin_metrics.csv`, and `handouts/lockin_findings_*.csv`; a full audit record in `handouts/29_lockin_au_tmpe_audit.md`.

Provenance. The corpus comprises 549 lock-in sweeper exports (plus 439 binary siblings) taken between 2025-03-04 and 2025-03-26 on a single Au/TMPE/triflate batch: the device-age field in every filename equals the measurement date minus 2025-03-03, so the batch has a single, self-consistent fabrication (top-electrode) date and is tracked from day 1 to day 23. Twelve pixels on three substrates were measured, one substrate per triflate salt (Li, Na, K). The substrates are not catalogued in the device library, the experimenter’s own labels carry an “unknown” prefix, but by chemistry, recipe, and date they are most plausibly the electrode-free UV-Vis trio of 2025-02-26 (fig. 4.8) after Au evaporation, or unlogged siblings of that batch; this identification is unproven, the same caveat class as the February campaigns of section C.7.

Measurement and decode. A Zurich Instruments MFLI drives $V_{\text{off}} + V_{\text{pk}} \sin(2\pi ft)$ through the junction into an external series capacitor to ground ($C_{\text{ext}} = 4 \text{ nF}$ standard; a 0.286 nF – 2 nF series on one pixel; a “ 2.5 pF ” day-1/2 configuration with no external capacitor, loaded by

the instrument’s 10 M Ω input); the voltage at the junction–capacitor node is demodulated at the drive frequency, and the transfer ratio is $|H| = V_{\text{out,RMS}}/(V_{\text{pk}}/\sqrt{2})$, the normalisation used by the corpus’ own visualisation script. Sweeps descend from 500 kHz to 1 Hz in three captures (100+97+200 points, ~ 1 s per point); the 110 “500k–1Hz” files are *derived* merges of the three captures (produced by the corpus’ `combine_freq.py`; frequency-sorted, timer running backwards) and are excluded from all statistics to avoid double counting. Of the four recorded demodulator channels, demodulator 4 is the output fundamental used throughout; demodulator 2 duplicates it through a different demodulator filter; demodulator 3 behaves as a DC monitor of the output node (magnitude = DC value, sign carried by its phase); demodulator 1 is an auxiliary current channel, unused here. A 50 Hz mains glitch is visible in most captures and lies on none of the probe frequencies used below.

L1, divider validation. On unkK2-L5 (day 3, 0.5 V_{pk}, 10 kHz), stepping C_{ext} through 0.286, 0.5, 0.667, 1, 1.3, 1.66, 2 and 4 nF moves $|H|$ from 0.830 to 0.423, and the implied device capacitance $C_{\text{dev}} = C_{\text{ext}}|H|/(1 - |H|)$ stays within 1.4 nF–2.9 nF (median 2.35 nF). The residual monotone trend of C_{dev} with C_{ext} is the expected signature of a lossy (constant-phase-like) dielectric, not an ideal one; the divider model is therefore taken as validated at the factor level, not as an exact calibration.

L2, the passive band is linear. At zero offset and $f \gtrsim 10$ Hz the junction is a passive dielectric: across the 0.5–2.4 V_{pk} amplitude series the plateau $|H|$ at 1 kHz and 10 kHz changes by a median 0.17 %–0.29 % (maximum 8 %; 9 device-days). The effective capacitance at 1 kHz is 2.32 nF (Li), 2.93 nF (Na), and 3.19 nF (K) by substrate, $\approx 28 \text{ nF cm}^{-2}$ – 39 nF cm^{-2} at the $8.25 \times 10^{-2} \text{ cm}^2$ junction area, i.e. $\epsilon_r \approx 5$ – 7 at the 150 nm–190 nm that profilometry measures on sibling substrates of the same recipe, with

a pixel-to-pixel spread *within* each substrate of $\approx 1\%$.

A candidate cation-ordered capacitance. The capacitance is itself cation-ordered, $K > Na > Li$ with a span of $\approx 1.4\times$, two orders of magnitude above the within-substrate pixel spread, on three substrates fabricated in a single batch. With one substrate per salt this cannot be fully separated from substrate-level film variation, but the archive’s profilometry of the same recipe (TMPE 0.3/salt 0.09/3000 rpm; `handouts/lockin_findings_thickness.csv`) bounds the one plausible nuisance channel. Explaining the span by thickness alone would require the Li film $\approx 37\%$ thicker than the K film; the two profiled same-day cation trios differ by only 6% – 12% across salts, and in the anion-matched 2024-10-29 pair the K film is the *thicker* one, the direction that would weaken, not produce, the observed ordering. Salt-uncorrelated solution-to-solution scatter does reach $\approx 30\%$ in the archive, so a thickness origin is not excluded, merely unprecedented in cation-correlated form. The ordering is therefore recorded as a *candidate* cation dielectric effect, an effective permittivity rising from Li^+ to K^+ , the direction qualitative coordination-strength reasoning would suggest (the most weakly coordinated cation the most polarizable), and it is the one cleanly cation-ordered observable in this archive. It is not in tension with the driven retention, which also moves with the cation, the silver TMPE/triflate series runs in the same $K>Na>Li$ direction, but which cannot be fixed to a host- and anion-independent ordering at one or two devices per cell. The flat probes (ion association by ATR, the optical gap by UV-Vis) address different physics (ion pairing, electronic structure) and are expected to be flat, so there is no contradiction. It is not advanced as demonstrated. The decisive, cheap test is profilometry on these three substrates, which still exist, followed by the replicated cation series of section 4.10. The gain magnitudes of the next paragraph are deliberately *not* given the same reading: unlike the capacitance plateau (offset-insensitive, week-stable,

1 % uniform), the low-frequency gain is sensitive to conditioning history and session protocol, both of which differ across the three substrates.

L3, bias-activated low-frequency gain. Table C.8 gives the day-23 offset series on unkLi2-L3 (11 offsets, 0.25 Vpk): the fundamental-frequency transfer at 1 Hz–2 Hz rises monotonically with $|V_{\text{off}}|$, from $|H| \approx 0.5$ unbiased to ≈ 18 at ± 2.5 V, while $|H|$ at 10 Hz stays near unity and the ≥ 1 kHz plateau does not move (offset and zero-offset files give plateau medians 0.422 and 0.423). Across the corpus the prevalence of gain ($|H|_{2\text{ Hz}} > 1$) rises from 7 % of zero-offset captures ($n = 46$; the band is noisy but not amplifying) through 72 % at $|V_{\text{off}}| = 1.4$ V ($n = 18$) to 100 % at $|V_{\text{off}}| \geq 2.4$ V ($n = 20$), with medians 0.46, 1.25, 2.78, 6.69, and 17.3 at $|V_{\text{off}}| = 0, 1.4, 2.0, 2.4, 2.5$ V. The 200 Hz boundary point, captured twice per condition minutes apart, repeats to a median 1.2 % ($n = 95$), so the gain values are not acquisition noise. Because $|H| > 1$ at the drive frequency is impossible for a passive divider into a passive load, the biased junction is an *active* element at these frequencies: the drive slowly modulates the ionic conductance, and that modulation mixes the DC bias up to the fundamental, transferring power from the bias into the signal band. The conclusion is robust to the demodulator amplitude convention (a worst-case $\sqrt{2}$ would reduce ≈ 18 to ≈ 13). Across 32 matched $\pm$ pairs the gain is close to polarity-symmetric, with a median 11 % positive-offset excess (Wilcoxon $p = 0.013$), far weaker than the DC potentiation asymmetry of section C.7.

L4, ageing. The passive plateau is stable within pixel where days repeat: K pixels move from 0.444 (day 3) to 0.450–0.451 (day 9), Na pixels stay within ± 2 % over the same week, while the Li pixels rise by ≈ 15 % from day 4 (0.35–0.37) to day 10 (≈ 0.42) and the day-23 pixel sits at 0.39. One pixel-day deviates (unkK2-L5 day 4, plateau 0.55 at -17 to -21° phase):

Table C.8: Day-23 offset series on unkLi2-L3 (0.25 Vpk, $C_{\text{ext}} = 4 \text{ nF}$, 200–1 Hz capture): fundamental-frequency transfer ratio $|H|$ at four probe frequencies and the DC monitor at the output node. Offsets were measured in the order $0 \rightarrow +2.5$ then $-0.5 \rightarrow -2.5$; the later-measured negative branch is not systematically elevated, so the near-symmetry is not a session-drift artefact.

V_{off} (V)	$ H _{1 \text{ Hz}}$	$ H _{2 \text{ Hz}}$	$ H _{5 \text{ Hz}}$	$ H _{10 \text{ Hz}}$	DC out (V)
−2.5	17.2	16.7	7.5	1.28	−3.46
−2.0	13.1	13.4	5.9	1.01	−2.82
−1.5	12.9	9.3	3.9	1.23	−1.93
−1.0	6.1	6.6	3.1	1.07	−1.38
−0.5	5.4	2.1	1.6	0.90	−0.45
0.0	1.7	0.5	1.0	0.97	+0.02
+0.5	3.7	1.9	2.3	1.12	+0.41
+1.0	8.6	6.9	1.9	1.07	+1.44
+1.5	9.7	10.0	5.2	1.03	+2.10
+2.0	16.1	13.8	4.6	0.83	+2.89
+2.5	17.1	17.9	6.1	1.82	+3.71

that pixel had served as the capacitor-series workhorse the previous day, so its elevated, resistive plateau is read as a conditioned (potentiated) state, not an artefact, it is excluded from the plateau statistics above. Bias-activated gain is present at every age at which offsets were applied (days 4, 9/10, and 23), but the offset–amplitude protocol changed between sessions, so no matched-condition ageing comparison of the gain itself is possible.

L5, DC monitor. The DC component at the output node follows the offset linearly in magnitude and sign across the day-23 series (table C.8, last column; $\approx 1.4\times$ the offset in the export’s convention), confirming that under bias the junction DC-conducts well below the 10 M Ω input impedance, a $\sim \text{M}\Omega$ -scale biased state, the same order as the constant-bias end states of the DC corpus (section C.7). At zero offset the rectified DC is not significantly different from zero (median -0.18 V , 61 % of $n = 46$ captures negative, binomial $p = 0.18$).

Bounds. One batch, one substrate per salt (cation and substrate fully confounded), and one pixel for the day-23 series. The sweeps descend in frequency, so every low-frequency point follows minutes of conditioning under the same offset, the gain states are conditioned steady states, not pristine ones, and the low-frequency points at *zero* offset are visibly non-stationary (state drift during the point). Absolute $|H|$ and DC values inherit the demodulator amplitude convention (at most a factor $\sqrt{2}$; all gain and symmetry statements are ratios at fixed channel and unaffected); C_{dev} is correspondingly approximate. The substrate identities are inferred, as stated under provenance.

C.9 Significance of the Composition Gradient

The composition trends of section 4.4 are reported in the main text as cell medians. This section attaches a significance test and a non-parametric uncertainty to them, on the replicated Ag/Li *Super Yellow*/PEO/LiOTf series. For each descriptor we compute the Spearman rank correlation with the PEO fraction across individual devices, with a two-sided permutation p -value (20 000 relabellings; table C.9). The fading-memory time $t_{1/2}$, the descriptor that matters for chapter 5, carries a clearly significant PEO gradient ($\rho = -0.52$, $p = 0.009$, $n = 25$). The remaining descriptors fall in the same direction but are only marginal at this sample size ($p \approx 0.06$ – 0.07): the per-device gradient is genuinely weaker and noisier than the cell medians alone suggest, and we report it as suggestive, not established. This is the honest counterpart to the median heatmaps, the timescale trend is statistically secure, the window and potentiation trends are indicative.

A non-parametric bootstrap (20 000 device-level resamples per cell) gives a

Table C.9: Per-device Spearman rank correlation of each descriptor with the PEO mass fraction (Ag/Li composition series), with a two-sided permutation p -value. Only the fading-memory time clears $p < 0.05$.

Descriptor	n	Spearman ρ (vs PEO)	permutation p
Fading-memory time $t_{1/2}$	25	−0.52	0.009
Growth exponent α	28	−0.36	0.065
Peak conductance ratio	28	−0.34	0.073
On–off ratio	47	−0.28	0.060

95 % confidence interval on every cell median; the full set is in `handouts/ch4_cell_ci.csv`. The intervals are tight where the cell is well replicated and consistent (e.g. PEO 0.3/salt 0.09: $t_{1/2} = 19.2 [18.6, 22.0]$ s) and honestly wide where a cell mixes a long- and a short-retention device (e.g. PEO 0.3/salt 0.045: $15.8 [3.3, 36.7]$ s), which is the right-skew already flagged in section 4.4 made quantitative.

Two composition axes act near-orthogonally. To test the main-text reading that PEO sets the strength and memory while salt sets the saturation, each log-descriptor was regressed on $\log PEO$, $\log$ salt, and their (centred) interaction, with permutation p -values per term (table C.10). The fading-memory time is governed by PEO alone ($b_{PEO} = -0.92$, $p = 0.001$) with no significant salt or interaction term, so the two axes do not interact detectably at this power: a clean, near-orthogonal control. The potentiation descriptors are noisier ($R^2 \leq 0.2$) and no term is individually significant, consistent with the salt axis acting on *saturation* (the turnover fraction and N_{peak} of table C.4, a categorical effect not captured by a log-linear slope), not on the growth magnitude.

Table C.10: Two-way decomposition of each log-descriptor into log *PEO*, log salt, and their interaction (Ag/Li series; centred predictors; permutation p). PEO is the only significant axis, on the fading-memory time, and no interaction term is significant, the two axes act near-orthogonally.

Descriptor	$b_{PEO}(p)$	$b_{\text{salt}}(p)$	$b_{\text{int}}(p)$	R^2	n
$\log t_{1/2}$	-0.92 (0.001)	+0.38 (0.26)	+0.61 (0.22)	0.47	25
α	-0.15 (0.14)	-0.10 (0.38)	+0.19 (0.30)	0.19	28
$\log(\text{peak})$	-0.89 (0.14)	-0.13 (0.84)	+1.02 (0.33)	0.15	28

C.10 Absolute Conductance Tests the Dilution Picture

The working mechanism of section 4.9 initially read the window contraction at high PEO as a *dilution* of the *Super Yellow* electronic-percolation network. That picture makes a direct, falsifiable prediction on a quantity the peak-normalised dynamics discard: if added ion-transport polymer simply interrupted the electronic network, the *absolute* on-state conductance should *fall* with PEO. It does the opposite. The on-state conductance (HYST conductance at the sweep maximum, device medians, $n = 47$ Ag series devices) *rises* steeply and monotonically with PEO, Spearman $\rho = +0.65$ ($p < 10^{-4}$), a log-log slope of $\approx +3.6$, and more than three orders of magnitude from PEO 0.15 to 1.2 (fig. C.2). Because film thickness rises with PEO (section C.2), a geometric effect would push the conductance the other way, so the rise is if anything understated.

The dilution reading is therefore wrong in its literal form, and the corrected mechanism is the one carried in the revised section 4.9: the ion-rich phase contributes a parallel, weakly switchable conduction path that *raises the conductance floor*. Since the on-off ratio nonetheless falls with PEO, the off-state conductance must rise even faster than the on-state, so the window contracts not because the network is diluted but because its baseline is lifted toward the on-level. The fading-memory and potentiation

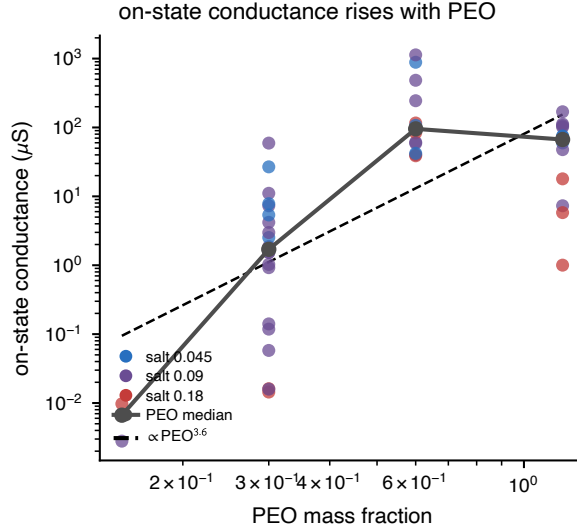


Figure C.2: Absolute on-state conductance (HYST, conductance at the sweep maximum; device medians, Ag series, $n = 47$, coloured by salt) against PEO fraction, log-log. The conductance rises steeply with PEO ($\rho = +0.65$, $p < 10^{-4}$, slope $\approx +3.6$, dashed), the opposite of the prediction of an electronic-percolation *dilution*, and the basis for the rising-floor reading of the window contraction in section 4.9.

trends are unaffected by this correction, they were always read in terms of ionic relaxation, not electronic percolation.

C.11 Reconciling the Impedance and Fading-Memory Timescales

Section 4.7 notes that the EIS resistance *scale* tracks the fading-memory time across composition, while the small-signal apex *frequency* does not: the Nyquist-apex RC time is $\tau_{RC} = 1/(2\pi f_{\text{apex}}) \approx 0.12\text{ s}$ (Ag series, 0 V DC, $n = 26$), more than an order of magnitude shorter than the seconds-long $t_{1/2}$. This section shows the gap is not a contradiction: it is a small-signal/large-signal capacitance ratio, so the two land at the same

magnitude.

The apex RC is set by the geometric capacitance $C_{\text{geo}} = 1/(2\pi f_{\text{apex}} Z_{\text{real}}^{\text{apex}})$. The forgetting of a *written* state, by contrast, is the relaxation of a displaced ionic profile and is governed by the much larger chemical/double-layer capacitance C_{chem} of the ion-rich film. Measuring C_{chem} independently from the low-frequency branch of the same spectra ($C_{\text{lf}} = -1/(2\pi f Z_{\text{imag}})$ at the lowest measured frequency) gives a capacitance enhancement $C_{\text{lf}}/C_{\text{geo}} \approx 33$ (median, $n = 26$), and since the lowest measured frequency does not reach the full DC plateau, this is a *lower bound*. The predicted large-signal relaxation is then $\tau_{\text{large}} \approx \tau_{RC} (C_{\text{chem}}/C_{\text{geo}}) \gtrsim 3.9$ s, squarely inside the observed $t_{1/2}$ range of 2 s–22 s and within a factor of two of the series median (6.6 s). The independently measured capacitance enhancement thus accounts for the timescale separation quantitatively, upgrading the impedance corroboration of section 4.7 from same-direction to same-magnitude.

C.12 Within-Device Reproducibility

A physical reservoir requires elements that are diverse *across* the bank but *individually reproducible*. The chapter documents the diversity at length; this section quantifies the reproducibility. The fading-memory protocol measured a single junction per device, so a population-level repeat-*decay* reproducibility cannot be estimated, only one series device carries two junctions, and there the model-free $t_{1/2}$ agrees to $\sigma(\ln t_{1/2}) \approx 0.19$; broader repeat-decay statistics are a stated gap (section 4.10). The steady-state switching window, however, is recorded over several sweeps per junction (median ≈ 5), giving a genuine population-level within-device reproducibility: the sweep-to-sweep scatter of the on-off ratio is $\sigma(\ln R_{\text{on/off}}) \approx 0.14$ across 165 series junctions, i.e. each junction reproduces its switching window to $\approx 15\%$. This within-device scatter is

far smaller than the across-composition span the bank exploits ($\Delta \ln t_{1/2} \approx 2.1$, section C.13), so a device’s behaviour is a stable signature relative to the diversity the substrate spans, the premise chapter 5 relies on.

C.13 Heterogeneity as a Computational Resource

The chapter treats the spread of fading-memory times as a resource, not noise. This section makes the claim a number. The cell-median $t_{1/2}$ spans $\Delta \ln t_{1/2} \approx 2.1$ across the composition series (from ≈ 2.7 s to ≈ 22 s), while the device-to-device scatter *within* a cell is $\sigma_{\text{within}}(\ln t_{1/2}) \approx 0.29$ on the robust model-free timescale, rising to ≈ 0.85 on the fit-noisier single-exponential τ cited in section D.3. Taking a two- σ separation as the criterion for two timescales to be distinguishable, the bank provides $N_{\text{eff}} = 1 + \Delta \ln t_{1/2} / (2\sigma_{\text{within}}) \approx 2.2$ (using the conservative $\sigma = 0.85$) to ≈ 4.6 (using the model-free $\sigma = 0.29$) resolvably distinct memory bands selectable by composition. The resource is therefore modest but real: a handful of cleanly separated timescales, not a continuum, which is why chapter 5 assembles a *bank* of compositions, not a single optimum.

Single-device and inter-device disorder are linked. The stretched exponent provides an independent read on dispersion: the median single-device stretch corresponds to a barrier-distribution width $1 - \beta \approx 0.36$, of the same order as the within-cell device-to-device $\sigma(\ln t_{1/2})$. The disorder seen *within* one device’s relaxation and the spread *between* sibling devices are thus comparable in magnitude, consistent with both arising from the same heterogeneous distribution of local activation barriers, not from independent causes.

Appendix D

Supporting Information for Chapter 5

This appendix documents how the behavioural-model parameter cards of section 5.2 are sourced from the measured archive, and lists the simulation and read-out settings used across the three evidence tiers. Together with the per-cell tables of section C.3, it makes the in-silico results reproducible without consulting the analysis scripts (`scripts/ch5_figures.py`, `scripts/ch5_physio_context.py`, `scripts/ch5_onset.py`, `scripts/ch5_deployment.py`, `scripts/ch5_noneeg.py`, `scripts/ch5_case.py`).

D.1 Parameter-Card Sourcing

Each composition cell is reduced to a parameter card $(\tau_c, \beta_c, \alpha_c, \text{peak}_c)$ that drives one reservoir node type. The values are taken directly from the comparative fits, with no free re-tuning:

- the fading-memory law $\lambda_c(\Delta t) = \exp[-(\Delta t/\tau_c)^{\beta_c}]$ uses the per-cell

(τ_c, β_c) of table C.3; for the few cells without an identified stretched fit, a single exponential reproducing the model-free half-life is used instead ($\tau_c = t_{1/2}/\ln 2$, $\beta_c = 1$);

- the write nonlinearity uses the per-cell growth exponent α_c and peak enhancement of table C.4;
- the device-to-device spread injected into each bank is the within-cell scatter of those same tables.

The heterogeneous banks of sections 5.4 and 5.6 span the replicated PEO/LiOTf composition cells of table C.3 (τ from ≈ 3 to ≈ 26 s); the homogeneous control banks are jittered copies of the lead cell PEO 0.3/0.09 alone.

D.2 Simulation and Read-out Settings

Table D.1 collects the settings used throughout chapter 5. All read-outs are a single linear ridge regression; no nonlinear decoder is used, so any demonstrated computation is attributable to the device nodes rather than the read-out. The read-out ridge penalty ($\lambda = 10^{-3}$), the causal label-smoothing horizon (15 s), the bank size N , and the reservoir step Δt are fixed *a priori* and held identical across all banks and all held-out subjects; none is selected on the held-out subject, so the leave-one-subject-out protocol carries no hyperparameter leakage.

D.3 Measured Device-to-Device Scatter

The robustness analysis of section 5.10 (fig. 5.9) injects device-to-device variation as a multiplicative log-normal *jitter* on each node’s timescale,

Table D.1: Simulation and read-out settings across the three evidence tiers of chapter 5. All banks are spatial banks of independent (non-recurrent) leaky nodes driven by randomly-masked input; all read-outs are linear ridge regression.

Setting	Value
Node model	$x_n^{(i)} = \lambda_i x_{n-1}^{(i)} + [\max(w_i u_n, 0)]^{\alpha_i}$ (section 5.2)
Read-out	single linear ridge regression
Bank size N	24 (benchmarks); 48 (physiology/affect)
Algorithmic variability	10 random input/bank seeds; means $\pm$ SD and sign consistency reported; seeds are not inferential replicates
Inferential unit	independent held-out subject for WESAD/CASE; independently fabricated device for experimental comparisons
Memory capacity	MC_k summed over lags, random uniform input [155]
NARMA	NARMA-10 system identification, NRMSE [156]
Information-processing capacity	degree-1 (linear) + degree-2 (nonlinear), bounded by N [80]
Affective corpus	WESAD chest signals, leave-one-subject-out over 15 subjects [161]
Streaming cadence	$\Delta t = 1$ s; reconstruction delays 1, 3, 8, 20, 45 s
Affect scoring	class-balanced macro-F1

$\tau_i = \tau e^\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$; the jitter magnitude σ is, by construction, the standard deviation of $\ln \tau$ across nominally identical devices. That quantity is measured directly here, so the dotted “measured scatter” line in fig. 5.9 is data-derived rather than assumed.

The estimate uses the fading-memory time constants of every silver PEO/LiOTf device ($n = 31$). A naive spread taken across the whole SY, PEO, LiTr group conflates the *deliberate* PEO $\times$ salt tuning sweep of chapter 4 with random scatter and over-states it ($\sigma \approx 1.59$); the meaningful quantity is the spread *within* each fixed PEO/salt cell, $\sigma(\ln \tau) \approx 0.85$ (table D.2). This residual is genuine device-to-device variation: it is essentially unchanged after regressing out spin-coating speed and annealing (≈ 0.91), and within-cell residuals show no monotonic dependence on film thickness (Spearman $\rho \approx 0.14$, $n = 31$, complete profilometry) or on device age ($\rho \approx 0.05$). The measured value is roughly seven times the

Table D.2: Variance decomposition of $\ln \tau$ for the silver PEO/LiOTf series ($n = 31$ devices). The within-cell value is the measured jitter σ ; the group-level value is reported only to show the confound it would introduce.

Estimate of $\sigma(\ln \tau)$	Value
Across SY, PEO, LiTr group (conflates designed sweep)	1.59
Within fixed PEO/salt cell (= measured jitter)	0.85
after regressing out spin speed + anneal	0.91
Within-cell residual vs. film thickness (Spearman ρ)	0.14
Within-cell residual vs. device age (Spearman ρ)	0.05
Value previously annotated on fig. 5.9	0.12

figure previously annotated on the robustness plot, and the heterogeneous bank retains its memory-capacity advantage at it (fig. 5.9), so the realised material diversity is, if anything, larger than the simulations assume. The audit is reproduced by `scripts/ch5_scatter_audit.py`.

D.4 Continuous-Monitoring Stress Test

The deployment-robustness result of section 5.9 (`scripts/ch5_onset.py`) is a continuous binary stress detector: at every $\Delta t = 1$ s step the read-out predicts stress (WESAD label 2) versus not-stress (baseline and amusement pooled), leave-one-subject-out, with the same single linear ridge read-out and 15-second causal label smoothing used in the streaming task. Four matched banks of $N = 24$ nodes are compared on identical input masks: the instantaneous input (no memory), a memoryless bank (leak set to zero, dimensionality only), the homogeneous lead-only bank, and the heterogeneous composition bank.

Sensor corruption is modelled by adding, to the per-subject scaled streams (channels in $[0, 1]$), independent Gaussian noise of standard deviation σ at every sample, plus sparse motion-artefact bursts (short windows of amplified noise at a 1% per-sample rate), clipped back to $[0, 1]$; the same

realisation is applied to the training and held-out streams so the comparison isolates each architecture’s intrinsic noise tolerance. Robustness curves are seed-averaged over three bank/noise seeds; the per-subject significance test at $\sigma = 0.4$ is a paired Wilcoxon signed-rank over the fifteen subjects. Reported metrics are the binary macro-F1, the pooled false-alarm rate (fraction of not-stress steps called stress), and, for completeness (though they do not separate the banks), the transition-window macro-F1 (steps within ± 20 s of a label change) and the onset detection latency (time to the first ≥ 5 s sustained stress call after each onset).

The cross-corpus replication (table 5.5, `scripts/ch5_noneeg.py`) applies the *same* detector and noise model to the PhysioNet Non-EEG corpus [164] (open access; obtainable with `wfdb.dl_database('noneeg', 'data/noneeg')`). The 8 Hz electrodermal and temperature record and the 1 Hz heart-rate record are resampled onto a common timeline and per-subject scaled exactly as the WESAD streams; the phase annotations map cognitive and emotional stress to the stress class and relaxation to not-stress, with the physical-stress (exercise) phase excluded.

Real-motion corruption. To confirm the robustness is not an artefact of the Gaussian noise model, the corruption is also driven by the *measured* chest accelerometer (`add_motion_noise`, `load_motion` in the scripts). A per-step motion index is the magnitude of the gravity-removed acceleration, RMS-smoothed over 1 s and robustly scaled to $[0, 1]$ per subject; the injected noise standard deviation at each step is $\text{scale} \times (0.05 + \text{motion})$, with an amplified artefact burst where the index exceeds 0.6, so corruption lands where the wearer actually moves. Across a corruption scale of 0 to 0.8 the instantaneous detector falls from macro-F1 0.914 to 0.898 while the memory bank holds near 0.92; at the strongest gating the memory-minus-instantaneous advantage is +0.032 (paired Wilcoxon $p = 0.04$, $r_{\text{rb}} = +0.6$) and the single- τ moving average matches it (+0.036). The

advantage is smaller than under the synthetic injection because the chest strap is mechanically stable and the protocol largely seated; the wrist site (table 5.6) is where motion artefact dominates.

D.5 Control Experiments

Three controls underpin the claims of chapter 5 and are tabulated here. They are reproduced by `scripts/ch5_reservoir.py` (the generic-ESN and coupling controls) and `scripts/ch5_onset.py` (the classical-filter control). For seed-only controls, matched means, standard deviations, effect differences, and sign consistency describe algorithmic sensitivity; no population-level p -value is assigned. Paired Wilcoxon tests and rank-biserial effects are retained only where the pairs are independent held-out subjects.

Classical-filter control. Table D.3 is the control behind the deployment-robustness reframing of section 5.9: an ordinary exponential moving average of the raw channels, fed to the same linear read-out with no reservoir, recovers the noise robustness, so the robustness is the generic property of a fading-memory low-pass rather than anything specific to the device or its nonlinearity. The device’s contribution is to *be* that filter, passively and at the budget of section D.6. The cross-corpus and wrist robustness results (tables 5.5 and 5.6) share this character: there too the gain is the low-pass advantage of temporal memory over an instantaneous read.

Generic-ESN control (matching thesis). Table D.4 asks whether the *particular* measured timescales matter or only their spread, by replacing the device bank with an echo-state network of the same size whose leak times are drawn at random. A random bank over the same range ties the

Table D.3: Classical-filter control for the continuous stress monitor (binary macro-F1, leave-one-subject-out, σ the injected sensor noise of fig. 5.8). A single-time-constant moving average of the raw channels recovers the robustness an instantaneous read loses, and a linear moving-average bank spread over the device timescale range adds little; the device reservoir’s compressive nonlinearity adds nothing further on this single-timescale task. Paired at $\sigma = 0.4$ over the fifteen subjects: EMA(single)–instantaneous = +0.091 ($p < 10^{-4}$, $r_{\text{rb}} = +1.0$); EMA(bank)–EMA(single) = +0.005 (*ns*); reservoir–EMA(bank) = −0.008 (*ns*).

Detector (linear read-out on ...)	clean	$\sigma = 0.4$
Instantaneous channels (no memory)	0.914	0.840
EMA, single τ (lead cell)	0.937	0.927
EMA bank, τ spread (linear reservoir)	0.926	0.930
Device reservoir (nonlinear, heterogeneous)	0.920	0.916

Table D.4: Total linear memory capacity ($N = 24$, ten matched seeds, means $\pm$ SD) of the measured device heterogeneous bank versus random echo-state networks of equal size. Relative to the device bank, the same-range ESN differs by +0.28 on average and the broad-range ESN by +1.14 (positive in all ten seed pairs). These are algorithmic-sensitivity comparisons, not population-level significance tests.

Reservoir ($N = 24$)	total MC
Measured device heterogeneous bank	6.4 ± 0.3
Random ESN, same τ range (3–26 s)	6.7 ± 0.3
Random ESN, broad τ range (0.5–60 s)	7.6 ± 0.3

device bank, and a broader random range exceeds it: the resource is the timescale *spread*, which the composition grid supplies intrinsically, and extending the range (the drive-amplitude variable of chapter 4) is where further capacity lies.

Coupling-sensitivity control. Table D.5 bounds the composition assumption of section 5.2: each node’s retention time is rescaled with its own write-nonlinearity exponent by a factor $e^{\kappa z}$ (z the standardised exponent), modelling the drive–retention coupling the separate-amplitude

measurement cannot exclude. The heterogeneous bank stays above the homogeneous one for every coupling strength swept, so the heterogeneity conclusion does not rest on the assumption being exact. The sweep is then *anchored to data*. The gold/TMPE negative-polarity corpus of chapter 4 supplies the only direct measurement of this coupling: across 28 single-run potentiation + relaxation traces the write-depth and the forgetting time are rank-correlated at $\rho = +0.46$ ($p = 0.015$; 95% CI 0.10–0.71 by Fisher’s z). Calibrating κ so the simulated across-node write–retention correlation matches that figure fixes the data-anchored operating point. The nominal composed cards already carry a baseline correlation of $\rho = +0.32$ (cells that write more strongly also retain longer), so reproducing the measurement needs only a modest extra coupling, $\kappa^* = +0.18$ (CI 0.00–0.63), at which the heterogeneous advantage is undiminished (capacity ratio 1.45, positive in all ten seed pairs; the ratio stays above 1.18 across the whole measured CI). The figure is transferred as an order-of-magnitude anchor only – a different host, electrode, and polarity – but it converts the bound from an arbitrary range into a data-referenced point and independently confirms the *sign* of the coupling the composed model assumes. The calibration (κ -for- ρ bisection) is reproduced by `scripts/ch5_reservoir.py`.

Chance baselines. Under the streaming three-class protocol of section 5.7, a majority-class predictor scores macro-F1 0.23 and a stratified-random predictor 0.33 (leave-one-subject-out, per step), so the reservoir’s 0.758 is well above chance. Significance of the bank comparisons is assessed across the fifteen independent held-out subjects, so within-subject temporal autocorrelation does not inflate it.

Drive-boosted tonic-node extension. The minutes-scale tonic gap of fig. 5.4 is probed by extending the heterogeneous bank with drive-

Table D.5: Memory capacity of the heterogeneous and homogeneous banks under an injected drive–retention coupling of strength κ ($N = 24$, ten seeds; $\kappa = 0$ is the nominal composed model, which already carries a baseline across-node write–retention rank correlation $\rho = +0.32$). *Upper block*: an arbitrary symmetric sweep; the heterogeneous advantage is positive in all ten seed pairs at every κ . *Lower block* ($\dagger$): the data-anchored point, $\kappa^* = +0.18$, calibrated so the simulated correlation reproduces the measured Au/TMPE coupling ($\rho = +0.46$; chapter 4); the advantage is undiminished there. Seed consistency describes sensitivity, not experimental replication.

κ	het MC	hom MC	ratio
−0.50	6.31	5.50	1.15
−0.25	6.22	4.65	1.34
0.00	6.43	4.20	1.53
+0.25	6.57	4.64	1.42
+0.50	6.75	5.43	1.24
+0.18 †	6.49	4.46	1.45

boosted nodes at $\tau = 120$ and 300 s, a labelled extrapolation of the drive-amplitude variable, copying the lead cell’s write nonlinearity and stretch and overriding only τ (`tonic_cards` in `scripts/ch5_model.py`). On memory capacity (DT = 5 s, k to 40) the extension nearly doubles the long-lag tail beyond 75 s (MC $_{k>15}$ from 0.73 to 1.37, positive in all ten seed pairs) and raises total capacity from 6.4 to 7.0. On the physiological-context reconstruction, with the target delays extended to 90 and 120 s so the genuinely tonic context is present, the slow-band held-out R^2 rises from 0.672 to 0.690 (+0.019, positive in all ten seed pairs) and the overall R^2 by +0.009 (`tonic_extension` in `scripts/ch5_physio_context.py`). The extension is not part of the measured composition banks anywhere else in the chapter.

Table D.6: Deployment envelope for an N -node device reservoir read out by a linear classifier of K classes, from the measured device constants of chapter 2, contrasted with a typical gradient-trained deep model. Parameter counts are exact; energy figures are order-of-magnitude estimates (see text).

Quantity	Device reservoir (this work)	Deep model (typical)
Trainable parameters	$(N+1)K$: 75 ($N=24$) / 147 ($N=48$), $K=3$	10^4 – 10^6
Reservoir dynamics	physical, untrained (0 parameters)	—
Training method	one closed-form ridge solve	gradient descent / backpropagation
Energy per update	$\approx N \times 50$ nJ (≈ 1.2 μ J at $N=24$)	—
Average power (1 Hz, always-on)	≈ 1.2 μ W– 2.4 μ W ($N=24$ – 48)	—
Read-out compute	NK MACs/step (< 1 nJ)	—
Operating regime	sub-kHz (affect is sub-Hz)	—

D.6 Deployment Envelope

Table D.6 collects the operating budget quoted in section 5.10 (`scripts/ch5_deployment.py`). The trainable-parameter count is exact: the reservoir dynamics are fixed physical devices, and the read-out is a single $(N+1) \times K$ ridge-regression weight matrix. The energy figures are order-of-magnitude estimates built from the *measured* proof-of-concept device of chapter 2 (chapter A): an energy per write event of ≈ 50 nJ (a 1 V, 0.1 s pulse), equivalently ≈ 6 fJ μm^{-2} over the 0.0825 cm^2 junction (the two agree to within 1%, 6 fJ $\mu\text{m}^{-2} \times 0.0825$ $\text{cm}^2 \approx 49.5$ nJ). The estimate conservatively charges each node one full write event per one-second update; the sub-threshold, state-preserving read at 0.5 V is cheaper, and these are unoptimised large-area figures, so the absolute energy should be read as an upper bound that integration would lower. The read-out compute is NK multiply–accumulates per step at an assumed 1 pJ per operation.

D.7 Continuous Affect: CASE Protocol and Results

Table D.7 gives the full continuous valence/arousal result behind section 5.8 (`scripts/ch5_case.py`). The CASE corpus [162] (open access; figshare 10.6084/m9.figshare.8869157) provides, for thirty subjects, a continuous joystick annotation of valence and arousal (range [0.5, 9.5], 20 Hz) synchronised with eight physiological channels at 1 kHz. We use the four slow channels that match the device band and the WESAD set – electrodermal activity, respiration, skin temperature, and a heart rate recovered from the electrocardiogram by the same robust R-peak detector used for WESAD – resampled to the $\Delta t = 1$ s reservoir cadence over the eight emotional videos (the blue baseline clips are excluded), per-subject robustly scaled exactly as the WESAD streams. The reservoir state is reset and a 10 s washout dropped at each video boundary. The read-out is a single linear ridge regression predicting valence and arousal jointly, leave-one-subject-out, scored by Lin’s concordance correlation coefficient (CCC) [163], $\text{CCC} = 2 \text{cov}(y, \hat{y}) / [\sigma_y^2 + \sigma_{\hat{y}}^2 + (\mu_y - \mu_{\hat{y}})^2]$.

Two classical controls bound the device. The *EMA* rows replace the reservoir with an exponential moving average of the raw channels fed to the same read-out: a single time constant (lead cell), and a bank spread over the device timescale range (3 s–26 s). The *per-channel* device bank gives each node a single-channel input mask instead of the random multi-channel mask – the apples-to-apples device analogue of the linear EMA bank – so the device-minus-EMA gap isolates the compressive nonlinearity rather than the mask. Paired Wilcoxon tests over the thirty held-out subjects (mean CCC per subject, seed-averaged): τ -spread – single- τ (EMA) = +0.176 (27/30, $p = 1.6 \times 10^{-7}$, $r_{\text{rb}} = +0.94$); memory – instantaneous = +0.079 (24/30, $p = 1.2 \times 10^{-4}$); per-channel device – EMA bank = −0.047 ($p = 6.6 \times 10^{-3}$); heterogeneous – homogeneous

Table D.7: Held-out concordance correlation coefficient (CCC) for continuous valence and arousal on CASE [162] ($N = 48$ nodes, leave-one-subject-out, linear ridge read-out; reservoir/memoryless rows are means over ten bank seeds). A single timescale barely beats the instantaneous read; a *spread* of timescales (the linear τ -spread EMA bank, bold) nearly doubles it. The device banks capture much of that spread but, on this smooth regression, trail the clean linear filter; the per-channel device recovers part of the gap that the random cross-channel mask costs.

Read-out features	CCC valence	CCC arousal	CCC mean
Instantaneous (no memory)	0.178	0.320	0.249
Memoryless bank	0.232	0.375	0.304
EMA, single τ (classical)	0.170	0.290	0.230
EMA bank, τ spread (classical)	0.311	0.516	0.413
Device, homogeneous (lead τ)	0.240	0.433	0.337
Device, heterogeneous (mixed mask)	0.247	0.440	0.344
Device, heterogeneous (per-channel)	0.258	0.458	0.358

device = +0.005 ($p = 0.18$, *ns*).

References

1. Horowitz, M. *1.1 Computing's energy problem (and what we can do about it)* in *2014 IEEE International Solid-State Circuits Conference Digest of Technical Papers (ISSCC)* (2014), 10–14. doi:10.1109/ISSCC.2014.6757323.
2. LeCun, Y., Bengio, Y. & Hinton, G. Deep learning. *Nature* **521**, 436–444. doi:10.1038/nature14539 (2015).
3. Hennessy, J. L. & Patterson, D. A. A new golden age for computer architecture. *Communications of the ACM* **62**, 48–60. doi:10.1145/3282307 (2019).
4. Mead, C. Neuromorphic electronic systems. *Proceedings of the IEEE* **78**, 1629–1636. doi:10.1109/5.58356 (1990).
5. Tanaka, G., Yamane, T., Héroux, J. B., *et al.* Recent advances in physical reservoir computing: A review. *Neural Networks* **115**, 100–123. doi:10.1016/j.neunet.2019.03.005 (2019).
6. Prado-Socorro, C. D., Giménez-Santamarina, S., Mardegan, L., *et al.* Polymer-Based Composites for Engineering Organic Memristive Devices. *Advanced Electronic Materials* **8**, 2101192. doi:10.1002/aelm.202101192 (2022).
7. Abbott, L. F. & Regehr, W. G. Synaptic computation. *Nature* **431**, 796–803. doi:10.1038/nature03010 (2004).

8. Hennessy, J. L. & Patterson, D. A. *Computer Architecture: A Quantitative Approach* 6th ed. (Morgan Kaufmann, Cambridge, MA, 2017).
9. Wulf, W. A. & McKee, S. A. Hitting the memory wall: implications of the obvious. *ACM SIGARCH Computer Architecture News* **23**, 20–24. doi:10.1145/216585.216588 (1995).
10. Von Neumann, J. First Draft of a Report on the EDVAC. *IEEE Annals of the History of Computing* **15**, 27–75. doi:10.1109/85.238389 (1993).
11. Editorial. Big data needs a hardware revolution. *Nature* **554**, 145–146. doi:10.1038/d41586-018-01683-1 (2018).
12. Moore, G. E. Cramming more components onto integrated circuits. *Electronics* **38**, 114–117 (1965).
13. Dennard, R. H., Gaensslen, F. H., Yu, H.-N., Rideout, V. L., Bassous, E. & LeBlanc, A. R. Design of ion-implanted MOSFET's with very small physical dimensions. *IEEE Journal of Solid-State Circuits* **SC-9**, 256–268. doi:10.1109/JSSC.1974.1050511 (1974).
14. Esmailzadeh, H., Blem, E., St. Amant, R., Sankaralingam, K. & Burger, D. *Dark silicon and the end of multicore scaling in Proceedings of the 38th Annual International Symposium on Computer Architecture (ISCA '11)* (2011), 365–376. doi:10.1145/2000064.2000108.
15. Prezioso, M., Merrih-Bayat, F., Hoskins, B., Adam, G. C., Likharev, K. K. & Strukov, D. B. Training and operation of an integrated neuromorphic network based on metal-oxide memristors. *Nature* **521**, 61–64. doi:10.1038/nature14441 (2015).
16. Ielmini, D. & Wong, H.-S. P. In-memory computing with resistive switching devices. *Nature Electronics* **1**, 333–343. doi:10.1038/s41928-018-0092-2 (2018).

REFERENCES

17. Jaeger, H. *The “echo state” approach to analysing and training recurrent neural networks – with an erratum note* tech. rep. GMD Report 148 (German National Research Center for Information Technology (GMD), Bonn, Germany, 2001).
18. Maass, W., Natschläger, T. & Markram, H. Real-Time Computing Without Stable States: A New Framework for Neural Computation Based on Perturbations. *Neural Computation* **14**, 2531–2560. doi:10.1162/089976602760407955 (2002).
19. Koelmans, W. W., Sebastian, A., Jonnalagadda, V. P., Krebs, D., Dellmann, L. & Eleftheriou, E. Projected phase-change memory devices. *Nature Communications* **6**, 8181. doi:10.1038/ncomms9181 (2015).
20. Sebastian, A., Le Gallo, M. & Krebs, D. Crystal growth within a phase change memory cell. *Nature Communications* **5**, 4314. doi:10.1038/ncomms5314 (2014).
21. Fuller, E. J., Keene, S. T., Melianas, A., *et al.* Parallel programming of an ionic floating-gate memory array for scalable neuromorphic computing. *Science* **364**, 570–574. doi:10.1126/science.aaw5581 (2019).
22. Van de Burgt, Y., Lubberman, E., Fuller, E. J., *et al.* A non-volatile organic electrochemical device as a low-voltage artificial synapse for neuromorphic computing. *Nature Materials* **16**, 414–418. doi:10.1038/nmat4856 (2017).
23. Van de Burgt, Y., Melianas, A., Keene, S. T., Malliaras, G. & Salleo, A. Organic electronics for neuromorphic computing. *Nature Electronics* **1**, 386–397. doi:10.1038/s41928-018-0103-3 (2018).
24. Yao, P., Wu, H., Gao, B., *et al.* Fully hardware-implemented memristor convolutional neural network. *Nature* **577**, 641–646. doi:10.1038/s41586-020-1942-4 (2020).

-
25. Vaswani, A., Shazeer, N., Parmar, N., *et al.* *Attention Is All You Need* in *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)* (2017), 5998–6008.
doi:10.48550/arXiv.1706.03762.
 26. Roy, K., Jaiswal, A. & Panda, P. Towards spike-based machine intelligence with neuromorphic computing. *Nature* **575**, 607–617.
doi:10.1038/s41586-019-1677-2 (2019).
 27. Indiveri, G. & Liu, S.-C. Memory and Information Processing in Neuromorphic Systems. *Proceedings of the IEEE* **103**, 1379–1397.
doi:10.1109/JPROC.2015.2444094 (2015).
 28. Boyd, S. & Chua, L. O. Fading memory and the problem of approximating nonlinear operators with Volterra series. *IEEE Transactions on Circuits and Systems* **32**, 1150–1161.
doi:10.1109/TCS.1985.1085649 (1985).
 29. Jaeger, H. & Haas, H. Harnessing Nonlinearity: Predicting Chaotic Systems and Saving Energy in Wireless Communication. *Science* **304**, 78–80. doi:10.1126/science.1091277 (2004).
 30. Lukoševičius, M. & Jaeger, H. Reservoir computing approaches to recurrent neural network training. *Computer Science Review* **3**, 127–149. doi:10.1016/j.cosrev.2009.03.005 (2009).
 31. Verstraeten, D., Schrauwen, B., D’Haene, M. & Stroobandt, D. An experimental unification of reservoir computing methods. *Neural Networks* **20**, 391–403. doi:10.1016/j.neunet.2007.04.003 (2007).
 32. Du, C., Cai, F., Zidan, M. A., Ma, W., Lee, S. H. & Lu, W. D. Reservoir computing using dynamic memristors for temporal information processing. *Nature Communications* **8**, 2204.
doi:10.1038/s41467-017-02337-y (2017).

REFERENCES

33. Midya, R., Wang, Z., Asapu, S., *et al.* Reservoir Computing Using Diffusive Memristors. *Advanced Intelligent Systems* **1**, 1900084. doi:10.1002/aisy.201900084 (2019).
34. Zhong, Y., Tang, J., Li, X., Gao, B., Qian, H. & Wu, H. Dynamic memristor-based reservoir computing for high-efficiency temporal signal processing. *Nature Communications* **12**, 408. doi:10.1038/s41467-020-20692-1 (2021).
35. Cucchi, M., Gruener, C., Petrauskas, L., *et al.* Reservoir computing with biocompatible organic electrochemical networks for brain-inspired bio-signal classification. *Science Advances* **7**, eabh0693. doi:10.1126/sciadv.abh0693 (2021).
36. Drubach, D. *The Brain Explained* (Prentice Hall, Upper Saddle River, NJ, 2000).
37. *Principles of Neural Science* 5th ed. (eds Kandel, E. R., Schwartz, J. H., Jessell, T. M., Siegelbaum, S. A. & Hudspeth, A. J.) (McGraw-Hill, New York, 2013).
38. Hodgkin, A. L. & Huxley, A. F. A quantitative description of membrane current and its application to conduction and excitation in nerve. *The Journal of Physiology* **117**, 500–544. doi:10.1113/jphysiol.1952.sp004764 (1952).
39. Rieke, F., Warland, D., de Ruyter van Steveninck, R. & Bialek, W. *Spikes: Exploring the Neural Code* (MIT Press, Cambridge, MA, 1997).
40. Südhof, T. C. The synaptic vesicle cycle. *Annual Review of Neuroscience* **27**, 509–547. doi:10.1146/annurev.neuro.26.041002.131412 (2004).
41. Hebb, D. O. *The Organization of Behavior: A Neuropsychological Theory* (Wiley, New York, 1949).

-
42. Zucker, R. S. Short-term synaptic plasticity. *Annual Review of Neuroscience* **12**, 13–31.
doi:10.1146/annurev.ne.12.030189.000305 (1989).
 43. Zucker, R. S. & Regehr, W. G. Short-term synaptic plasticity. *Annual Review of Physiology* **64**, 355–405.
doi:10.1146/annurev.physiol.64.092501.114547 (2002).
 44. Bliss, T. V. P. & Lømo, T. Long-lasting potentiation of synaptic transmission in the dentate area of the anaesthetized rabbit following stimulation of the perforant path. *Journal of Physiology* **232**, 331–356. doi:10.1113/jphysiol.1973.sp010273 (1973).
 45. Bliss, T. V. P. & Collingridge, G. L. A synaptic model of memory: long-term potentiation in the hippocampus. *Nature* **361**, 31–39.
doi:10.1038/361031a0 (1993).
 46. Abraham, W. C. How long will long-term potentiation last? *Philosophical Transactions of the Royal Society B: Biological Sciences* **358**, 735–744. doi:10.1098/rstb.2002.1222 (2003).
 47. Mayer, M. L., Westbrook, G. L. & Guthrie, P. B. Voltage-dependent block by Mg^{2+} of NMDA responses in spinal cord neurones. *Nature* **309**, 261–263. doi:10.1038/309261a0 (1984).
 48. Nowak, L., Bregestovski, P., Ascher, P., Herbet, A. & Prochiantz, A. Magnesium gates glutamate-activated channels in mouse central neurones. *Nature* **307**, 462–465.
doi:10.1038/307462a0 (1984).
 49. Malinow, R. & Malenka, R. C. AMPA receptor trafficking and synaptic plasticity. *Annual Review of Neuroscience* **25**, 103–126.
doi:10.1146/annurev.neuro.25.112701.142758 (2002).

REFERENCES

50. Lüscher, C. & Malenka, R. C. NMDA receptor-dependent long-term potentiation and long-term depression (LTP/LTD). *Cold Spring Harbor Perspectives in Biology* **4**, a005710. doi:10.1101/cshperspect.a005710 (2012).
51. Dudek, S. M. & Bear, M. F. Homosynaptic long-term depression in area CA1 of hippocampus and effects of N-methyl-D-aspartate receptor blockade. *Proceedings of the National Academy of Sciences* **89**, 4363–4367. doi:10.1073/pnas.89.10.4363 (1992).
52. Mulkey, R. M. & Malenka, R. C. Mechanisms underlying induction of homosynaptic long-term depression in area CA1 of the hippocampus. *Neuron* **9**, 967–975. doi:10.1016/0896-6273(92)90248-C (1992).
53. Malenka, R. C. & Bear, M. F. LTP and LTD: an embarrassment of riches. *Neuron* **44**, 5–21. doi:10.1016/j.neuron.2004.09.012 (2004).
54. Markram, H., Lübke, J., Frotscher, M. & Sakmann, B. Regulation of synaptic efficacy by coincidence of postsynaptic APs and EPSPs. *Science* **275**, 213–215. doi:10.1126/science.275.5297.213 (1997).
55. Bi, G.-q. & Poo, M.-m. Synaptic Modifications in Cultured Hippocampal Neurons: Dependence on Spike Timing, Synaptic Strength, and Postsynaptic Cell Type. *Journal of Neuroscience* **18**, 10464–10472. doi:10.1523/JNEUROSCI.18-24-10464.1998 (1998).
56. Herculano-Houzel, S. The Human Brain in Numbers: A Linearly Scaled-up Primate Brain. *Frontiers in Human Neuroscience* **3**, 31. doi:10.3389/neuro.09.031.2009 (2009).

-
57. Attwell, D. & Laughlin, S. B. An Energy Budget for Signaling in the Grey Matter of the Brain. *Journal of Cerebral Blood Flow & Metabolism* **21**, 1133–1145.
doi:10.1097/00004647-200110000-00001 (2001).
58. Kuzum, D., Yu, S. & Wong, H.-S. P. Synaptic electronics: materials, devices and applications. *Nanotechnology* **24**, 382001.
doi:10.1088/0957-4484/24/38/382001 (2013).
59. Chua, L. O. Memristor – The Missing Circuit Element. *IEEE Transactions on Circuit Theory* **18**, 507–519.
doi:10.1109/TCT.1971.1083337 (1971).
60. Adhikari, S. P., Sah, M. P., Kim, H. & Chua, L. O. Three Fingerprints of Memristor. *IEEE Transactions on Circuits and Systems I: Regular Papers* **60**, 3008–3021.
doi:10.1109/TCSI.2013.2256171 (2013).
61. Chua, L. O. If it's pinched it's a memristor. *Semiconductor Science and Technology* **29**, 104001.
doi:10.1088/0268-1242/29/10/104001 (2014).
62. Chua, L. O. & Kang, S. M. Memristive devices and systems. *Proceedings of the IEEE* **64**, 209–223.
doi:10.1109/PROC.1976.10092 (1976).
63. Hickmott, T. W. Low-frequency negative resistance in thin anodic oxide films. *Journal of Applied Physics* **33**, 2669–2682.
doi:10.1063/1.1702530 (1962).
64. Argall, F. Switching phenomena in titanium oxide thin films. *Solid-State Electronics* **11**, 535–541.
doi:10.1016/0038-1101(68)90092-0 (1968).

REFERENCES

65. Dearnaley, G., Stoneham, A. M. & Morgan, D. V. Electrical phenomena in amorphous oxide films. *Reports on Progress in Physics* **33**, 1129–1191. doi:10.1088/0034-4885/33/3/306 (1970).
66. Strukov, D. B., Snider, G. S., Stewart, D. R. & Williams, R. S. The missing memristor found. *Nature* **453**, 80–83. doi:10.1038/nature06932 (2008).
67. Yang, J. J., Pickett, M. D., Li, X., Ohlberg, D. A. A., Stewart, D. R. & Williams, R. S. Memristive switching mechanism for metal/oxide/metal nanodevices. *Nature Nanotechnology* **3**, 429–433. doi:10.1038/nnano.2008.160 (2008).
68. Yang, J. J., Strukov, D. B. & Stewart, D. R. Memristive devices for computing. *Nature Nanotechnology* **8**, 13–24. doi:10.1038/nnano.2012.240 (2013).
69. Valov, I., Waser, R., Jameson, J. R. & Kozicki, M. N. Electrochemical metallization memories—fundamentals, applications, prospects. *Nanotechnology* **22**, 254003. doi:10.1088/0957-4484/22/25/254003 (2011).
70. Waser, R. & Aono, M. Nanoionics-based resistive switching memories. *Nature Materials* **6**, 833–840. doi:10.1038/nmat2023 (2007).
71. Sun, W., Gao, B., Chi, M., *et al.* Understanding memristive switching via in situ characterization and device modeling. *Nature Communications* **10**, 3453. doi:10.1038/s41467-019-11411-6 (2019).
72. Sawa, A. Resistive switching in transition metal oxides. *Materials Today* **11**, 28–36. doi:10.1016/S1369-7021(08)70119-6 (2008).

-
73. Le Gallo, M., Sebastian, A., Mathis, R., *et al.* Mixed-precision in-memory computing. *Nature Electronics* **1**, 246–253. doi:10.1038/s41928-018-0054-8 (2018).
 74. Chappert, C., Fert, A. & Van Dau, F. N. The emergence of spin electronics in data storage. *Nature Materials* **6**, 813–823. doi:10.1038/nmat2024 (2007).
 75. Mizrahi, A., Hirtzlin, T., Fukushima, A., *et al.* Neural-like computing with populations of superparamagnetic basis functions. *Nature Communications* **9**, 1533. doi:10.1038/s41467-018-03963-w (2018).
 76. Torrejon, J., Riou, M., Araujo, F. A., *et al.* Neuromorphic computing with nanoscale spintronic oscillators. *Nature* **547**, 428–431. doi:10.1038/nature23011 (2017).
 77. Xiao, Z. & Huang, J. Energy-Efficient Hybrid Perovskite Memristors and Synaptic Devices. *Advanced Electronic Materials* **2**, 1600100. doi:10.1002/aelm.201600100 (2016).
 78. Bichler, O., Querlioz, D., Thorpe, S. J., Bourgoin, J.-P. & Gamrat, C. Extraction of temporally correlated features from dynamic vision sensors with spike-timing-dependent plasticity. *Neural Networks* **32**, 339–348. doi:10.1016/j.neunet.2012.02.022 (2012).
 79. Sangwan, V. K. & Hersam, M. C. Neuromorphic nanoelectronic materials. *Nature Nanotechnology* **15**, 517–528. doi:10.1038/s41565-020-0647-z (2020).
 80. Dambre, J., Verstraeten, D., Schrauwen, B. & Massar, S. Information Processing Capacity of Dynamical Systems. *Scientific Reports* **2**, 514. doi:10.1038/srep00514 (2012).

REFERENCES

81. Ohno, T., Hasegawa, T., Tsuruoka, T., Terabe, K., Gimzewski, J. K. & Aono, M. Short-term plasticity and long-term potentiation mimicked in single inorganic synapses. *Nature Materials* **10**, 591–595. doi:10.1038/nmat3054 (2011).
82. Ambrogio, S., Narayanan, P., Tsai, H., *et al.* Equivalent-accuracy accelerated neural-network training using analogue memory. *Nature* **558**, 60–67. doi:10.1038/s41586-018-0180-5 (2018).
83. Wong, H.-S. P., Raoux, S., Kim, S., *et al.* Phase Change Memory. *Proceedings of the IEEE* **98**, 2201–2227. doi:10.1109/JPROC.2010.2070050 (2010).
84. Bhatti, S., Sbiaa, R., Hirohata, A., Ohno, H., Fukami, S. & Piramanayagam, S. N. Spintronics based random access memory: a review. *Materials Today* **20**, 530–548. doi:10.1016/j.mattod.2017.07.007 (2017).
85. Ikeda, S., Miura, K., Yamamoto, H., *et al.* A perpendicular-anisotropy CoFeB–MgO magnetic tunnel junction. *Nature Materials* **9**, 721–724. doi:10.1038/nmat2804 (2010).
86. Friend, R. H., Gymer, R. W., Holmes, A. B., *et al.* Electroluminescence in conjugated polymers. *Nature* **397**, 121–128. doi:10.1038/16393 (1999).
87. Gray, F. M. *Polymer Electrolytes* (Royal Society of Chemistry, Cambridge, 1997).
88. Armand, M. Polymer solid electrolytes – an overview. *Solid State Ionics* **9–10**, 745–754. doi:10.1016/0167-2738(83)90083-8 (1983).
89. Pearson, R. G. Hard and Soft Acids and Bases. *Journal of the American Chemical Society* **85**, 3533–3539. doi:10.1021/ja00905a001 (1963).

-
90. Kim, Y., Chortos, A., Xu, W., *et al.* A bioinspired flexible organic artificial afferent nerve. *Science* **360**, 998–1003. doi:10.1126/science.aao0098 (2018).
 91. Park, H.-L., Lee, Y., Kim, N., Seo, D.-G., Go, G.-T. & Lee, T.-W. Flexible Neuromorphic Electronics for Computing, Soft Robotics, and Neuroprosthetics. *Advanced Materials* **32**, 1903558. doi:10.1002/adma.201903558 (2020).
 92. Berthier, C., Gorecki, W., Minier, M., Armand, M. B., Chabagno, J. M. & Rigaud, P. Microscopic investigation of ionic conductivity in alkali metal salts-poly(ethylene oxide) adducts. *Solid State Ionics* **11**, 91–95. doi:10.1016/0167-2738(83)90068-1 (1983).
 93. Lei, Y., Liu, Y., Xia, Y., *et al.* Memristive learning and memory functions in polyvinyl alcohol polymer memristors. *AIP Advances* **4**, 077105. doi:10.1063/1.4887010 (2014).
 94. Liu, G., Wang, C., Zhang, W., *et al.* Organic Biomimicking Memristor for Information Storage and Processing Applications. *Advanced Electronic Materials* **2**, 1500298. doi:10.1002/aelm.201500298 (2016).
 95. Goswami, S., Matula, A. J., Rath, S. P., *et al.* Robust resistive memory devices using solution-processable metal-coordinated azo aromatics. *Nature Materials* **16**, 1216–1224. doi:10.1038/nmat5009 (2017).
 96. Goswami, S., Rath, S. P., Thompson, D., *et al.* Charge disproportionate molecular redox for discrete memristive and memcapacitive switching. *Nature Nanotechnology* **15**, 380–389. doi:10.1038/s41565-020-0653-1 (2020).

REFERENCES

97. Zakhidov, A. A., Jung, B., Slinker, J. D., Abruña, H. D. & Malliaras, G. G. A light-emitting memristor. *Organic Electronics* **11**, 150–153. doi:10.1016/j.orgel.2009.09.015 (2010).
98. Pei, Q., Yu, G., Zhang, C., Yang, Y. & Heeger, A. J. Polymer Light-Emitting Electrochemical Cells. *Science* **269**, 1086–1088. doi:10.1126/science.269.5227.1086 (1995).
99. De Mello, J. C., Tessler, N., Graham, S. C. & Friend, R. H. Ionic space-charge effects in polymer light-emitting diodes. *Physical Review B* **57**, 12951–12963. doi:10.1103/PhysRevB.57.12951 (1998).
100. De Mello, J. C. Interfacial feedback dynamics in polymer light-emitting electrochemical cells. *Physical Review B* **66**, 235210. doi:10.1103/PhysRevB.66.235210 (2002).
101. Matyba, P., Maturova, K., Kemerink, M., Robinson, N. D. & Edman, L. The dynamic organic p-n junction. *Nature Materials* **8**, 672–676. doi:10.1038/nmat2478 (2009).
102. Van Reenen, S., Matyba, P., Dzwilewski, A., Janssen, R. A. J., Edman, L. & Kemerink, M. A Unifying Model for the Operation of Light-Emitting Electrochemical Cells. *Journal of the American Chemical Society* **132**, 13776–13781. doi:10.1021/ja1045555 (2010).
103. Wågberg, T., Hania, P. R., Robinson, N. D., Shin, J.-H., Matyba, P. & Edman, L. On the Limited Operational Lifetime of Light-Emitting Electrochemical Cells. *Advanced Materials* **20**, 1744–1749. doi:10.1002/adma.200702595 (2008).
104. Mardegan, L., Dreessen, C., Sessolo, M., Tordera, D. & Bolink, H. J. Stable Light-Emitting Electrochemical Cells Using Hyperbranched Polymer Electrolyte. *Advanced Functional Materials* **31**, 2104249. doi:10.1002/adfm.202104249 (2021).

-
105. Gkoupidenis, P., Schaefer, N., Garlan, B. & Malliaras, G. G. Neuromorphic Functions in PEDOT:PSS Organic Electrochemical Transistors. *Advanced Materials* **27**, 7176–7180. doi:10.1002/adma.201503674 (2015).
 106. Fang, J., Matyba, P., Robinson, N. D. & Edman, L. Identifying and alleviating electrochemical side-reactions in light-emitting electrochemical cells. *Journal of the American Chemical Society* **130**, 4562–4568. doi:10.1021/ja7113294 (2008).
 107. Scott, J. C. & Bozano, L. D. Nonvolatile Memory Elements Based on Organic Materials. *Advanced Materials* **19**, 1452–1463. doi:10.1002/adma.200602564 (2007).
 108. Yang, Y., Ouyang, J., Ma, L., Tseng, R. J.-H. & Chu, C.-W. Electrical Switching and Bistability in Organic/Polymeric Thin Films and Memory Devices. *Advanced Functional Materials* **16**, 1001–1014. doi:10.1002/adfm.200500429 (2006).
 109. Erokhin, V., Berzina, T. & Fontana, M. P. Hybrid electronic device based on polyaniline-polyethyleneoxide junction. *Journal of Applied Physics* **97**, 064501. doi:10.1063/1.1861508 (2005).
 110. Erokhin, V., Berzina, T., Camorani, P. & Fontana, M. P. Non-equilibrium electrical behaviour of polymeric electrochemical junctions. *Journal of Physics: Condensed Matter* **19**, 205111. doi:10.1088/0953-8984/19/20/205111 (2007).
 111. Smerieri, A., Berzina, T., Erokhin, V. & Fontana, M. P. Polymeric electrochemical element for adaptive networks: Pulse mode. *Journal of Applied Physics* **104**, 114513. doi:10.1063/1.3033399 (2008).
 112. Berzina, T., Smerieri, A., Bernabò, M., *et al.* Optimization of an organic memristor as an adaptive memory element. *Journal of Applied Physics* **105**, 124515. doi:10.1063/1.3153944 (2009).

REFERENCES

113. Gkoupidenis, P., Koutsouras, D. A. & Malliaras, G. G. Neuromorphic device architectures with global connectivity through electrolyte gating. *Nature Communications* **8**, 15448. doi:10.1038/ncomms15448 (2017).
114. Xu, W., Min, S.-Y., Hwang, H. & Lee, T.-W. Organic core-sheath nanowire artificial synapses with femtojoule energy consumption. *Science Advances* **2**, e1501326. doi:10.1126/sciadv.1501326 (2016).
115. Seo, D.-G., Lee, Y., Go, G.-T., *et al.* Versatile neuromorphic electronics by modulating synaptic decay of single organic synaptic transistor: From artificial neural networks to neuro-prosthetics. *Nano Energy* **65**, 104035. doi:10.1016/j.nanoen.2019.104035 (2019).
116. Heremans, P., Gelinck, G. H., Müller, R., Baeg, K.-J., Kim, D.-Y. & Noh, Y.-Y. Polymer and Organic Nonvolatile Memory Devices. *Chemistry of Materials* **23**, 341–358. doi:10.1021/cm102006v (2011).
117. Yang, M., Zhao, X., Tang, Q., *et al.* Stretchable and conformable synapse memristors for wearable and implantable electronics. *Nanoscale* **10**, 18135–18144. doi:10.1039/C8NR05336G (2018).
118. Kim, M.-K. & Lee, J.-S. Ultralow Power Consumption Flexible Biomemristors. *ACS Applied Materials & Interfaces* **10**, 10280–10286. doi:10.1021/acsami.8b01781 (2018).
119. Shih, C.-C., Lee, W.-Y. & Chen, W.-C. Nanostructured materials for non-volatile organic transistor memory applications. *Materials Horizons* **3**, 294–308. doi:10.1039/C6MH00049E (2016).
120. Marković, D., Mizrahi, A., Querlioz, D. & Grollier, J. Physics for neuromorphic computing. *Nature Reviews Physics* **2**, 499–510. doi:10.1038/s42254-020-0208-2 (2020).

-
121. Liang, X., Tang, J., Zhong, Y., Gao, B., Qian, H. & Wu, H. Physical reservoir computing with emerging electronics. *Nature Electronics* **7**, 193–206. doi:10.1038/s41928-024-01133-z (2024).
 122. Gao, C., Liu, D., Xu, C., *et al.* Toward grouped-reservoir computing: organic neuromorphic vertical transistor with distributed reservoir states for efficient recognition and prediction. *Nature Communications* **15**, 740. doi:10.1038/s41467-024-44942-8 (2024).
 123. Han, C., Kim, M., Park, S. J., *et al.* Reconfigurable Organic Electrochemical Transistors with High Dynamic Ranges for Fully Integrated Physical Reservoir Computing. *Advanced Functional Materials* **35**, 2423814. doi:10.1002/adfm.202423814 (2025).
 124. Lee, S. H., Kim, S., Han, J., *et al.* Electrically tunable physical reservoir computing on two-terminal devices. *npj Unconventional Computing* **2**, 27. doi:10.1038/s44335-025-00043-3 (2025).
 125. Fenton, D. E., Parker, J. M. & Wright, P. V. Complexes of alkali metal ions with poly(ethylene oxide). *Polymer* **14**, 589. doi:10.1016/0032-3861(73)90146-8 (1973).
 126. Wright, P. V. Electrical conductivity in ionic complexes of poly(ethylene oxide). *British Polymer Journal* **7**, 319–327. doi:10.1002/pi.4980070505 (1975).
 127. Lightfoot, P., Mehta, M. A. & Bruce, P. G. Crystal Structure of the Polymer Electrolyte Poly(ethylene oxide)₃:LiCF₃SO₃. *Science* **262**, 883–885. doi:10.1126/science.262.5135.883 (1993).
 128. Froehling, P. Development of DSM's Hybrane hyperbranched polyesteramides. *Journal of Polymer Science Part A: Polymer Chemistry* **42**, 3110–3115. doi:10.1002/pola.20113 (2004).

REFERENCES

129. Bruce, P. G. & Vincent, C. A. Polymer electrolytes. *Journal of the Chemical Society, Faraday Transactions* **89**, 3187–3203. doi:10.1039/ft9938903187 (1993).
130. Kreuer, K.-D., Rabenau, A. & Weppner, W. Vehicle Mechanism, A New Model for the Interpretation of the Conductivity of Fast Proton Conductors. *Angewandte Chemie International Edition in English* **21**, 208–209. doi:10.1002/anie.198202082 (1982).
131. Agmon, N. The Grotthuss mechanism. *Chemical Physics Letters* **244**, 456–462. doi:10.1016/0009-2614(95)00905-J (1995).
132. Druger, S. D., Nitzan, A. & Ratner, M. A. Dynamic bond percolation theory: A microscopic model for diffusion in dynamically disordered systems. I. Definition and one-dimensional case. *The Journal of Chemical Physics* **79**, 3133–3142. doi:10.1063/1.446144 (1983).
133. Ratner, M. A. & Shriver, D. F. Ion transport in solvent-free polymers. *Chemical Reviews* **88**, 109–124. doi:10.1021/cr00083a006 (1988).
134. Angell, C. A. Relaxation in liquids, polymers and plastic crystals — strong/fragile patterns and problems. *Journal of Non-Crystalline Solids* **131–133**, 13–31. doi:10.1016/0022-3093(91)90266-9 (1991).
135. Gadjourova, Z., Andreev, Y. G., Tunstall, D. P. & Bruce, P. G. Ionic conductivity in crystalline polymer electrolytes. *Nature* **412**, 520–523. doi:10.1038/35087538 (2001).
136. Voit, B. I. & Lederer, A. Hyperbranched and Highly Branched Polymer Architectures—Synthetic Strategies and Major Characterization Aspects. *Chemical Reviews* **109**, 5924–5973. doi:10.1021/cr900068q (2009).

-
137. Tang, S. & Edman, L. Light-Emitting Electrochemical Cells: A Review on Recent Progress. *Topics in Current Chemistry* **374**, 40. doi:10.1007/s41061-016-0040-4 (2016).
138. Smith, D. L. Steady state model for polymer light-emitting electrochemical cells. *Journal of Applied Physics* **81**, 2869–2880. doi:10.1063/1.363966 (1997).
139. Sandström, A., Dam, H. F., Krebs, F. C. & Edman, L. Ambient fabrication of flexible and large-area organic light-emitting devices using slot-die coating. *Nature Communications* **3**, 1002. doi:10.1038/ncomms2002 (2012).
140. Hohertz, D. & Gao, J. How Electrode Work Function Affects Doping and Electroluminescence of Polymer Light-Emitting Electrochemical Cells. *Advanced Materials* **20**, 3298–3302. doi:10.1002/adma.200800068 (2008).
141. AlTal, F. & Gao, J. Long-term testing of polymer light-emitting electrochemical cells: Reversible doping and black spots. *Organic Electronics* **18**, 1–7. doi:10.1016/j.orgel.2014.12.028 (2015).
142. Tordera, D., Frey, J., Vonlanthen, D., *et al.* Low Current Density Driving Leads to Efficient, Bright and Stable Green Electroluminescence. *Advanced Energy Materials* **3**, 1338–1343. doi:10.1002/aenm.201300284 (2013).
143. Munar, A., Sandström, A., Tang, S. & Edman, L. Shedding Light on the Operation of Polymer Light-Emitting Electrochemical Cells Using Impedance Spectroscopy. *Advanced Functional Materials* **22**, 1511–1517. doi:10.1002/adfm.201102687 (2012).
144. Shannon, R. D. Revised effective ionic radii and systematic studies of interatomic distances in halides and chalcogenides. *Acta Crystallographica Section A* **32**, 751–767. doi:10.1107/S0567739476001551 (1976).

REFERENCES

145. Zhang, M., Höfle, S., Czolk, J., Mertens, A. & Colsmann, A. All-solution processed transparent organic light emitting diodes. *Nanoscale* **7**, 20009–20014. doi:10.1039/C5NR05820A (2015).
146. Kim, H., Gilmore, C. M., Piqué, A., *et al.* Electrical, optical, and structural properties of indium–tin–oxide thin films for organic light-emitting devices. *Journal of Applied Physics* **86**, 6451–6461. doi:10.1063/1.371708 (1999).
147. Ramsdale, C. M., Barker, J. A., Arias, A. C., MacKenzie, J. D., Friend, R. H. & Greenham, N. C. The origin of the open-circuit voltage in polyfluorene-based photovoltaic devices. *Journal of Applied Physics* **92**, 4266–4270. doi:10.1063/1.1506385 (2002).
148. Frech, R., Chintapalli, S., Bruce, P. G. & Vincent, C. A. Crystalline and Amorphous Phases in the Poly(ethylene oxide)–LiCF₃SO₃ System. *Macromolecules* **32**, 808–813. doi:10.1021/ma9812682 (1999).
149. Harikesh, P. C., Tu, D. & Fabiano, S. Organic electrochemical neurons for neuromorphic perception. *Nature Electronics* **7**, 525–536. doi:10.1038/s41928-024-01200-5 (2024).
150. Krauhausen, I., Griggs, S., McCulloch, I., den Toonder, J. M. J., Gkoupidenis, P., van de Burgt, Y., *et al.* Bio-inspired multimodal learning with organic neuromorphic electronics for behavioral conditioning in robotics. *Nature Communications* **15**, 4765. doi:10.1038/s41467-024-48881-2 (2024).
151. Xia, Y., Zhang, C., Xu, Z., *et al.* Organic iontronic memristors for artificial synapses and bionic neuromorphic computing. *Nanoscale* **16**, 1471–1489. doi:10.1039/D3NR06057H (2024).
152. Sturman, B., Podivilov, E. & Gorkunov, M. Origin of Stretched Exponential Relaxation for Hopping-Transport Models. *Physical*

- Review Letters* **91**, 176602. doi:10.1103/PhysRevLett.91.176602 (2003).
153. Vallée, A., Besner, S. & Prud'homme, J. Comparative study of poly(ethylene oxide) electrolytes made with LiN(CF₃SO₂)₂, LiCF₃SO₃ and LiClO₄: Thermal properties and conductivity behaviour. *Electrochimica Acta* **37**, 1579–1583. doi:10.1016/0013-4686(92)80115-3 (1992).
154. Appeltant, L., Soriano, M. C., Van der Sande, G., *et al.* Information processing using a single dynamical node as complex system. *Nature Communications* **2**, 468. doi:10.1038/ncomms1476 (2011).
155. Jaeger, H. *Short term memory in echo state networks* tech. rep. GMD Report 152 (German National Research Center for Information Technology (GMD), Bonn, Germany, 2002).
156. Atiya, A. F. & Parlos, A. G. New results on recurrent network training: unifying the algorithms and accelerating convergence. *IEEE Transactions on Neural Networks* **11**, 697–709. doi:10.1109/72.846741 (2000).
157. Picard, R. W. *Affective Computing* doi:10.7551/mitpress/1140.001.0001 (The MIT Press, Cambridge, MA, 1997).
158. Boucsein, W. *Electrodermal Activity* 2nd ed. ISBN: 978-1-4614-1126-0. doi:10.1007/978-1-4614-1126-0 (Springer, New York, NY, 2012).
159. Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology. Heart rate variability: Standards of measurement, physiological interpretation, and clinical use. *Circulation* **93**, 1043–1065. doi:10.1161/01.CIR.93.5.1043 (1996).

REFERENCES

160. Chourpiliadis, C. & Bhardwaj, A. *Physiology, Respiratory Rate* In: StatPearls [Internet]. Last updated September 12, 2022. <https://www.ncbi.nlm.nih.gov/books/NBK537306/> (2026).
161. Schmidt, P., Reiss, A., Duerichen, R., Marberger, C. & Van Laerhoven, K. *Introducing WESAD, a Multimodal Dataset for Wearable Stress and Affect Detection* in *Proceedings of the 20th ACM International Conference on Multimodal Interaction (ICMI '18)* (ACM, 2018), 400–408. doi:10.1145/3242969.3242985.
162. Sharma, K., Castellini, C., van den Broek, E. L., Albu-Schäffer, A. & Schwenker, F. A Dataset of Continuous Affect Annotations and Physiological Signals for Emotion Analysis. *Scientific Data* **6**, 196. doi:10.1038/s41597-019-0209-0 (2019).
163. Lin, L. I.-K. A Concordance Correlation Coefficient to Evaluate Reproducibility. *Biometrics* **45**, 255–268. doi:10.2307/2532051 (1989).
164. Birjandtalab, J., Cogan, D., Pouyan, M. B. & Nourani, M. A *Non-EEG Biosignals Dataset for Assessment and Visualization of Neurological Status* in *2016 IEEE International Workshop on Signal Processing Systems (SiPS)* (IEEE, 2016), 110–114. doi:10.1109/SiPS.2016.27.
165. Williams, M. L., Landel, R. F. & Ferry, J. D. The Temperature Dependence of Relaxation Mechanisms in Amorphous Polymers and Other Glass-forming Liquids. *Journal of the American Chemical Society* **77**, 3701–3707. doi:10.1021/ja01619a008 (1955).
166. Hung, L. S., Tang, C. W. & Mason, M. G. Enhanced electron injection in organic electroluminescence devices using an Al/LiF electrode. *Applied Physics Letters* **70**, 152–154. doi:10.1063/1.118344 (1997).

167. Lewis, J. S. & Weaver, M. S. Thin-Film Permeation-Barrier Technology for Flexible Organic Light-Emitting Devices. *IEEE Journal of Selected Topics in Quantum Electronics* **10**, 45–57. doi:10.1109/JSTQE.2004.824072 (2004).